

CONTEMPORARY MATHEMATICS

560

Topology and Geometry in Dimension Three Triangulations, Invariants, and Geometric Structures

Conference in Honor of William Jaco's 70th Birthday
June 4–6, 2010

Oklahoma State University, Stillwater, Oklahoma

Weiping Li, Loretta Bartolini, Jesse Johnson,
Feng Luo, Robert Myers, J. Hyam Rubinstein
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Preface

For many young topologists, their introduction to three-manifolds is marked by the blue volume *Lectures on Three-Manifold Topology*. This CBMS publication, which captures a series of lectures by William ‘Bus’ Jaco in the Fall of 1977, has become a classic reference for students and researchers. Progressing into the field of three-manifolds, one discovers the JSJ-decomposition; this result of Bus Jaco and Peter Shalen (discovered independently by Klaus Johannson) on the theory of characteristic varieties underlies the geometrization conjecture of Thurston. Bus’ many research achievements also include crucial contributions to the development of normal surface theory, triangulations and algorithms in 3-dimensional geometry and topology.

Bus grew up in Grafton, West Virginia, planning at Fairmont College to become a school teacher. However, his outstanding talents in mathematics and drive for learning led to graduate school. A student of R H Bing and D. R. McMillan, Bus received his Ph.D. from the University of Wisconsin in 1968. Graduation was followed by a postdoctoral position at the University of Michigan, before taking a permanent position at Rice University, where he achieved rapid promotion to Full Professor. A highly active scholar, Bus held a variety of visiting positions thereafter, including terms at the Institute for Advanced Study, Columbia University, University of Melbourne, Mathematical Sciences Research Institute, American Institute of Mathematics and University of Michigan.

Moving to Oklahoma State University in 1982, as Head of the Department of Mathematics, marked the start of a highly influential term. Bus’ far-sighted leadership and tireless work ethic saw a boom in research and scholarly activities in the Department, accompanied by his own ongoing research achievements. His strong leadership and commitment to the profession were to rise to national prominence, with a term as Executive Director of the American Mathematical Society from 1988 to 1995. This outstanding contribution was followed with honors both scientific and professional: elected as Fellow of the American Association for Advancement of Science, Regents Professor at Oklahoma State University and Trustee of the American Mathematical Society. Along with his accomplishments in mathematics, Bus’ contributions to the Department, University, Profession and American Sciences are both remarkable and continuing.

The Jacofest conference, held in Stillwater, Oklahoma, June 4-6 2010, brought together over 80 topologists and geometers from around the globe. There were 15 plenary talks attended by a wide range of participants: from long-standing collaborators to the latest generation of graduate students. This group produced an atmosphere rich in ideas and energy, most fitting to celebrate a career with such

qualities in abundance. We hope this volume captures the mathematical endeavours and warm recognition of Bus at the Conference.

We would like to acknowledge the support received from the National Science Foundation under Grant No. 0900229 and Grant No. 1005383, the American Institute of Mathematics, the College of Arts and Sciences at Oklahoma State University and the Department of Mathematics at Oklahoma State University. We thank them for their generous financial and administrative support. We also thank D. Alspach, B. Conrey, M. Denzler, S. Downing, M. Gordon and A. M. McFarlin for their help at various stages of the Conference, and C. M. Thivierge for her assistance in preparing this volume.

The Editors
July 2011

Jacofest Talks

Ian Agol <i>University of California at Berkeley</i> “Ideal triangulations of bundles”	Joseph Maher <i>CUNY – College of Staten Island</i> “Random Heegaard splittings”
Francis Bonahon <i>University of Southern California</i> “Kauffman brackets, character varieties, and triangulations of surfaces”	Jessica Purcell <i>Brigham Young University</i> “State surfaces, polyhedra, and guts of knots”
Danny Calegari <i>California Institute of Technology</i> “Faces of the scl norm ball”	J. Hyam Rubinstein <i>University of Melbourne</i> “Normal 3-manifolds in triangulated 4-manifolds”
Nathan Dunfield <i>University of Illinois at Urbana-Champaign</i> “Twisted Alexander polynomials, hyperbolic geometry, and knot genus”	Saul Schleimer <i>University of Warwick</i> “On train track splitting sequences”
David Futer <i>Temple University</i> “Angled triangulations and Dehn surgery”	Peter Shalen <i>University of Illinois at Chicago</i> “Generic Margulis numbers”
Stavros Garoufalidis <i>Georgia Institute of Technology</i> “The Slope Conjecture”	Stephan Tillmann <i>University of Queensland</i> “Straightening, spinning and the recognition of closed hyperbolic 3-manifolds”
Cameron Gordon <i>University of Texas at Austin</i> “Seifert fibered Dehn filling”	
Elisenda Grigsby <i>Boston College</i> “On sutured Khovanov homology and sutured Floer homology”	
Feng Luo <i>Rutgers University</i> “Minimally triangulated 3-manifolds with special normal surfaces”	

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Ideal triangulations of pseudo-Anosov mapping tori

Ian Agol

Dedicated to Bus Jaco on the occasion of his 70th birthday

ABSTRACT. We show how to construct an ideal triangulation of a mapping torus of a pseudo-Anosov map punctured along the singular fibers. This gives rise to a new conjugacy invariant of mapping classes, and a new proof of a theorem of Farb-Leininger-Margalit. The approach in this paper is based on ideas of Hamenstädt.

1. Introduction

Recently, Farb, Leininger and Margalit [2] proved that the mapping tori of small dilatation mapping classes of closed surfaces are obtained by Dehn filling on finitely many cusped hyperbolic 3-manifolds. Recent results of Hamenstädt [8] suggested another approach to this theorem using splitting sequences of train tracks. In discussions with Hamenstädt, she suggested that her complex of train track splitting sequences should give a new solution to the conjugacy problem for pseudo-Anosov mapping classes, somewhat similar to work of Mosher on “circular expansion complexes” [16, 17, 18]. In this paper we give an exposition of these results from the perspective of measured train track splitting sequences. Instead of a complex of splittings, we obtain a taut ideal triangulation associated to a mapping class. The fact that the reverse of splitting sequences (folding sequences) give rise to Perron-Frobenius maps of the weight spaces of train tracks enables us to give a new proof of the main result of [2].

What Hamenstädt proved is that if one is given a minimal lamination $\mathcal{L}$ on a surface, there is a complex of splitting sequences of train tracks carrying $\mathcal{L}$ which forms a CAT(0) cube complex in a natural way, with vertices corresponding to train tracks carrying the lamination $\mathcal{L}$, directed edges corresponding to $\mathcal{L}$ -splits of the train tracks, and cubes corresponding to commuting collections of $\mathcal{L}$ -splits [8, 7, Section 5]. This is analogous to the complex constructed by Mosher [18], however we remark that the definition of “splitting sequence” in that manuscript is distinct from the usage in this paper (Mosher allows shifts/slides as well). If one takes the stable lamination associated to a pseudo-Anosov map φ , then one may find a bi-infinite cube complex of train tracks which is invariant under φ , and this gives rise to an invariant of the conjugacy class of φ . Instead of considering all sequences

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of splittings in this paper, we consider “maximal splitting” sequences, where the train tracks have a measure, and the splittings occur at the branches of the train track of maximal weight. We prove that these sequences are eventually periodic for a pseudo-Anosov stable lamination in Section 3 (this is somewhat analogous to the approach of [19, 20] and [18, Lemma 10.2.6]). These sequences give something like a continued fraction expansion for pA mapping classes, analogous to the case of Anosov maps of the 2-torus (Mosher observed the analogy between continued fractions and his circular expansion complexes [18]). This approach allows us to reprove the main result of [2] in Section 6. We also obtain a layered ideal triangulation of the mapping torus punctured at the singular fibers, from Whitehead moves on the triangulations dual to the train tracks in the splitting sequence in Section 4. The natural structure of these triangulations is a taut ideal triangulation, introduced by Lackenby [14].

Acknowledgements: We thank Ursula Hamenstädt, Marc Lackenby, Chris Leininger, and Saul Schleimer for helpful conversations. We thank Matthias Görner, Yi Liu, and Lee Mosher for making comments on an earlier draft.

2. Definitions

We review some background definitions and establish some notation in this section.

Let $\Sigma = \Sigma_{g,n}$ be an orientable surface of genus g with n punctures, and let $\varphi : \Sigma \rightarrow \Sigma$ be a homeomorphism. By the Nielsen-Thurston classification of mapping classes, φ is conjugate to either an element of finite order, a reducible mapping class, or to a pseudo-Anosov map [25, 23]. In the third case, there are the stable and unstable measured laminations $\mathcal{L}^s$ and $\mathcal{L}^u \in \mathcal{ML}(\Sigma)$ associated to φ such that for any curve $c \subset \Sigma$, $[\varphi^n(c)] \rightarrow [\mathcal{L}^s] \in \mathcal{PL}(\Sigma)$, and $[\varphi^{-n}(c)] \rightarrow [\mathcal{L}^u] \in \mathcal{PL}(\Sigma)$ (see [24, Chapter 8] for the notation). Moreover, $\mathcal{L}^s$ and $\mathcal{L}^u$ meet every essential closed curve in Σ which is not isotopic into a neighborhood of a puncture. Further, there exists a dilatation $\lambda(\varphi) \in (1, \infty)$, $\varphi(\mathcal{L}^s) = \lambda(\varphi)\mathcal{L}^s$ and $\varphi(\mathcal{L}^u) = \lambda(\varphi)^{-1}\mathcal{L}^u$, up to isotopy. These laminations are only unique up to scaling and isotopy. If we choose a complete finite area hyperbolic metric on Σ , then we may isotope the measured laminations $\mathcal{L}^s$ and $\mathcal{L}^u$ to have totally geodesic leaves. The closure of each complementary region will be isometric to an ideal polygon with at least three sides, or to a punctured disk with at least one puncture on the boundary. If we puncture Σ at a point in each ideal polygon complementary region, we obtain a surface Σ° , and we have a well-defined (up to isotopy) restriction map $\varphi^\circ = \varphi|_{\Sigma^\circ} : \Sigma^\circ \rightarrow \Sigma^\circ$. Moreover, $\lambda(\varphi^\circ) = \lambda(\varphi)$, and we may assume $[\varphi^\circ(\mathcal{L}^{u,s})] = [\mathcal{L}^{u,s}] \in \mathcal{PL}(\Sigma^\circ)$.

To encode a lamination combinatorially, we use train tracks (after Thurston [24, Ch. 8]). A train track $\tau \subset \Sigma$ is a 1-complex with trivalent vertices which is locally modeled on a switch (see Figure 1(a)) and satisfying some extra conditions. Edges of the train track are called *branches*, and vertices are called *switches*. Each branch $e \subset \tau$ is C^1 embedded, and at each switch of τ , there is a well-defined tangent space to the branches coming into the switch. A *half-branch* is an end of the interior of a branch. Each switch is in the closure of three half-branches. A *large half-branch* is the end of a half-branch meeting a switch on the the side of the tangent space to the switch opposite of the other two half-branches incident with the switch, and a *small half-branch* is a half-branch meeting the switch on the same side of the tangent space to the switch as another branch (in Figure 1(b), the

half-branch labelled a is large, whereas the half-branches labeled b and c are small). For each component R of $\Sigma - \tau$, the boundary ∂R is a piecewise smooth curve. A non-smooth point of ∂R is a *cuspl*. For each component R of $\Sigma - \tau$, the double of R along ∂R with the cusps removed must have negative Euler characteristic. We will follow the notation and conventions of Penner-Harer [22]. A *large branch* is a branch with both ends being large half-branches, and a *small branch* has both ends small half-branches. Otherwise, a branch is *mixed*. When all of the complementary regions of the train track τ are punctured disks, then it is convenient to label the branches of the track by the isotopy class of the arc type of an ideal edge dual to the branch. This gives an unambiguous way of labeling branches in differing train tracks on the same surface.

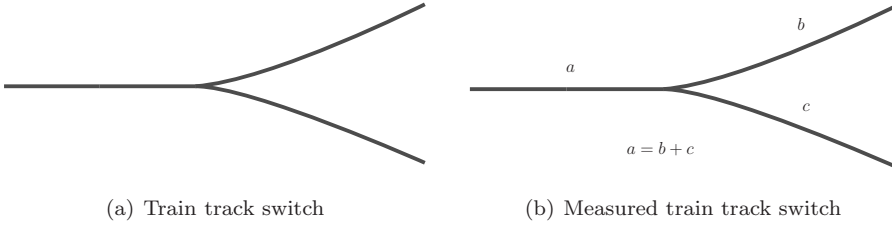
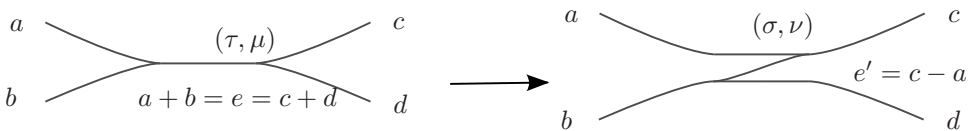


FIGURE 1. Train track switches

A *measured train track* is a train track τ together with a transverse measure μ which is a function assigning a weight to each branch of τ . We will denote a measured train track as a pair (τ, μ) . At each switch of τ with incoming branches b, c and outgoing branch a , the weights of the adjacent branches must satisfy $\mu(a) = \mu(b) + \mu(c)$ (Figure 1(b)). In some pictures, we will abuse notation and say $a = b + c$ when we really mean $\mu(a) = \mu(b) + \mu(c)$. This is especially convenient when the measured train track changes by a move, but most of the labeled branches do not. The *weight space* $W(\tau)$ is the convex space of all positive measures on τ . The lamination $\mathcal{L}$ is *suited to* the train track τ if there is a differentiable map $f : \Sigma \rightarrow \Sigma$ homotopic to the identity such that $f(\mathcal{L}) = \tau$, and f is non-singular on the tangent spaces to the leaves of $\mathcal{L}$. If $\mathcal{L}$ is a measured lamination with transverse measure ν (giving a measure on arcs transverse to $\mathcal{L}$, up to transverse isotopy), and $\mathcal{L}$ is suited to τ , then $\mathcal{L}$ induces a measure μ on τ . For a branch $e \subset \tau$, take a transverse arc c to e , then $f^{-1}(c)$ is an arc transverse to $\mathcal{L}$, and let $\mu(e) = \nu(f^{-1}(c))$.

A *splitting* of a measured train track (τ, μ) is a move at a large branch which splits it according to the weights of the neighboring branches (see Figure 2). We use the notation $(\tau, \mu) \rightarrow_e (\sigma, \nu)$.

FIGURE 2. Splitting a large branch when $a < c$.

If the measured lamination $\mathcal{L}$ is suited to (τ, μ) , then $\mathcal{L}$ will also be suited to (σ, ν) when $(\tau, \mu) \rightarrow_e (\sigma, \nu)$. If the weights $\mu(a) = \mu(c)$, then the branch e' will

not appear in the train track σ , but we do not consider this to be a split in this paper. Conversely, any measured train track (τ, μ) gives rise to a unique measured lamination $\mathcal{L}$ suited to (τ, μ) .

Two other moves on train tracks do not depend on the measure. A *shift* changes the track at a mixed branch (Figure 3) (these are also called *slides* [18]). A *fold* changes the track at a small branch (Figure 4). The shift is self-inverse, whereas the fold is inverse to a split. However, the split depends on the measure, whereas the fold does not, which is why it is important to make a distinction.

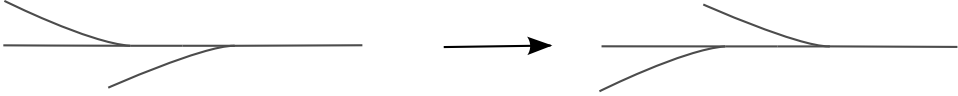


FIGURE 3. Shifting a mixed branch.

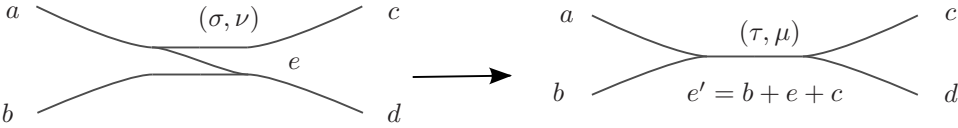


FIGURE 4. Folding a small branch.

Two measured train tracks $(\tau, \mu), (\tau', \mu')$ are *equivalent* if there is a sequence of train tracks obtained by splits, shifts, folds and isotopies which interpolates between the two.

A *maximal splitting* of (τ, μ) is a splitting along all of the large branches of τ with maximal μ weight. We denote $(\tau_0, \mu_0) \rightarrow (\tau_1, \mu_1)$ if (τ_1, μ_1) is obtained from (τ_0, μ_0) by a single maximal splitting. If we have n maximal splittings $(\tau_0, \mu_0) \rightarrow (\tau_1, \mu_1) \cdots \rightarrow (\tau_n, \mu_n)$, then we denote $(\tau_0, \mu_0) \rightarrow^n (\tau_n, \mu_n)$, or if we don't want to specify n , we will use $\rightarrow^*$.

A *trainpath* on a train track τ is a C^1 -immersion $\rho : [m, n] \rightarrow \tau \subset \Sigma$ which maps each interval $[k, k+1]$ ($m \leq k \leq n-1$) onto a branch of τ . The integer $n-m$ is then called the *length* of ρ . Each complementary region of τ is bounded by a finite number of trainpaths which either are simple closed curves or terminate at the cusps of the region. A lamination is *minimal* if every half-leaf is dense.

LEMMA 2.1. *Let $\mathcal{L}$ be a minimal lamination, and let $\mathcal{L}$ be suited to (τ, μ) . Let $e \subset \tau$ be a branch of τ , and let $(\tau, \mu) \rightarrow (\tau_1, \mu_1) \rightarrow (\tau_2, \mu_2) \rightarrow \cdots \rightarrow (\tau_n, \mu_n) \rightarrow \cdots$ be an infinite sequence of maximal splittings of (τ, μ) . Then there exists n such that $(\tau_n, \mu_n) \rightarrow (\tau_{n+1}, \mu_{n+1})$ splits the branch e (so that $\mu_n(e)$ is maximal weight for τ_n).*

PROOF. We may assume that we have a map $f : \Sigma \rightarrow \Sigma$ such that $f(\mathcal{L}) = \tau$ since $\mathcal{L}$ is suited to τ . For any half-leaf $l \subset \mathcal{L}$, since $\mathcal{L}$ is minimal, $f(l)$ must eventually cross the branch e . For each cusp of τ , there are two half-leaves of $\mathcal{L}$ which are parallel for all time (since $\mathcal{L}$ is suited to τ) whose start is adjacent to the cusp (these leaves are the ideal arcs corresponding to the cusp in the complementary ideal polygon region of $\mathcal{L}$). For each cusp c of τ , let $\rho_c : [n, m] \rightarrow \tau$ be a trainpath

which is parallel to the path $f(l)$ emanating from the cusp c , such that $\rho_c([m-1, m]) = e$. Each time we split $\tau_k \rightarrow \tau_{k+1}$ at a branch adjacent to a cusp c , the trainpath ρ_c shrinks to $\rho_c : [n-1, m] \rightarrow \tau$. Thus, at each stage of splitting, the total length of such cusp paths decreases by 2 (since each splitting branch is adjacent to two cusps), and thus we see that eventually we must split the branch e . $\square$

3. Splitting sequences

We state some basic results about measured train tracks and measured laminations.

THEOREM 3.1. [24, Prop. 8.9.2], [19, Theorem 4.1] *If $\mathcal{L}$ is a measured lamination, then $\mathcal{L}$ is suited to a measured train track (τ, μ) .*

THEOREM 3.2. [22, Theorem 2.8.5] *Measured train tracks (τ_i, μ_i) give rise to the same measured lamination $\mathcal{L}$ (up to isotopy) if and only if they are equivalent.*

Together, these theorems imply that equivalence classes of measured train tracks are in one-to-one correspondence to measured laminations. Equivalence classes of measured train tracks are generated by splitting, shifting, and folding. The next theorem implies that shifting is not needed.

THEOREM 3.3. [22, Theorem 2.3.1] *If (τ_i, μ_i) are equivalent positively measured train tracks, then there is a train track (τ, μ) such that (τ_i, μ_i) splits (and isotopes) to (τ, μ) for $i = 1, 2$.*

This theorem is proven by demonstrating that two train tracks which are equivalent by a split, shift, or fold, have common splittings (having a common splitting is not hard to show is an equivalence relation using the technique of proof of Corollary 3.4). For splits and folds this is clear. For two train tracks related by a shift, one performs the same sequence of splits of maximal branches to a sequence of two train tracks related by shifts to see that once the middle incoming half-branch b becomes large, the two resulting train tracks are the same (see Figure 5).

COROLLARY 3.4. *Suppose $\mathcal{L}$ is suited to (τ, μ) , (τ', μ') . Then there exists (τ'', μ'') such that $(\tau, \mu) \rightarrow^* (\tau'', \mu'')$ and $(\tau', \mu') \rightarrow^* (\tau'', \mu'')$.*

PROOF. The property of having a common maximal split is transitive, so by Theorem 3.3, we need only assume that $(\tau, \mu) \rightarrow_e (\tau', \mu')$ for some large branch $e \subset \tau$. If e is the sole branch of maximal weight of (τ, μ) , then we have $(\tau, \mu) \rightarrow (\tau', \mu') = (\tau'', \mu'')$ (a maximal split). If there are other branches of (τ, μ) of maximal weight $= \mu(e)$, then we have $(\tau, \mu) \rightarrow (\tau'', \mu'')$ and $(\tau', \mu') \rightarrow (\tau'', \mu'')$, where this maximal splitting is along the branches of τ' which have the same μ' weight as $\mu(e)$.

Otherwise, assume that $\mu(e)$ is not a maximal weight of μ . Let $(\tau, \mu) = (\tau_1, \mu_1) \rightarrow (\tau_2, \mu_2) \rightarrow \dots$ be the maximal splitting sequence. By Lemma 2.1, there exists n such that $\mu_n(e)$ is a maximal weight of μ_n .

We claim that $(\tau_1, \mu_1) \rightarrow^n (\tau_{n+1}, \mu_{n+1})$ and $(\tau', \mu') \rightarrow^{n-1} (\tau_{n+1}, \mu_{n+1})$. We'll assume that $(\tau'_i, \mu'_i) \rightarrow (\tau'_{i+1}, \mu'_{i+1})$, for $i = 2, \dots, n$, where $\tau'_2 = \tau'$. By induction, we prove that $(\tau_i, \mu_i) \rightarrow_e (\tau'_{i+1}, \mu'_{i+1})$ for $i < n$. Once e becomes a maximal weight branch of (τ_n, μ_n) , there may be other maximal branches of (τ_n, μ_n) of the same weight as $\mu_n(e)$, but we will have $(\tau_n, \mu_n) \rightarrow (\tau_{n+1}, \mu_{n+1}) = (\tau'_{n+1}, \mu'_{n+1})$.

$$\begin{array}{ccccccc} (\tau_1, \mu_1) & \rightarrow & (\tau_2, \mu_2) & \rightarrow \cdots \rightarrow & (\tau_n, \mu_n) \\ \downarrow_e & & \downarrow_e & & \downarrow_e \\ (\tau'_2, \mu'_2) & \rightarrow & (\tau'_3, \mu'_3) & \rightarrow \cdots \rightarrow & (\tau_{n+1}, \mu_{n+1}) \end{array}$$

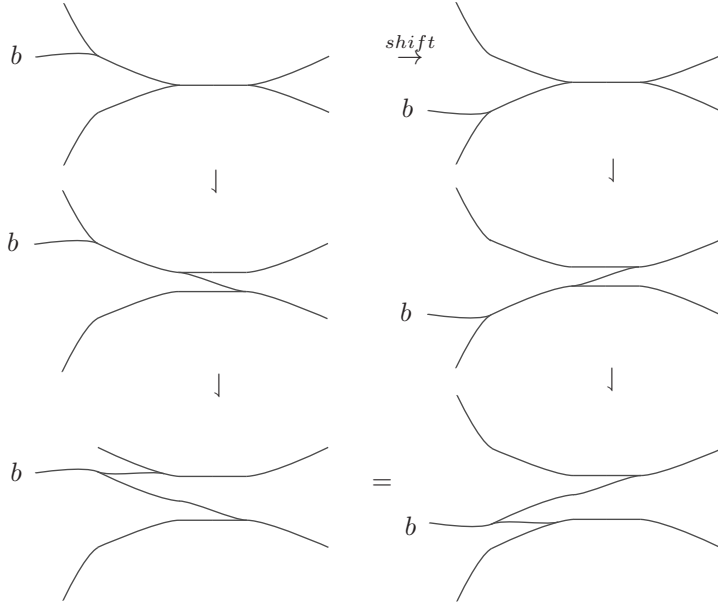
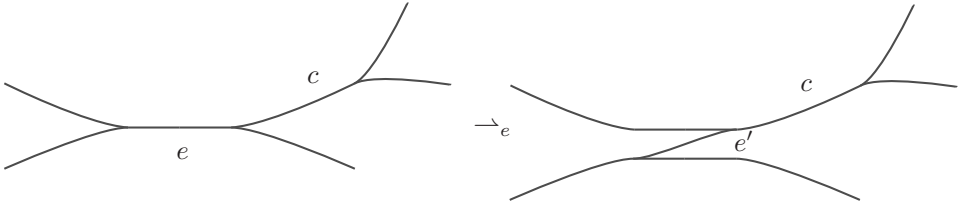


FIGURE 5. The effect of shifting then splitting.

Let e' be the new branch of τ'_2 . To see the induction, we prove that if $\mu_i(e)$ is

FIGURE 6. Splitting a large branch to get a new large branch $c < e$.

not maximal weight, then τ_i and τ'_{i+1} agree outside of a neighborhood of e and e' consisting of the branches incident with e . The point is that after splitting e , the only new large branches c that may appear must be incident with e' (see Figure 6). However, the weight of these branches is $\mu_i(c) < \mu_i(e)$, and therefore they will not be maximal weight in $(\tau'_{i+1}, \mu'_{i+1})$ if e is not maximal weight in (τ_i, μ_i) . Thus, the maximal splittings of τ_i, τ'_{i+1} will occur at the same branches, which will be disjoint from the branches adjacent to e and e' , respectively. So we will have $\tau_{i+1} \rightarrow_e \tau'_{i+2}$. $\square$

Now, we observe that if $\mathcal{L}$ is suited to (τ, μ) , then it will also be suited to (τ'', μ'') . The argument in Corollary 3.4 can be used to show that any sequence of splits of a measured train track may be arranged by a sequence of commuting splits to be in maximal order.

THEOREM 3.5. *If $\varphi : \Sigma \rightarrow \Sigma$ is a pA map, with stable lamination $\mathcal{L}^s$, and (τ, μ) is suited to $\mathcal{L}^s$, then there exists n, m such that*

$$(\tau, \mu) \rightharpoonup^n (\tau_n, \mu_n) \rightharpoonup^m (\tau_{n+m}, \mu_{n+m}),$$

and $\tau_{n+m} = \varphi(\tau_n)$ and $\mu_{n+m} = \lambda(\varphi)^{-1}\varphi_*(\mu_n)$.

PROOF. We have $\varphi(\mathcal{L}^s) = \lambda(\varphi)\mathcal{L}^s$. Therefore, (τ, μ) and $(\varphi(\tau), \lambda(\varphi)^{-1}\varphi_*(\mu))$ are equivalent measured train tracks by Theorem 3.2.

Let

$$(\tau, \mu) \rightharpoonup (\tau_1, \mu_1) \rightharpoonup (\tau_2, \mu_2) \rightharpoonup \cdots \rightharpoonup (\tau_n, \mu_n) \rightharpoonup \cdots$$

be the sequence of maximal splits. Then we have

$$(\varphi(\tau), \varphi_*(\mu)) \rightharpoonup (\varphi(\tau_1), \varphi_*(\mu_1)) \rightharpoonup \cdots \rightharpoonup (\varphi(\tau_n), \varphi_*(\mu_n)) \rightharpoonup \cdots$$

is also a sequence of maximal splits. Since $(\tau, \mu) \sim (\varphi(\tau), \lambda(\varphi)^{-1}\varphi_*(\mu))$, by Corollary 3.4 there exists n, m such that $\varphi(\tau_n) = \tau_{n+m}$ and $\varphi_*(\mu_n) = \lambda(\varphi)\mu_{n+m}$. $\square$

To summarize, the splitting sequence (τ_n, μ_n) is eventually periodic, modulo the action of φ and rescaling the measure. One can recognize this periodicity combinatorially. See the example in Figure 17.

REMARK 3.6. We make a remark about the kind of train tracks which can arise in such periodic splitting sequences. There can be no branch of the train track τ which is a small branch with two branches coming off of the same side, or a monogon (see Figure 7). If τ has such a branch, then it is stable under any sequence of folds, since the vertices involved in a fold can never involve the interior of the small branch of one of these excluded configurations, so that the branch always remains small. So this branch will never disappear, contradicting Lemma 2.1. Note that the isolated monogon is the same as a one-sided small branch when pulled back to the universal cover.

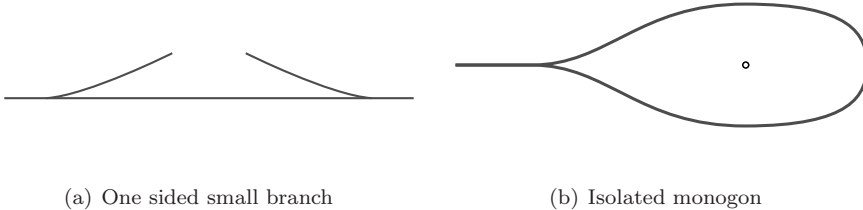


FIGURE 7. Excluded configurations

4. Triangulations

An ideal triangulation T of a punctured surface Σ is a decomposition along ideal arcs into ideal triangles. A *Whitehead move* takes any arc in T which is adjacent to two distinct triangles, removes it, and replaces it with the other diagonal of the quadrilateral (Figure 8).

Any two ideal triangulations of Σ are related by Whitehead moves (see [14, Lemma 6] for a proof).

The *mapping torus* of $\varphi : \Sigma \rightarrow \Sigma$ is the manifold $\mathcal{T}(\varphi) = \Sigma \times [0, 1] / \{(x, 0) \sim (\varphi(x), 1)\}$. Given a pseudo-Anosov homeomorphism $\varphi : \Sigma \rightarrow \Sigma$, let $\Sigma_\varphi^\circ = \Sigma^\circ \subset \Sigma$

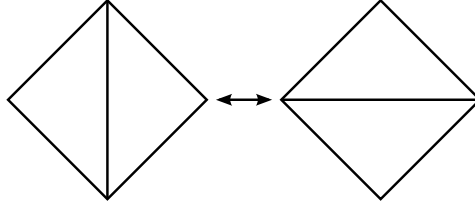


FIGURE 8. A Whitehead move

be the surface obtained by removing the singular points of the stable and unstable foliations for φ and let $\varphi^\circ : \Sigma^\circ \rightarrow \Sigma^\circ$ denote the restriction.

An *ideal triangulation* of a 3-manifold N with boundary is obtained by taking a CW complex with affinely identified tetrahedra, such that removing the vertices gives a manifold homeomorphic to the interior of N . A *taut tetrahedron* is a tetrahedron such that each face is assigned a coorientation, such that two faces are pointing inwards and two are pointing outwards. Each dihedral edge of a taut tetrahedron may be assigned an angle of either 0 or π , such that the sum of the angles around each corner of a vertex is π , and so that each face is co-oriented in such a way that the orientations of adjacent faces change only along an edge of angle 0. There is only one taut tetrahedron up to combinatorial equivalence (see Figure 9(a)). A *taut ideal triangulation* is an ideal triangulation of N such that each triangle is assigned a coorientation in such a way that every tetrahedron is taut, and the sum of the angles around every edge is 2π (Figure 9(b)). One may “pinch” the triangles together along an edge to obtain a branched surface, so that there are cusps along the angle zero dihedral corners, and a smooth surface between the angle π faces of the tetrahedra.

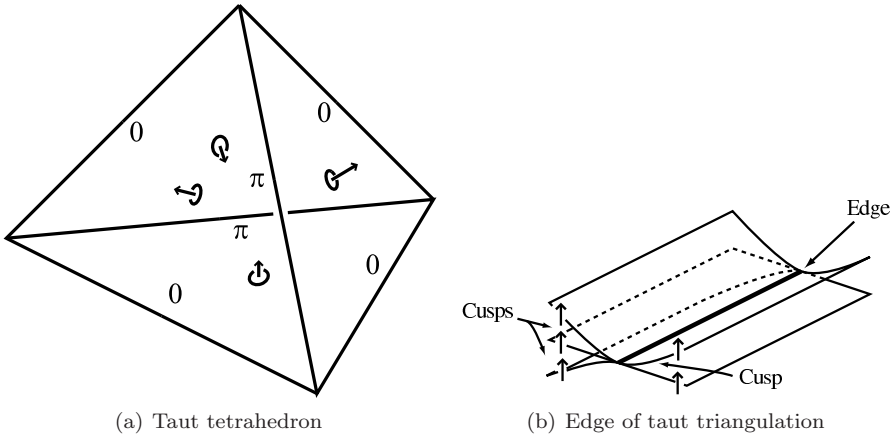


FIGURE 9. Conditions for a taut ideal triangulation

If we are given $\mathcal{T}(\varphi^\circ)$ a mapping torus with ideal triangulation T of Σ° , and a periodic sequence of Whitehead moves $T \rightarrow T_1 \rightarrow \dots \rightarrow T_m = \varphi^\circ(T)$, we may form a taut ideal triangulation of $\mathcal{T}(\varphi^\circ)$. Start with the triangulation T of Σ° . We

attach a tetrahedron to Σ° along the two triangles of T which are removed under the Whitehead move to make T_1 (see Figure 10).

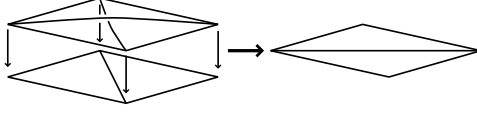


FIGURE 10. Attaching a taut tetrahedron.

The triangles are cooriented in a consistent fashion with Σ° . We repeat this process until we get T_m , which then may be glued to T by φ . The fact that φ° is pseudo-Anosov guarantees that we get a triangulation of $\mathcal{T}(\varphi^\circ)$. This triangulation is called a *layered triangulation*.

Main Construction:

We obtain a layered triangulation from the periodic sequence of train tracks given by Theorem 3.5. Each train track τ_i gives a spine of Σ° , which is dual to a unique ideal triangulation T_i of Σ° . For each split $\tau_i \rightarrow_e \tau_{i+1}$, one has a corresponding dual Whitehead move $T_i \rightarrow T_{i+1}$ (see Figure 11). Later on, we will also consider the reverse Whitehead move associated to a fold. It's also clear that

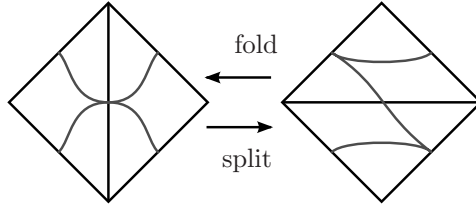


FIGURE 11. Splitting and folding

if we have a maximal split $\tau_i \rightarrow \tau_{i+1}$, then this breaks up into a sequence of splits along the maximal weight branches of τ_i , and that the order in which we attach the tetrahedra does not matter since they are attached along disjoint pairs of ideal triangles in T_i . Thus, we obtain a canonical taut ideal triangulation of $\mathcal{T}(\varphi^\circ)$ associated to the mapping class φ .

We would now like to obtain an intrinsic characterization of the taut ideal triangulations that arise from this process. We would like to be able to draw pictures of a taut ideal triangulation. In particular, we may take the preimage of the ideal triangulation in the universal cover $\widetilde{\mathcal{T}(\varphi^\circ)}$, and flatten it out into the universal cover of $\tilde{\Sigma}^\circ \cong \mathbb{H}^2$, in such a way that each ideal triangle is projected so that its coorientation agrees with that of $\tilde{\Sigma}^\circ$. We will draw local pictures of triangulations circumscribed by a circle corresponding to $\partial_\infty \mathbb{H}^2$, with the convention that the coorientation is pointing toward the reader.

DEFINITION 4.1. An edge e of a taut ideal triangulation is *left-veering* if the sequence of oriented triangles move to the left on both sides of the edge e , when viewed from the edge e and ordered by the coorientation of the triangles. Moreover, on each side of e there is at least one left-veering move. Similarly, e is right-veering

if the triangles move to the right on both sides of the edge e . In particular, the degree of e must be at least 4 (see Figure 12). A taut triangulation is called *veering* if every edge is left- or right-veering.

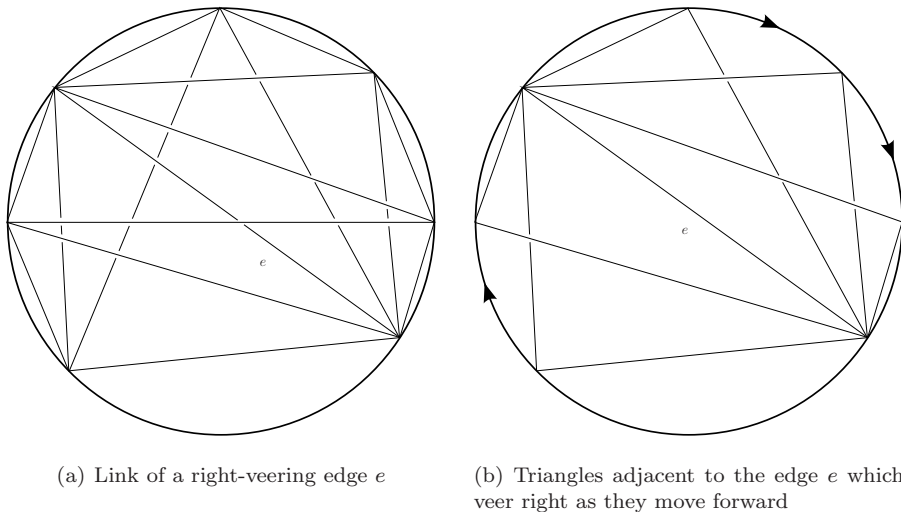


FIGURE 12

Remark: This criterion is determined purely in terms of the link of an edge in the taut ideal triangulation, so it makes sense even if the taut triangulation is not layered.

PROPOSITION 4.2. *A taut ideal triangulation of a fibered manifold coming from Whitehead moves is associated to a periodic splitting sequence if and only if it is veering.*

PROOF. The veering condition follows directly from the combinatorics of a splitting sequence, together with the condition on excluded branches in Remark 3.6. In order to see these conditions, it turns out it is necessary to work with folding sequences instead of splitting sequences. The first time an edge e appears in the sequence of Whitehead moves associated to a folding sequence, it will be dual to a large branch. In order for another Whitehead move to be applied to e , the dual branch must become small, and therefore foldings must occur involving both switches of the branch, corresponding to Whitehead moves on both sides of the edge e (viewed in the universal cover $\tilde{\Sigma}^\circ$). If one of these moves is left-veering and the other right-veering, then one sees that the branch dual to e becomes a one-sided small branch (Figure 13), which is excluded from a folding sequence (Figure 7). Thus, the initial Whitehead moves adjacent to e must be veering in the same direction.

The subsequent Whitehead moves on one side of e must veer in the same direction as the initial move. Initially, when the branch dual to e is large, there are two edges f, g in a triangle containing e . When we perform a Whitehead move, say along the edge g , then the edge g' replacing g is dual to a large branch. Thus, any subsequent move adjacent to e must be a Whitehead move dual to a fold along an

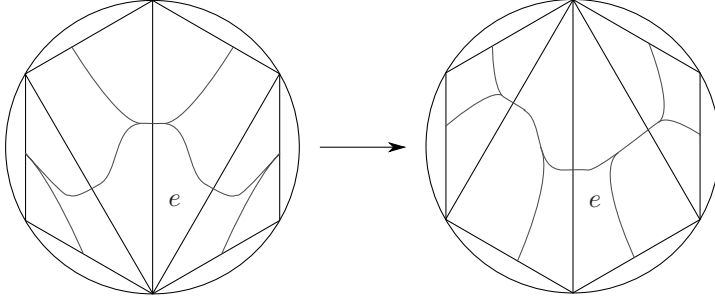


FIGURE 13. Left/right Whitehead moves give an excluded configuration

edge h which is not g' , and the edge h' replacing h must be dual to a large branch after such a move (see Figure 14). By induction, we see that the sequence of moves will always veer in the same direction.

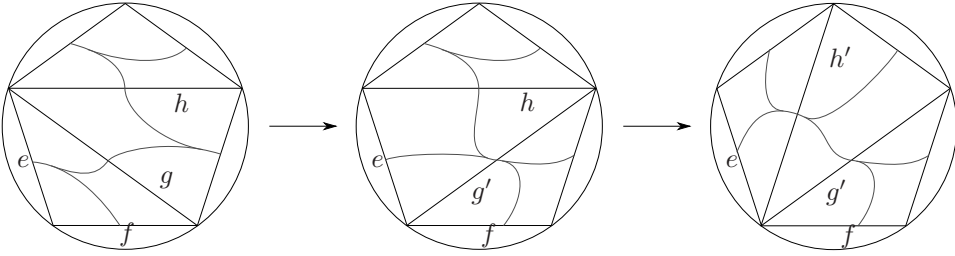


FIGURE 14. Left Whitehead moves must be followed by left moves

Conversely, suppose we have a layered taut ideal triangulation such that the link of every edge is veering. We want to show how to associate to each triangle of the triangulation a train track switch. These switches need to have the property that the induced branched surface on the boundary of a taut tetrahedron T corresponds to a fold, with the front edge e' of T corresponding to the large branch, and the back edge e of T corresponding to the small branch. We note that the train track on the front two faces of T is uniquely determined by the fact that the front edge e' is dual to a large branch. So we choose the switch in each triangle by the choice consistent with the taut tetrahedron in back of the triangle. Now we check that the veering condition implies that this choice is globally consistent.

For each taut tetrahedron T , the switches in the front two faces are determined by our convention so that e' is a large branch, so we need only check what happens on the back two faces of T . Take the two taut tetrahedra adjacent to these back faces. The veering property implies that the back edge e of T is a small branch of the induced train track (see Figure 15). Thus, we see that for any closed loop of Whitehead moves corresponding to the layered taut veering triangulation, we get a closed folding sequence. To finish, we need to see that the folding sequence gives rise to a maximal splitting sequence. The fact that the folding sequence has an invariant class of projective measures follows from a Perron-Frobenius argument since the

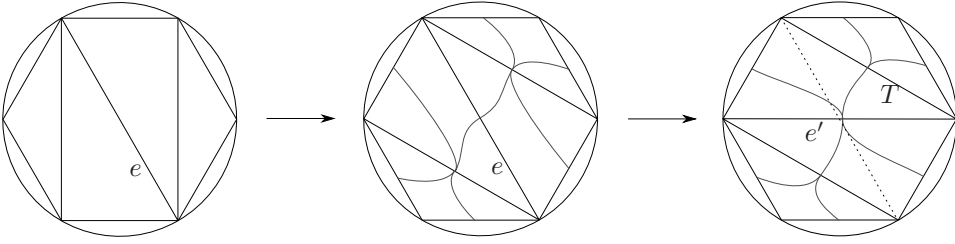


FIGURE 15. Veering determines the train track on the boundary of T

induced map of weight spaces is a Perron-Frobenius matrix, as in [21]. Thus, the invariant measure gives rise to a lamination invariant under the monodromy φ . Reversing the folding sequence gives a splitting sequence invariant under φ . Now we apply a result of Hamenstädt which implies that any two splitting sequences are related by commutations [8, Section 5]. The proof of Hamenstädt's result may be made similarly to the proof of Corollary 3.4. This implies that the layered veering triangulation is the same as the one obtained by a maximal splitting sequence. $\square$

COROLLARY 4.3. *A periodic splitting sequence of a pseudo-Anosov mapping torus coming from $\mathcal{L}^s$ gives rise to a sequence of Whitehead moves, which when reversed corresponds to a periodic splitting sequence associated to $\mathcal{L}^u$.*

PROOF. The induced triangulation of the punctured mapping torus is veering. Changing the orientation of all the triangles also gives a veering triangulation. So the reversed sequence of Whitehead moves must also be associated to a periodic splitting sequence, by Proposition 4.2. $\square$

This shows that the layered taut ideal triangulation associated to a pseudo-Anosov map is intrinsic, in that it does not depend on $\mathcal{L}^s$ or $\mathcal{L}^u$.

5. Example

For mapping classes of the punctured torus or the 4-punctured sphere, the main construction will produce the canonical layered ideal triangulations considered in [12, 3]. For concreteness, we present the results of the main construction for the case of a 4-strand pseudo-Anosov braid of minimal dilatation. This braid was proven to be minimal dilatation among 4-strand pseudo-Anosov braids by Ko, Los, and Song [13]. The associated pseudo-Anosov map $\varphi : \Sigma_{0,5} \rightarrow \Sigma_{0,5}$ has dilatation $\lambda = \lambda(\varphi) = 2.29663\dots$, where λ is the maximal root of the polynomial $x^4 - 2x^3 - 2x + 1$. The invariant laminations have 5 complementary regions which are punctured monogons, and one complementary region which is a triangle. So $\Sigma_\varphi^\circ = \Sigma_{0,6}$, and the mapping torus $\mathcal{T}(\varphi^\circ)$ is shown in Figure 16. The manifold is a 3-component link complement, which has hyperbolic volume 5.33... (conjecturally this is the minimal volume 3-cusped hyperbolic 3-manifold). Given the lamination data computed by Ko-Los-Song, we found a periodic maximal splitting sequence by hand, shown in Figure 17. Since there are six splits, the manifold has a 6 tetrahedron layered veering triangulation. If we represent the tetrahedra by ideal hyperbolic simplices, then they form an ideal hyperbolic tetrahedron triangulation

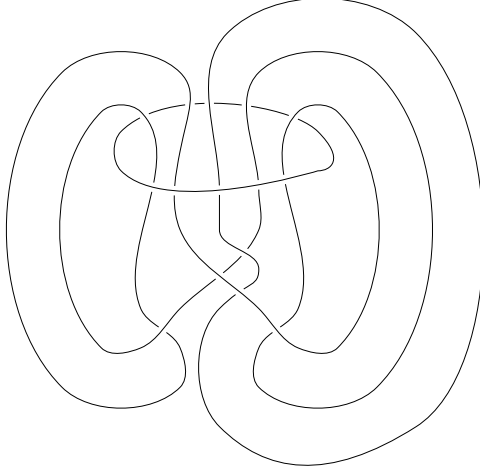


FIGURE 16. Braid representing $\mathcal{T}(\varphi^\circ)$ (the middle strand is the singular fiber)

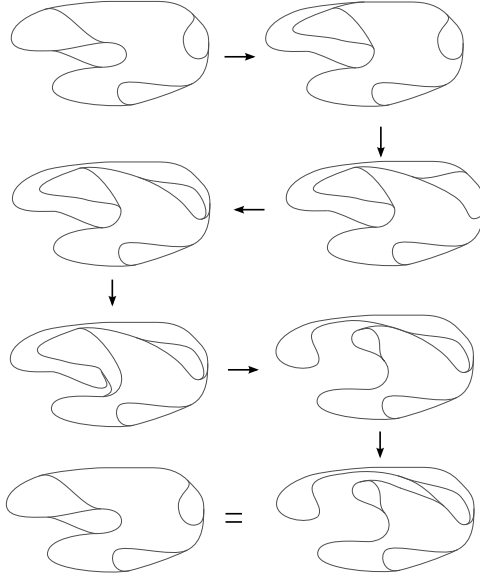


FIGURE 17. A maximal splitting sequence for φ

of the manifold, since if one of the tetrahedra had reversed orientation, the volume would be < 5.3 .

6. Dilatation bounds

This section gives an alternative approach to the paper [2]. Given a pseudo-Anosov homeomorphism φ of a surface $\Sigma_{g,n}$, $\chi(\Sigma_{g,n}) < 0$, let $\lambda(\varphi)$ denote its

dilatation. For any $P > 1$, we define

$$\Psi_P = \left\{ \varphi : \Sigma_{g,n} \rightarrow \Sigma_{g,n} \mid \varphi \text{ pseudo-Anosov, and } \lambda(\varphi) \leq P^{\frac{1}{2g-2+\frac{2}{3}n}} \right\}.$$

It follows from work of Penner [21] that for P sufficiently large, and for each closed surface Σ_g of genus $g \geq 2$, there exists $\varphi_g : \Sigma_g \rightarrow \Sigma_g$ so that

$$\{\varphi_g : \Sigma_g \rightarrow \Sigma_g\}_{g=2}^\infty \subset \Psi_P.$$

Given a pseudo-Anosov homeomorphism $\varphi : \Sigma \rightarrow \Sigma$, let $\Sigma_\varphi^\circ = \Sigma^\circ \subset \Sigma$ be the surface obtained by removing the singularities of the stable and unstable foliations for φ and let $\varphi|_{\Sigma^\circ} : \Sigma^\circ \rightarrow \Sigma^\circ$ denote the restriction. The set of pseudo-Anosov homeomorphisms

$$\Psi_P^\circ = \{\varphi|_{\Sigma^\circ} : \Sigma^\circ \rightarrow \Sigma^\circ \mid (\varphi : \Sigma \rightarrow \Sigma) \in \Psi_P\}$$

is therefore also infinite.

LEMMA 6.1. *A maximal train track on $\Sigma_{g,n}$ has at most $18g - 18 + 6n$ branches.*

PROOF. Assume every complementary region of τ is a punctured monogon or a trigon. Then the number of monogons is at most n , and let t denote the number of trigons. Let e denote the number of branches, and let v denote the number of switches of the train track. Since each switch is in one to one correspondence with a cusp of a complementary region, we get $v = n + 3t$. Moreover, $3v = 2e$ counts the number of pairs of incidences between branches and switches. The euler characteristic is given by $\chi(S) = 2 - 2g - n = v - e + t$. Solving for e , using the equations $t = (v - n)/3$ and $v = \frac{2}{3}e$ to eliminate v, t , we get $e = 18g - 18 + 6n$. $\square$

The following theorem gives a refinement of [2, Theorem 1.1].

THEOREM 6.2. *Let $M \in \mathcal{T}(\Psi_P^\circ)$ be a mapping torus of a punctured pseudo-Anosov class $M = \mathcal{T}(\varphi^\circ)$, $\varphi \in \Psi_P$. Then M has a taut ideal triangulation with at most $\frac{1}{2}(P^9 - 1)$ tetrahedra.*

PROOF. Let e be the number of branches of a train track τ fully carrying a stable lamination for φ . By the previous lemma, $e \leq 18g - 18 + 6n$. Then Σ° has an ideal triangulation with e edges. Let $\tau = \sigma_0 \rightarrow \sigma_1 \rightarrow \cdots \rightarrow \sigma_m = \varphi(\tau)$ be an invariant sequence of train track foldings coming from Theorem 3.5. The weight space $W(\sigma_i)$ has positive coordinates in a space of dimension e . Folding changes the weight on the folded branch by the sum of the weights of the two edges folded onto it (see Figure 4). Since there is a 1-1 correspondence between the branches of σ_i and σ_{i+1} , we can think of the map $W(\sigma_i) \rightarrow W(\sigma_{i+1})$ as a unipotent matrix, with the sum of entries $e + 2$. Mutiplying such matrices at least adds the off-diagonal entries. So the induced map $W(\tau) \rightarrow W(\varphi(\tau))$ is a matrix with the sum of the entries at least $2m + e$. Then from [6, Lemma 3.1], we get $2m + 1 \leq \lambda(\varphi)^e$. By the hypothesis that $\varphi \in \Psi_P$, we have $\lambda(\varphi) \leq P^{\frac{9}{e}}$, so we conclude that $2m + 1 \leq P^9$. Thus, $\mathcal{T}(\varphi^\circ)$ has a taut ideal layered triangulation with at most $\frac{1}{2}(P^9 - 1)$ tetrahedra. $\square$

In particular, the collection of mapping tori $\{\mathcal{T}(\varphi^\circ) \mid \varphi \in \Psi_P\}$ is finite. Let's apply this result to the case of the closed surface of genus g . Let δ_g be the minimal dilatation of a pseudo-Anosov map of a closed surface Σ_g of genus g . By a result of Hironaka and Kin, $\delta_g^{g-1} \leq 2 + \sqrt{3}$. Thus, if φ is a pA map of Σ_g with dilatation δ_g , then $\lambda(\varphi) = \delta_g \leq (2 + \sqrt{3})^{\frac{1}{g-1}} = (2 + \sqrt{3})^{\frac{2}{-x(S)}}$. So we may take $P = (2 + \sqrt{3})^2$

in the above theorem, and we obtain the number of tetrahedra of a taut ideal triangulation is bounded by $\frac{1}{2}((2 + \sqrt{3})^{18} - 1) \leq 10^{10}$.

7. Conclusion

There are several natural questions arising from this paper.

First, in the previous section we gave an upper bound on the number of tetrahedra needed for a taut ideal layered triangulation of the punctured mapping torus of a minimal dilatation pA map of a closed surface Σ_g . One ought to be able to improve this estimate from an improvement of the upper bound on δ_g in the papers [10, 1]. Moreover, one ought to be able to improve the lower bound given in Theorem 6.2 by getting a better estimate of the Perron-Frobenius eigenvalues of the transformations of weight spaces. It is an interesting question to compute the numbers δ_g or more generally $\delta_{g,n}$. The splitting sequences in this paper give a possible approach to this for a given g, n . One may form finitely many automata of folding sequences for a given surface $\Sigma_{g,n}$, one for each possible collection of indices of the singular points. Then one can compute the minimal dilatations for closed paths in these folding automata. Unfortunately, there appears to be many more train tracks which appear in these automata than in those considered by Ko-Los-Song [13, 6].

Secondly, the main construction of a taut ideal layered triangulation associated to a pseudo-Anosov mapping class gives a new classification of conjugacy classes of pA maps. To a pA mapping class, one may associate the taut layered triangulation of the mapping torus, together with an encoding of the homology class of the fiber. This homology class may be represented canonically as a harmonic simplicial 2-cycle for the taut triangulation. This amounts to assigning weights to the faces of the triangulation such that the boundary is zero, and the signed sum of the weights on the faces of a taut tetrahedron is zero (corresponding to the dual 1-chain being a 1-cycle). Then two pA mapping classes are conjugate if and only if there is a combinatorial equivalence between the two triangulations which preserves the harmonic 2-cycles. This gives a new way of encoding pseudo-Anosov conjugacy classes, which seems to be somewhat simpler than previous methods (see [9, 16, 17, 18]). There is also a conjugacy invariant implicit in the geometrization theorem for mapping tori [26]. In particular, our approach simplifies Hamenstädt's conjugacy invariant coming from her study of cube complexes of train track splitting sequences [8], but unpublished. It would be interesting to implement this classification algorithmically, and analyze the computational complexity.

Thirdly, it would be interesting to understand the implications of a veering taut ideal triangulation which is not layered. There are two branched surfaces naturally associated to such a triangulation as constructed in the proof of Proposition 4.2, which should carry essential laminations. It would be interesting to find an example of a veering triangulation which does not come from a fibration (that is, is not layered).

Finally, it is an interesting question whether the veering triangulations studied in Section 4 can be realized in the hyperbolic metric as ideal hyperbolic triangulations with all positively oriented tetrahedra. This is true in the punctured torus case [12, 15, 5], and in the example investigated in Section 5. It's possible that the approach of [5] may be generalized to deal with this question for veering triangulations.

Added in proof: Some progress on this fourth question has been made in the preprints [11, 4] which prove the existence of angle structures on veering triangulations. Moreover, they give interesting combinatorial characterizations and properties of veering taut ideal triangulations. We briefly indicate the connection between the definition of veering given in [11] and the derivation from the property of train tracks given in our paper. Each branch of a train track occurring in a splitting sequence may be labelled with an L or R (standing for Left or Right) if the branch goes into a switch from the left branch or the right branch, respectively (in Figure 1(b), the b and c branches are labeled R and L respectively). As a train track is split, the excluded configurations in Figure 7 imply that the labeling is independent of which switch is used for the label when the branch is a small branch, and the labeling of corresponding branches in the two train tracks is consistent. Thus, a consistent label may be given to each branch, and therefore to the faces of the branched surface and its dual taut ideal triangulation arising from the Main Construction. In the invariant splitting sequence of train tracks, every split branch is large, and thus the labels of the four adjacent branches alternate between L and R . Reinterpreting this in terms of the dual taut triangulation, the edge labeling has the property that in every taut ideal tetrahedron, the labels alternate around the four cusp edges (see Figure 11). This is the definition given in [11], which they demonstrate is equivalent to definition of a taut ideal veering triangulation given in Definition 4.2 (and where they use blue for L and red for R). However, the connection with train track splitting sequences is more direct. They also give an example of a veering taut ideal triangulation which is not layered, answering our third question.

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A note on complete hyperbolic structures on ideal triangulated 3-manifolds

Feng Luo

Dedicated to Bus Jaco on the occasion of his 70th birthday

ABSTRACT. It is a theorem of Casson and Rivin that the complete hyperbolic metric on a cusp end ideal triangulated 3-manifold maximizes volume in the space of all positive angle structures. We show that the conclusion still holds if some of the tetrahedra in the complete metric are flat.

1. Introduction

1.1. Epstein and Penner [1] proved that a non-compact finite volume complete hyperbolic 3-manifold has a geodesic triangulation in which each 3-simplex is a (possibly flat) ideal hyperbolic tetrahedron. Here a flat ideal hyperbolic tetrahedron is a tetrahedron with all dihedral angles being 0 and π . The purpose of this paper is to show that a geodesic ideal triangulation of a complete hyperbolic 3-manifolds with some flat tetrahedra maximizes the volume in the closure of the space of non-negative angle structures. In the case all tetrahedra are non-flat, this was proved by Casson and Rivin (see [2], [3] for a written proof). We remark that the corresponding result also holds for hyper-ideal geodesic triangulations of compact hyperbolic 3-manifolds with totally geodesic boundary. This will be addressed in another paper.

1.2. Recall that a triangulated closed pseudo 3-manifold (M^*, T^*) is the quotient of a disjoint union of tetrahedra so that co-dimension-1 faces are identified in pairs by affine homeomorphisms. The simplices in the triangulation T^* are the quotients of simplices in the disjoint union. If we remove all vertices of T^* from M^* , the result is an *ideal triangulated non-compact 3-manifold* $(M, \mathbf{T})$. We say M has *cuspidal ends* if the link of each vertex in T^* is a torus or a Kleinbottle. We will deal with ideal triangulated cusp end manifolds $(M, \mathbf{T})$ in this paper. An *angle structure* on $(M, \mathbf{T})$, introduced by Casson, Rivin and Lackenby ([7]), assigns each edge of each tetrahedron a positive number, called the angle, so that

(1) the sum of three angles at edges from each vertex of each tetrahedron is π , and

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(2) the sum of angles around each edge is 2π .

An *angled tetrahedron* is a tetrahedron so that each edge has assigned a positive number called the angle so that condition (1) above holds. Given an angled tetrahedron, there is a unique ideal hyperbolic tetrahedron, call the *geometric realization*, whose dihedral angles are the assigned angles. The volume of an angled tetrahedron is defined to be the volume of its geometric realization. The volume of an angle structure is the sum of the volumes of its angled tetrahedra.

For an ideal triangulated 3-manifold $(M, \mathbf{T})$ with n tetrahedra, let $\mathbf{A}(\mathbf{T}) \subset \mathbf{R}^{6n}$ be the space of all angle structures on $\mathbf{T}$ and let $\text{vol} : \mathbf{A}(\mathbf{T}) \rightarrow \mathbf{R}$ be the volume function. By the Lobachevsky-Milnor's formula for volume, the volume function extends continuously to $\text{vol} : \overline{\mathbf{A}(\mathbf{T})} \rightarrow \mathbf{R}$ where $\overline{\mathbf{A}(\mathbf{T})}$ is the (compact) closure of $\mathbf{A}(\mathbf{T})$ in $\mathbf{R}^{6n}$.

THEOREM 1.1 (Casson-Rivin). *For an ideal triangulated cusped 3-manifold $(M, \mathbf{T})$ so that $\mathbf{A}(\mathbf{T}) \neq \emptyset$, a point $p \in \mathbf{A}(\mathbf{T})$ corresponds to a complete hyperbolic structure on M if and only if p is the maximum point of the volume function $\text{vol} : \mathbf{A}(\mathbf{T}) \rightarrow \mathbf{R}$.*

Since the space $\mathbf{A}(\mathbf{T})$ is non-compact, there is no guarantee that the maximum point of vol exists. Our main theorem generalizes theorem 1.1 in one direction.

THEOREM 1.2. *For an ideal triangulated cusp end 3-manifold $(M, \mathbf{T})$ so that $\overline{\mathbf{A}(\mathbf{T})} \neq \emptyset$, if a point $p \in \overline{\mathbf{A}(\mathbf{T})}$ corresponds to a complete hyperbolic structure on M , then p is the maximum point of the volume function $\text{vol} : \overline{\mathbf{A}(\mathbf{T})} \rightarrow \mathbf{R}$. Furthermore, the point p is the unique maximum volume point in $\overline{\mathbf{A}(\mathbf{T})}$.*

In [4], Futer and Gueritaud found an example of $(M, \mathbf{T})$ so that $\mathbf{A}(\mathbf{T}) \neq \emptyset$ and the maximum point of vol on $\overline{\mathbf{A}(\mathbf{T})}$ does not correspond to a complete hyperbolic metric. (A similar example was also found by Luo-Zheng [5] by making five 2-3 moves on the standard triangulation of the figure-8 knot complement). This shows that theorem 1.2 cannot be improved to be a necessary and sufficient condition. We are curious if the uniqueness of the maximum volume on $\overline{\mathbf{A}(\mathbf{T})}$ is the sufficient condition for the completeness of the metric.

Theorem 1.2 will be proved in §2.

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2. Proof of Theorem 1.2

Let us begin with some notations and conventions. Let $\mathbf{R}_{>0} = \{x \in \mathbf{R} | x > 0\}$ and $\mathbf{R}_{\geq 0} = \{x \in \mathbf{R} | x \geq 0\}$. Given a set X , let $\mathbf{R}^X = \{f : X \rightarrow \mathbf{R}\}$ be the vector space of all functions from X to $\mathbf{R}$. The function $x \ln x : (0, \infty) \rightarrow \mathbf{R}$ is extended continuously to $[0, \infty) \rightarrow \mathbf{R}$ by setting $0 \ln 0 = 0$. Suppose $(M, \mathbf{T})$ is an ideal triangulated 3-manifold so that V , E and T are the sets of all (ideal) vertices, edges and tetrahedra. Let $I = \{(e, \sigma) \in E \times T | \text{edge } e \text{ is adjacent to the tetrahedron } \sigma\}$. An angle structure is a vector in the space $\mathbf{R}^I$ satisfying a set of linear equations and linear inequalities. If $x \in \mathbf{R}^I$, we use x_i to denote $x(i)$. If $i = (e, \sigma) \in I$, we use $i > e$ and $i < \sigma$ to indicate the incident relation. Three distinct $i = (e_1, \sigma)$, $j = (e_2, \sigma)$ and $k = (e_3, \sigma)$ in I so that e_1, e_2, e_3 are three

edges from the same vertex in σ will be denoted by $\{i, j, k\} \in \Delta$. Finally, we use $(e, \sigma) \sim (e', \sigma)$ to indicate that e, e' are two opposite edges in the same tetrahedron σ . Using these notations, we have

$$A(\mathbf{T}) = \{x \in \mathbf{R}^I \mid (1), (2), (3) \text{ hold}\}$$

where

- (1) if $\{i, j, k\} \in \Delta$, $x_i + x_j + x_k = \pi$;
- (2) if $e \in E$, $\sum_{i \in I, i > e} x_i = 2\pi$;
- (3) $x_i > 0$ for all $i \in I$.

Note that condition (1) implies that $x_i = x_j$ when $i \sim j$. The closure $\overline{\mathbf{A}(\mathbf{T})}$ of $\mathbf{A}(\mathbf{T})$ is given by $\{x \in \mathbf{R}_{\geq 0}^I \mid (1) \text{ and } (2) \text{ hold, and } x_i \geq 0 \text{ for all } i\}$. Theorem 1.2 does not assume $\mathbf{A}(\mathbf{T}) \neq \emptyset$, but it assumes $\overline{\mathbf{A}(\mathbf{T})} \neq \emptyset$. Elements in $\overline{\mathbf{A}(\mathbf{T})}$ will be called *non-negative angle structures*.

Suppose σ is an angled tetrahedron with three angles x_1, x_2, x_3 at three edges from a vertex. Then the Lobachevsky-Milnor volume formula says the volume $\text{vol}(\sigma)$ of σ is $\Lambda(x_1) + \Lambda(x_2) + \Lambda(x_3)$ where $\Lambda(t) = -\int_0^t \ln |2 \sin(u)| du$ is the Lobachevsky function. The function $\Lambda(t)$ is continuous on $[0, \pi]$. In particular, the volume function $\text{vol} : \overline{\mathbf{A}(\mathbf{T})} \rightarrow \mathbf{R}$ is given by

$$(2.1) \quad \text{vol}(x) = \frac{1}{2} \sum_{i \in I} \Lambda(x_i).$$

Note that $\text{vol}(x) = \sum_{\sigma \in T} \text{vol}(\sigma)$ where σ is the angled tetrahedron with angles given by x .

2.1. Setting up the proof. Assume that $p \in \overline{\mathbf{A}(\mathbf{T})}$ corresponds to the complete hyperbolic metric on M , i.e., there is a geometric ideal triangulation (isotopic to $\mathbf{T}$) of a complete hyperbolic metric on M so that p corresponds to the dihedral angles. If $p \in \mathbf{A}(\mathbf{T})$, then Casson-Rivin's theorem implies that p is the maximum point of the volume. It remains to deal with $p \in \partial \mathbf{A}(\mathbf{T}) = \overline{\mathbf{A}(\mathbf{T})} - \mathbf{A}(\mathbf{T})$. Take $q \in \overline{\mathbf{A}(\mathbf{T})}$ so that $q \neq p$. The goal is to show that $\text{vol}(p) > \text{vol}(q)$. To this end, let $f(t) = \text{vol}((1-t)p + tq)$ for $t \in [0, 1]$. We will show that $f(0) > f(1)$, i.e., $\text{vol}(p) > \text{vol}(q)$.

LEMMA 2.1. *The function $f(t)$ is concave in $[0, 1]$ and is strictly concave in the open interval $(0, 1)$.*

PROOF. By a result of Rivin [9], the volume function $\Lambda(t_1) + \Lambda(t_2) + \Lambda(t_3)$ is strictly concave in the set $\{(t_1, t_2, t_3) \in \mathbf{R}^3 \mid t_1 + t_2 + t_3 = \pi, t_i > 0 \text{ for } i = 1, 2, 3\}$. In particular, this implies that the function $\text{vol}(x)$ is concave in $x \in \overline{\mathbf{A}(\mathbf{T})}$. Thus $f(t)$ is concave in $[0, 1]$. To see the strict concavity, note that $\text{vol}(p) > 0$ since it is the volume of a complete hyperbolic structure. In particular, there is tetrahedron $\sigma \in T$ so that its angles in p are all positive. This implies that for $t \in (0, 1)$, the angles of σ in $(1-t)p + tq$ are positive. By Rivin's theorem, the volume $\sum_{i \in I, i < \sigma} \Lambda((1-t)p_i + tq_i)$ is strictly concave in $t \in (0, 1)$. Since $f(t)$ is the sum of concave functions in t so that one of them is strictly concave, it follows that $f(t)$ is strictly concave in $(0, 1)$. $\square$

For $t \in (0, 1)$, by the definition of the volume (2.1), we have

$$(2.2) \quad f'(t) = -\frac{1}{2} \sum_{i \in I} (q_i - p_i) \ln |2 \sin((1-t)p_i + tq_i)|$$

Note that we have used the convention that $0 \ln 0 = 0$ in (2.2). Indeed, if $p_i = q_i$ is 0 or π , then the term in (2.2) corresponding to i is defined to be 0. (This is due to the fact that $\Lambda(x) + \Lambda(y) + \Lambda(\pi - x - y) = 0$ if $x \in \pi \mathbf{Z}$.)

The goal is to show that

$$(2.3) \quad \lim_{t \rightarrow 0^+} f'(t) \leq 0.$$

Note that (2.3) and lemma 2.1 imply that $f(0) > f(1)$. In the rest of the subsections, we will focus on proving (2.3).

2.2. Let $J = \{i \in I | p_i = 0 \text{ or } p_i = \pi\}$. By the assumption, the angle structure p comes from a geodesic ideal triangulation of a complete hyperbolic metric. Thus, each tetrahedron σ in p either has all positive angles or its angles are $0, 0, \pi$, i.e., if $(e, \sigma) \in J$, then $(e', \sigma) \in J$ for all other edges e' in σ by the definition of flat tetrahedron, i.e., all its dihedral angles are 0 or π . Let $a = q - p \in \mathbf{R}^I$.

- LEMMA 2.2. (1) If $\{i, j, k\} \in \Delta$, then $a_i + a_j + a_k = 0$, i.e., for each tetrahedron σ , $\frac{1}{2} \sum_{i < \sigma} a_i = 0$.
 (2) For each edge $e \in E$, $\sum_{i > e} a_i = 0$.
 (3) $\sum_{i \in I} a_i = 0$.

Indeed, the first two conditions follows from the definition of angle structures (1) and (2). The last condition follows from part (1) by summing over all tetrahedra and then divided by 2.

By lemma 2.2(3), we can rewrite $f'(t)$ in (2.2) as

$$(2.4) \quad f'(t) = -\frac{1}{2} \sum_{i \in I} a_i \ln |\sin((1-t)p_i + tq_i)|$$

The following was proved in [6]. It can also be found in [4].

LEMMA 2.3.

$$(2.5) \quad \lim_{t \rightarrow 0^+} f'(t) = -\frac{1}{2} \left(\sum_{i \notin J} a_i \ln |\sin(p_i)| - \sum_{i \in J} a_i \ln |a_i| \right).$$

2.3. Penner's decorated ideal simplexes. To understand the right-hand-side of (2.5), we need a proposition about the geometry of decorated ideal hyperbolic tetrahedra. Following Penner [8], a *decorated ideal n -simplex* is an ideal hyperbolic n -simplex so that each vertex is assigned a horosphere centered at the vertex. If σ is a decorated ideal n -simplex and e is an edge of it, the length $L(e)$ of e is defined to be the signed distance between the two horospheres centered at the end points of e (the distance is negative if the horospheres intersect). More precisely, suppose p, p' are the two points of intersection of e with these two horospheres. Then $L(e)$ is $\text{dist}(p, p')$ if these two horospheres are disjoint and is $-\text{dist}(p, p')$ if they intersect.

LEMMA 2.4. Suppose σ is a decorated ideal hyperbolic tetrahedron with edge length $L(e)$ and dihedral angle $\theta(e)$ at the edge e . Assume that $\theta(e) \in (0, \pi)$ for all edges. Then there is a constant $c(\sigma)$ depending only on σ so that for any pairs of opposite edges e, e' in σ ,

$$(2.6) \quad \frac{1}{2}(L(e) + L(e')) = \ln |\sin(\theta(e))| + c(\sigma)$$

PROOF. The proof is based on the cosine law for decorated ideal triangles first discovered by Penner [8]. Namely, give a decorated ideal triangle of lengths l_1, l_2, l_3 , the "angles" of the triangle, denoted by a_1, a_2, a_3 , are the lengths of the portion of the horocycle inside the triangle. Indices are arranged so that the angle a_i is facing the edge of length l_i . Then the cosine law says

$$(2.7) \quad l_i = -(\ln a_j + \ln a_k) \quad \{i, j, k\} = \{1, 2, 3\}.$$

For the edge e (respectively e'), there are two face triangles of the tetrahedron σ having e (resp. e') as an edge. These face triangles are naturally decorated ideal hyperbolic triangles. Let a_1, a_2, a_3, a_4 (resp. a'_1, a'_2, a'_3, a'_4) be the inner angles of these decorated face ideal triangles so that a_i 's are adjacent to e (resp. e'). Let the rest of the four face angles (of the four decorated ideal triangles) be $b_1, \dots, b_4$. Here vertices of angles b_i are either in e or e' . Then by the cosine law, we have

$$(2.8) \quad L(e) = -\frac{1}{2} \sum_{i=1}^4 \ln a_i \quad \text{and} \quad L(e') = -\frac{1}{2} \sum_{i=1}^4 \ln a'_i$$

This shows

$$\begin{aligned} L(e) + L(e') &= -\frac{1}{2} \sum_{i=1}^4 (\ln a_i + \ln a'_i) \\ &= \frac{1}{2} c_0(\sigma) + \frac{1}{2} \sum_{i=1}^4 \ln b_i \end{aligned}$$

where $c_0(\sigma) = \sum_{i=1}^4 (\ln a_i + \ln a'_i + \ln b_i)$ is the sum over all twelve face angles.

Consider the Euclidean triangles obtained by intersecting the horospheres with the ideal tetrahedron. The dihedral angles $\theta(e)$'s are the inner angles and the face angles b_i 's are the edge lengths of the Euclidean triangles. Thus, by the Sine law for Euclidean triangles, we can write

$$\ln b_i = c_i(\sigma) + \ln |\sin(\theta(e))|$$

where b_i has its vertex at e . Putting these together and using the fact that $\theta(e) = \theta(e')$, we obtain (2.6) where $c(\sigma) = \frac{1}{2} \sum_{i=0}^4 c_i(\sigma)$. $\square$

For a decorated ideal triangle of edge lengths $L(e)$, we define the *average edge length* of e to be $W(e) = \frac{1}{2}(L(e) + L(e'))$ where e, e' are opposite edges.

LEMMA 2.5. *For a decorated ideal tetrahedron σ , if e_1, e_2, e_3 are three edges from a vertex v , then*

$$(2.9) \quad e^{W(e_1)} + e^{W(e_2)} \geq e^{W(e_3)}$$

so that equality holds if and only if $\theta(e_3) = \pi$, $\theta(e_1) = \theta(e_2) = 0$.

Indeed, consider the Euclidean triangle obtained by intersecting the horosphere centered at the vertex v with the ideal tetrahedron. The inner angles of the Euclidean triangle are $\theta(e_i)$'s and the edge lengths of it are $R \sin(\theta(e_i))$ where R is the radius of the circumcircle. Now by lemma 2.4 that $\sin(\theta(e_i)) = c' e^{W(e_i)}$, the

lengths of the edges in the triangle are $ce^{W(e_i)}$ for some constant c . Thus the lemma follows from the triangular inequality for edge lengths of triangles.

2.4. A proof of theorem 1.2. Recall that the maximum volume point $p \in \overline{\mathbf{A}(\mathbf{T})}$ corresponds to a complete hyperbolic metric, i.e., there exists a geodesic triangulation of a complete hyperbolic metric on M so that the triangulation is isotopic to T and the dihedral angles coincide with the angles given by p . Choose small horospheres at the cusp ends of M so that each tetrahedron becomes an ideal decorated hyperbolic tetrahedron. In particular, each edge e in $\mathbf{T}$ has the edge length $L(e)$ (in the decorated tetrahedra). For each $i = (e, \sigma) \in I$, we define the average length w_i (of e in σ) to be

$$(2.10) \quad w_i = \frac{1}{2}(L(e) + L(e'))$$

where e, e' are opposite edges in σ .

LEMMA 2.6. *We have*

$$(2.11) \quad \sum_{i \notin J} a_i \ln |\sin(p_i)| = \sum_{i \notin J} a_i w_i.$$

PROOF. A tetrahedron σ is called flat (in p) if its dihedral angles in p are $0, 0, \pi$, i.e., there is $i \in I$ with $i < \sigma$. By our assumption that the angle structure p comes from a geodesic ideal triangulation, each tetrahedron σ in p is either flat, or has positive angles at its edges. Thus, if σ is not flat, then by lemma 2.4, there is a constant $c(\sigma)$ so that (2.6) holds for each pair of opposite edges e, e' . This is the same as

$$(2.12) \quad w_i = \ln |\sin(p_i)| + c(\sigma)$$

for $i < \sigma$. Multiply (2.12) by a_i and sum over all not flat tetrahedra, we obtain

$$(2.13) \quad \sum_{i \notin J} a_i w_i = \sum_{i \notin J} a_i \ln |\sin(p_i)| + \sum_{i \notin J, i < \sigma} a_i c(\sigma).$$

But

$$\sum_{i \notin J, i < \sigma} a_i c(\sigma) = \sum_{\sigma \text{ not flat}} c(\sigma) \left(\sum_{i < \sigma} a_i \right) = 0$$

due to lemma 2.2 (1). This ends the proof. $\square$

On the other hand, we have

$$(2.14) \quad \sum_{i \in I} a_i w_i = 0.$$

Indeed, if $i = (e, \sigma)$ and $j = (e', \sigma)$ where e, e' are opposite edges in σ , then $a_i = a_j$ and $w_i = w_j$. Furthermore by (2.10), $a_i w_i + a_j w_j = a_i L(e) + a_j L(e')$. Thus

$$\begin{aligned} \sum_{i \in I} a_i w_i &= \sum_{i \in I} a_i \left(\sum_{e < i} L(e) \right) \\ &= \sum_{e \in E} L(e) \left(\sum_{i < e} a_i \right) \\ &= 0 \end{aligned}$$

due to lemma 2.2 (2).

By combining (2.11) and (2.14), we obtain

$$(2.15) \quad \sum_{i \notin J} a_i \ln |\sin(p_i)| = - \sum_{i \in J} a_i w_i.$$

By (2.15), we can rewrite (2.5) as,

$$\begin{aligned} \lim_{t \rightarrow 0^+} f'(t) &= -\frac{1}{2} \left(\sum_{i \in J} a_i \ln |a_i| + \sum_{i \in J} a_i w_i \right) \\ &= \sum_{\sigma \text{ is flat}} (-a_i \ln |a_i| - a_j \ln |a_j| - a_k \ln |a_k| + a_i w_i + a_j w_j + a_k w_k) \end{aligned}$$

where $i, j, k < \sigma$, $\{i, j, k\} \in \Delta$.

Since σ is flat, we may assume that $p_k = \pi$, $p_i = p_j = 0$. Then by lemma 2.5, three average lengths w_i, w_j, w_k satisfy the triangular equality, i.e., $e^{w_k} = e^{w_i} + e^{w_j}$. Furthermore, $a_i \geq 0$, $a_j \geq 0$ and $a_i + a_j + a_k = 0$.

We claim

$$(2.16) \quad -a_i \ln |a_i| - a_j \ln |a_j| - a_k \ln |a_k| + a_i w_i + a_j w_j + a_k w_k \leq 0.$$

Evidently, (2.16) implies that $\lim_{t \rightarrow 0^+} f'(t) \leq 0$. Now (2.16) follows from the following simple lemma on a convex function where we take $x = a_i, y = a_j, z = a_k, a = w_i, b = w_j, c = w_k$.

LEMMA 2.7. *Suppose $x, y, a, b, c \in \mathbf{R}_{\geq 0}$, $x + y + z = 0$ and $e^c \geq e^a + e^b$. Then*

$$(2.17) \quad -x \ln x - y \ln y - z \ln z + ax + by + cz \leq 0.$$

We remark that if $e^c = e^a + e^b$, then the inequality becomes equality for some non-zero x, y, z .

PROOF. Replacing $z = -x - y$, we obtain the equivalent form of (2.17) as

$$(x + y) \ln(x + y) - x \ln x - y \ln y \leq (c - a)x + (c - b)y.$$

The above inequality is homogeneous in (x, y) , i.e., it is equivalent if we replace (x, y) by $(\lambda x, \lambda y)$ where $\lambda > 0$. Thus we may assume further that $y = 1$. Thus it remains to prove,

$$(2.18) \quad (x + 1) \ln(x + 1) - x \ln x \leq (c - a)x + (c - b)$$

for all $x \geq 0$. Let $g(x) = (x + 1) \ln(x + 1) - x \ln x$. Then $g'(x) = \ln(1 + \frac{1}{x})$ and $g''(x) = -\frac{1}{x(1+x)}$. It follows that $g''(x) \leq 0$, i.e., g is concave in $[0, \infty)$. The equation of the tangent line to g at the point $x_0 = \frac{e^a}{e^c - e^a}$ is $y = h(x)$ where

$$h(x) = (c - a)x + (c - \ln(e^c - e^a)).$$

Since g is concave, we have $g(x) \leq h(x)$. Now use $b \geq \ln(e^c - e^a)$, we obtain $h(x) \leq (c - a)x + (c - b)$. Thus lemma follows. $\square$

We end the paper by note that if p is a maximum volume point in $\overline{\mathbf{A}(\mathbf{T})}$, then each tetrahedron in p is either flat or has positive angles at its edges. This was known to [4].

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A linear bound on the tetrahedral number of manifolds of bounded volume (after Jørgensen and Thurston)

Tsuyoshi Kobayashi and Yo'av Rieck

To Bus Jaco, on the event of his 70th birthday

1. Introduction

The purpose of this note is to prove Theorem 1.1 below, due to Jørgensen and Thurston and known to experts in the field; the earliest reference we found to this theorem and a sketch of its proof appears in Thurston's notes [6, Chapter 5]. For definitions and notation see the next section. We study the triangulation of the thick part. For another geometric study of the triangulation of the thick part see Breslin's [2]. We prove:

THEOREM 1.1 (Jørgensen, Thurston). *Let $\mu > 0$ be a Margulis constant for $\mathbb{H}^3$. Then for any $d > 0$ there exists a constant $K > 0$, depending on μ and d , so that for any complete finite volume hyperbolic 3-manifold M , $N_d(M_{[\mu, \infty)})$ can be triangulated using at most $K \text{Vol}(M)$ tetrahedra.*

The manifold $N_d(M_{[\mu, \infty)})$ is the closed d -neighborhood of the μ -thick part of M and is denoted by X throughout this paper. By the Margulis Lemma, $M \setminus X$ consists of disjoint cusps and open solid tori, and each of these solid tori is a regular neighborhood of an embedded closed geodesic. We refer to removing such an open solid torus neighborhood of a geodesic γ as *drilling out* γ . Thus X is obtained from M by drilling out short geodesics. Note that any preimage of any component of $M \setminus X$ in the universal cover $\mathbb{H}^3$ is convex (for an explicit description see Definition 4.1). In the next proposition we bring the basic facts about X ; this proposition is independent of Theorem 1.1. For a complete Riemannian manifold A and a point $a \in A$, we denote the radius of injectivity of A at a by $\text{inj}_A(a)$.

PROPOSITION 1.2. *Fix the notation of Theorem 1.1. Then the following hold:*

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- (1) *There exists $R = R(\mu, d) > 0$, independent of M , so that for any $x \in X$, $\text{inj}_M(x) > R$.*
- (2) *$M_{[\mu, \infty)} \subset X \subset M$, and X is obtained from M by drilling out short geodesics and truncating cusps. In particular:*
 - (a) *If $\gamma \subset M$ is a geodesic of length less than $2R$ then γ is drilled out.*
 - (b) *If $\gamma \subset M$ is a simple geodesic of length at least 2μ then γ is not drilled out.*

PROOF OF PROPOSITION 1.2. It follows from the decay of radius of injectivity (see, for example, [3, Proposition 4.19]) that there exists $R > 0$, depending only on μ and d , so that for any $x \in X$, $\text{inj}_M(x) > R$. This establishes (1).

By construction $M_{[\mu, \infty)} \subset X \subset M$. Let U be a component of $M_{(0, \mu]}$. The set of points removed from U is:

$$\{x \in U \mid d(x, M \setminus U) > d\}.$$

When U is a solid torus neighborhood of a closed short geodesic γ , the set of points removed is $\{x \in U \mid d(x, \gamma) \leq d(X, \gamma) - d\}$, and is either empty or an open solid torus neighborhood of γ . In the first case $U \subset X$ and in the second case we remove a neighborhood of γ . When U is a cusp, $(M \setminus X) \cap U$ is isotopic to U . This establishes (2).

Let $\gamma \subset M$ be a geodesic of length less than $2R$. Then for every $p \in \gamma$, $\text{inj}_M(p) < R$. By (1) above γ is drilled out. This establishes (2)(a).

Let $\gamma \subset M$ be a simple geodesic of length at least 2μ . It is clear from the definitions that if γ is drilled out then $\gamma \subset M_{[0, \mu]}$. A geodesic is contained in $M_{(0, \mu]}$ if and only if it covers a short geodesic (that is, has the form δ^n for some geodesic δ with $l(\delta) < 2\mu$ and some $n > 0$). Such a geodesic is simple if and only if $n = 1$; we conclude that simple geodesics in $M_{(0, \mu]}$ are shorter than 2μ . Thus $\gamma \not\subset M_{(0, \mu]}$, and it is not drilled out. This establishes 2(b). $\square$

We denote by $t_C(M)$ the minimal number of tetrahedra required to triangulate a link exterior in M , that is, the minimal number of tetrahedra required to triangulate $M \setminus \mathring{N}(L)$, where the minimum is taken over all links $L \subset M$ (possibly, $L = \emptyset$) and all possible triangulations. Similarly we define $t_{HC}(M)$ to be the minimal number of tetrahedra necessary to triangulate $M \setminus \mathring{N}(L)$, where $L \subset M$ ranges over all possible hyperbolic links. As a consequence of Theorem 1.1 we get the following corollary, showing that $\text{Vol}(M)$, $T_C(M)$, and $T_{HC}(M)$ are the same up-to linear equivalence:

COROLLARY 1.3. *Let $\mu > 0$ be a Margulis constant for $\mathbb{H}^3$ and fix $d > 0$. Let $K > 0$ be the constant given in Theorem 1.1 and v_3 be the volume of a regular ideal tetrahedron in $\mathbb{H}^3$. Then for any complete finite volume hyperbolic 3-manifold M we have:*

$$t_C(M) \leq t_{HC}(M) \leq K\text{Vol}(M) \leq Kv_3t_C(M) \leq Kv_3t_{HC}(M).$$

PROOF. The first and last inequalities are obvious.

By Proposition 1.2 (2), $X = N_d(M_{\geq \mu})$ is obtained from M by drilling out geodesics; hence by Kojima [5] it is a hyperbolic. Thus the second inequality follows directly for Theorem 1.1.

The proof of third inequality is well known (see, for example, Chapter C of [1]); we sketch its argument for the reader's convenience. Let $L \subset M$ be a link and $\mathcal{T}$ a triangulation of $M \setminus \mathring{N}(L)$ using $t_C(M)$ tetrahedra. Let Δ be a 3-simplex in $\mathbb{R}^3$ and denote the characteristic maps of the tetrahedra in $\mathcal{T}$ by $\delta_i : \Delta \rightarrow M \setminus \mathring{L}$ ($i = 1, \dots, t_C(M)$). Let $f : M \setminus \mathring{N}(L) \rightarrow M$ be the degree 1 map obtained by crushing each torus of $\partial(M \setminus \mathring{N}(L))$ to a circle¹. Denote $f \circ \delta_i : \Delta \rightarrow M$ by f_i ($i = 1, \dots, t_C(M)$).

With the notation as in the previous paragraph, we now prove the third inequality. Note that $\sum_{i=1}^n f_i$ represents a generator of $H_3(M) \cong \mathbb{Z}$. Let $\tilde{f}_i$ be a lift of f_i to the universal cover $\mathbb{H}^3$. We construct a map $\tilde{f}_i : \delta \rightarrow \mathbb{H}^3$ by “pulling $\tilde{f}$ tight”². Note that $\tilde{f}_i$ is homotopic to $\tilde{f}_i$; denote this homotopy by $\tilde{F}_{i,t}(p)$. Since $\sum_{i=1}^n f_i$ defines an element of $H_3(M)$, faces must cancel in pairs. Let F and F' be such a pair, that is, $f_i(F) = -f_j(F')$, and let $p \in F$ and $p' \in F'$ be corresponding points, that is, points with the same barycentric coordinates. By performing the homotopy at constant speed we obtain, for any $t \in [0, 1]$:

$$\pi \circ \tilde{F}_{i,t}(p) = \pi \circ \tilde{F}_{j,t}(p').$$

Here π is the universal cover projection. This implies that for any $t \in [0, 1]$, $\sum_{i=1}^{t_C(M)} \pi \circ \tilde{F}_{i,t}(p)$ is homologous to $\sum_{i=1}^n f_i$ and therefore represents a generator of $H_3(M)$. For $t = 1$, we see that $\sum_{i=1}^{t_C(M)} \pi \circ \tilde{f}_i$ represents a generator for $H_3(M)$; in particular, every point of M is in the image of at least one $\pi \circ \tilde{f}_i$. Hence the sum of the volumes of the images of $\pi \circ \tilde{f}_i$ is no less than $\text{Vol}(M)$. Using this, the fact that volumes do not increase under π , and the fact that the volume of a hyperbolic tetrahedron is less than v_3 we get:

$$\begin{aligned} \text{Vol}(M) &\leq \sum_{i=1}^{t_C(M)} \text{Vol}(\pi \circ \tilde{f}_i) \\ &\leq \sum_{i=1}^{t_C(M)} \text{Vol}(\tilde{f}_i) \\ &< t_C(M)v_3. \end{aligned}$$

The third inequality follows. □

The proposition below is the key to the proof of Theorem 1.1 and is very useful in its own right. For this proposition we need the following notation, that we will use throughout this paper. Fix a Margulis constant $\mu > 0$ and $d > 0$, and let $R > 0$ be as in Proposition 1.2 (1). We define $D = \min\{R, d\}$. A set A in a metric space is called *D-separated* if for any $p, q \in A$, $p \neq q$, we have that $d(p, q) > D$. Fix $\{x_1, \dots, x_N\} \subset X$ a generic set of N points (*a-priori* N may be infinite) fulfilling the following conditions:

- (1) $\{x_1, \dots, x_N\} \subset X$ is *D-separated*.
- (2) $\{x_1, \dots, x_N\}$ is maximal (with respect to inclusion) subject to this constraint.

¹That is, f is obtained as following: foliate each component of $\partial M \setminus \mathring{N}(L)$ by circles, where each leaf is isotopic to a meridian on M . Then f is obtained by identifying each leaf to a point.

²By “pulling $\tilde{f}$ tight” we mean: for $p \in \Delta$ a vertex, then $\tilde{f}(p) = \tilde{f}(p)$. Next, for a general point $p \in \Delta$, $\tilde{f}(p)$ is the unique point of $\mathbb{H}^3$ that has the same barycentric coordinates as p (for more on barycentric coordinate see, for example, [4, Page 103]).

Let $V_1, \dots, V_N$ be the *Voronoi cells* in M corresponding to $\{x_1, \dots, x_N\}$, that is,

$$V_i = \{p \in M \mid d_M(p, x_i) \leq d_M(p, x_j) \ (j = 1, \dots, N)\}.$$

We emphasize that *a-priori* a Voronoi cell need not be “nice”; for example, it need not be a ball and may have infinite diameter. Consider the following simple example: given any metric space and a single point in it, the Voronoi cell corresponding to that point is the entire space.

PROPOSITION 1.4. *With the notation of Theorem 1.1, there exists a constant $C = C(\mu, d)$ so that the following holds:*

- (1) M is decomposed into $N \leq C \text{Vol}(M)$ Voronoi cells.
- (2) $V_i \cap X$ is triangulated using at most C tetrahedra ($i = 1, \dots, N$) ($V_i \cap X$ may not be connected).
- (3) For any i, i' ($i, i' = 1, \dots, N$), the triangulations of $V_i \cap X$ and $V_{i'} \cap X$ given in (2) above coincide on $(V_i \cap X) \cap (V_{i'} \cap X)$.

We note that Theorem 1.1 follows easily from Proposition 1.4 by setting $K = C^2$.

Structure of this paper. In Section 2 we cover some basic preliminary notion. In Section 3 we describe the decomposition of M into Voronoi cells. In an attempt to make this paper self-contained and accessible to all we provide proofs for many elementary facts about Voronoi cells. In Section 4 we study the intersection of the Voronoi cells with X . In Section 5 we prove Proposition 1.4.

Strategy. As mentioned, our approach is based on Thurston’s original work. However, as discussed in [1] pp 190–192, to make this work requires control over $V_i \cap X$. We now briefly explain our strategy for obtaining this control.

We first decompose M into the N Voronoi cells described above. An easy volume argument shows that N is bounded above linearly in terms of the volume of M . We then show the following:

- (1) Every component of $V_i \cap X$ is a handlebody, as we will show it deformation retracts onto a surface contained in ∂X .
- (2) There is a universal bound on the number of components of $V_i \cap X$.
- (3) There is a universal bound on the genus of each component of $V_i \cap X$.

We obtain a certain cell decomposition of $V_i \cap X$.

Of course it is possible to triangulate $V_i \cap X$ with a bounded number of tetrahedra, but that is not quite enough: the triangulations must agree on intersection in order to yield a triangulation of X . (Consider a lens space: it is the union of two solid tori, but as there are infinitely many distinct lens spaces, they require arbitrarily many tetrahedra.) We triangulate $V_i \cap X$ in a way that agrees on intersections using the cell decomposition mentioned in the previous paragraph.

To get a bound on the number of tetrahedra, we observe that the faces of the cell decomposition mentioned above are totally geodesic. This is used to bound the number of vertices, which turns out to be the key for bounding the number of tetrahedra in our setting.

By contrast, when considering the cell decomposition of the lens space $L(p, q)$ obtained by taking two solid tori and a meridian disk for each, the number of vertices is not bounded; it equals the number of intersections between the disks, which is p .

A note on notation. Objects in $\mathbb{H}^3$ are denoted using tilde (for example, $\tilde{s}$ or $\tilde{C}$) or using script lettering (for example, $\mathcal{A}$). Constants denoted by C are universal (namely, C as defined in Proposition 1.4, C_3 as define in Lemma 3.1, C_2 as define in Lemma 4.3, C_1 as define in Lemma 4.4, C_0 as define in Lemma 4.5, and $\tilde{C}_0$ as defined in Section 5). Once defined they are fixed for the remainder of the paper. The constants μ , d , R , and D that were introduced in This section are fixed throughout this paper.

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2. preliminaries

The notation of Section 1 is fixed for the remainder of this paper.

We assume familiarity with hyperbolic space $\mathbb{H}^3$ and its isometries, as well as the Margulis lemma. The model of $\mathbb{H}^3$ we use is upper half space. Given $\tilde{x}, \tilde{y} \in \mathbb{H}^3$, we denote the closed geodesic segment connecting them by $[\tilde{x}, \tilde{y}]$. All manifolds considered are assumed to be orientable. In a metric space, $N_d(\cdot)$ denotes the set of all points of distance at most d from a given object. The ball of radius r centered at x is denoted $B(x, r)$. The volume of a ball of radius r in $\mathbb{H}^3$ is denoted by $\text{Vol}(B(r))$. We use the notation $\text{int}(\cdot)$ and $\text{cl}(\cdot)$ for *interior* and *closure*.

We fix $\mu > 0$ a Margulis constant for $\mathbb{H}^3$. By *hyperbolic manifold* M we mean a complete, finite volume Riemannian 3-manifold locally isometric to $\mathbb{H}^3$. The universal covering of a hyperbolic manifold M is denoted $\pi : \mathbb{H}^3 \rightarrow M$; π is called *the universal cover projection*, or simply *the projection*, from $\mathbb{H}^3$ to M . The *thick part* of M is

$$M_{[\mu, \infty)} = \{p \in M \mid \text{inj}_M(p) \geq \mu\}.$$

The *thin part* of M is

$$M_{(0, \mu]} = \text{cl}\{p \in M \mid \text{inj}_M(p) < \mu\} = \text{cl}(M \setminus M_{[\mu, \infty)}).$$

It is well known that $M = M_{(0, \mu]} \cup M_{[\mu, \infty)}$, $M_{(0, \mu]}$ is a disjoint union of closed solid torus neighborhood of short geodesics and closed cusps, and $M_{(0, \mu]} \cap M_{[\mu, \infty)}$ consists of tori.

3. Voronoi Cells

Keep all notation as in the pevious sections, and recall that N was the number of Voronoi cells and $D = \min\{R, d\}$. Since $\{x_1, \dots, x_N\}$ was chosen generically, we may assume that the Voronoi cells $\{V_i\}$ are transverse to each other and to ∂X (note that the Voronoi cells are a decomposition of M , not X , and $\partial X \subset \text{int}(M)$). In the remainder of the paper, all our constructions are generic and allow for small perturbation, and we always assume transversality (usually without explicit mention). We bound N in term of the volume of M :

LEMMA 3.1. *There exists a constant C_3 so that $N \leq C_3 \text{Vol}(M)$.*

PROOF. For each i , $x_i \in X$, and hence by Proposition 1.2 (1), $\text{inj}_M(x_i) > R \geq D$. Since $\{x_1, \dots, x_N\}$ is D -separated, for $i \neq j$, $d(x_i, x_j) > D$. Hence

$\{B(x_i, D/2)\}_{i=1}^N$ is a set of balls disjointly embedded in M , each of volume $\text{Vol}(B(D/2))$. Thus $N \leq \text{Vol}(M)/\text{Vol}(B(D/2))$; the lemma follows by setting

$$C_3 = 1/\text{Vol}(B(D/2)).$$

□

The preimages of $\{x_1, \dots, x_N\}$ in $\mathbb{H}^3$ gives rise to a Voronoi cell decomposition of $\mathbb{H}^3$ in a similar manner to the cells in M . It is convenient to fix one of these cells for each i :

NOTATION 3.2. (1) For each i , fix a preimage of x_i , denoted $\tilde{x}_i$.

(2) $\tilde{V}_i$ is the Voronoi cell corresponding to $\tilde{x}_i$, that is:

$$\tilde{V}_i = \{\tilde{p} \in \mathbb{H}^3 \mid d(\tilde{p}, \tilde{x}_i) \leq d(\tilde{p}, \tilde{q}), \forall \tilde{q} \text{ so that } \pi(\tilde{q}) \in \{x_1, \dots, x_N\}\}$$

(3) For each i , the components of $V_i \cap X$ are denoted by $V_{i,j}$ ($j = 1, \dots, n_i$), where n_i is the number of the components of $V_i \cap X$.

(4) The preimage of $V_{i,j}$ in $\tilde{V}_i$ is denoted $\tilde{V}_{i,j}$, that is:

$$\tilde{V}_{i,j} = \{\tilde{p} \in \tilde{V}_i \mid \pi(\tilde{p}) \in V_{i,j}\}.$$

LEMMA 3.3. *If $\tilde{p}, \tilde{p}' \in \tilde{V}_i$ project to the same point $p \in V_i$ then $d(\tilde{p}, \tilde{x}_i) = d(\tilde{p}', \tilde{x}_i)$.*

PROOF. Let $\tilde{p}, \tilde{p}'$ be points in $\tilde{V}_i$ that project to the same point and assume that $d(\tilde{x}_i, \tilde{p}) \neq d(\tilde{x}_i, \tilde{p}')$; say $d(\tilde{x}_i, \tilde{p}) < d(\tilde{x}_i, \tilde{p}')$. Since $\tilde{p}$ and $\tilde{p}'$ project to the same point, there is an isometry $\phi \in \pi_1(M)$ so that $\phi(\tilde{p}) = \tilde{p}'$. Let $\tilde{x}'_i = \phi(\tilde{x}_i)$, for some $\tilde{x}'_i \in \pi^{-1}(x_i)$. Since ϕ acts freely $\tilde{x}'_i \neq \tilde{x}_i$. We get: $d(\tilde{x}'_i, \tilde{p}') = d(\phi^{-1}(\tilde{x}'_i), \phi^{-1}(\tilde{p}')) = d(\tilde{x}_i, \tilde{p}) < d(\tilde{x}_i, \tilde{p}')$. Hence $\tilde{p}' \notin \tilde{V}_i$, contradicting our assumption. The lemma follows. □

In general, the distance between points in V_i may be smaller than the distance between their preimages in $\tilde{V}_i$. However this is not the case when one of the points is x_i :

LEMMA 3.4. *For any $\tilde{V}_i$ and any $\tilde{p} \in \tilde{V}_i$, $d(\tilde{x}_i, \tilde{p}) = d(x_i, p)$ (here $p = \pi(\tilde{p})$).*

PROOF. Of all paths from x_i to p in M , let β be one that minimizes length (note that β need not be unique). First we claim that $\beta \subset V_i$. Suppose, for a contradiction, that this is not the case and let $q \in \beta$ be a point not in V_i . Then for some $j \neq i$, $d(q, x_j) < d(q, x_i)$. By connecting the shortest path from p to q to the shortest path from q to x_j we obtain a path strictly shorter than β , showing that $d(p, x_j) < l(\beta) = d(p, x_i)$. Thus $p \notin V_i$, a contradiction. Hence $\beta \subset V_i$.

Let $\tilde{\beta}$ be the lift of β to $\mathbb{H}^3$ starting at $\tilde{x}_i$. Then $\tilde{\beta}$ is a geodesic segment, say $[\tilde{x}_i, \tilde{p}']$, for some $\tilde{p}'$ that projects to p . Fix $\tilde{q} \in \{\pi^{-1}(x_1), \dots, \pi^{-1}(x_n)\}$. Then $[\tilde{p}', \tilde{q}]$ projects to $\pi([\tilde{p}', \tilde{q}])$, a path that connects p to some point of $\{x_1, \dots, x_n\}$. By choice of β (and since paths have the same length as their projections), $d(\tilde{x}_i, \tilde{p}') = l(\tilde{\beta}) = l(\beta) \leq l(\pi([\tilde{p}', \tilde{q}])) = l([\tilde{p}', \tilde{q}]) = d(\tilde{p}', \tilde{q})$. We conclude that $\tilde{p}' \in \tilde{V}_i$.

We see that $d(x_i, p) = l(\beta) = l(\tilde{\beta}) = d(\tilde{x}_i, \tilde{p}')$. Since $\tilde{p}, \tilde{p}' \in \tilde{V}_i$, by Lemma 3.3 $d(\tilde{x}_i, \tilde{p}) = d(\tilde{x}_i, \tilde{p}')$; the lemma follows. □

A *convex polyhedron* is the intersection of half spaces in $\mathbb{H}^3$. Note that a convex polyhedron is not required to be of bounded diameter or finite sided (that is, the intersection of finitely many half spaces).

LEMMA 3.5. $\tilde{V}_i$ is a convex polyhedron that projects onto V_i

PROOF. It is immediate that $\tilde{V}_i$ is a convex polyhedron.

Given any $\tilde{p} \in \tilde{V}_i$, $[\tilde{x}_i, \tilde{p}]$ is the shortest geodesic from $\tilde{p}$ to any preimage of $\{x_1, \dots, x_N\}$. The projection of $[\tilde{x}_i, \tilde{p}]$ is the shortest geodesic from the projection of $\tilde{p}$ to $\{x_1, \dots, x_N\}$. It follows easily that $\pi(\tilde{p}) \in V_i$. As $\tilde{p}$ was an arbitrary point of $\tilde{V}_i$, we see that $\tilde{V}_i$ projects into V_i .

Conversely, given any $p \in V_i$, let β be the shortest geodesic from $\{x_1, \dots, x_N\}$ to p . Then β connects x_i to p . Let $\tilde{\beta}$ be the unique lift of β that starts at $\tilde{x}_i$, and denote its terminal point by $\tilde{p}$. Similar to the argument of the proof of Lemma 3.4, $\tilde{\beta}$ is the shortest geodesic connecting any preimage of $\{x_1, \dots, x_N\}$ to $\tilde{p}$, showing that $\tilde{p} \in \tilde{V}_i$. Hence p is in the image is $\tilde{V}_i$; As p was an arbitrary point of V_i , we see that $\tilde{V}_i$ projects onto V_i . $\square$

Decomposition of V_i . By Lemma 3.5, the boundary of $\tilde{V}_i$ is decomposed into faces, edges and vertices. By the same lemma, it projects into V_i . The images of this faces, edges and vertices from the decomposition of $\tilde{V}_i$ that is the basis for our work in the next section. Note that some faces of $\tilde{V}_i$ are identified, and the corresponding faces of V_i are contained in the interior, not boundary, of V_i . (We will show in Lemma 4.9 (3) that faces in the interior of V_i are contained in $M \setminus X$, and they will play no role in our construction.) We remark that this is not the final decomposition: in the next section we will add more faces, edges and vertices to the decomposition.

4. Decomposing X

We first define:

DEFINITION 4.1. Fix $r > 0$ and a geodesic $\tilde{\gamma} \subset \mathbb{H}^3$. Let $\tilde{C} = \{p \in \mathbb{H}^3 | d(p, \tilde{\gamma}) \leq r\}$. We call $\tilde{C}$ a *cone* about γ , or simply a cone³, and $\tilde{\gamma}$ the *axis* of $\tilde{C}$. The set $\{p \in \mathbb{H}^3 | d(p, \gamma) \geq r\}$ is called *exterior* of $\tilde{C}$, denoted $\tilde{E}$.

The reason we look at cones is that if V is a solid torus component of $\text{cl}(M \setminus X)$ and γ its core geodesic, then $\pi^{-1}(V)$ is a cone and $\pi^{-1}(\gamma)$ its axis. It can be seen directly that the intersection of a geodesic and a cone is a (possibly empty) connected set; hence cones are convex. If V is a cusp component of $\text{cl}(M \setminus X)$, then its preimage is a horoball which is also convex. Below, we often use the fact that the every component of the preimage of $\text{cl}(M \setminus X)$ is convex.

LEMMA 4.2. For any i , $V_i \cap \text{cl}(M \setminus X)$ is connected.

PROOF. The number of times V_i intersects $\text{cl}(M \setminus X)$ is at most the number of times $\tilde{V}_i$ intersects the preimage of $\text{cl}(M \setminus X)$. Since $\tilde{V}_i$ and any component of the preimage of $\text{cl}(M \setminus X)$ are both convex, their intersection is connected. Thus

³In the upper half space model, if $\tilde{\gamma}$ is a Euclidean vertical straight ray from $\tilde{p}_\infty$ in the xy -plane, then $\tilde{C}$ is the cone of all Euclidean straight rays from $\tilde{p}_\infty$ that form angle at most α (for some α) with $\tilde{\gamma}$. If $\tilde{\gamma}$ is a semicircle then $\tilde{C}$ looks more like a banana.

all we need to show is that $\tilde{V}_i$ intersects only one component of the preimage of $\text{cl}(M \setminus X)$.

Suppose this is not the case, and let $\tilde{\alpha}$ be the shortest arc in $\tilde{V}_i$ that connects distinct components of the preimage of $\text{cl}(M \setminus X)$. Since $\tilde{V}_i$ is convex, $\tilde{\alpha}$ is a geodesic. Since $\tilde{\alpha}$ connects *distinct* components of the preimage of $\text{cl}(M \setminus X)$, some point on $\tilde{\alpha}$ projects into $M_{[\mu, \infty)}$. Let $\alpha = \pi(\tilde{\alpha})$. Recall that the distance from ∂X to $M_{[\mu, \infty)}$ is d . We conclude that $l(\tilde{\alpha}) = l(\alpha) > 2d \geq 2D$. Thus the distance between the endpoints of $\tilde{\alpha}$ is greater than $2D$, and by the triangle inequality, for some point $\tilde{p} \in \tilde{\alpha}$, $d(\tilde{x}_i, \tilde{p}) > D$. Let p be the image of $\tilde{p}$. By Lemma 3.4, $d(x_i, p) = d(\tilde{x}_i, \tilde{p})$ and by Lemma 3.5 $p \in V_i$. Hence by construction of the Voronoi cells, for any j , $d(x_j, p) \geq d(x_i, p) > D$. Thus $\{x_1, \dots, x_n, p\} \subset X$ is a D -separated set, contradicting maximality of $\{x_1, \dots, x_N\}$. The lemma follows. $\square$

In the next lemma we bound the number of faces of V_i that intersect X and study that intersection:

LEMMA 4.3. *The following two conditions hold:*

- (1) *There exists a constant C_2 so that for every i , $1 \leq i \leq N$, the number of faces of V_i that intersect X is at most C_2 .*
- (2) *For each i and every face F of V_i , $F \cap X$ is either empty, or a single annulus, or a collection of disks.*

PROOF. Let V_i be a Voronoi cell, F a face of V_i so that $F \cap X \neq \emptyset$, and $p \in F \cap X$. Let $\tilde{p} \in \tilde{V}_i$ be a preimage of p ($\tilde{p}$ exists by Lemma 3.5) and let $\tilde{F}$ be a face of $\tilde{V}_i$ containing $\tilde{p}$. Let $\tilde{x}$ be the preimage of $\{x_1, \dots, x_N\}$ that is contained in the cell adjacent to $\tilde{F}$ on the opposite side from $\tilde{V}_i$. By Lemma 3.4, $d(\tilde{x}_i, \tilde{p}) = d(x_i, p)$. Similar to the argument of the proof of Lemma 4.2, maximality of $\{x_1, \dots, x_N\}$ implies that $d(x_i, p) < D$. We conclude that $d(\tilde{x}_i, \tilde{p}) < D$, and similarly $d(\tilde{x}, \tilde{p}) < D$. By the triangle inequality, $d(\tilde{x}_i, \tilde{x}) < 2D$.

For each face F of V_i with $F \cap X \neq \emptyset$, consider the cell adjacent to $\tilde{V}_i$ along $\tilde{F}$ as constructed above. The balls of radius $D/2$ centered at the preimages of $\{x_1, \dots, x_N\}$ in these cells are disjointly embedded and their centers are no further than $2D$ from $\tilde{x}_i$, so these balls are contained in $B(\tilde{x}_i, 2.5D)$. Thus (1) follows by setting

$$C_2 = \text{Vol}(B(2.5D)) / \text{Vol}(B(D/2)).$$

For (2), fix V_i and F a face of V_i . Let $\tilde{F}$ be the face of $\tilde{V}_i$ that projects to F . Since $\tilde{V}_i$ is a convex polyhedron, $\tilde{F}$ is a totally geodesic convex polygon. By Lemma 4.2, $\tilde{F}$ intersects at most one component of the preimage of $M \setminus X$, and by convexity of that component and of $\tilde{F}$, the intersection is either empty or a disk. We see that one of the following holds:

- (1) When the intersection is empty: then the intersection of $\tilde{F}$ with the preimage of X is $\tilde{F}$ (and hence a disk).
- (2) When the intersection is a disk contained in $\text{int}(\tilde{F})$: then the intersection of $\tilde{F}$ with the preimage of X is an annulus.
- (3) When the intersection is a disk not contained in $\text{int}(\tilde{F})$: then the intersection of $\tilde{F}$ with the preimage of X is a collection of disks.

We claim that the intersection of $\tilde{F}$ with the preimage of X projects homeomorphically onto its image. Otherwise, there are two points $\tilde{p}_1, \tilde{p}_2 \in \tilde{F}$ that project

to the same point $p \in F \cap X$. By maximality of $\{x_1, \dots, x_N\}$, $d(x_i, p) < D$. By Lemma 3.4, $d(\tilde{x}_i, \tilde{p}_1), d(\tilde{x}_i, \tilde{p}_2) = d(x_i, p)$. The shortest path from $\tilde{p}_1$ to $\tilde{p}_2$ that goes through $\tilde{x}_i$ projects to an essential closed path that contains x_i and has length less than $2D$. But then $\text{inj}_M(x_i) < D \leq R$, contradicting Proposition 1.2 (1). Thus the intersection of the preimage of X with $\tilde{F}$ projects homeomorphically and (2) follows. $\square$

We consider the intersection of an edge e of V_i with X . We call the components of $e \cap X$ *segments*. In the next lemma we bound the number of segments:

LEMMA 4.4. *There exists a constant C_1 so that for every i , $1 \leq i \leq N$, the number of segments from the intersection of edges of V_i with X is at most C_1 .*

PROOF. Fix i and e an edge of V_i . We first show that e contributes at most two segment. If $e \subset X$ then it contributes exactly one segment and if $e \cap X = \emptyset$ then it contributes no segment. Otherwise, let $\tilde{e}$ be a lift of e that is in $\tilde{V}_i$. By Lemma 4.2, $\tilde{e}$ intersects at most one component of the preimage of $\text{cl}(M \setminus X)$. Since $\tilde{e}$ and any component of the preimage of $\text{cl}(M \setminus X)$ are both convex, their intersection is convex and hence connected. Thus the intersection of $\tilde{e}$ and the preimage of $\text{cl}(M \setminus X)$ is connected, and projecting to M we see that the intersection of e and $\text{cl}(M \setminus X)$ is connected as well. Thus e contributes at most 2 segments.

Since $\tilde{V}_i$ is convex, the intersection of 2 faces of $\partial\tilde{V}_i$ is at most one edge. Hence the number of edges is bounded above by the the number of pairs of faces, $\frac{1}{2}C_2(C_2 - 1)$ (using Lemma 4.3). The number of edges of V_i is no larger; Lemma 4.4 follows by setting

$$C_1 = C_2(C_2 - 1).$$

$\square$

LEMMA 4.5. *There exists a constant C_0 so that for every i , $1 \leq i \leq N$, the number of vertices of V_i that lie in X is at most C_0 .*

PROOF. Each segment contributes at most 2 vertices. Lemma 4.5 follows by setting

$$C_0 = 2C_1.$$

$\square$

DEFINITION 4.6. Let $\tilde{C}$ be a cone, $\tilde{\gamma}$ its axis, and $\tilde{E}$ its exterior (recall Definition 4.1). Fix $\tilde{s} \in \tilde{C}$. We say that a set $\tilde{K} \subset \tilde{E}$ is *$\tilde{s}$ -convex* if for any $\tilde{p} \in \tilde{K}$, $[\tilde{p}, \tilde{s}] \cap \tilde{E}$ is contained in $\tilde{K}$.

LEMMA 4.7. *Let $\tilde{K} \subset \tilde{E}$ be an $\tilde{s}$ -convex set (for some $\tilde{s} \in \tilde{C}$). Then there exists a deformation retract from $\tilde{K}$ onto $\tilde{K} \cap \partial\tilde{E}$.*

PROOF. Fix $\tilde{p} \in \tilde{K}$. Since $\tilde{C}$ is convex, $[\tilde{p}, \tilde{s}] \cap \tilde{C}$ is an interval, say $[\tilde{r}, \tilde{s}]$, and $\tilde{E} \cap [\tilde{r}, \tilde{s}] = [\tilde{p}, \tilde{r}]$. Since $\tilde{K}$ is $\tilde{s}$ -convex, $[\tilde{p}, \tilde{r}] \subset \tilde{K}$. We move $\tilde{p}$ along $[\tilde{p}, \tilde{r}]$ from its original position to $\tilde{r} \in \tilde{K} \cap \partial\tilde{E}$ in constant speed. It is easy to see that this is a deformation retract.

$\square$

NOTATION 4.8. With the notation of the previous lemma, we define $f : \tilde{K} \rightarrow \tilde{K} \cap \partial\tilde{E}$ to be $f(\tilde{p}) = \tilde{r}$.

Recall the definition of $\tilde{V}_i$, $V_{i,j}$, and $\tilde{V}_{i,j}$ from Notation 3.2. Recall also that n_i was the number of components of $V_i \cap X$ from Notation 3.2 (3).

LEMMA 4.9. *The following conditions hold:*

- (1) *For each i , $n_i \leq C_0$.*
- (2) *For each i, j , if $\tilde{V}_{i,j} \neq \tilde{V}_i$, there is a cone $\tilde{C}$, a component of the preimage of $\text{cl}(M \setminus X)$, so that $\tilde{V}_{i,j}$ is $\tilde{s}$ convex for any point $\tilde{s} \in \tilde{V}_i \cap \tilde{C}$.*
- (3) *The projection of $\cup_{j=1}^{n_i} \tilde{V}_{i,j}$ to $\cup_{j=1}^{n_i} V_{i,j}$ is a diffeomorphism.*
- (4) *For each i, j , $V_{i,j}$ is a handlebody.*

PROOF. Fix i . It is easy to see that each component of $V_i \cap X$ must contain a vertex of V_i . Applying Lemma 4.5 we see that there are at most C_0 such components. This establishes (1).

Any component of the preimage of $\text{cl}(M \setminus X)$ is a cone. Assuming that $V_{i,j} \neq V_i$, by Lemma 4.2 there exists a unique such component, say $\tilde{C}$, that intersects $\tilde{V}_i$. Fix a point $\tilde{s} \in \tilde{C} \cap \tilde{V}_i$. Fix $\tilde{p} \in \tilde{V}_{i,j}$. Convexity of $\tilde{V}_i$ implies that $[\tilde{p}, \tilde{s}] \subset \tilde{V}_i$. Convexity of $[\tilde{p}, \tilde{s}]$ and $\tilde{C}$ implies that $[\tilde{p}, \tilde{s}] \cap \tilde{C}$ is an interval, say $[\tilde{r}, \tilde{s}]$, and therefore $[\tilde{p}, \tilde{s}] \cap (\tilde{V}_i \setminus \text{int}\tilde{C}) = [\tilde{p}, \tilde{r}]$. Since $\tilde{V}_{i,j}$ is connected, $[\tilde{p}, \tilde{s}] \cap \tilde{V}_{i,j} = [\tilde{p}, \tilde{r}] \subset \tilde{V}_{i,j}$. This establishes (2).

For (3), it is easy to see that all we need to show is that the projection $\cup_{j=1}^{n_i} \tilde{V}_{i,j} \rightarrow \cup_{j=1}^{n_i} V_{i,j}$ is one-to-one. Assume not (this is similar to Lemma 4.3 (2)); then there exist $\tilde{p}_1, \tilde{p}_2 \in \cup_{j=1}^{n_i} \tilde{V}_{i,j}$ that project to the same point $p \in \cup_{j=1}^{n_i} V_{i,j}$. Then the shortest path from $\tilde{p}_1$ to $\tilde{p}_2$ that goes through $\tilde{x}_i$ projects to an essential closed path that contains x_i , and has length less than $2D$. But then $\text{inj}_M(x_i) < D \leq R/2$, contradicting Proposition 1.2 (1). This establishes (3).

If $V_i \subset X$ then $V_i \cap X$ is a ball and (4) follows. Otherwise, (4) follows from (2), Lemma 4.7, and (3). $\square$

We denote $V_{i,j} \cap \partial X$ by $P_{i,j}$. We bound $g(V_{i,j})$, the genus of $V_{i,j}$:

LEMMA 4.10. $g(V_{i,j}) \leq C_1$.

PROOF. If $V_{i,j} = V_i$ then it is a ball and there is nothing to show. Assume this is not the case. Then by Lemmas 4.9 and 4.7 $V_{i,j}$ deformation retracts onto $P_{i,j}$. Hence $\text{cl}(\partial V_{i,j} \setminus P_{i,j})$ is homeomorphic to $P_{i,j}$, and is a $g(V_{i,j})$ -times punctured disk. The faces of V_i induce a decomposition on $\text{cl}(\partial V_{i,j} \setminus P_{i,j})$. By Lemma 4.3 (2), each face of $\text{cl}(\partial V_{i,j} \setminus P_{i,j})$ is a disk or an annulus; in particular the Euler characteristic of each such component is non-negative. Denote the faces of $\text{cl}(\partial V_{i,j} \setminus P_{i,j})$ by $\{F_k\}_{k=1}^{k_0}$, the number of edges by e , and the number of vertices by v . Note further, that the edges of $\text{cl}(\partial V_{i,j} \setminus P_{i,j})$ come in two types, edges in the interior of $\text{cl}(\partial \tilde{V}_{i,j} \setminus P_{i,j})$ (say e_{int} of them) and edges on its boundary (say e_{∂} of them). Similarly, v_{int} (resp. v_{∂}) denotes the number of vertices in the interior (resp. boundary) of $\text{cl}(\partial \tilde{V}_{i,j} \setminus P_{i,j})$. Since the boundary of $\text{cl}(\partial V_{i,j} \setminus P_{i,j})$ consists of circles, $e_{\partial} = v_{\partial}$. Since e_{int} is the number of segments on $V_{i,j} \cap X$, by Lemma 4.4, $e_{\text{int}} \leq C_1$. An Euler characteristic calculation gives:

$$\begin{aligned}
1 - g(V_{i,j}) &= \chi(\text{cl}(\partial V_{i,j} \setminus P_{i,j})) && \text{as } \text{cl}(\partial V_{i,j} \setminus P_{i,j}) \text{ is a } g(V_{i,j})\text{-times disk} \\
&= (\Sigma_{k=1}^{k_0} \chi(F_k)) - e + v \\
&= (\Sigma_{k=1}^{k_0} \chi(F_k)) - (e_{\text{int}} + e_{\partial}) \\
&\quad + (v_{\text{int}} + v_{\partial}) \\
&= (\Sigma_{k=1}^{k_0} \chi(F_k)) - e_{\text{int}} + v_{\text{int}} && e_{\partial} = v_{\partial} \\
&\geq -e_{\text{int}} + 1 && \Sigma_{k=1}^{k_0} \chi(F_k) \geq 0 \text{ and } v_{\text{int}} > 0 \\
&\geq -C_1 + 1. && \text{Lemma 4.4}
\end{aligned}$$

The lemma follows. $\square$

We use the notation ∂_{∞} for the limit points at infinity. In particular, if $\tilde{L}$ is a totally geodesic plane then $\partial_{\infty}\tilde{L}$ is a simple closed curve and if $\tilde{C}$ is a cone then $\partial_{\infty}\tilde{C}$ consists of two points.

LEMMA 4.11. *Let $\tilde{C}$ be a cone, $\tilde{s} \in \tilde{C}$ a point not on the axis of $\tilde{C}$. Let $\tilde{\beta} \subset \partial\tilde{C}$ be an arc and $\tilde{p} \in \tilde{\beta}$, $\tilde{q} \in \partial\tilde{C}$ points. Assume that the totally geodesic plane that contains $\tilde{s}$ and $\partial_{\infty}\tilde{C}$ intersects $\tilde{\beta}$ in a finite set of points.*

Then after an arbitrarily small perturbation of $\tilde{p}$ in $\tilde{\beta}$, there exists a closed arc $\tilde{\alpha} \subset \partial\tilde{C}$ connecting $\tilde{p}$ and $\tilde{q}$, so that $\tilde{\alpha}$ and $\tilde{s}$ are contained in a totally geodesic plane.

REMARK. The condition on $\tilde{\beta}$ is generic, and since we allow for small perturbations in our construction we will always assume it holds.

PROOF. Let $\tilde{L}$ be a totally geodesic plane containing $\tilde{p}$, $\tilde{q}$, and $\tilde{s}$. We prove Lemma 4.11 in two cases:

Case One. $\partial_{\infty}\tilde{C} \not\subset \partial_{\infty}\tilde{L}$. The reader can easily verify that in this case $\tilde{L} \cap \partial\tilde{C}$ is connected. Then we take the arc $\tilde{\alpha}$ to be a component of $\tilde{L} \cap \partial\tilde{C}$ that connects $\tilde{p}$ to $\tilde{q}$.

Case Two. $\partial_{\infty}\tilde{C} \subset \partial_{\infty}\tilde{L}$. Equivalently, $\tilde{\gamma} \subset \tilde{L}$, where $\tilde{\gamma}$ denotes the axis of $\tilde{C}$. Since $\tilde{s} \notin \tilde{\gamma}$, $\tilde{L}$ is the *unique* totally geodesic plane that contains both $\tilde{s}$ and $\tilde{\gamma}$. By assumption, $\tilde{\beta} \cap \tilde{L}$ is a finite set. By perturbing $\tilde{p}$ slightly in $\tilde{\beta}$ we reduce the problem to Case One. The lemma follows. $\square$

In the following lemma we construct the main tool we will use for cutting $V_{i,j}$ into balls. In that lemma, $\tilde{C}$ is a cone and $\tilde{s} \in \tilde{C}$ a point. A collection of simple closed curves on $\partial\tilde{C}$ is called *generic* if it intersects any totally geodesic plane containing $\tilde{s}$ in a finite collection of points. As remarked after Lemma 4.11, we will always assume it holds.

LEMMA 4.12. *Let $\tilde{C}$ be a cone, $\tilde{s} \in \tilde{C}$ a point not on the axis of $\tilde{C}$, and $\mathcal{C} \subset \partial\tilde{C}$ a collection of $n + 1$ disjoint generic simple closed curves, for some $n \geq 0$.*

Then there exists a graph $\mathcal{A} \subset \partial\tilde{C}$ with the following properties:

- (1) $\mathcal{A}$ has at most $2n - 1$ edges.
- (2) For every edge $\tilde{e}$ of $\mathcal{A}$, $\tilde{e}$ and $\tilde{s}$ are contained in a single totally geodesic plane.
- (3) $\mathcal{C} \cup \mathcal{A}$ is a connected trivalent graph.

(4) *The graph obtained by removing any edge of $\mathcal{A}$ from $\mathcal{A} \cup \mathcal{C}$ is disconnected.*

PROOF. We induct on n . If $n = 0$ there is nothing to prove.

Assume $n > 0$ and let $\tilde{c}$ be a component of $\mathcal{C}$. Let $\mathcal{C}' = \mathcal{C} \setminus \tilde{c}$. By the induction hypothesis, there exists a graph $\mathcal{A}' \subset \partial\tilde{C}$ with at most $2n - 3$ edges, so that every edge of $\mathcal{A}'$ is contained in a totally geodesic plane that contains $\tilde{s}$, and $\mathcal{C}' \cup \mathcal{A}'$ is a connected trivalent graph.

Case One. $\tilde{c} \cap (\mathcal{C}' \cup \mathcal{A}') = \emptyset$. Fix $\tilde{p} \in \tilde{c}$ and $\tilde{q} \in \mathcal{C}' \cup \mathcal{A}'$ so that $\tilde{q}$ is not a vertex. Since $\tilde{c}$ is generic, by Lemma 4.11 after a small perturbation of $\tilde{p}$ in $\tilde{c}$, there exists an arc $\tilde{\alpha}'$ connecting $\tilde{p}$ and $\tilde{q}$ so that $\tilde{\alpha}'$ and $\tilde{s}$ are contained in a totally geodesic plane. Since the perturbation was generic we may assume that $\tilde{\alpha}'$ is transverse to $\mathcal{C}' \cup \mathcal{A}' \cup \tilde{c}$.

Let $\tilde{\alpha}$ be a component of $\tilde{\alpha}'$ cut open along the points of $\tilde{\alpha}' \cap (\mathcal{C}' \cup \mathcal{A}' \cup \tilde{c})$ that connects $\tilde{c}$ to $\mathcal{C}' \cup \mathcal{A}'$. Since the perturbation was generic we may assume that the endpoint of $\tilde{\alpha}$ on $\mathcal{A}' \cup \mathcal{C}'$ is not a vertex, so that $\mathcal{C} \cup \mathcal{A}' \cup \tilde{\alpha}$ is a connected trivalent graph. The lemma follows in Case One by setting $\mathcal{A} = \mathcal{A}' \cup \tilde{\alpha}$.

Case Two. $\tilde{c} \cap (\mathcal{C}' \cup \mathcal{A}') \neq \emptyset$. Let $\mathcal{A}''$ be the graph obtained from $\mathcal{A}'$ by adding a vertex at every point of $\mathcal{A}' \cap \tilde{c}$. Note that there is no bound on the number of edges of $\mathcal{A}''$, and the vertices of $\mathcal{C} \cup \mathcal{A}''$ have valence 3 or 4 (the vertices of valence 4 are $\mathcal{A}' \cap \tilde{c}$). Clearly, $\mathcal{C} \cup \mathcal{A}''$ is connected.

Step One. Let $\tilde{e}$ be an edge of $\mathcal{A}''$ so that the graph obtained by removing $\tilde{e}$ from $\mathcal{C} \cup \mathcal{A}''$ is connected. We remove $\tilde{e}$.

Step Two. Note that as after Step One we may have a vertex, say $\tilde{v}$, of valence 2. Let $\tilde{e}_i$ ($i = 1, 2$) be the other two edges incident to $\tilde{v}$; denote the endpoints of $\tilde{e}_i$ by $\tilde{v}$ and $\tilde{v}_i$. We remove $\tilde{e}_1$ and $\tilde{e}_2$. If the graph obtained is disconnected, then it consists of two components, one containing $\tilde{v}_1$ and one containing $\tilde{v}_2$. As in Case One, we construct an arc to connect the two components.

Step Two may produce a new vertex of valence 2. We iterate Step Two. This process reduces the number of edges and so terminates; when it does, we obtain a connected graph with no vertices of valence 2.

We now repeat Step One (if possible). After every application of Step One we repeat Step Two (if necessary). Step One also reduces the number of edges, so it terminates. When it does, we obtain a graph (still denoted $\mathcal{A}''$) so that $\mathcal{C} \cap \mathcal{A}''$ is connected, but removing any edge of $\mathcal{A}''$ disconnects it.

By construction, the vertices of $\mathcal{C} \cup \mathcal{A}''$ have valence 3 or 4. If a vertex has valence 4, we choose an edge adjacent to it from $\mathcal{A}''$. Perturbing the endpoint of this edge and applying Lemma 4.11, we obtain two vertices of valence 3. We iterate this process. Since this process reduces the number of vertices of valence 4, it will terminate. The graph obtained is denoted $\mathcal{A}$.

By construction, conditions (2), (3) and (4) of Lemma 4.12 hold. All that remains is proving:

Claim. $\mathcal{A}$ has at most $2n - 1$ edges.

Proof of claim: Let Γ be the graph obtained from $\mathcal{C} \cup \mathcal{A}$ by identifying every component of $\mathcal{C}$ to a single point. Note that the vertex set of Γ has $n + 1$ vertices that correspond to the components of Γ , and extra vertices from the vertices of $\mathcal{A}$ that are disjoint from $\mathcal{C}$; these vertices all have valence 3. The edges of Γ are

naturally in 1-1 correspondence with the edges of $\mathcal{A}$; thus to prove the claim all we need to show is that Γ has at most $2n - 1$ edges.

It is easy to see that Γ is connected because $\mathcal{C} \cup \mathcal{A}$ is. Moreover, if there is any edge e of Γ so that the graph obtained from Γ by removing e is connected, then the graph obtained from $\mathcal{C} \cup \mathcal{A}$ by removing the corresponding edge is connected as well; this contradicts our construction. Hence Γ is a tree, with $n + 1$ vertices of arbitrary valence, and all other vertices have valence 3. In particular, Γ has at most $n + 1$ vertices of valence 1 or 2. We will use the following claim:

Claim. Let $G = (V, E)$ be a finite tree with vertex set V and edge set E and with $k \geq 2$ vertices of valence 1 or 2. Then G has at most $2k - 3$ edges.

We prove the claim by induction on k . If $k = 2$, it is easy to see that G is a single edge, and indeed $1 = 2 \cdot 2 - 3$. Assume from now on that $k > 2$.

It is well known that every finite tree has a leaf (that is, a vertex of valence 1). Let $v \in V$ be a leaf and $(v, v') \in E$ the only edge containing v . Consider $G' = (V - \{v\}, E - \{(v, v')\})$. There are four cases, depending on the valence of v' as a vertex of G' :

- (1) The valence of v' is zero: then G is a single edge, contrary to our assumption.
- (2) The valence of v' is one: that is, v' is a leaf of G' . Note that in this case both v and v' have valence 1 or 2 in G , and we see that G' has exactly $k - 1$ vertices of valence 1 or 2. By induction G' has at most $2(k - 1) - 3 = 2k - 5$ edges. Since G has exactly one more edge than G' , G has at most $2k - 4$ edges in this case.
- (3) The valence of v' is two: In this case, the number of vertices of valence 1 or 2 in G' is exactly k (note that v' has valence 3 in G). Let (v', v'') and (v', v''') be the two edges adjacent to v' . Let G'' be the graph obtained from G' by removing v' from the vertex set and (v', v'') , (v', v''') from the edge set, and add the edge (v'', v''') . It is easy to see that G'' is a tree with exactly $k - 1$ vertices of valence 1 or 2. By induction G'' has at most $2(k - 1) - 3$ edges. Since G' has one more edge than G'' and G has one more edge than G' , G has at most $2k - 3$ edges as desired.
- (4) The valence of v' is at least three: then G' has exactly $k - 1$ vertices of valence 1 or 2. Similar to the above, we see that G has at most $2k - 4$ vertices in this case.

This proves the claim.

To establish (1), we use the claim and the fact that Γ is a tree with at most $n + 1$ vertices of valence 1 or 2, and see that the number of edges in Γ is at most $2(n + 1) - 3 = 2n - 1$.

This completes the proof of Lemma 4.12. □

Next, we prove the existence of totally geodesic disks that cut $V_{i,j}$ into balls. We note that the disks may not be disjoint. The precise statement is:

LEMMA 4.13. *For any $V_{i,j}$ there exists a 2-complex $K_{i,j} \subset V_{i,j}$ so that the following hold:*

- (1) $V_{i,j}$ cut open along $K_{i,j}$ is a single ball.

- (2) *The faces of $K_{i,j}$ are totally geodesic disks. The edges of $K_{i,j}$ have valence 3.*
 (3) *All the vertices are on $\partial V_{i,j}$.*
 (4) *$K_{i,j}$ has at most $2C_1 - 1$ faces and $4C_1 - 2$ edges in the interior of $V_{i,j}$.*

PROOF. If $V_{i,j} = V_i$ then it is a ball and there is nothing to prove. Assume this is not the case. Then $V_i \neq V_{i,j}$, and hence $V_i \cap \text{cl}(M \setminus X) \neq \emptyset$; by Lemma 4.2, $\tilde{V}_i$ intersects exactly one preimage of $\text{cl}(M \setminus X)$, say $\tilde{C}$. Recall that $\tilde{C}$ is a cone. We first establish conditions analogous to (1)–(4) for $\tilde{V}_{i,j}$.

Let $\tilde{P}_{i,j}$ denote $\tilde{V}_{i,j} \cap \partial \tilde{C}$. By Lemma 4.7, $\tilde{V}_{i,j}$ deformation retracts onto $\tilde{P}_{i,j}$; hence $\tilde{P}_{i,j} \subset \partial \tilde{C}$ is a connected, planar surface.

Let $\tilde{s} \in \tilde{V}_i \cap \tilde{C}$ be a point not on the axis of $\tilde{C}$. The Voronoi cells were constructed around generic points $\{x_i\}$. Therefore, after perturbing $\tilde{s}$ slightly if necessary, $\partial \tilde{P}_{i,j} \subset \partial \tilde{C}$ is a generic collection of circles, and Lemma 4.12 applies to give a graph, denoted $\mathcal{A}_{i,j}$, so that $\partial \tilde{P}_{i,j}$ and $\mathcal{A}_{i,j}$ fulfill the conditions of Lemma 4.12. It follows easily from Lemma 4.12 (4) that $\mathcal{A}_{i,j} \subset \tilde{P}_{i,j}$.

Set $\mathcal{K}_{i,j}$ to be $f^{-1}(\mathcal{A}_{i,j})$, where the function f is described in Notation 4.8. By construction, for every edge $\tilde{e}$ of $\mathcal{A}_{i,j}$, $f^{-1}(\tilde{e})$ is the intersection of the totally geodesic plane containing $\tilde{e}$ and $\tilde{s}$ with $\tilde{V}_{i,j}$. Since f is a deformation retract, $f^{-1}(\tilde{e})$ is a disk. These disks are the *faces* of $\mathcal{K}_{i,j}$; thus the faces of $\mathcal{K}_{i,j}$ are totally geodesic disks. By Lemma 4.12, $\partial \tilde{P}_{i,j} \cup \mathcal{A}_{i,j}$ is a trivalent graph. The edges of $\mathcal{K}_{i,j}$ correspond to the preimage of vertices of $\mathcal{A}_{i,j}$, and hence have valence 3. This establishes (2) for $\tilde{V}_{i,j}$.

There are 3 type of vertices: vertices of $\mathcal{K}_{i,j}$, intersection of edges of $\mathcal{K}_{i,j}$ with faces of $\tilde{V}_{i,j}$, and intersection of faces of $\mathcal{K}_{i,j}$ with edges of $\tilde{V}_{i,j}$. By construction, $\mathcal{K}_{i,j}$ has no vertices. By Lemma 4.9 (3) the faces and edges of $\tilde{V}_{i,j}$ are contained in its boundary. Condition (3) follows.

Denote the genus of $\tilde{V}_{i,j}$ by n . Then $|\partial \tilde{P}_{i,j}| = n + 1$. By Lemma 4.12, $\mathcal{A}_{i,j}$ has at most $2n - 1$ edges. By construction, each edge of $\mathcal{A}_{i,j}$ corresponds to exactly one face of $\mathcal{K}_{i,j}$. Hence $\mathcal{K}_{i,j}$ has at most $2n - 1$ faces. By Lemma 4.10, $n = g(\tilde{V}_{i,j}) \leq C_1$; thus $K_{i,j}$ has at most $2C_1 - 1$ faces. Similarly, every vertex of $\mathcal{A}_{i,j}$ corresponds to exactly one edge of $\mathcal{K}_{i,j}$ in the interior of $\tilde{V}_{i,j}$. Since the number of vertices of $\mathcal{A}_{i,j}$ is at most twice the number of its edges, we see that the number of edges of $\mathcal{K}_{i,j}$ in the interior of $\tilde{V}_{i,j}$ is at most $4C_1 - 2$. This establishes (4) for $\tilde{V}_{i,j}$.

By construction, the components of $\tilde{V}_{i,j}$ cut open along $\mathcal{K}_{i,j}$ deformation retract onto $\tilde{P}_{i,j}$ cut open along $\mathcal{A}_{i,j}$. It follows from Lemma 4.12 (3) that $\mathcal{P}_{i,j}$ cut open along $\mathcal{A}_{i,j}$ consists of disks, and from Lemma 4.12 (4) that this is a single disk. We conclude that $\tilde{V}_{i,j}$ cut open along $\mathcal{K}_{i,j}$ is a single ball, establishing (1) for $\tilde{V}_{i,j}$.

By Lemma 4.9 (3) the projection of $\tilde{V}_{i,j}$ to $V_{i,j}$ is a diffeomorphism. Setting $K_{i,j}$ to be the image of $\mathcal{K}_{i,j}$ under the universal covering projection we obtain a complex fulfilling the requirements of Lemma 4.13 $\square$

5. Proof of Proposition 1.4

We use the notation of the previous sections.

We begin with the decomposition of X given by $V_i \cap X = \{V_{i,j}\}_{j=1}^{n_i}$.

Fix one $V_{i,j}$ and consider its decomposition obtained by projecting the faces of $\tilde{V}_{i,j}$ to $V_{i,j}$ (as discussed in Lemma 3.5). Recall that all the faces of this decomposition are totally geodesic by construction. We decompose $V_{i,j}$ further using the faces of $K_{i,j}$, as described in Lemma 4.13. By Lemma 4.13 (2), these faces are totally geodesic as well. By Lemma 4.13 (4), all the vertices of this decomposition are on $\partial V_{i,j}$. We first bound the number of these vertices:

Claim. There is a universal $\bar{C}_0$ so that the number of vertices in $V_{i,j}$ is at most $\bar{C}_0$.

PROOF OF CLAIM. By Lemma 4.9 (3), the universal covering projection induces a diffeomorphism between $V_{i,j}$ and $\tilde{V}_{i,j}$. It follows that a totally geodesic disk and a geodesic segment in $V_{i,j}$ intersect at most once; this will be used below several times. By Lemma 4.13 (2) all the vertices are contained in $\partial V_{i,j}$.

We first bound the number of vertices that lie in the interior of X . There are three types of vertices:

- (1) *The intersection of three faces of ∂V_i :* By Lemma 4.5 there are at most C_0 such vertices. (By transversality the intersection of more than three faces of ∂V_i does not occur.)
- (2) **The intersection of an edge of ∂V_i with a face of $K_{i,j}$:** Since every face of $\partial V_{i,j}$ is totally geodesic and every edge of $K_{i,j}$ is a geodesic segment, every face meets every edge at most once. By Lemma 4.4 there are at most C_1 edges on $\partial V_{i,j}$, and by Lemma 4.13 there are at most $2C_1 - 1$ faces in $K_{i,j}$. It follows that there are at most $C_1(2C_1 - 1)$ vertices of this type.
- (3) **The intersection of a face of ∂V_i and an edge of $K_{i,j}$:** Since every edge of $\partial V_{i,j}$ is a geodesic segment and every face of $K_{i,j}$ is totally geodesic, every edge meets every face at most once. By Lemma 4.3 there are at most C_2 edges on $\partial V_{i,j}$. It is clear that we are discussing only edges of $K_{i,j}$ that lie in the interior of $V_{i,j}$. By Lemma 4.13 (4) there are at most $4C_1 - 2$ such edges. It follows that there are at most $C_2(4C_1 - 2)$ such vertices.

Next we bound the number of vertices on ∂X . There are two cases to consider.

- (1) **An endpoint of an edge of ∂V_i :** Each such vertex is an endpoint of a segment (as defined before Lemma 4.4) and hence by that lemma there are at most $2C_1$ such vertices.
- (2) **The intersection of a face of $K_{i,j}$ with $\partial(V_i \cap \partial X)$ and the intersection of an edge of $K_{i,j}$ with ∂X :** Every face of $K_{i,j}$ contributes at most two such vertices. By Lemma 4.13 (4), there are at most $8C_1 - 4$ such points.

The claim follows by setting (the different contributions are in brackets)

$$\bar{C} = [C_0] + [C_1(2C_1 - 1)] + [C_2(4C_1 - 2)] + [2C_1] + [8C_1 - 4].$$

□

Next, we subdivide each face into triangles by adding edges (note that this does not require faces to be disks). This is done in X , so the subdivision agrees on adjacent cells (including a cell that is adjacent to itself). Note that the new edges have valence 2. Since the edges of $K_{i,j}$ have valence 3 and edges on the boundary have valence at most 3, all edges have valence at most 3.

By Lemma 4.13 (1) $V_{i,j}$ cut open along $K_{i,j}$ is a single ball. Therefore there is a map from the closed ball B onto $V_{i,j}$ that is obtained by identifying disks on ∂B that correspond to the disks of $K_{i,j}$. Since edges have valence at most 3, no point of $V_{i,j}$ has more than 3 preimages. The preimages of the triangulated faces induce a triangulation of ∂B with at most $3\bar{C}_0$ vertices. Denote the number of faces, edges, and vertices in this triangulation by f , v , and e , respectively. Note that $3f = 2e$, or $e = \frac{3}{2}f$. Euler characteristic gives: $2 = f - e + v = -\frac{1}{2}f + v$, or $f = 2v - 4$. Thus, $f \leq 6\bar{C}_0 - 4$. We obtain a triangulation of B by adding a vertex in the center of B , and coning every vertex, edge, and triangle in ∂B . By construction there are exactly f tetrahedra in this triangulation. The image of this triangulation gives a triangulation of $V_{i,j}$ that has at most $6\bar{C}_0 - 4$ tetrahedra. By Lemma 4.9 (1), $n_i \leq C_0$. Since $\{V_{i,j}\}_{j=1}^{n_i}$ are mutually disjoint, by considering their union we obtain a triangulation of $V_i \cap X$ with at most $(6\bar{C}_0 - 4)C_0$ tetrahedra.

By construction the triangulation of $V_i \cap X$ agrees with that of $V_{i'} \cap X$ on $(V_i \cap X) \cap (V_{i'} \cap X)$.

Proposition 1.4 follows from this and Lemma 4.5 by setting $C = \max\{C_3, (6\bar{C}_0 - 4)C_0\}$.

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LAYERED MODELS FOR CLOSED 3-MANIFOLDS

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ABSTRACT. We define a combinatorial structure on 3-manifolds that combines the model manifolds constructed in Minsky's proof of the ending lamination conjecture with the layered triangulations defined by Jaco and Rubinstein.

Minsky's recent proof of the ending lamination conjecture [3] introduced a construction in which a path in the pants complex of a boundary component S of a hyperbolic 3-manifold M is used to build a combinatorial model of a hyperbolic structure on $S \times [0, \infty)$. This structure is analogous in many ways to a triangulation: It consists of a one-dimensional object (a link) spanned by two-dimensional pieces (embedded pairs of pants) that cut the manifold into relatively simple pieces (which we will describe below).

The construction of these models is closely related to the layering construction used to build ideal triangulations of punctured torus bundles in [1]. In particular, a model is determined by a path hierarchy in the curve complex, or (equivalently) a path in the pants complex for the boundary. The layered triangulations in [1] correspond to path hierarchies in the arc complex for the punctured surface.

Jaco and Rubinstein [2] have studied layered triangulations of handlebodies, which can be glued together to form layered triangulations of closed 3-manifolds. Applying the analogy between Minsky models and layered triangulations suggests a way to construct Minsky models for closed 3-manifolds. In the present paper, we study this construction, applying results from the theory of Heegaard splittings to prove the following:

1. Theorem. *Every compact, hyperbolic 3-manifold admits a model decomposition in which every component of the 1-skeleton is a knotted loop.*

By a *knotted* loop in an arbitrary 3-manifold, we mean a loop that does not bound an embedded disk. Such a loop may still be homotopy trivial in M .

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This construction can be readily extended to 3-manifolds with atoroidal boundary. It should also be possible to extend it to 3-manifolds with toroidal boundary by modifying the definition of model blocks slightly to allow them to “wrap” around the torus boundary, which is more in line with the construction in [3]. However, to simplify the exposition, we will not consider these generalizations here.

We also note the following property of any model of a closed 3-manifold. This result does not require that the 3-manifold be hyperbolic, or that the model structure satisfy the conclusions of Theorem 1.

2. Theorem. *The 1-skeleton of any model structure for a compact, closed, orientable 3-manifold is a hyperbolic link.*

The outline of the paper is as follows: We define the model structure in Section 1, then review the theory of generalized Heegaard splittings in Section 2. We define a general construction of models for compression bodies (the building blocks of generalized Heegaard splittings) in Section 3, then in Section 4 we refine this construction to produce models for compression bodies with no unknotted loops. Finally, we use this refined construction to prove Theorem 1 in Section 5.

1. MODEL STRUCTURES

A *pants decomposition* for a compact, closed, orientable surface S is a set P of pairwise disjoint, essential simple closed curves in S such that each component of $S \setminus P$ is a thrice-punctured sphere (i.e. a pair of pants).

Given a pants decomposition P and a loop $\ell \subset P$, the complement $S \setminus (P \setminus \ell)$ consists of a collection of pairs of pants and either a four-punctured sphere or a once-punctured torus $F_\ell \subset S$ that contains ℓ . When F_ℓ is a punctured torus and ℓ' is a second essential loop in F_ℓ not parallel to ℓ , we will say that ℓ' intersects ℓ minimally if $\ell \cap \ell'$ is a single point. When F_ℓ is a four-punctured sphere and $\ell' \subset F_\ell$ is an essential, non-parallel loop we will say ℓ' intersects ℓ minimally if $\ell \cap \ell'$ is exactly two points. In both cases, replacing ℓ with a new loop that intersects ℓ minimally produces a new pants decomposition for S . (The condition that ℓ' intersects ℓ minimally is not necessary to produce a new pants decomposition, but it is useful for other reasons.)

3. Definition. The *pants complex* $\mathcal{P}(S)$ for a compact, closed surface S is the graph whose vertices are isotopy classes of pants decompositions for S and whose edges connect each pants decomposition P to each pants decomposition that results from replacing a loop of P with a new essential loop in F_ℓ that intersects ℓ minimally.

An edge path in the pants complex for S determines a sequence of pants decompositions for S . We will use this sequence of pants decompositions to construct a combinatorial structure on $S \times [0, 1]$ as follows:

4. Definition. A *pants block* is a handlebody with a collection of essential loops in its boundary, forming a pants decomposition of one of two forms: The first type can be identified with a once punctured torus cross an interval $T \times [0, 1]$ so that one loop is essential in the annulus $\partial T \times [0, 1]$ and the other two loops are contained in $T \times \{0\}$ and $T \times \{1\}$ so that their projections into T intersect in a single point. The second type can be identified with a four-times punctured sphere cross an interval $T \times [0, 1]$ so that one loop is essential in each component of $\partial T \times [0, 1]$ and the remaining two loops are contained in $T \times \{0\}$ and $T \times \{1\}$ so that their projections into T intersect in exactly two points.

In the case when T is a once punctured torus, the complement of the loops is two pairs of pants. For T a four punctured sphere, there are four pairs of pants in the complement. We will call the one or two pairs of pants that intersect $T \times \{0\}$ the *bottom pants* and the one or two remaining pairs of pants the *top pants*.

Consider a sequence of pants decompositions $P_1, \dots, P_n$ corresponding to a path in $\mathcal{P}(S)$. The first pants decomposition P_1 determines a decomposition of the surface $S \times \{0\} \subset S \times [0, 1]$. Let $\ell \subset S \times \{0\}$ be the loop that is replaced in the first edge of the path. We can embed a pants block in $S \times [0, 1]$ so that the union of its bottom faces are identified with $F_\ell \times \{0\}$.

The union of $(S \setminus F_\ell) \times \{0\}$ and the top faces of this pants block will be a surface homeomorphic to S with an induced pants decomposition. Moreover, because the top and bottom loops of a pants block project to loops that intersect minimally, we can choose the embedding of the pants block so that the projection of this pants decomposition onto $S \times \{0\}$ will be isotopic to P_2 . We can continue in this way, embedding a pants block for each edge in the path. If P_1 and P_n have no loops in common then the union of these pants blocks will form a regular neighborhood of $S \times \{0\}$, and this can be isotoped so that their union is $S \times [0, 1]$.

This construction produces a model of a surface cross an interval. We would like to apply a similar layering construction to produce models of closed 3-manifolds.

5. Definition. A *model decomposition* of a 3-manifold M is a triple (L, P, B) where L is a link in M , P is a collection of immersed pairs of

pants with pairwise disjoint, embedded interiors and boundaries contained in L . The set B is a collection of embedded pants blocks with boundaries in $\bigcup P$ such that the marked loops in each pants block are sent into L .

Proof of Theorem 2. To show that the 1-skeleton L of a model decomposition (L, P, B) is a hyperbolic, we will show that the complement is irreducible and atoroidal. Each model block B_i is a handlebody, and thus boundary compressible. However, it has the structure of a surface cross and interval $F \times [0, 1]$, so every compressing disk must intersect the annuli $\partial F \times [0, 1]$. Because $L \cap B_i$ contains an essential loop in each of these annuli, there is no compressing disk for B_i disjoint from $L \cap B_i$. Similarly, any incompressible annulus in B_i is either boundary parallel or vertical in $F \times [0, 1]$. In the second case, it must be parallel to one of the annuli in $\partial F \times [0, 1]$, and thus to a loop in $L \cap B_i$.

Let $S \subset (M \setminus L)$ be an embedded sphere, and assume that we have isotoped S so as to minimize its intersection with the pairs or pants P . If $S \cap P$ is not empty then every loop bounds a disk in S , so some loop of $S \cap P$ will bound a disk in a block B_i . However, this disk must be boundary parallel in B_i , so there is an isotopy of S that eliminates this loop of intersection. Thus $S \cap P$ is empty, S is contained in a block B and thus bounds a ball (since handlebodies are irreducible.)

Given an embedded torus $T \subset (M \setminus L)$, a similar argument implies we can isotope T to intersect P in a collection of essential loops. Thus its intersection with each block B_i will be a collection of annuli. As noted above, each annulus must be parallel to a loop in $L \cap B_i$. As we follow the annuli around T , we see that they must all be parallel to the same component of L , and so T must be boundary parallel. Thus $M \setminus L$ contains no essential tori, and is thus hyperbolic. $\square$

The construction above determines a model decomposition for $S \times [0, 1]$. By choosing a model so that the pants decompositions for $S \times \{0\}$ and $S \times \{1\}$ are related by a given homeomorphism $\phi : S \rightarrow S$, one can construct a model structure for the surface bundle with monodromy ϕ . By a similar construction for surfaces with boundary, one can construct model decompositions for any closed 3-manifold from an open decomposition. However, in order to get the control needed to prove Theorem 1, we will construct a model decomposition from a generalized Heegaard splitting.

2. GENERALIZED HEEGAARD SPLITTINGS

6. Definition. A handlebody is a 3-manifold homomorphic to the regular neighborhood of a graph embedded in S^3 . A *strict compression*

body H is a connected 3-manifold with boundary that results from attaching 1-handles to the $S \times \{1\}$ boundary component of $S \times [0, 1]$ for a compact, closed (not necessarily connected) surface S . The image in H of $S \times \{0\}$ is the *negative boundary* $\partial_- H$ and the complement $\partial H \setminus \partial_- H$ is the *positive boundary*, $\partial_+ H$. A *compression body* is a disjoint union of zero or more handlebodies and zero or more strict compression bodies.

Note that we use a very general definition of compression body here. In particular, a compression body may not be connected.

7. Definition. A *generalized Heegaard splitting* for a closed 3-manifold M is a collection of compression bodies $H_1^-, H_1^+, \dots, H_k^-, H_k^+ \subset M$ with pairwise disjoint interiors whose union is M and whose boundary components coincide as follows: For each i , $\partial_+ H_i^- = \partial_+ H_i^+$ and $\partial_- H_i^+ = \partial_- H_{i+1}^-$.

For each i , we will define the submanifold $M_i = H_i^- \cup H_i^+ \subset M$. The pair of handlebodies H_i^-, H_i^+ determines a Heegaard splitting for M_i along the Heegaard surface $\Sigma_i = \partial_+ H_i^- = \partial_+ H_i^+$. Each Σ_i is called a *thick surface* of the generalized Heegaard splitting. The surfaces that come from the negative boundary components of the compression bodies are called *thin surfaces*.

If for some i there is an essential, properly embedded disk in H_i^- whose boundary is disjoint from an essential disk in H_i^+ then following Scharlemann-Thompson [5], we can replace H_i^-, H_i^+ with a generalized Heegaard splitting for M_i whose intermediate surfaces have lower genus. This process is called a *weak reduction*.

If it is not possible to carry out a weak reduction then we will have the condition that for each i , every essential disk in H_i^- must intersect every essential disk in H_i^+ in a non-empty set in Σ_i . A generalized Heegaard splitting that satisfies this property will be called *strongly irreducible*. The main results of Scharlemann and Thompson's paper can be summarized as follows:

8. Theorem (Scharlemann-Thompson [5]). *Every 3-manifold admits a strongly irreducible generalized Heegaard splitting and the thin surfaces of any strongly irreducible generalized Heegaard splitting are incompressible.*

We will also need the following property of strongly irreducible generalized Heegaard splittings. (Scharlemann's proof is for Heegaard splittings rather than generalized splittings, but the result can be generalized directly due to the fact that the thin surfaces in a strongly irreducible generalized Heegaard splitting are incompressible.)

9. Lemma (Scharlemann's no-nesting Lemma [4]). *If ℓ is a simple closed curve in a thick surface Σ_i of a strongly irreducible generalized Heegaard splitting such that ℓ bounds a disk in M then ℓ bounds a disk contained in H_i^- or H_i^+ .*

Note that if M is hyperbolic then every compression body in a strongly irreducible generalized Heegaard splitting of M will have no boundary component of genus one.

3. COMPRESSION BODIES AND FAT SPINES

We will construct a model decomposition for a 3-manifold M by choosing a strongly irreducible generalized Heegaard splitting for M and constructing a model decomposition for each compression body in the splitting so that they match up along their boundary surfaces. Our construction will proceed by induction, starting with models for the simplest types of compression bodies.

10. Definition. Given a compact, closed, orientable surface S , a *fat spine* for $S \times [0, 1]$ is a pants decomposition for $S \times \{\frac{1}{2}\}$. A *fat spine* for a genus-two handlebody is an embedded pair of pants with a pants decomposition consisting of its boundary loop and a loop in its interior, whose complement in H is homeomorphic to $\partial H \times [0, 1)$.

Each type of fat spine is a union of pairs-of-pants whose boundary loops form a link in H . Moreover, the complement of each spine is a regular neighborhood of the boundary of H . Consider a pants block B that intersects a fat spine K in its bottom pants and has interior disjoint from K . Then the complement of $K \cup B$ will also be a regular neighborhood of ∂H . After attaching this first pants block, we can attach a second pants block along its bottom pairs of pants and so on for any number of pants blocks.

11. Definition. A model complex in the interior of H constructed in this way will be called a *layered model* for H .

Every compression body with no tori in its boundary can be constructed by attaching 1-handles to one or more of these initial types of compression bodies. More precisely, given any compression body H that is not one of these two initial types, there is a properly embedded, essential disk $D \subset H$ such that the complement of H is one or two compression bodies whose positive boundary or boundaries are atoroidal and have lower genera than that of H .

12. Definition. A *fat spine* for H is the union of a model for each component of $H \setminus D$ and a pair of pants F with one boundary loop in

the boundary of each model such that the complement of the fat spine is homeomorphic to $\partial H \times [0, 1)$.

A fat spine can be constructed by induction for any compression body with atoroidal boundary. The pair of pants F has either one or two boundary components in the models for the smaller compression bodies, depending on whether D is separating or non-separating.

4. MODELS FOR COMPRESSION BODIES

An *interior loop* of a fat spine K is a loop ℓ in the 1-skeleton of K such that some regular neighborhood of ℓ is contained in the pants blocks of K . Any non-interior loop in the 1-skeleton is called a *boundary loop*. As with the fat spines for our initial types of compression bodies, we can layer pants blocks onto any fat spine, and the resulting model decomposition will be called a *model* for H . Each model for H induces a pants decomposition for ∂H as follows:

13. Definition. A pants decomposition for ∂H is *induced* by a model K for H if for each loop ℓ in the pants decomposition, there is an annulus $A_\ell \subset H$ with one boundary component coinciding with ℓ , the other component coinciding with a boundary loop in K , and interior disjoint from K . The annulus A_ℓ will be called a *vertical annulus*. A pants decomposition of ∂H is *spinal* if it is induced by a fat spine for H .

In many cases, the union of the pants blocks in the model for H will be isotopic onto H , and the induced pants decomposition comes from the image after this isotopy. However, in some models (such as fat spines) the union of the pants blocks and any loose pair of pants, will not be a 3-manifold.

14. Lemma. *Each model K for H induces a unique pants decomposition for ∂H .*

Proof. Let ℓ be a loop in K that is the boundary of one or more pairs of pants. A regular neighborhood of ℓ is a solid torus T such that $K \cap T$ consists of a collection of annuli, each with one boundary component on ℓ and the other in ∂T . Let A'' be a collection of annuli with interior in the complement $T \setminus K$ such that each annulus in A'' has a boundary component on ℓ and a boundary loop in ∂T , and such that each component of $T \setminus K$ contains exactly one annulus of A'' . Construct such a collection for each loop of K and let A' be their union.

The complement $H \setminus K$ is homeomorphic to $\partial H \times [0, 1)$ and the image of A' in this product is a collection of annuli, whose closures

in $\partial H \times [0, 1]$ have essential boundary components in $\partial H \times \{1\}$. Let $A \subset \partial H \times [0, 1]$ be the result of extending each annulus of A' to an annulus with one boundary component in $\partial H \times \{0\}$ and one in $\partial H \times \{1\}$. By construction, the intersection of A with $\partial H \times \{1\}$ is a pants decomposition. Because the annuli are all essential, they determine the same isotopy classes of loops in $\partial H \times \{0\}$, so the intersection of A with this surface is a pants decomposition. Moreover, the images in H of the annuli in A show that this pants decomposition is induced by K .

Any annulus with a boundary component on a loop of K and the other in ∂H is isotopic into an annulus of A within a regular neighborhood of the loop. After this isotopy the two loops will determine the same loop in $\partial H \times \{1\}$, so the two annuli will be isotopic in $\partial H \times [0, 1]$ and thus in H . Thus every spine of H determines a unique spinal pants decomposition. $\square$

By definition, every model for a compression body H contains a fat spine K for H . The *layer number* of the model is the smallest number of model blocks not contained in K , over all models for H and fat spines K in those models.

15. Lemma. *Let H be a compression body and let P be a pants decomposition for $\partial_+ H$. If K is a model for H that induces P on $\partial_+ H$, and has the minimal layer number among all such models then any loop in K that bounds a disk in H is in the interior of the fat spine.*

Proof. Let K be a fat spine for H contained in the model. The model blocks not in K determine a path in the pants complex for $\partial_+ H$ from a spinal decomposition to P . Let $P_1, \dots, P_n$ be the vertices of this path, where $P = P_1$ and P_n is induced by K . Let i be the index such that P_i is the first pants decomposition containing a loop bounding a disk in H . Let $\ell \subset P_i$ be the loop bounding a disk.

By assumption, P does not contain such a loop so $i > 1$. We will show that P_{i-1} is spinal, implying that the model does not have minimal layer number.

Note that if F_ℓ is a punctured torus then ℓ is non-separating in F_ℓ . Moreover, if one takes two parallel copies of a disk bounded by ℓ and attaches them by a band in the complement of the annulus between the boundaries of two disks, the resulting disk has boundary parallel to the puncture in the torus. Thus the loop in the boundary of F_ℓ already bounds a disk in H . Since we chose i to be the earliest edge creating such a loop, we can assume that F_ℓ is a four-punctured sphere.

Let D be a disk with $\partial D = \ell$ and let $H' \subset H$ be the complement in H of an open regular neighborhood N of D . Then H' is either a compression body or a disjoint union of two compression bodies (depending on whether or not D is separating).

The intersection of the closure of N with H' is a pair of disks $E_1, E_2 \subset \partial H'$. The restriction of the pants decomposition P_i to $\partial H'$ forms a pants decomposition for the punctured surface $\partial H' \setminus (E_1 \cup E_2)$. The two pairs of pants containing the boundary loops of this surface are the restrictions of F_ℓ . Each of these pairs of pants has one boundary component that bounds a disk in $\partial H'$, and the union of this pair of pants and the disk is an annulus. Let A_1, A_2 be these two annuli in $\partial H'$.

If A_1 is contained in a solid torus component of H' then we will define ℓ_1 to be a core of the solid torus. Otherwise, we will crush A_1 down to a single loop ℓ_1 in $\partial H'$. Define ℓ_2 similarly. Crushing A_1 and A_2 to loops (if they are not in solid tori components) induces a pants decomposition P' on the boundary of the non-solid-torus component(s) of H' .

For each component of H' that is not a solid torus, choose a model that induces the pants decomposition P' on $\partial H'$. Consider the result of attaching a pair of pants to this model for H' , sending two of the boundary components to the loops ℓ_1, ℓ_2 . If ℓ_1 or ℓ_2 is in a solid torus then the pair of pants will be attached to H' along only one boundary loop. If both ℓ_1 and ℓ_2 are in solid tori then H is a genus-two handle-body and the resulting fat spine consists of a single pair of pants.

The induced pants decomposition of ∂H is identical to P_i outside of F_ℓ . Within F_ℓ , the induced pants decomposition consists of a loop that intersects ℓ in two points. By twisting around the disk D , we can make this loop into any essential loop in F_ℓ that intersects ℓ in two points. In particular, we can make it into the loop that preceded ℓ in P_{i-1} . Thus P_{i-1} is spinal and the model for H does not have the minimal layer number. $\square$

16. Corollary. *Let H be a compression body and let P be a pants decomposition for $\partial_+ H$ such that no loop in P bounds a disk in H . Then there is a model K for H that induces P on $\partial_+ H$, has minimal layer number among all such models and such that no loop in K bounds a disk in H .*

Proof. We will prove this by induction on the genus of H , starting with the base case of genus two. In this case a fat spine is a single pair of pants embedded in H and it is straightforward to check that no loop bounds a disk. Moreover, if we extend this to a model that minimizes

the layer number then Lemma 15 implies that none of the new loops will bound disks in H .

For the induction step, assume that the Corollary holds for every genus strictly less than g . Choose a model for H that minimizes the layer number. Let K be the fat spine and let $H' \subset H$ be the handlebody contained in K (i.e. we get the fat spine for H by attaching a pair of pants to H' .) The model for H determines a pants decomposition for $\partial H'$ such that no loop in this pants decomposition bounds a disk in H . Therefore, no loop bounds a disk in H' . By the induction assumption, this implies that we can choose a model for H' such that no loop bounds a disk in H' . Since this model for H' agrees with our original model for H , we can assume that the restriction of K to H' is such a model.

The boundary of H' is incompressible into $H \setminus H'$, so if a loop in H' bounds a disk in H then it bounds a disk in H' . By the contrapositive, no loop in the model can bound a disk in H because it does not bound a disk in H' . Thus the model we've constructed contains no loop bounding a disk in H . $\square$

5. CONSTRUCTING THE MODEL

Proof of Theorem 1. By Scharlemann-Thompson's Theorem (Theorem 8), every hyperbolic 3-manifold M admits a strongly irreducible generalized Heegaard splitting $\{H_1^-, H_1^+, \dots, H_k^-, H_k^+\}$ whose thin surfaces are incompressible. Because M is closed and hyperbolic, each thin surface must have genus at least two. Because M is not a lens space or a connect sum containing a lens space, each thick surface must also have genus at least two.

For each i , choose a fat spine K^+ for H^+ and a model K^- for H^- so that the pants decomposition of $\Sigma_i = \partial_+ H_i^-$ is isotopic to the decomposition induced by K^- on $\Sigma_i = \partial_+ H_i^+$. Moreover, choose K_i^-, K_i^+ so that no loop in the fat spines of K_i^+, K_i^- bounds a disk in $H_i^- \cup H_i^+$, respectively and so that K_i^- has minimal layer number over all such pairs of models.

Because K_i^+ is a fat spine for H_i^+ , every boundary loop in K_i^+ is either disjoint from a compressing disk for K_i^+ or is the loop added with the final pair of pants. Because Σ_i is strongly irreducible, if a boundary loop of K^+ bounds a disk in H_i^- then it must be this final loop ℓ . If this is the case then there is a meridian disk D for H_i^+ that intersects the final pair of pants in an arc and intersects ℓ in two points. Applying a half-twist along D to K_i^+ replaces ℓ with a new loop ℓ' that intersects ℓ in two points. Since ℓ bounds a disk in H_i^- , the construction in Lemma 15 can be used to find a fat spine K_i^+ for

H_i^+ in which ℓ' is the loop added with the final pair of pants. In this case the layer number is zero, so this pair K_i^- , K_i^+ will minimize the layer number.

Thus we can assume no loop in K_i^+ bounds a disk in H_i^- . In particular, no loop in the induced pants decomposition of Σ_i bounds a disk in H_i^- , so by Lemma 16, we can choose K_i^- so that no loop bounds a disk in H_i^- . The last block layered into H_+^+ has top pants that project to a pair of pants in the projection of K_i^+ . If we remove this pants block from K_i^- and layer it onto K_i^+ , the projections of the resulting models for H_i^- , H_i^+ will still coincide (up to isotopy) in Σ_i . We can transfer each subsequent model block outside the fat spine in H_i^- in a similar fashion until we have a fat spine for H_i^- and a model for H_i^+ . If this model does not have minimal layer number then we could choose a model with lower layer number, then transfer the blocks back into H_i^- , to reduce its layer number. Thus this model for H_i^+ must have minimal layer number. Thus by Lemma 15, no loop added by the layering (i.e. outside the fat spine) will bound a disk in H^+ .

Any loop in the interior of the fat spine of H_i^- or H_i^+ is disjoint from a compressing disk for the compression body. Thus if such a loop bounds a disk in the opposite compression body, the induced Heegaard splitting of M_i will be weakly reducible. Thus no loop in K_i^- or K_i^+ bounds a disk in either compression body. Moreover, by Scharlemann's no-nesting Lemma (Lemma 9) these loops cannot bound disks in M_i .

We would like to turn the models for the compression bodies in the generalized Heegaard splitting for M into a single model for M . For each thin surface $\partial_- H_i^+ = \partial_- H_{i+1}^-$, choose a path in the pants complex between the projections of the models we have chosen for the two compression bodies. If there is a loop that appears in both compression bodies, choose a non-minimal path that passes through a pants decomposition that does not contain any such loop. This path will induce a model structure for the neighborhood of $\partial_- H_i^+ = \partial_- H_{i+1}^-$ between the two compression body models that matches up with the models along the intersection. Because each thin surface is incompressible, these loops will not bound disks in M .

To extend the models along the thick surfaces, recall that on each Σ_i there is a pants decomposition such that each loop is connected to a loop in H_i^- and H_i^+ by an embedded annulus with interior disjoint from $K_i^\pm$ and its second boundary loop in $K_i^\pm$. Let A be the union of these annuli. If A contains an embedded torus $T \subset M_i$ then this embedded torus must be compressible (since M is atoroidal). If T bounds a solid

torus then we will throw away all the model cells inside the solid torus and crush the solid torus down to a loop.

Otherwise, assume for contradiction that T does not bound a solid torus. Let D be a compressing disk for T and let $X \subset M_i$ be the component of $M_i \setminus T$ that does not contain D . Let $\Sigma \subset M_i$ be a regular neighborhood of the fat spine for H_i^+ . Let S be the result of isotoping T in the complement of X until there is a compressing disk for T disjoint from Σ and compressing. Then the complement of S will be a ball B containing X . Because Σ is strongly irreducible and S is incompressible in the complement of Σ , Scharlemann's characterization of how a strongly irreducible Heegaard surface intersects a ball [4] implies that $\Sigma \cap B$ must be planar and unknotted. In particular, any loop in $\Sigma \cap B$ bounds an embedded disk in the complement of Σ . However this is impossible because $\Sigma \cap B$ contains one or more loops in a spinal pants decomposition for Σ , which by construction do not bound a disks in H_i^+ or H_i^- .

Thus we can crush each torus in A down to a loop. The resulting set A' is homeomorphic to the product of S^1 with a disjoint union of trees. Thus we can shrink each component of A' down to a loop. The annuli in A' bound regions of $M_i \setminus (K_i^+ \cup K_i^-)$ with the structure of a pair of pants cross an interval. When we shrink the annuli to loops, we can isotope K_i^- and K_i^+ into these regions, to shrink each one down to a pair of pants. After this isotopy, the union of the models for the compression bodies is a model decomposition for M such that no loop bounds a disk in M . $\square$

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Triangulations and Nonorientable Incompressible Surfaces

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Dedicated to my advisor, William Jaco, in honor of his 70th birthday.

ABSTRACT. In this paper, we classify all nonorientable incompressible surfaces, up to isotopy, in the generalized quaternion spaces S^3/Q_{4k} , which are $M_k = (S^2 : (2, 1), (2, 1), (k, -k + 1))$, $k \geq 1$. The techniques used can also be expanded to give the classification of nonorientable incompressible surfaces in the minimal layered chain pair triangulations of Seifert fibered spaces $M_{r,s} = (S^2 : (2, -1), (r + 1, 1), (s + 1, 1))$, $r, s \geq 1$.

1. Introduction

Ben Burton, in [1], constructed a layered chain triangulation of the solid torus. Furthermore, he used this special triangulation to construct the twist layered loop triangulation of the generalized quaternion spaces S^3/Q_{4k} , $k \geq 1$, which are $M_k = (S^2 : (2, 1), (2, 1), (k, -k + 1))$, $k \geq 1$, and the layered chain pair triangulation of Seifert fibered spaces $M_{r,s} = (S^2 : (2, -1), (r + 1, 1), (s + 1, 1))$, $r, s \geq 1$.

Recently in [9, 10] William Jaco, J. Hyam Rubinstein and Stephan Tillmann together proved that the twisted layered loop triangulation of S^3/Q_{4k} , $k \geq 1$, is minimal. Given the existence of certain $\mathbb{Z}_2$ -cohomology classes, they use covering spaces and the $\mathbb{Z}_2$ -Thurston norm to give a new lower bound for the complexity of a manifold, i.e. the minimal number of tetrahedra in a triangulation of a manifold. Determining the minimal genus nonorientable surfaces in homology classes plays an important role here.

In this paper we use these two special triangulations, the twist layered loop and the layered chain pair, as a tool to study and classify the nonorientable incompressible surfaces up to isotopy in spaces M_k and $M_{r,s}$. It is possible that the normal surface methods of this paper could be used in conjunction with techniques in [9, 10] to study the complexity of other 3-manifolds containing nonorientable surfaces. For readers interested in proofs of the existence and homeomorphism theorems for nonorientable incompressible surfaces in Seifert fibered spaces, especially in circle bundles, see C. Frohman, Richard Rannard, in [2, 13].

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2. Layered chain triangulation of the solid torus

In order to give the classification of all the normal surfaces in these two families of triangulations, we need to first focus on the layered chain triangulation of the solid torus, which is constructed by Ben Burton in [1].

Let's define the layered chain triangulation of the solid torus. Notice that the boundary of a solid torus is a torus that can be obtained by gluing two annuli along their corresponding boundary components, denoted by t and b . The layered chain triangulation of the solid torus starts from a triangulation of the bottom annulus, denoted by A_0 . Given a tetrahedron σ_1 , we can layer it on top of A_0 along the edge e_1 . See figure 1. The arrow indicates the gluing between two corresponding triangular faces.

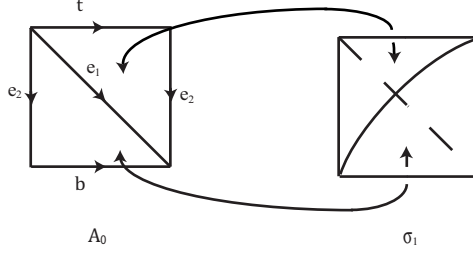


FIGURE 1. Layering the tetrahedron σ_1 on top of A_0 along the edge e_1 .

Thus, we get a one tetrahedron triangulation C_1 of a creased solid torus. See figure 2. The top annulus on the boundary is denoted by A_1 .

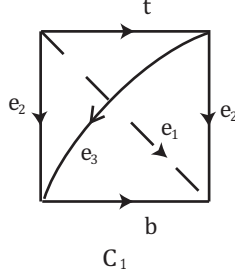


FIGURE 2. C_1 , a triangulation of the creased solid torus.

Now let's continue to layer the second tetrahedron σ_2 on top of A_1 along the edge e_2 . This gives us a triangulation of the solid torus. σ_1 and σ_2 together after face-pairing identification give us a 2-tetrahedron triangulation of the solid torus, denoted by C_2 .

Repeating this procedure, after layering the k^{th} tetrahedron σ_k on top of A_{k-1} along the edge e_k , $k \geq 2$, we get a k -tetrahedron triangulation of the solid torus, C_k . See figure 3. This special way of constructing a k -tetrahedron triangulation of

the solid torus is called *the layered chain triangulation of length k* . The annulus A_0 is called the bottom annulus of the triangulation C_k , and the annulus A_k is called the top annulus of C_k . This triangulation has two vertices, the base point of edge loop t and the base point of edge loop b .

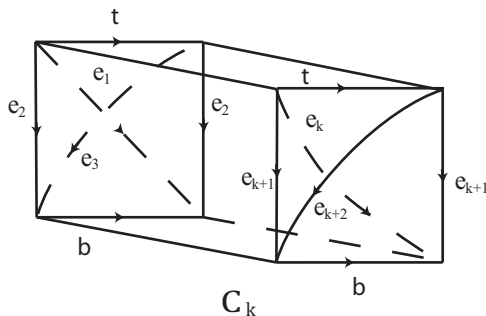


FIGURE 3. A layered chain triangulation C_k of a solid torus of length k .

Next, we try to classify the normal surfaces in the layered chain triangulation of the solid torus, C_k . The normal surface theory we use here can be found in the papers, [5, 7, 8]. The 3-manifold terminology mentioned here can be found in [3, 4, 6], and that of Seifert fibered spaces can be found in [11, 14].

We'll also set up some additional notation and terminology for the classification of normal surfaces.

DEFINITION 2.1. The edge-weight of a normal arc in the bottom or top annulus of a layered chain triangulation of the solid torus is an ordered 4-tuple $(wt_t, wt_b, wt_{e_1}, wt_{e_2})$ or $(wt_t, wt_b, wt_{e_{k+1}}, wt_{e_{k+2}})$, respectively, where wt_x is the number of intersections of a normal arc with the edge x . We call it the bottom or top edge-weight of a normal arc in the layered chain triangulation of the solid torus.

For a normal curve in a layered triangulation of the solid torus, it will intersect with the bottom and top annuli of the boundary of this triangulation. Therefore, we will use $(wt_t, wt_b, wt_{e_1}, wt_{e_2}); (wt_t, wt_b, wt_{e_{k+1}}, wt_{e_{k+2}})$ to represent the edge-weight of a normal curve. For the case of a normal surface, we consider the edge-weight of the surface to be the sum of the edge-weights of its boundary components.

DEFINITION 2.2. A normal surface in a triangulation T of a 3-manifold is called a quadrilateral splitting surface if it consists of precisely one quadrilateral disc within each tetrahedron of T and no other normal disc.

DEFINITION 2.3. In a triangulation, two surfaces are normally isotopic to each other if and only if these two surfaces are isotopic via an isotopy missing the vertices.

REMARK 2.4. If two surfaces are normally isotopic to each other, then they are isotopic to each other. The converse is not necessarily true.

DEFINITION 2.5. A thin edge in a triangulation is an edge which has a regular neighborhood with normal boundary in the triangulation.

DEFINITION 2.6. A thin edge-linking tube of an edge in a triangulation is the boundary of a small regular neighborhood of a thin interior edge in the triangulation consisting of normal disks.

REMARK 2.7. In the layered chain triangulation, every edge of degree 4 is an interior edge, therefore, it has a thin edge-linking tube as in figure 4.

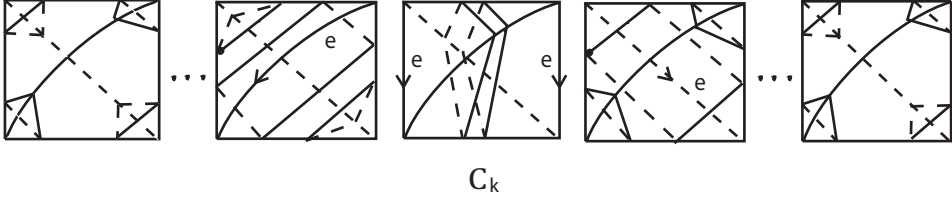


FIGURE 4. A thin edge-linking tube of edge e in C_k .

Suppose we have a layered chain triangulation C_k of the solid torus. Now if we layer a new tetrahedron σ_{k+1} on top of A_k along the edge e_{k+1} , we will get a new layered chain triangulation C_{k+1} of the solid torus with $k+1$ tetrahedra. The difference between C_{k+1} and C_k is that we just add a product structure of the top annulus of the triangulation C_k . Now suppose S_{k+1} is a normal surface in C_{k+1} , then $S_{k+1} \cap C_k = S_k$ is a normal surface in C_k . The only difference between S_{k+1} and S_k is a collection of normal triangles and quadrilaterals in the tetrahedron σ_{k+1} . Now we pay close attention to how we get these extra normal pieces from the normal surface S_k to form the new normal surface S_{k+1} . There are two possible ways to add normal disks.

Push-through: We extend every normal arc, the intersection of S_k with the two faces of the top annulus A_k of the triangulation C_k in σ_k , by adding one or more of the 4 types of normal triangles and/or possibly one of the 3 types of normal quadrilaterals in σ_{k+1} . These normal disks are completely determined by the arc types of the intersection of S_k with the top annulus A_k . Obviously, a push-through just adds the product structure to the normal surface. We get a new normal surface which is homeomorphic to the old one.

Banding: Instead of pushing through every normal arc, the intersections of S_k with A_k of C_k in σ_k , we allow the addition of a band connecting two parallel arcs, as long as the new band will not intersect with any other newly added normal disks in the tetrahedron σ_{k+1} . Sometimes, we can add more than one band in σ_{k+1} . These bands are of the same 3 possible types of quadrilaterals, according to the Quadrilateral Condition (the intersection of a normal surface with every tetrahedra in a triangulation must have no more than one quadrilateral type). Every time we add a band on the surface, the Euler characteristic will be decreased by 1, i.e. $\chi(\text{new surface}) = \chi(\text{old surface}) - 1$.

LEMMA 2.8. *All the possible bottom (top) edge-weights of a connected normal surface in a layered chain triangulation of the solid torus are $(2, 0, 1, 1)$, $(0, 2, 1, 1)$, $(2, 2, 2, 2)$, $(0, 0, 1, 1)$, $(1, 1, p + 1, p)$, $(1, 1, p, p + 1)$, $p \geq 0$ or at most 2 copies of the last two cases.*

PROOF. The proof of this lemma is not difficult but time consuming. We will just outline the main idea of the proof. The boundaries of a connected surface are circles. Any closed curve from the boundary of a connected normal surface on the boundary of the layered chain triangulation of a solid torus will intersect the bottom annulus A_0 in trivial arcs, essential arcs or essential simple closed curves. Notice a trivial closed curve cannot be normally isotopic to a normal curve on the bottom annulus, hence we will not pay attention to this case. There are only finitely many families of these arcs and curves. It's not hard to check that they are $(2, 0, 1, 1)$, $(0, 2, 1, 1)$, $(0, 0, 1, 1)$, $(1, 1, p + 1, p)$ and $(1, 1, p, p + 1)$, $p \geq 0$. Now if we want to obtain a normal surface, we can only start from these normal arcs, curves or a compatible combination of these arcs or curves on A_0 . We can either push through or band them in the first tetrahedron σ_1 . The generated normal surface will intersect the top annulus A_1 of σ_1 with a collection of normal arcs or essential curves. We can repeat the same procedure in each tetrahedron until we get one or more normal surfaces in the triangulation. Notice, push-through just adds the product structure to the normal surface. The new normal surface obtained by pushing through is homeomorphic to the old one. Every band can only connect at most two normal surfaces together. In this special layered chain triangulation, the only way we can add a band to a normal surface is when we see the normal arc with edge-weight $(2, 2, 2, 2)$ or $(1, 1, 0, 1)$ or the double of the latter case on the bottom annulus A_{i-1} in the tetrahedron σ_i , $i \leq k$. Furthermore, in this special triangulation once a band is added to get a connected normal surface, denoted by S , no other band can be added to connect S with any other normal surfaces but itself later in the procedure, i.e. bands can only connect at most two normal surfaces together in this special triangulation. Therefore, in order to get only one connected normal surface from disconnected normal arcs or curves in A_0 , we need a pair of trivial arcs, $(0, 2, 1, 1)$ and $(2, 0, 1, 1)$, which will give us the bottom edge-weight $(2, 2, 2, 2)$, or at most two copies of those essential curves with edge-weight $(1, 1, p + 1, p)$ or $(1, 1, p, p + 1)$, $p \geq 0$.

Combining all the observations above, we see that there are only finitely many combinations of normal arcs and curves on the bottom annulus A_0 which can give us a connected normal surface. The edge-weights are $(2, 0, 1, 1)$, $(0, 2, 1, 1)$, $(2, 2, 2, 2)$, $(0, 0, 1, 1)$, $(1, 1, p + 1, p)$, $(1, 1, p, p + 1)$, or at most 2 copies of $(1, 1, 0, 1)$ and $(1, 1, p + 1, p)$, $p \geq 0$.

As for the top edge-weights, we just need to consider all the possible top edge-weights we can get from pushing through or banding these normal arcs or circles with the bottom edge-weights described above. For the trivial normal arc, $(2, 0, 1, 1)$ or $(0, 2, 1, 1)$, on the bottom annulus A_0 , we can only push it through. On each layer, it will keep the same edge-weight $(2, 0, 1, 1)$ or $(0, 2, 1, 1)$. Therefore, we will get a vertex-linking disk at the end, and it will have the same edge-weight on the top annulus. Similarly for an essential curve with bottom edge-weight $(0, 0, 1, 1)$, we will get a vertical annulus in C_k with top edge-weight $(0, 0, 1, 1)$. For normal arcs with edge-weight $(1, 1, p + 1, p)$ or $(1, 1, p, p + 1)$, $p \geq 0$, if we push it through once, the last two coordinates of the edge-weight of the new surface will be changed

by 1. For the case $(1, 1, p+1, p)$, after one push-through in a tetrahedron the edge weight will become $(1, 1, p, p-1)$ when $p \geq 1$, otherwise, it will become $(1, 1, 0, 1)$, for $p = 1$. For the case $(1, 1, p, p+1)$, $p \geq 0$, it will become $(1, 1, p+1, p+2)$ after one push-through in a tetrahedron. All of these will give us meridian disks. If we have two copies of normal arcs with edge-weights $2 \times (1, 1, p+1, p)$, $0 \leq p \leq k-2$, then we can first push them through the first $p+1$ tetrahedra and S_{p+1} is two copies of a meridian disk with top edge-weights $2 \times (1, 1, 0, 1)$ on the top annulus A_{p+1} . Now if we want to get a connected normal surface in C_k , we need to add a band in σ_{p+2} . Then we will get a connected normal surface S_{p+2} , a connected normal disk, with edge-weight $(2, 2, 2, 2)$ on the top annulus A_{p+2} . Notice if we push through normal arcs with edge-weight $(2, 2, 2, 2)$, the edge-weight will remain the same. Therefore, if we push through S_{p+2} all the way, we will get a disk with edge-weight $2 \times (1, 1, p+1, p); (2, 2, 2, 2)$. Recall that we can add a band to normal arcs $(2, 2, 2, 2)$ in any σ_i , $i > p+2$, and the edge-weight will become $2 \times (1, 1, 1, 0)$ in σ_i . If we keep pushing it through, the edge-weight will become $2 \times (1, 1, 0, 1)$. We can either add one or two bands or push through as we proceed. One can see that if we add one or two bands to S_{p+2} in the later layering instead of only pushing through, we'll get a more complicated connected surface, with possible top edge-weight $2 \times (1, 1, 1, 0)$ or $2 \times (1, 1, p, p+1)$, $p \geq 0$. Therefore, all the possible top edge-weights are included in the lemma. $\square$

THEOREM 2.9. *A connected, embedded, normal surface in a layered chain triangulation, C_k , of the solid torus, $k \geq 2$, is normally isotopic to one of the model surfaces listed in the following:*

0. *Vertex-linking disk*, $(0, 2, 1, 1); (0, 2, 1, 1)$ or $(2, 0, 1, 1); (2, 0, 1, 1)$
1. *Vertex-linking disks, possibly with thin edge-linking tubes*, $(2, 2, 2, 2); (2, 2, 2, 2)$.
2. *Vertical annulus*, $(0, 0, 1, 1); (0, 0, 1, 1)$.
3. *Meridian disk*.
 - (a) *Meridian disk*, $(1, 1, p, p+1); (1, 1, p+k, (p+1)+k)$, $p \geq 0$ or
 - (b) *Meridian disk*, $(1, 1, p+1, p); (1, 1, |(p+1)-k|, |p-k|)$, $p \geq 0$.
4. *Upper edge-linking disk, possibly with thin edge-linking tubes*, $(2, 2, 2, 2); 2 \times (1, 1, 1, 0)$ or $(2, 2, 2, 2); 2 \times (1, 1, q, q+1)$, with $q \geq 0$.
5. *Lower edge-linking disk, possibly with thin edge-linking tubes*, $2 \times (1, 1, 0, 1); (2, 2, 2, 2)$ or $2 \times (1, 1, p+1, p); (2, 2, 2, 2)$, $p \geq 0$.
6. *nonorientable surface*.
 - (a) *nonorientable surface of genus c* , $(1, 1, 0, 1); (1, 1, |k-2c|, (k+1)-2c)$, $k \geq 2c-1$.
 - (b) *nonorientable surface of genus c* , $(1, 1, p+1, p); (1, 1, |(k-1)-(p+2c)|, k-(p+2c))$, $k \geq (p+2c)$.
7. *Annulus, possibly with thin edge-linking tubes*.
 - (a) *Annulus, possibly with thin edge-linking tubes, which is the double of surface 6(a), with the following two possible edge-weights*,
 - (a.1) $2 \times (1, 1, 0, 1); 2 \times (1, 1, 1, 0);$
 - (a.2) $2 \times (1, 1, 0, 1); 2 \times (1, 1, q, q+1)$, $q \geq 0$;
 - (b) *Annulus, possibly with thin edge-linking tubes, which is the double of surface 6(b), with the following two possible edge-weights*,
 - (b.1) $2 \times (1, 1, p+1, p); 2 \times (1, 1, 1, 0)$, $p \geq 0$; or
 - (b.2) $2 \times (1, 1, p+1, p); 2 \times (1, 1, q, q+1)$, $p, q \geq 0$.

(c) *Annulus, possibly with thin edge-linking tubes, with the following possible weights.*

- (c.1) $2 \times (1, 1, 0, 1); 2 \times (1, 1, 1, 0);$
- (c.2) $2 \times (1, 1, 0, 1); 2 \times (1, 1, q, q + 1), q \geq 0;$
- (c.3) $2 \times (1, 1, p + 1, p); 2 \times (1, 1, 1, 0), p \geq 0;$
- (c.4) $2 \times (1, 1, p + 1, p); 2 \times (1, 1, q, q + 1), p, q \geq 0.$

PROOF. We will prove it by induction.

In C_2 , we just need to consider how to push through or band the surface starting from those normal arcs or curves with possible bottom edge-weights on A_0 in a layered chain triangulation of the solid torus. According to Lemma 2.8, all the possibilities are $(2, 0, 1, 1)$, $(0, 2, 1, 1)$, $(2, 2, 2, 2)$, $(0, 0, 1, 1)$, $(1, 1, p, p + 1)$, $(1, 1, p + 1, p)$, $2 \times (1, 1, p, p + 1)$ or $2 \times (1, 1, p + 1, p)$ with $p \geq 0$.

Let's first study the $(2, 0, 1, 1)$ case. We can only push through this normal arc in the first tetrahedron, which will produce a normal disk with edge-weight $(2, 0, 1, 1)$ on the top annulus A_1 , which is the bottom annulus of the second tetrahedron. Again, we can only push through the normal arc $(2, 0, 1, 1)$, in the second tetrahedron, which will intersect the top annulus with the same edge-weight $(2, 0, 1, 1)$. Hence, we get a vertex-linking disk with edge-weight $(2, 0, 1, 1); (2, 0, 1, 1)$. This gives us case 0 of the theorem.

By the same method, one can check all the normal surfaces found in C_2 obtained from the possible edge-weights listed in Lemma 2.8 are among the list.

Now let's assume that all the normal surfaces in $C_k, k \geq 2$, are among those listed in the theorem.

Let's consider C_{k+1} . A connected normal surface S_{k+1} in C_{k+1} meets C_k in a normal surface S_k and meets the tetrahedron σ_{k+1} in a collection of normal triangles and normal quadrilaterals. There are two possibilities that determine the collection of triangles and quadrilaterals in σ_{k+1} : pushing S_k through or adding bands on S_k . Case A. The surface S_{k+1} is obtained from S_k by pushing S_k through σ_{k+1} .

In this case the surface S_{k+1} is homeomorphic to S_k ; hence, we only need to check the intersection numbers for components of S_{k+1} meeting the top annulus of C_{k+1} . We have the same top edge weight for $(2, 0, 1, 1)$, $(0, 2, 1, 1)$, $(2, 2, 2, 2)$ and $(0, 0, 1, 1)$. For the general case $(1, 1, p, p + 1)$, with $p \geq 0$ we have $(1, 1, p + 1, p + 2)$ and for $(1, 1, p + 1, p)$, we have $(1, 1, p, |p - 1|), p \geq 0$. This satisfies our induction hypothesis.

Case B. The surface S_{k+1} is obtained from S_k by banding in σ_{k+1} .

Recall that for a band to be added, the surface S_k must meet the top annulus of C_k with edge-weight $(2, 2, 2, 2)$ or $(1, 1, 0, 1)$ or the double of the latter case. Furthermore, since we are only interested in the case where the surface S_{k+1} is connected, then either S_k is connected or S_k is disconnected with edge-weight $(2, 2, 2, 2)$ or $2 \times (1, 1, 0, 1)$.

Our induction hypothesis is that we have classified the connected normal surfaces in C_k and they appear in the above list of surfaces. Hence, by running through the different types or combinations of surfaces 0 to 7 in the list, we can distinguish all possibilities having the edge-weights in the top annulus A_k of C_k either $(2, 2, 2, 2)$ or $(1, 1, 0, 1)$ or $2 \times (1, 1, 0, 1)$.

Obviously, the combination of two different surfaces in case 0 will give us a disconnected S_k with top edge-weight $(2, 2, 2, 2)$ in σ_k . We can add a band to this

S_k in σ_{k+1} , and we will get an upper edge-linking disk S_{k+1} , with edge-weight $(2, 2, 2, 2); 2 \times (1, 1, 1, 0)$ in C_{k+1} . This gives us the surface in case 4.

One can similarly check the other cases. We find that all possible surfaces S_{k+1} are among the list in the hypothesis. $\square$

3. Twisted layered loop triangulations of M_k

First let's see how Burton constructed this triangulation. We identify the four free faces on the top and bottom annuli of a layered chain triangulation of the solid torus by layering σ_1 onto σ_k along e_k with $e_1 \leftrightarrow -e_{k+1}$, $e_2 \leftrightarrow -e_{k+2}$, and $t \leftrightarrow -b$. See figure 5. The result is a closed 3-manifold denoted M_k , with the triangulation denoted $\hat{C}_k$. In [1], he also shows that for each $k \geq 2$, the twisted layered loop triangulation $\hat{C}_k$ is a one-vertex triangulation of the space S^3/Q_{4k} , or equivalently of the Seifert fibered space $SFS(S^2 : (2, 1), (2, 1), (k, -k + 1))$.

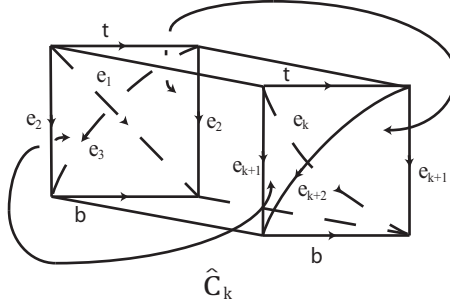


FIGURE 5. Twisted layered loop triangulation $\hat{C}_k$.

Recently in [9], it is proved that the generalized quaternion spaces S^3/Q_{4k} have complexity k , $k \geq 2$. The complexity of a 3-manifold M is the minimal number of tetrahedra in a triangulation of M . Therefore, a twisted layered loop triangulation $\hat{C}_k$ of M_k is a minimal triangulation of M_k .

3.1. Normal surfaces in twisted layered loop triangulations. We will discuss and classify normal surfaces in the twisted layered loop triangulations $\hat{C}_k$ of M_k , where $M_k = S^3/Q_{4k} = (S^2 : (2, 1), (2, 1), (k, 1 - k))$, $k \geq 1$. Notice when $k = 1$, M_k is a lens space. Otherwise, M_k is a Seifert fibered space.

If $\hat{S}$ is a normal surface in $\hat{C}_k$, then $\hat{S}$ determines a unique normal surface S in the layered chain triangulation C_k of the solid torus, and $\hat{S}$ is obtained from S by identifications along ∂S . Hence, it is necessary that the surface S has the same edge-weights on the identified edges e_{k+1} and e_1 , and similarly with edges e_{k+2} and e_2 , and edges t and b . In other words, S must have the ordered edge weights $(wt_t, wt_b, wt_{e_1}, wt_{e_2}) = (wt_t, wt_b, wt_{e_{k+1}}, wt_{e_{k+2}})$, i.e. the corresponding coordinates are the same. Furthermore, we realize that $wt_t = wt_b$, since the edge t is identified with $-b$.

THEOREM 3.1. *A connected, embedded normal surface in the twisted layered loop triangulation $\hat{C}_k$, $k \geq 1$ is normally isotopic to one of the following surfaces:*

1. *Vertex-linking 2-sphere, possibly with thin edge-linking tubes.*
2. *A Klein bottle, which is an incompressible, quadrilateral splitting surface, splitting the opposite edges $t = -b$ in each tetrahedron.*
3. *k is even, and there are two nonorientable surfaces:*
 - (a) *a surface of genus $\frac{k}{2} + 1$, which is an incompressible, quadrilateral splitting surface, splitting the odd index edges.*
 - (b) *a surface of genus $\frac{k}{2} + 1$, which is an incompressible, quadrilateral splitting surface, splitting the even index edges.*

PROOF. We only need to go through the list of embedded normal surfaces in the layered chain triangulation C_k of the solid torus as in Theorem 2.9 and consider those normal surfaces that have ordered edge weights $(wt_t, wt_b, wt_{e_1}, wt_{e_2}) = (wt_t, wt_b, wt_{e_{k+1}}, wt_{e_{k+2}})$, and $wt_t = wt_b$.

We will not show the details for checking each type of surface listed, but since we are interested in the nonorientable case, let's focus on these special cases:

Case A. From the vertical annulus, $(0, 0, 1, 1); (0, 0, 1, 1)$, in the layered chain triangulation we have a *Klein bottle* which is a quadrilateral splitting surface splitting the opposite edges $t = -b$ in each tetrahedron.

In this case the two boundaries of the vertical annulus have an orientation reversing identification, and by calculating the Euler characteristic we find that it is a Klein bottle.

Case B. If k is even we get two nonorientable surfaces each having genus $\frac{k}{2} + 1$.

- (a) From the nonorientable surfaces of genus c , $(1, 1, 0, 1); (1, 1, 0, 1)$, we get a nonorientable surface of genus $\frac{k}{2} + 1$, $\widehat{S}_1$, which is a quadrilateral splitting surface splitting the odd index edges.

In case 6(a) of Theorem 2.9 we have a family of nonorientable surfaces of genus c , $(1, 1, 0, 1); (1, 1, |k - 2c|, (k + 1) - 2c)$. For a surface from this family to give a surface in $\widehat{C}_k$, we must have the edge-weight $|k - 2c| = 0$, which happens if and only if $k = 2c$. We know that this surface is a quadrilateral splitting surface, splitting the odd index edges. This surface has connected boundary in C_k and upon identification of its boundary, we add another crosscap, giving a nonorientable surface with genus $c + 1 = \frac{k}{2} + 1$ in $\widehat{C}_k$.

- (b) From the nonorientable surface of genus $c = \frac{k}{2}$, $(1, 1, 1, 0); (1, 1, 1, 0)$, we get a nonorientable surface of genus $\frac{k}{2} + 1$, $\widehat{S}_2$, which is a quadrilateral splitting surface, splitting the even index edges.

In case 6(b) of Theorem 2.9 we have a family of nonorientable surfaces of genus c , $(1, 1, p + 1, p); (1, 1, |(k - 1) - (p + 2c)|, k - (p + 2c))$, $k \geq (p + 2c)$. For a surface from this family to give a surface in $\widehat{C}_k$, we must have the edge-weight $k - (p + 2c) = p$, which happens if and only if $k = 2p + 2c$. However, we also must have the edge-weight $|(k - 1) - (p + 2c)| = p + 1$ for $k = 2p + 2c$, which happens if and only if $p = 0$. So, from part 6(b) above, we must have the nonorientable surface of genus $c = \frac{k}{2}$, $(1, 1, 1, 0); (1, 1, 1, 0)$. We noted that this surface is a quadrilateral splitting surface, splitting the even index edges. It has connected boundary in C_k and upon identification of its boundary, we add another crosscap giving a nonorientable surface with genus $c + 1 = \frac{k}{2} + 1$ in $\widehat{C}_k$.

In any triangulation, at least one incompressible surface from each isotopy class must appear. Notice that the three possible nonorientable surfaces in case 2 and 3 are the only possible nonorientable surfaces in the twisted layered chain triangulation. No other nonorientable surfaces with smaller genus exist which these nonorientable surfaces can be compressed to, therefore they are incompressible.

When $k = 1$, M_k is a lens space. We can use the same argument about the edge-weights to classify all the normal surfaces. One can easily check that there are only two normal surfaces, one is vertex-linking 2-sphere, and the other is a nonorientable incompressible surface: a klein bottle, obtained by identifying along the boundary of the vertical annulus $(0, 0, 1, 1); (0, 0, 1, 1)$ in C_1 . $\square$

REMARK 3.2. The twisted layered loop triangulation of M_k is a 0-efficient triangulation, since the only embedded normal 2-sphere is vertex-linking according to Theorem 3.1.

REMARK 3.3. In Theorem 3.1, the double of the surface in case 2 is a thin edge-linking torus about the edge $t = -b$. The doubles of the surfaces in case 3 are vertex-linking 2-spheres with thin edge-linking tubes; in case 3(a), the thin edge-linking tubes are around the odd index edges and in case 3(b), the thin edge-linking tubes are around the even index edges. In all of these doubles, we have quadrilateral surfaces that fall into case 1.

3.2. Classification of nonorientable incompressible surfaces in M_k .

THEOREM 3.4. *Let $M_k = S^3/Q_{4k} = (S^2 : (2, 1), (2, 1)(k, 1 - k))$, $k \geq 1$. If k is even, then M_k has precisely three distinct (up to isotopy) embedded, connected, nonorientable, incompressible surfaces; one is a Klein bottle, each of the other two have genus $\frac{k}{2} + 1$. If k is odd, then M_k has precisely one (up to isotopy) embedded, connected, nonorientable, incompressible surface, a Klein bottle.*

PROOF. The fundamental group of M_k is

$$\pi_1 M_k = \langle f, s_1, s_2, s_3 | [s_i, f], s_1^2 f, s_2^2 f, s_3^k f, s_1 s_2 s_3 f^e = 1, i = 1, 2, 3 \rangle,$$

where the interger e is the usual Euler class representing the obstruction to extend a section given on the boundary components of regular neighborhoods of the exceptional fibers to the complement. Here $e = 1$. (c.f. [6, 11])

Therefore, the first homology group $H_1(M_k, \mathbb{Z}_2)$ is $\mathbb{Z}_2 \oplus \mathbb{Z}_2$ for k even, and is $\mathbb{Z}_2$ for k odd. For any nonzero class in $H_1(M_k, \mathbb{Z}_2) \cong H^1(M_k, \mathbb{Z}_2) \cong H_2(M_k, \mathbb{Z}_2)$, there is a nonorientable incompressible surface associated with it (See details in [12]). Furthermore, any two nonorientable incompressible surfaces associated with different nonzero classes in $H_1(M_k, \mathbb{Z}_2)$ cannot be isotopic to each other.

For k even there are 3 nonzero classes in $H_1(M_k, \mathbb{Z}_2)$. Hence, M_k has at least three distinct (up to isotopy), embedded, nonorientable incompressible surfaces.

By Theorem 3.1 the triangulation $\hat{C}_k$ has precisely three nonorientable incompressible surfaces, up to normal isotopy. Since normal isotopy is isotopy and the converse is not necessarily true, therefore, in the twisted layered loop triangulation, there are at most three nonorientable incompressible normal surfaces, up to isotopy. Since in any triangulation, at least one incompressible surface from each isotopy class must appear, therefore, there are at most three nonorientable incompressible normal surfaces, up to isotopy, in M_k .

Hence, there are exactly three nonorientable incompressible surfaces, up to isotopy, in M_k . In any triangulation, at least one incompressible surface from

each istopy class in the manifold must appear. Therefore, the three nonorientable, incompressible surfaces in $\widehat{C}_k$ we obtained in Theorem 3.1 are distinct, up to isotopy, and they are the only three nonorientable incompressible surfaces in M_k : a Klein bottle and two surfaces each with genus $\frac{k}{2} + 1$.

By the similar argument, for k is odd, there is only one incompressible surface up to isotopy, which is the Klein bottle. $\square$

4. Layered chain pair triangulations of $M_{r,s}$

The starting point for a layered chain pair triangulation is two layered chain triangulations C_r and C_s of the solid torus, of length r and s , respectively. The boundary of the solid torus in a layered chain triangulation is made up of four triangles. Two of them make up what we are calling the top annulus and the other two make up the bottom annulus. In the layered chain C_r we label the boundary edges $\tau, \beta, f_1, f_2, f_{r+1}$, and f_{r+2} and direct them as in Figure 3.24 in [1]. In the layered chain C_s we label the boundary edges t, b, e_1, e_2, e_{s+1} and e_{s+2} and direct them as in Figure 3.24 in [1]. A manifold with triangulation is then obtained by identifying the four boundary faces of C_r with the four boundary faces of C_s , using the following edge identifications. See Figure 6.

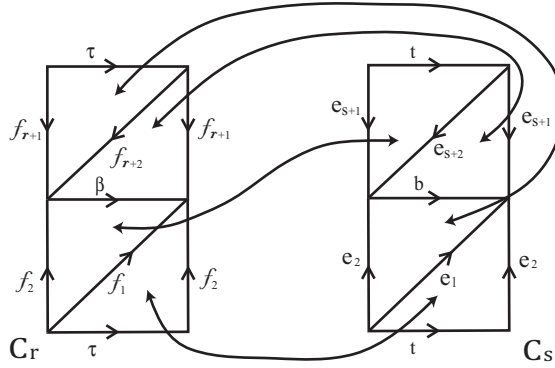


FIGURE 6. Face identifications of two layered chain triangulations, C_r and C_s , giving a layered chain pair triangulation, $C_{r,s}$.

$$\begin{aligned} b &\leftrightarrow f_{r+1} & t &\leftrightarrow -f_2 \\ e_2 &\leftrightarrow -\tau & e_{s+1} &\leftrightarrow \beta \\ e_1 &\leftrightarrow -f_1 \leftrightarrow -e_{s+2} \leftrightarrow f_{r+2}. \end{aligned}$$

The result is a closed 3-manifold, denoted $M_{r,s}$, and the triangulation, denoted $C_{r,s}$, is called an (r, s) *layered chain pair* after Ben Burton.

Ben Burton proved in [1] that for each $r, s \geq 1$, the layered chain pair $C_{r,s}$ is a triangulation of the Seifert fibred space $(S^2 : (2, -1), (r+1, 1), (s+1, 1))$. The layered chain pair $C_{r,1}$ is in fact simply the twisted layered loop $\widehat{C}_{r+1}$.

$C_{r,s}$ is symmetric in r and s , so $C_{r,s} = C_{s,r}$ for all r and s .

4.1. Normal surfaces in layered chain pair triangulations. If S is a normal surface in $C_{r,s}$, then S determines a unique normal surface S_r in C_r and a unique normal surface S_s in C_s . Notice, the normal surface S_r and S_s are not necessarily connected. Furthermore, S is obtained from S_r and S_s by identifications along their boundaries. Hence, it is necessary that the boundary of the surface S_r have the same edge-weights in C_r as the boundary of the surface S_s has in C_s on edges matching under the above face identifications to obtain $C_{r,s}$. In particular, we must have the edge-weights $wt_{f_2}(S_r) = wt_t(S_s)$, $wt_{f_{r+1}}(S_r) = wt_b(S_s)$, $wt_\tau(S_r) = wt_{e_2}(S_s)$, and $wt_\beta(S_r) = wt_{e_{s+1}}(S_s)$, and edge-weights $wt_{f_1}(S_r) = wt_{e_1}(S_s) = wt_{f_{r+2}}(S_r) = wt_{e_{s+2}}(S_s)$.

In our notation, the edge-weights for the boundary of the surface S_r are given as $(wt_\tau, wt_\beta, wt_{f_1}, wt_{f_2}); (wt_\tau, wt_\beta, wt_{f_{r+1}}, wt_{f_{r+2}})$, whereas those for the surface S_s are given as $(wt_t, wt_b, wt_{e_1}, wt_{e_2}); (wt_t, wt_b, wt_{e_{s+1}}, wt_{e_{s+2}})$. Hence, in order for S_r and S_s to match to give a normal surface in $C_{r,s}$, we must have pairs of 4-tuples:

$$(4.1) \quad (x, y, z, u); (x, y, v, z) \leftrightarrow (u, v, z, x); (u, v, y, z),$$

where the first pair $(x, y, z, u); (x, y, v, z)$ are the parameterizations for edge-weights of the boundary of S_r in the bottom annulus and the top annulus of C_r , respectively, and the second pair $(u, v, z, x); (u, v, y, z)$ are the parameterizations for the edge-weights of the boundary of S_s in the bottom annulus and the top annulus of C_s , respectively. From now on, we will identify the unique normal surface S obtained from S_r and S_s by the edge-weight matching equation $(x, y, z, u); (x, y, v, z) \leftrightarrow (u, v, z, x); (u, v, y, z)$, determined from S_r and S_s .

THEOREM 4.1. *A connected, embedded, normal surface S in the layered chain pair triangulation $C_{r,s}$, $r, s > 1$ is normally isotopic to one of the following surfaces: For the orientable case,*

- I. S is a vertex-linking S^2 , possibly with thin edge-linking tubes.

It has one of the following possible edge-weight matching equations,

1. $(2, 2, 2, 2); (2, 2, 2, 2) \leftrightarrow (2, 2, 2, 2); (2, 2, 2, 2)$
2. $2 \times (1, 1, 0, 1); 2 \times (1, 1, 1, 0) \leftrightarrow 2 \times (1, 1, 0, 1); 2 \times (1, 1, 1, 0)$
3. $[(0, 2, 1, 1); (0, 2, 1, 1)] + [(0, 0, 1, 1); (0, 0, 1, 1)] \leftrightarrow 2 \times (1, 1, 1, 0); (2, 2, 2, 2)$
4. $[(2, 0, 1, 1); (2, 0, 1, 1)] + [(0, 0, 1, 1); (0, 0, 1, 1)] \leftrightarrow (2, 2, 2, 2); 2 \times (1, 1, 0, 1)$
5. $2 \times (0, 0, 1, 1); 2 \times (0, 0, 1, 1) \leftrightarrow 2 \times (1, 1, 1, 0); 2 \times (1, 1, 0, 1)$

- II. S is an orientable normal surface, which is not a thin edge-linking tube surface. If S_r in C_r has genus g' , and S_s in C_s has genus g , then:

1. S is a nonseparating torus in $C_{2,5}$ with edge-weight matching equation $(2, 2, 3, 1); (2, 2, 1, 3) \leftrightarrow (1, 1, 3, 2); (1, 1, 2, 3)$.

It is a torus fiber in the fibration of $M_{2,5}$ over S^1 .

2. S is an orientable surface with genus $g + 2$ in $C_{2,n}$, $n \geq 7$ with edge-weight matching equation $2 \times (2, 2, 3, 1); 2 \times (2, 2, 1, 3) \leftrightarrow 2 \times (1, 1, 3, 2); 2 \times (1, 1, 2, 3)$.

3. S is a nonseparating torus in $C_{3,3}$ with edge-weight matching equation $(1, 1, 2, 1); (1, 1, 1, 2) \leftrightarrow (1, 1, 2, 1); (1, 1, 1, 2)$.

It is a torus fiber in the fibration of $M_{3,3}$ over S^1 .

4. S is an orientable surface with genus $g' + g + 2$ in $C_{3,5}$ or $C_{r,s}$, $r, s \geq 5$, with edge-weight matching equation $2 \times (1, 1, 2, 1); 2 \times (1, 1, 1, 2) \leftrightarrow 2 \times (1, 1, 2, 1); 2 \times (1, 1, 1, 2)$.

For the nonorientable case,

I. r even and s odd, $s = 2c + 1, c \geq 1$:

1. a nonorientable incompressible surface of genus c , if $r = 2, s = 2c + 1, s \geq 7$, i.e. $c \geq 3$. The edge-weight matching equation is $(2, 2, 3, 1); (2, 2, 1, 3) \leftrightarrow (1, 1, 3, 2); (1, 1, 2, 3)$.

This surface's double is an orientable surface with genus $c - 1$ in the orientable case II.2. Also, if $s = 5$ this surface is a nonseparating torus in $C_{2,5}$.

2. a nonorientable, quadrilateral, splitting surface of genus $c + 2$, if r is even, $s = 2c + 1, s \geq 3$. The edge-weight matching equation is $(0, 0, 1, 1); (0, 0, 1, 1) \leftrightarrow (1, 1, 1, 0)(1, 1, 0, 1)$.

In general, this nonorientable surface is incompressible except for $C_{2,2c+1}$, $c \geq 3$, i.e. $s \geq 7$, it is a compressible surface and can be compressed to a surface mentioned in the above I.1. In $C_{2,5}$, this nonorientable surface is incompressible. However, we can find a compressing disk and compress it to a nonseparating torus mentioned in the above II.1.

II. r, s both odd, $r = 2c' + 1, s = 2c + 1, c', c \geq 1$

1. a nonorientable, incompressible, quadrilateral splitting surface of genus $c + 2$, with edge-weight matching equation $(0, 0, 1, 1); (0, 0, 1, 1) \leftrightarrow (1, 1, 1, 0); (1, 1, 0, 1)$.
2. a nonorientable, incompressible, quadrilateral splitting surface of genus $c' + 2$. The edge-weight matching equation is $(1, 1, 1, 0)(1, 1, 0, 1) \leftrightarrow (0, 0, 1, 1); (0, 0, 1, 1)$.
3. a nonorientable, quadrilateral splitting surface of genus $c' + c + 2$. The edge-weight matching equation is $(1, 1, 0, 1); (1, 1, 1, 0) \leftrightarrow (1, 1, 0, 1); (1, 1, 1, 0)$.

For $c = c' = 1$, i.e. in $C_{3,3}$, this surface is incompressible. Otherwise this surface is compressible. They can be compressed to get a surface in the following case.

4. a nonorientable, incompressible surface of genus $c' + c$, except for $c' = c = 1$. The edge-weight matching equation is $(1, 1, 2, 1); (1, 1, 1, 2) \leftrightarrow (1, 1, 2, 1); (1, 1, 1, 2)$.

When $c' = c = 1$, this surface is a nonseparating torus in $C_{3,3}$ in the orientable case II.3.

PROOF. As noted above, a normal surface S in $C_{r,s}$ determines unique normal surfaces S_r in C_r and S_s in C_s and is formed by identification of the boundary of S_r with the boundary of S_s . Furthermore, in order for the boundaries of S_r and S_s to match to give a normal surface in $C_{r,s}$, we must have the pairs of 4-tuples of edge-weight match as in Equation 4.1, including possible multiplicities.

For the proof, we analyze the possible matches of these 4-tuples, using the classification of normal surfaces in a layered chain triangulation of the solid torus given in Theorem 2.9.

Notice all the families of normal surfaces in Theorem 2.9 have edge-weight 4-tuples satisfying the equation $wt_t = wt_b$, except for the vertex linking disk, $(0, 2, 1, 1); (0, 2, 1, 1)$ and $(2, 0, 1, 1); (2, 0, 1, 1)$ in case 0. Therefore, $|x - y| = 2m$, $m \geq 0$, holds for the edge-weights of the normal surface S_r , $(x, y, z, u); (x, z, v, y)$, in C_r , where m is the number of extra copies of $(0, 2, 1, 1); (0, 2, 1, 1)$ or $(2, 0, 1, 1); (2, 0, 1, 1)$;

$(2, 0, 1, 1)$ in case 0. If $x \geq y$, then m is the number of extra copies of $(2, 0, 1, 1)$; $(2, 0, 1, 1)$, and vice versa. Similarly, $|u - v| = 2n$, $n \geq 0$, holds for the the edge-weights of the boundary of S_s , $(u, v, z, x); (u, v, y, z)$, in C_s , where n is the the number of extra copies of $(0, 2, 1, 1); (0, 2, 1, 1)$ or $(2, 0, 1, 1); (2, 0, 1, 1)$ in case 0.

Since $C_{r,s}$ is symmetric in r and s , we only need to consider the following 3 cases.

- $x = y$ and $u = v$,
- $x = y$ and $u \neq v$,
- $x \neq y$ and $u \neq v$.

Then we can run through the list of normal surfaces in Theorem 2.9 and discuss each possibility by using the matching equation 4.1. We will not show the details for checking all the possibilities. The problem here is that we need to not only consider the possible multiplicities, but also the disconnected case for S_r and S_s , respectively, since we can get connected normal surface after identification of disconnected surface S_r and S_s . The reason is that the vertex-linking disks $(0, 2, 1, 1); (0, 2, 1, 1)$ and $(2, 0, 1, 1); (2, 0, 1, 1)$ are involved in the combination when $x \neq y$ or $u \neq v$.

Since here we are only interested in the nonorientable surfaces and most nonorientable surfaces are incompressible, we will just show one case to illustrate how we find a single nonorientable surface which is compressible.

In the case $x = y$ and $u = v$, If S_r is from 6(a) of Theorem 2.9, it is a nonorientable surface of genus c' , $(1, 1, 0, 1); (1, 1, |r - 2c'|, (r + 1) - 2c')$, $r \geq 2c' - 1$, $c' \geq 1$,. In this case, by Equation 4.1, we have the match $(1, 1, 0, 1); (1, 1, 1, 0) \leftrightarrow (1, 1, 0, 1); (1, 1, 1, 0)$. Hence, the surface S_r has edge-weight $(1, 1, 0, 1); (1, 1, 1, 0)$, where $r + 1 - 2c' = 0$, i.e. $r = 2c' - 1$. Moreover, the only possibility for the surface S_s in this case with edge-weight $(1, 1, 0, 1); (1, 1, 1, 0)$ is a nonorientable surface of genus c from 6(a) with $s = 2c - 1$.

Now, changing the index to match that above, we have for $r = 2c' + 1$ with $c' \geq 0$ and $s = 2c + 1$ with $c \geq 0$, nonorientable surfaces, S_r and S_s , of genus $c' + 1$ and $c + 1$, respectively. Considering how these surfaces are attached along their boundaries, we have in $C_{2c'+1, 2c'+1}$ a *nonorientable, compressible, quadrilateral splitting surface of genus $c' + c + 2 = \frac{r+s+2}{2}$* . Notice that this surface compresses about the valence 4 edge $e_1 \leftrightarrow -f_1 \leftrightarrow -e_{s+2} \leftrightarrow f_{r+2}$. After compressing, we will get a nonorientable, incompressible surface of genus $c' + c = \frac{r+s-2}{2}$,. It has the edge-weight matching equation $(1, 1, 2, 1); (1, 1, 1, 2) \leftrightarrow (1, 1, 2, 1); (1, 1, 1, 2)$, which is a surface in case II.4.

One can check all the other possibilities and find that all possible surfaces are in the list.

In any triangulation at least one incompressible surface from each isotopy class should appear. Moreover, for any nonorientable normal surface, if there is no smaller genus nonorientable surface that it can be compressed to in the triangulation, it is incompressible. Therefore, in the layered chain pair triangulation $C_{r,s}$ of $M_{r,s}$, $r, s > 1$, up to normal isotopy, when r and s have opposite parity, there is precisely one nonorientable incompressible surface: in $C_{2,3}$ and $C_{2,5}$, it is an incompressible surface in the nonorientable case I.2; In $C_{2,s}$, $s \geq 7$ and $s = 2c + 1$, it is an incompressible surface in the nonorientable case I.1; otherwise, it is again an incompressible surface in the nonorientable case I.2. When r and s are both odd, there are exactly three nonorientable incompressible surfaces. In $C_{3,3}$, the three incompressible surfaces are the ones in nonorientable case II.1-3; In $C_{r,s}$, $r, s > 3$, the three incompressible surfaces are the ones in nonorientable case II.1, 2 and 4.

When r and s are both even, there exist no nonorientable incompressible surfaces. □

REMARK 4.2. The layered chain pair triangulation of $M_{r,s}$ is a 0-efficient triangulation, since the only embedded normal 2-sphere is vertex-linking according to Theorem 4.1.

4.2. Classification of nonorientable incompressible surfaces in $M_{r,s}$.

THEOREM 4.3. *Let $M_{r,s} = (S^2 : (2, -1), (r+1, 1), (s+1, 1))$. If r and s have opposite parity, then $M_{r,s}$ has precisely one embedded, connected, nonorientable incompressible normal surface, up to isotopy. If r and s are both odd, then $M_{r,s}$ has precisely three distinct embedded, connected, nonorientable, incompressible surfaces, up to isotopy. If r and s are both even, then $M_{r,s}$ has no embedded connected, nonorientable, incompressible surfaces, up to isotopy.*

PROOF. $M_{r,s} = M_{r+1}$ if $s = 1$, and $M_{1,s} = M_{s+1}$ if $r = 1$. By Theorem 3.4, the conclusion holds.

Now let's focus on general case when $r > 1$ and $s > 1$. The fundamental group of $M_{r,s}$ is

$$\pi_1 M_{r,s} = \langle f, s_1, s_2, s_3 | [s_i, f], s_1^2 f, s_2^{r+1} f, s_3^{s+1} f, s_1 s_2 s_3 f^e = 1, i = 1, 2, 3 \rangle,$$

where the integer e is the usual Euler class representing the obstruction to extend a section given on the boundary components of regular neighborhoods of the exceptional fibers to the complement. Here $e = 1$. (c.f. [6, 11])

The first homology group $H_1(M_{r,s}, \mathbb{Z}_2)$ is $\mathbb{Z}_2 \oplus \mathbb{Z}_2$ for r, s both odd, $\mathbb{Z}_2$ for exactly one of r and s even, and is trivial for r and s both even. Therefore, there are 3 nonzero classes in $H_1(M_{r,s}, \mathbb{Z}_2)$ for r, s both odd, one nonzero class for exactly one of r and s even, and no nonzero classes for r and s both even.

For any nonzero class in $H_1(M_{r,s}, \mathbb{Z}_2) \cong H^1(M_{r,s}, \mathbb{Z}_2) \cong H_2(M_{r,s}, \mathbb{Z}_2)$, there is a nonorientable incompressible surface associated with it (See details in [12]). Therefore, in $M_{r,s}$, up to isotopy, there are at least 3 nonorientable incompressible surfaces for r, s both odd, at least one nonorientable incompressible surface for r, s both even.

By Theorem 4.1, in the layered chain pair triangulation $C_{r,s}$ of $M_{r,s}$, $r, s > 1$, up to normal isotopy, when r and s are both odd, there are exactly three nonorientable incompressible surfaces; when r and s have opposite parity, there is precisely one nonorientable incompressible surface; when r and s are both even, there is no nonorientable incompressible surface. By the same argument in Theorem 3.4, in $M_{r,s}$, up to isotopy, there are three distinct nonorientable incompressible surfaces when r and s are both odd; there is at most one nonorientable incompressible surface when r and s have opposite parity; there is no nonorientable incompressible surface.

Therefore, in $M_{r,s}$, up to isotopy, there are precisely three nonorientable incompressible surfaces when r and s are both odd, there is precisely one nonorientable incompressible surface when r and s have opposite parity, and there are no nonorientable incompressible surfaces when r and s are both even. These nonorientable incompressible surfaces are listed in Theorem 4.1 □

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Introduction to the theory of Haken n -manifolds

Bell Foozwell and Hyam Rubinstein

ABSTRACT. We define the class of Haken n -manifolds, following Johansson [14]. A number of basic results are proved and some examples given. A key property is that these manifolds have universal coverings $\mathbf{R}^n$ and so are aspherical. The latter is established here and the former is proved in [8]. Some problems are given in the final section. In particular, there is a natural Haken cobordism category and computing this would provide many interesting examples.

Dedicated to Bus Jaco for his 70th birthday.

1. Introduction

A Haken n -manifold is a generalisation of a Haken 3-manifold to higher dimensions. The basic feature of Haken 3-manifolds is that of a *hierarchy*. Let M be a compact¹ 3-manifold. A hierarchy for M is a sequence:

$$(M_0, F_0), (M_1, F_1), \dots, (M_k, F_k)$$

where

- (1) $M_0 = M$,
- (2) F_j is an incompressible and boundary-incompressible compact orientable surface properly embedded in M_j ,
- (3) M_{j+1} is obtained by splitting M_j open along F_j ,
- (4) M_{k+1} is a disjoint union of 3-cells.

To generalise this idea to higher dimensions, we need to define incompressible and boundary-incompressible hypersurfaces in all dimensions. To obtain the appropriate structures, these hypersurfaces must also be Haken $(n - 1)$ -manifolds. Suppose that M is an orientable n -manifold and that F is a *closed* orientable $(n - 1)$ -manifold embedded in M . We say that F is *essential* in M if the inclusion map $F \rightarrow M$ induces an injection $\pi_1(F) \rightarrow \pi_1(M)$ of fundamental groups. However, as in 3-manifold theory, we need to consider submanifolds G of M with boundary that are properly embedded in M in an essential way. This can be done by requiring that the inclusion map of pairs $(G, \partial G) \rightarrow (M, \partial M)$ induces an injection

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¹We will assume that all manifolds are compact, unless explicitly stated otherwise, or obvious from context.

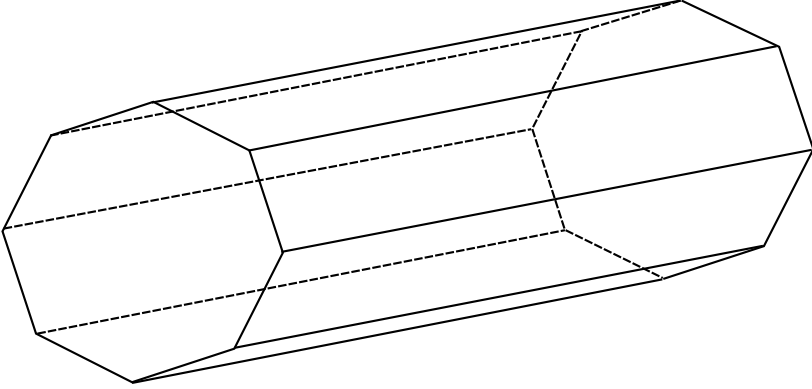


FIGURE 1. A three-dimensional manifold with boundary-pattern.

$\pi_1(G, \partial G) \rightarrow \pi_1(M, \partial M)$ of relative homotopy groups. This generalises the concept of boundary-incompressibility. To obtain a hierarchy with similar properties to dimension 3, we use Johansson's idea of boundary-patterns [15]. The reason is that conditions of incompressibility and boundary incompressibility for the hypersurfaces are not sufficient; we also require that the boundaries of the hypersurfaces cut the hypersurfaces up into hierarchies and similarly for the boundaries of the hierarchies and their intersections. Using boundary-patterns allows us to obtain desirable hierarchy properties in all codimensions simultaneously.

2. Basic Definitions

DEFINITION 2.1. Let M be a compact n -manifold with boundary. A *boundary-pattern* $\underline{m}$ for M is a finite collection of compact, connected $(n-1)$ -manifolds in ∂M such that the intersection of any i of them is either empty or consists of $(n-i)$ -manifolds, for $i = 1, \dots, n+1$.

Such a manifold is called a *manifold with boundary-pattern*. We use the notation $(M, \underline{m})$ when we wish to emphasize that M is a manifold with boundary-pattern. The elements of $\underline{m}$ are called *faces* of the boundary-pattern.

A boundary-pattern is *complete* if $\partial M = \bigcup \{A : A \in \underline{m}\}$. If the boundary-pattern is not complete, a complete boundary-pattern can be formed by including the components of closure $(\partial M \setminus \bigcup \{A : A \in \underline{m}\})$ together with the elements of $\underline{m}$. This complete boundary-pattern is called the *completion* of $\underline{m}$ and is denoted $\overline{\underline{m}}$.

The tetrahedron, the cube and the dodecahedron are examples of 3-manifolds with boundary-pattern. The octahedron and the icosahedron are not manifolds with boundary-pattern, for we can find a pair of faces (triangles) that intersect in a point (a 0-manifold). The n -cube and n -simplex are n -manifolds with boundary-pattern. Figure 1 shows a 3-manifold with boundary-pattern of the form $P \times [0, 1]$, where P is an octagon. The boundary-pattern is $\{P \times 0, P \times 1, A_1, \dots, A_8\}$, where each A_i is a rectangle. Examples in four dimensions can be formed by taking the product of a closed interval with any 3-manifold with boundary-pattern.

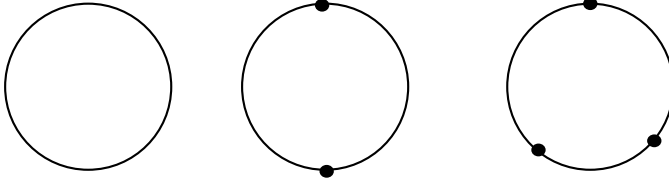


FIGURE 2. Preimages of intersection complex for short loops.

The *intersection complex* $K = K(\underline{\underline{m}})$ of a manifold with boundary-pattern is

$$K = \bigcup_{A \in \underline{\underline{m}}} \partial A.$$

For 3-manifolds, the intersection complex is called the *graph* of the boundary-pattern.

DEFINITION 2.2. Let $(M, \underline{\underline{m}})$ and $(N, \underline{\underline{n}})$ be manifolds with boundary-patterns. An *admissible map* between M and N is a continuous proper map $f: M \rightarrow N$ such that

$$\underline{\underline{m}} = \bigsqcup_{A \in \underline{\underline{n}}} \{B : B \text{ is a component of } f^{-1}(A)\}.$$

Furthermore, f must be transverse to the boundary-patterns.

Consider a disk with complete boundary-pattern consisting of i components. If $i = 1$, then such a disk has zero vertices, and we call such a disk a *zerogon*. For $i \geq 2$, such a disk has i vertices. A *bigon* has two vertices and a *triangle* has three vertices. Collectively, zerogons, bigons and triangles are called *small disks*².

DEFINITION 2.3. Let K be the intersection complex of an n -manifold $(M, \underline{\underline{m}})$. Suppose that for any admissible map $f: (\Delta, \underline{\underline{\delta}}) \rightarrow (M, \underline{\underline{m}})$ of a small disk, there is a map $g: \Delta \rightarrow \partial M$, homotopic to f rel $\partial \Delta$, such that $g^{-1}(K)$ is the cone on $g^{-1}(K) \cap \partial \Delta$. Then the boundary-pattern $\underline{\underline{m}}$ of M is called a *useful boundary-pattern*.

An alternative way of thinking about definition 2.3 is as follows. Suppose that σ is a trivial short loop³ in the boundary of $(M, \underline{\underline{m}})$. Then $\sigma^{-1}(K)$ looks like one of the pictures in figure 2. The condition that $(M, \underline{\underline{m}})$ is useful means that σ extends to a map $g: \Delta \rightarrow \partial M$ of a disk such that $g^{-1}(K)$ looks like one of the corresponding pictures in figure 3.

Figure 1 is an example of a 3-manifold with useful boundary-pattern. A cube and a simplex (in any dimension) have useful boundary-patterns, while a triangular prism does not have a useful boundary-pattern.

DEFINITION 2.4. Let $(J, \underline{\underline{j}})$ be a compact one-dimensional manifold with boundary-pattern and let $(M, \underline{\underline{m}})$ be an n -dimensional manifold with boundary-pattern. An admissible map $\sigma: (J, \underline{\underline{j}}) \rightarrow (M, \underline{\underline{m}})$ is an *inessential curve* if there is a disk $(\Delta, \underline{\underline{\delta}})$ and an admissible map $g: (\Delta, \underline{\underline{\delta}}) \rightarrow (M, \underline{\underline{m}})$ such that:

²Monogons are disks with one vertex in the boundary. However, these do not play a role here, because they are not manifolds with boundary-pattern.

³A loop is trivial if it can be extended to a map of a disk. A loop in ∂M is short if it intersects at most three faces of the boundary-pattern.

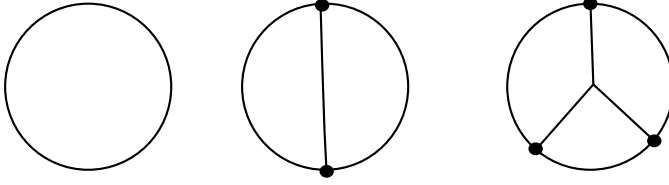


FIGURE 3. Preimages of intersection complex for disks bounding short loops.

- (1) $J = \text{closure}(\partial\Delta \setminus \bigcup\{A : A \in \underline{\underline{\delta}}\})$,
- (2) the completion $(\Delta, \underline{\underline{\delta}})$ is a small disk,
- (3) $g|_J = \sigma$.

Otherwise σ is called an *essential* curve.

DEFINITION 2.5. An admissible map $\varphi: (F, \underline{\underline{f}}) \rightarrow (M, \underline{\underline{m}})$ is called *essential* if for any essential curve $\sigma: (J, \underline{\underline{j}}) \rightarrow (F, \underline{\underline{f}})$ the composition $\varphi \circ \sigma: (J, \underline{\underline{j}}) \rightarrow (M, \underline{\underline{m}})$ is also essential. In particular, an essential submanifold F of M is a submanifold such that the inclusion map is essential.

DEFINITION 2.6. For $n = 1$, a Haken n -cell is an arc with complete (and useful) boundary-pattern. For $n > 1$, if $(M, \underline{\underline{m}})$ is an n -cell with complete and useful boundary-pattern and each face $A \in \underline{\underline{m}}$ is a Haken $(n - 1)$ -cell, then $(M, \underline{\underline{m}})$ is a Haken n -cell.

Thus a Haken 1-cell is of the form $([a, b], \{a, b\})$ for $a, b \in \mathbf{R}$. A Haken 2-cell is a disk with at least four sides.

DEFINITION 2.7. Let $(M, \underline{\underline{m}})$ be a compact n -manifold with boundary-pattern. Let F be a codimension-one, properly embedded, two-sided, essential submanifold of M . A boundary-pattern is *induced* on F if it is obtained by taking all the intersections of ∂F with the elements of the boundary-pattern $\underline{\underline{m}}$ on M . Equivalently, the boundary-pattern $\underline{\underline{f}}$ is the induced pattern if the inclusion $(F, \underline{\underline{f}}) \rightarrow (M, \underline{\underline{m}})$ is an admissible map.

DEFINITION 2.8. Let $(M, \underline{\underline{m}})$ be a compact n -manifold with complete and useful boundary-pattern. Let F be a codimension-one, properly embedded, two-sided, essential submanifold of M whose boundary-pattern is induced from the boundary-pattern on M and is complete and useful. Suppose that F is not admissibly boundary-parallel. Then the pair (M, F) is called a *good pair*.

Suppose that $(M, \underline{\underline{m}})$ is a manifold with boundary-pattern and that $(F, \underline{\underline{f}})$ is a codimension-one submanifold. Let N be the manifold obtained by splitting M open along F . There is an obvious map $q: N \rightarrow M$ that glues parts of the boundary of N together to regain M . We define a boundary-pattern $\underline{\underline{n}}$ on N by

$$B \in \underline{\underline{n}} \text{ if and only if } B \text{ is a component of } q^{-1}(A)$$

for $A \in \underline{\underline{m}}$ or $A = F$. This is the boundary-pattern that we will use whenever splitting situations arise.

DEFINITION 2.9. A finite sequence

$$(M_0, F_0), (M_1, F_1), \dots, (M_k, F_k)$$

of good pairs is called a *hierarchy of length k* for M_0 if the following conditions are satisfied:

- (1) M_{i+1} is obtained by splitting M_i open along F_i and,
- (2) M_{k+1} is a finite disjoint union of Haken n -cells.

A manifold with a hierarchy is called a *Haken n -manifold*. A Haken n -cell is a Haken manifold with a hierarchy of length zero.

More details are provided in [8]. Note that all the hypersurfaces F_i are either incompressible and boundary incompressible or are properly embedded $(n-1)$ -cells which are not homotopic into ∂M_i , by the argument in [14] (see also [2]).

We regard two Haken n -manifolds M and N as equivalent if there is an admissible homeomorphism $\varphi: (M, \underline{m}) \rightarrow (N, \underline{n})$.

We can construct simple examples of Haken n -manifolds as follows: let $(S, \underline{s})$ be a Haken k -manifold, let $(T, \underline{t})$ be a Haken $(n-k)$ -manifold, and let M be a fibre bundle with fibre S and base space T . Then it is easy to show that M is a Haken n -manifold (see [8]).

Observe that if M is a Haken n -manifold with non-empty boundary, then ∂M is a Haken $(n-1)$ -manifold. Furthermore, the hierarchy of M gives a hierarchy of ∂M , since at the end of the splitting process we obtain a manifold M_{k+1} , which is a disjoint union of Haken n -cells. Each face of a Haken n -cell is a Haken $(n-1)$ -cell, so it is easy to see that we have a hierarchy for ∂M .

Three-manifold topologists usually regard a hierarchy as a two-dimensional subcomplex of a 3-manifold. The subcomplex is a union of surfaces $\{S_0, S_1, \dots, S_k\}$ with $\partial S_{i+1} = S_{i+1} \cap (S_0 \cup S_1 \cup \dots \cup S_i)$. This is a useful viewpoint. However, we regard a hierarchy as a sequence of pairs of manifolds with boundary-patterns. The two viewpoints are clearly related. We have chosen the boundary-pattern approach here because of the ease with which it generalizes to higher dimensions.

3. Basic properties of Haken n -manifolds

The following theorems may be found in [8]. We outline the proof of the first theorem which uses standard methods.

THEOREM 3.1. *A Haken n -manifold is aspherical.*

PROOF. The argument is by induction on the lexicographically ordered pair consisting of the dimension of the manifold and the number of components in the hierarchy. We can start the induction using the easy case $n = 2$, where all surfaces which are Haken have non-positive Euler characteristic and so are aspherical, as is well-known.

Next, suppose that all Haken manifolds with smaller dimension or length of hierarchy than M are aspherical. If M is cut open along the first hypersurface F then the result is M_1 which is a Haken manifold with shorter hierarchy. Therefore by the induction hypothesis, both F and M_1 are aspherical. We also note that $\pi_1(M)$ is either a free product with amalgamation or an HNN extension over $\pi_1(F)$, using the Seifert van Kampen theorem, plus an easy argument that the inclusion map $F \rightarrow M$ is π_1 -injective or F is a properly embedded $(n-1)$ -cell.

To establish the former, suppose that there is a loop in F which is contractible in M and F is not an $(n-1)$ -cell. We can use a standard innermost loop argument to assume the contracting disk has interior disjoint from F . Next perturb the map

of the disk to be transverse to the hierarchy. The pull back to the domain of the disk of the hierarchy is a proper degree 3 graph, where each vertex can be viewed as a ‘T’ intersection. An Euler characteristic argument then establishes the existence of an innermost face of this graph with at most three vertices. But the properties of a hierarchy imply that we can push such a face into the hierarchy. Exactly as in [18], after a finite sequence of such moves, the disk can be pushed off the hierarchy and then into F . This shows the boundary loop is contractible in F which is π_1 -injective as asserted.

Next, let $\widetilde{M}$ be the universal cover of the Haken n -manifold M with covering projection $p: \widetilde{M} \rightarrow M$, and let F be the first essential hypersurface in the hierarchy of M . The closure of the complement in $\widetilde{M}$ of the preimage $p^{-1}R(F)$ of a closed regular neighbourhood of F is a countable collection of copies of the universal cover of $N := \text{closure}(M \setminus R(F))$. Label these components $N_1, N_2, N_3, \dots$, so that each N_i is a copy of the universal cover of N . Label the components of the preimage of F by $F_1, F_2, \dots$ so that each component of $p^{-1}R(F)$ is homeomorphic to $F_i \times I$ for some i . We use the technique of Waldhausen [19] to build the universal cover $\widetilde{M}$:

$$\begin{aligned} N^1 &= N_1; \\ N^j \cap (F_j \times I) &= F_j \times 0; \\ (F_j \times I) \cap N_{j+1} &= F_j \times 1; \\ N^{j+1} &= N^j \cup (F_j \times I) \cup N_j \end{aligned}$$

Then every compact subset of $\widetilde{M}$ can be contained in some N^j , so the natural map $\varinjlim H_k(N^j) \rightarrow H_k(\widetilde{M})$ is an isomorphism. We assume (using induction on the number of splitting hypersurfaces) that $H_k(N^1) = 0$. Assuming also that $H_k(N^j) = 0$, we will show that $H_k(N^{j+1}) = 0$. We obtain a Mayer-Vietoris sequence

$$\cdots \rightarrow H_k(F_j) \rightarrow H_k(N^j) \oplus H_k(N_{j+1}) \rightarrow H_k(N^{j+1}) \rightarrow H_{k-1}(F_j) \rightarrow \cdots$$

By induction on dimension, $H_k(F_j) = 0$, so we have an isomorphism $H_k(N^j) \oplus H_k(N_{j+1}) \rightarrow H_k(N^{j+1})$. Now $H_k(N^j) = 0$ and $H_k(N_{j+1}) = 0$, since N_{j+1} is the universal cover of N . Since $H_k(N^j) = 0$ for $k \geq 1$, it follows that $\varinjlim H_k(N^j) = 0$, and therefore $H_k(\widetilde{M}) = 0$. By the Hurewicz theorem, $\pi_k(\widetilde{M}) = 0$ for all $k \geq 1$. $\square$

THEOREM 3.2. *Let $\widetilde{M}$ be the universal cover of a Haken n -manifold M . Then the interior of $\widetilde{M}$ is homeomorphic to $\mathbf{R}^n$.*

For the proof see [8] or [9].

The proof of the theorem that the interior of the universal cover of a Haken n -manifold is $\mathbf{R}^n$ is considerably more difficult than the first theorem that a Haken n -manifold is aspherical. Waldhausen’s technique in dimension 3 [19] does not generalise to higher dimensions. Instead, a new approach that applies in all dimensions except three was found in [8], [9] using a theorem of Doyle [6].

THEOREM 3.3. *Let M be a Haken n -manifold. Then there is an algorithm to solve the word problem in the fundamental group of M .*

The proof (see [8] or [10]) is a straightforward generalisation of Waldhausen’s [18]. This is an encouraging result since it is known that the word problem for a

general compact 4-manifold is not solvable. Are other important problems algorithmically solvable in the class of Haken n -manifolds? For example, is there an algorithm to determine whether or not two Haken n -manifolds are homeomorphic? This is likely to be an extremely difficult problem.

4. Some Questions

Since Haken n -manifolds are aspherical, the first question is the Borel conjecture.

QUESTION 4.1. Suppose that M and N are homotopy equivalent n -manifolds and that M is a closed Haken n -manifold. Is N then homeomorphic to M ?

Results of Waldhausen [19] together with the Poincaré Conjecture, proved by Perelman, show that the answer to this question is “yes” for 3-manifolds. It is also true for 2-manifolds (and trivially true for 1-manifolds). In higher dimensions, much is known about the Borel conjecture due to work, for example, of Farrell and Jones [7], but in dimension 4, we are not aware of any general results.

However, there are major difficulties in applying Waldhausen’s approach to the case of dimensions greater than three. The basic problem is that while compressing a surface S in a Haken 3-manifold M until an incompressible surface is obtained, is relatively straightforward, to “compress” a hypersurface in a Haken n -manifold is very challenging.

This leads us to another question:

QUESTION 4.2. Let F be a 3-manifold embedded in a Haken 4-manifold M so that the image of $\pi_1(F)$ in $\pi_1(M)$ is a subgroup G so that the pair $(\pi_1(M), G)$ has more than one end. Suppose that F is not π_1 -injective in M . How can we modify F so that the result is an essential submanifold? Can the modification be done in a finite number of surgeries?

Next a very interesting question which would provide a large collection of examples of Haken n -manifolds is the following;

QUESTION 4.3. Suppose that M is a closed Haken n -manifold. Is there a Haken $(n+1)$ -manifold $(W, \underline{w})$ with $\partial W = M$? Also if M and N are Haken n -manifolds, is there a Haken $(n+1)$ -manifold $(W, \underline{w})$ with $\partial W = M \sqcup N$ and $\underline{w} = \{M, N\}$?

Notice that there is a cobordism category of Haken n -manifolds defined in this way. More details are given in the final section.

Finally a version of JSJ theory, (see [13], [15]) for Haken n -manifolds is a natural problem;

QUESTION 4.4. Suppose that a Haken n -manifold M has a π_1 -injective map of a flat $(n-1)$ -manifold T . Is there either an embedded π_1 -injective map of a flat hypersurface $T' \rightarrow M$ or is M a characteristic variety, i.e a singular fibre-bundle over a suitable base with fibre a flat manifold? More generally is there a canonical π_1 -injective region which contains all π_1 -injective maps of flat hypersurfaces?

5. Examples

We begin this section with a general construction and then apply it to several different types of examples. Suppose that W is a compact n -manifold with a Riemannian or a polyhedral metric of non-positive curvature (for the latter, see [3]).

Suppose also that there is a collection of embedded totally geodesic (and hence π_1 -injective) hypersurfaces in W that intersect orthogonally, and that the complement of the hypersurfaces is a collection of Haken n -cells. Then W is Haken. To show this we construct a hierarchy.

Pick a maximal subcollection of disjoint hypersurfaces. The union M_0 of these hypersurfaces forms our first submanifold which we split along to obtain W_1 . Now choose a maximal subcollection of disjoint hypersurfaces that meet the boundary of W_1 orthogonally. Such hypersurfaces exist; they are of the form $F \cap (W \setminus R(M_0))$, where F is a maximal subcollection of disjoint hypersurfaces in W that meet M_0 orthogonally. Call this union M_1 . It is clear that M_1 is essential in W_1 because of the conditions of totally geodesic, orthogonality and non-positive curvature of the metric. Let W_2 be the result of splitting W_1 open along M_1 .

To choose the next subcollection of hypersurfaces, let us recap what we have done so far. We have constructed our hypersurfaces from M_0 and F . Now we choose hypersurfaces G that meet both M_0 and F orthogonally. We choose a maximal disjoint subcollection and let M_2 be the intersection of this subcollection with W_2 . The three conditions again imply that M_2 is essential in W_2 .

We continue the construction in a similar manner until we obtain a finite collection of Haken n -cells. This completes the explanation of the construction of Haken n -manifolds.

5.1. Ratcliffe-Tschantz hyperbolic 4-manifolds. We refer to the 1171 hyperbolic 4-manifolds constructed by Ratcliffe and Tschantz in [17] as Ratcliffe-Tschantz manifolds. These manifolds are obtained by face-pairing of the ideal hyperbolic 24-cell. We argue that all of the Ratcliffe-Tschantz hyperbolic 4-manifolds are virtually Haken. By this we mean that they have finite sheeted covering spaces which are Haken 4-manifolds. The idea is to colour the faces of the ideal 24-cell P in $\mathbf{H}^4$ with four colours so that no two adjacent faces have the same colouring. If W is a Ratcliffe-Tschantz manifold, then $G = \pi_1(W)$ acts on $\mathbf{H}^4$. The coloured faces form coloured totally geodesic hyperplanes in $\mathbf{H}^4$, because all the dihedral angles between the faces are right-angles. The group G permutes these coloured hyperplanes. It is very easy to pick a subgroup of finite index Γ in G so that Γ is colour preserving. (In fact, there is a homomorphism from G onto the permutations of the four colours and we can take Γ as the kernel of this mapping). Then the coloured hyperplanes project to complete properly embedded totally geodesic hypersurfaces in $W' = W/\Gamma$. Clearly these hypersurfaces are orthogonal to each other.

We can now build a hierarchy of W' as above, since the metric has sectional curvature -1 and the hypersurfaces are totally geodesic and orthogonal. Note that these examples are complete finite volume hyperbolic 4-manifolds. From a combinatorial viewpoint, the Haken 4-cells resulting from cutting open the hierarchy are obtained by using the totally geodesic hypersurfaces and also truncating the cusps along quotients of horospheres, as is easy to check. So these examples can be viewed as Haken 4-manifolds with non-empty boundary, and hence give simple examples of Haken cobordisms with one or two boundary components. In fact, gluing together pairs of homeomorphic boundary components using finite sheeted covering spaces of examples from tables in [17] will produce many examples of Haken null cobordisms and Haken cobordisms between flat 3-manifolds. Compare with section 6.

In another paper of Ivansic, Ratcliffe and Tschantz [12], examples of embeddings of several Klein bottles and tori in S^4 are given, where the complements are hyperbolic. These examples also have the property that the complements are ‘virtually Haken’. That is, they have finite sheeted covering spaces which are Haken manifolds.

5.2. Cubings of non-positive curvature with even degree codimension two faces. Consider any manifold M obtained by gluing n -cubes, which has a polyhedral metric of non-positive curvature, and all the faces of codimension 2 have even order (see [3]). Then as in [1] there is a monodromy representation into the permutations of n pairs of opposite faces of the n -cube. The kernel of this representation gives a finite sheeted covering $\widetilde{M}$ of M for which the canonical hypersurface of the cubing splits into n embedded components. Equivalently there is a colouring of opposite pairs of the n -cube which extends consistently over all the cubes in the covering space $\widetilde{M}$. See [1] for more details.

It is now easy to check that these n hypersurfaces in $\widetilde{M}$, are totally geodesic and intersect orthogonally. Since the polyhedral metric has non-positive curvature, the construction gives Haken n -manifolds as above.

Note that the Ratcliffe-Tschantz examples have finite sheeted coverings of this type, but the only difference is that these have a cubing with some ideal vertices, rather than all finite vertices.

5.3. Closed hyperbolic 4-manifolds, including the Davis manifold.

Another class of examples is formed from the Davis manifold, which is a closed hyperbolic 4-manifold obtained by identifying opposite faces of the 120-cell [5]. The 120-cell is a right-angled polytope. Hence it is easy to see as in [16] that there is a finite sheeted covering where the faces of the 120-cell glue together to form embedded totally geodesic hypersurfaces meeting at right angles. This again gives a class of Haken 4-manifolds.

Note that the Davis example does admit a cubing of non-positive curvature. However the codimension one faces are not all of even order so one needs to use the method of [16].

5.4. Gromov-Thurston construction. In [11] a construction is given of a class of manifolds with negative curvature which is pinched arbitrarily closely to negative one, but most of the manifolds do not admit hyperbolic metrics. We observe that some of these manifolds are Haken and focus on the special case of the hyperbolic 120-cell as used in the Davis manifold. Consider the right-angled 120-cell in hyperbolic 4-space. There is an axis of symmetry which is a disk which is the fixed set of a rotation of order 5. We can then perform the following construction. Firstly take the quotient of the 120-cell under this rotation action. Next take an arbitrary cyclic branched cover over the disk of order $n \geq 5$. This gives a new 4-cell with 120 faces. Each face is either a dodecahedron, or the result of taking the quotient of the dodecahedron under rotation under a five fold rotation through an opposite pair of faces and then taking the n -fold branched cover over the fixed set. Such a 3-cell clearly has $2n$ pentagonal faces and two faces of order n .

We can now form a tiling of $\mathbf{R}^4$ by reflecting this 4-cell in its faces. Note that the cell has a natural lifted Riemannian metric, which has a cone structure at the disk with cone angle $2n\pi/5$. By the argument in [11], this metric can be modified to form a Riemannian metric with curvature arbitrarily pinched close to -1 . Moreover

this can be done so that the faces of the cell are still totally geodesic and meet at right angles. Therefore the tiling can be viewed as a Coxeter group construction with fundamental cell having a metric arbitrarily close to the hyperbolic metric in curvature. But now any torsion free subgroup H of finite index of the Coxeter group G has quotient a 4-manifold M . The same argument as in [16] applies, namely we can find a reflection r in one of the totally geodesic hypersurfaces formed by faces of the cells of the tiling in $\mathbf{R}^4$. Replacing H by $H \cap rHr$ gives a new torsion free subgroup H' which is r -invariant. Hence the fixed set of r is an embedded totally geodesic hypersurface in the resulting manifold $\mathbf{R}^4/H'$. Now keep doing this for each face in M . Passing to a bounded sequence of smaller and smaller subgroups of H , eventually we reach a subgroup with the property that every face of our 4-cells are in embedded orthogonal totally geodesic hypersurfaces and hence as above, such a manifold is Haken.

6. Haken cobordism category

We describe in more detail the ideas behind Haken cobordism theory and prove a simple h-cobordism type result. We use the convention that a boundary-pattern of a Haken manifold is simply the disjoint union of its boundary components, unless otherwise indicated.

DEFINITION 6.1. We say that a Haken $(n+1)$ -manifold $(W, \underline{w})$ is a *Haken cobordism between Haken n -manifolds* $(M, \underline{m})$ and $(N, \underline{n})$ if

- (1) Either M, N, W are orientable or M, N, W are non orientable.
- (2) $\partial W = M \cup N \cup P$ where $P \approx \partial M \times [0, 1]$,
- (3) $\partial M = P \cap M$ and $\partial N = P \cap N$,
- (4) $\underline{w} = \{M, N, A_1, \dots, A_j\}$ where $\cup A_i = P$ and each $A_i = B \times [0, 1]$ for some $B \in \underline{m}$.

Note that if M and N are closed Haken n -manifolds, then M is Haken cobordant to N if there is a Haken $(n+1)$ -manifold W with $\partial W = M \sqcup N$.

It seems reasonable to speculate that all closed Haken n -manifolds are null Haken cobordant, any two closed Haken n -manifolds are Haken cobordant and any two Haken n -manifolds with boundaries which are homeomorphic by a mapping preserving the boundary-patterns, are Haken cobordant, but these seem challenging to prove or disprove. The existence of obstructions to such cobordisms would be very interesting.

THEOREM 6.2. *Suppose that a Haken $(n+1)$ -manifold $(W, \underline{w})$ is a Haken cobordism between Haken n -manifolds $(M, \underline{m})$ and $(N, \underline{n})$ and the inclusion $M \hookrightarrow W$ is a homotopy equivalence. Then W is homeomorphic to $M \times [0, 1]$ and N is homeomorphic to M .*

PROOF. The proof is by induction using the lexicographic ordering given by the pair of integers consisting of dimension and number of hypersurfaces in the hierarchy. We can start the induction at $n+1 = 3$ using Stallings or Waldhausen's result [19]. For the induction step, suppose that we have a Haken cobordism W of dimension $n+1$ between Haken n -manifolds M, N so that the inclusion $i: M \rightarrow W$ is a homotopy equivalence. Moreover the induction hypothesis is that $n \geq 3$ and for any Haken cobordism W^* with smaller dimension or the same dimension and

fewer hypersurfaces in its hierarchy than W , satisfying similar conditions, it follows that W^* is homeomorphic to a product.

Before we begin the induction argument, it is convenient to prove that the inclusion $j: N \rightarrow W$ is also a homotopy equivalence. Assume first that M and N are both closed and that M, N and W are orientable. Clearly the inclusions i and j induce isomorphisms $H_n(i): H_n(M) \rightarrow H_n(W)$ and $H_n(j): H_n(N) \rightarrow H_n(W)$. But a degree one map is an epimorphism on fundamental groups. Furthermore, $\pi_1(j): \pi_1(N) \rightarrow \pi_1(W)$ is one-to-one, because N is essential in W . It follows that $\pi_1(j): \pi_1(N) \rightarrow \pi_1(W)$ is an isomorphism. Hence we see that the inclusion j is also a homotopy equivalence, since a mapping between aspherical spaces inducing an isomorphism on fundamental groups is a homotopy equivalence. If M and N are closed and M, N and W are non-orientable, it is easy to deduce the same result, by first passing to the orientable double coverings $\tilde{M}, \tilde{N}$ and $\tilde{W}$ and noting that the elements in $\pi_1(W)$ and $\pi_1(N)$ which are orientation-preserving, form index two subgroups in $\pi_1(\tilde{W})$ and $\pi_1(\tilde{N})$ respectively. Since the inclusion $\tilde{j}: \tilde{N} \rightarrow \tilde{W}$ is a homotopy equivalence, it immediately follows that $\pi_1(j): \pi_1(N) \rightarrow \pi_1(W)$ is an isomorphism and hence j is a homotopy equivalence.

If M and N have non-empty boundary, then $\partial W = M \cup N \cup P$ with $P \approx \partial M \times I$. As in the previous paragraph, it suffices to consider the case when M, N and W are all orientable, by passing to the orientable double covering if necessary. The inclusion of pairs $i': (M, \partial M) \rightarrow (W, P)$ clearly satisfies $H_n(i'): H_n(M, \partial M) \rightarrow H_n(W, P)$ is an isomorphism, using the homology exact sequence for pairs. Therefore the inclusion of pairs $j': (N, \partial N) \rightarrow (W, P)$ also induces an isomorphism $H_n(j'): H_n(N, \partial N) \rightarrow H_n(W, P)$. We can again deduce that $\pi_1(j): \pi_1(N) \rightarrow \pi_1(W)$ is an epimorphism, since otherwise j would factor through a covering map of W and so $H_n(j')$ would not be an isomorphism. Since $\pi_1(j)$ is one-to-one because N is essential in W we conclude that $\pi_1(j)$ is an isomorphism. Hence j is also a homotopy equivalence since N and W are aspherical.

We are now ready to start the induction argument. This is organised into a number of cases, depending on whether M and N are closed or have non-empty boundary and also depending on the boundary of the first hypersurface F_0 for the hierarchy for W .

Possibility (1): M and N are closed and $F_0 \cap M \neq \emptyset$. We claim that if λ is an arc in F_0 with ends on $\partial F_0 \cap M$, then λ is homotopic rel ends in F_0 to an arc in $\partial F_0 \cap M$. The reason is that such an arc is clearly homotopic rel ends into M since the inclusion $M \rightarrow W$ is a homotopy equivalence. But now we can make the homotopy transverse to F_0 , so that the inverse image is a collection of arcs in the domain disk D , the boundary ∂D consists of two arcs α and β with common end points, α mapped to λ and β mapped into M . Any loops in the inverse image can be removed since we know that F_0 is essential in W . The arcs of the inverse image of F_0 all have ends on β . Choose an innermost disk cut off by one of these arcs. Now by the useful boundary-pattern property of hierarchies, it is easy to see that this disk must be homotopic into a disk in F_0 . By homotoping this disk into and then off F_0 we remove one of the arcs in the inverse image of F_0 . So eventually there are no such arcs in the interior of the disk. Finally it follows that the mapping of the disk D is homotopic into F_0 as claimed, since F_0 is essential. Note that this immediately implies that $\partial F_0 \cap M$ is connected.

But then it follows from the exact sequence for the pair $(F_0, \partial F_0 \cap M)$, that the map $\pi_1(\partial F_0 \cap M) \rightarrow \pi_1(F_0)$ is an isomorphism. Note that the higher homotopy groups of these Haken manifolds are trivial, as proved previously. So the isomorphism follows from the homotopy sequence of a pair plus the fact that $\pi_1(F_0, \partial F_0 \cap M) = \{1\}$. But for aspherical manifolds, such an isomorphism of fundamental groups induces a homotopy equivalence. Consequently we can proceed by induction - we have a lower dimensional Haken h-cobordism which by induction hypothesis is homeomorphic to a product. Hence $F_0 \approx (\partial F_0 \cap M) \times [0, 1]$.

To complete the proof of this case, we split W_0 open along F_0 to obtain $(W_1, \underline{w}_1)$. If M_1 is the result of splitting M open along F_0 then M_1 has two or one components, depending on whether F_0 separates W_0 or does not separate W_0 respectively. In either case, the components of M_1 are elements of the boundary-pattern $\underline{w}_1$. We need to show that the inclusion of the components of M_1 into the components of W_1 are homotopy equivalences. Note that the components of M_1 are π_1 -injective in W_1 because M is essential in W_0 , so we must show that essential loops in W_1 can be pushed into M_1 . This is equivalent to showing that essential loops in W_0 that do not intersect F_0 can be pushed into M without intersecting F_0 . But this follows from the usual argument about pulling loops and arcs away from essential hypersurfaces. So we see that W_1 satisfies the conditions of the theorem, but it has a shorter hierarchy than W_0 , and so again by induction we can assume that $W_1 \approx M_1 \times [0, 1]$. Gluing back up along F_0 then gives a product structure on W and the proof is complete.

Possibility (2): Assume that M and N are closed and the first hypersurface F_0 satisfies $F_0 \cap N \neq \emptyset$. This is similar to the first case, using the observation above that $N \rightarrow W$ is a homotopy equivalence.

Possibility (3): Suppose that M and N are closed and the first hypersurface F_0 is closed. The argument before the first case, where it is proved that the inclusion of $N \rightarrow W$ is a homotopy equivalence, shows that when W is split open along F_0 , the result is two Haken cobordisms, where the inclusion of the boundary components is a homotopy equivalence. Since each cobordism has fewer hypersurfaces in its hierarchy than W , by induction it follows that each is homeomorphic to a product. Gluing back along F_0 establishes that W is also homeomorphic to a product.

Possibility (4): Assume that M and N have non-empty boundaries and that W has boundary consisting of $M \cup N \cup P$, where $P \approx \partial M \times I$, $P \cap M = \partial M \approx \partial M \times \{0\}$ and $P \cap N = \partial N \approx \partial M \times \{1\}$. Consider the first hypersurface F_0 for the hierarchy for W . We claim that F_0 must have non-empty boundary. For otherwise F_0 would be closed. If in addition, F_0 was non-separating, if necessary by passing to the orientable double cover, $[F_0]$ would give a non-trivial class in $H_n(W)$, which contradicts the assumption that $M \rightarrow W$ is a homotopy equivalence and $\partial M \neq \emptyset$. If F_0 was separating, then it must bound a compact region R in W . We claim that the inclusion $f: F_0 \rightarrow R$ induces an isomorphism $\pi_1(f): \pi_1(F_0) \rightarrow \pi_1(R)$. For firstly, $\pi_1(f)$ is one-to-one since F_0 is essential in W . Next, by assumption $\pi_1(i): \pi_1(M) \rightarrow \pi_1(W)$ is an isomorphism and $M \subset W \setminus R$. So $\pi_1(f)$ must be onto by the Seifert van Kampen theorem. But f would then give a homotopy equivalence between two aspherical spaces F_0 and R , contradicting the fact that $H_n(F_0)$ is cyclic and the mapping $H_n(f): H_n(F_0) \rightarrow H_n(R)$ is the zero map, as usual passing to the orientable double covering in the non orientable case. So this completes the proof of the claim that F_0 must have non-empty boundary.

Subcase (a): ∂F_0 is disjoint from $P \subset \partial W$. Then the same argument as for the case (1) can be applied to conclude that F_0 is homeomorphic to a product and W can be cut open along F_0 to give one or two new Haken cobordisms W_1 of smaller complexity, with similar properties to W . Therefore by our induction hypothesis, the components of W_1 are homeomorphic to products and gluing back along F_0 shows that W is homeomorphic to a product also.

Subcase (b): ∂F_0 intersects both P and M . We claim that in this case, ∂F_0 can be decomposed in a similar way to ∂W into three non-empty regions, $\partial F_0 \cap M$, $\partial F_0 \cap N$ and $\partial F_0 \cap P$. Moreover we claim that the inclusions $\pi_1(\partial F_0 \cap N) \rightarrow \pi_1(F_0)$, $\pi_1(\partial F_0 \cap M) \rightarrow \pi_1(F_0)$ and $\pi_1(\partial F_0 \cap P) \rightarrow \pi_1(F_0 \cap P)$ are all isomorphisms.

The key to proving these claims is the property that useful boundary-patterns do not admit essential admissible maps of small disks. We summarise the argument. As in case (1), any arc in F_0 with both ends on M can be homotoped into M and since there are no essential bigons between F_0 and M , it follows that the inclusion $\partial F_0 \cap M \rightarrow F_0$ induces an isomorphism on fundamental groups. Similarly, $\partial F_0 \cap N \rightarrow F_0$ and $k: \partial F_0 \cap \partial M \rightarrow F_0 \cap P$ also induce isomorphisms on fundamental groups. We provide more details for the argument involving the map k , for this requires a slightly different approach. To begin with, it is not obvious that $\partial F_0 \cap \partial M \neq \emptyset$. To establish that there are indeed points in $\partial F_0 \cap \partial M$, choose any arc λ in F_0 with one end on M and the other end on P . A triangular disk can be formed with λ as one boundary arc with the other two boundary arcs γ and μ in P and M respectively. For the inclusion $i: M \rightarrow W$ is a homotopy equivalence and the resulting homotopy of λ to an arc μ in M can be chosen so that the end point on M remains fixed and the end point on P travels along P along the arc γ . By a similar argument to case (1), if the map of this triangular disk is assumed transverse to the hierarchy, we can homotop the map to remove innermost loops or arcs cutting off triangular disks or bigons, due to the useful boundary-pattern condition. But if there are no remaining arcs or loops, then the triangular disk has image in a Haken cell and so is inessential. It follows that the arc γ in P is homotopic to an arc δ in $\partial F_0 \cap P$. This immediately proves that $\partial F_0 \cap \partial M \neq \emptyset$ since δ has an end point on ∂M .

Now that $\partial F_0 \cap \partial M \neq \emptyset$ has been established, the same argument as in case (1) can be applied to the inclusion k . For any arc in $\partial F_0 \cap P$ with both ends on ∂M can be homotoped into ∂M and in this way we deduce that $k: \partial F_0 \cap P \rightarrow \partial F_0 \cap P$ induces an isomorphism on fundamental groups. Consequently k is a homotopy equivalence and again by the induction assumption, the lower dimensional Haken cobordism $\partial F_0 \cap P$ is homeomorphic to a product.

Finally, to split W open along F_0 and apply our induction hypothesis to complete the argument, we need to show that the product $G = \partial F_0 \cap P$ splits the boundary-pattern for P into smaller products. But this follows along the same lines as the previous paragraph. Note first that G is a hypersurface in P . The elements of the boundary-pattern $\underline{w}$ for W which are contained in P , are products $A = B \times [0, 1]$, where B is in the boundary-pattern $\underline{m}$ for M . We claim that $G \cap A \approx (G \cap B) \times [0, 1]$. For if this was not true, we could find a face $F = E \times [0, 1] \in \partial B \times [0, 1]$ so that $G \cap F$ is not homeomorphic to $(G \cap E) \times [0, 1]$. But the usual argument involving picking an arc in $G \cap F$ with ends on ∂F and using the useful boundary-pattern argument would then give a contradiction. So this completes the argument for subcase (b).

Subcase (c): If ∂F_0 intersects P and ∂F_0 intersects N , the argument is the same as subcase (c).

Subcase (d): Here $\partial F_0 \subset P$. In this case, if we split W open along F_0 , the claim is that two new Haken cobordisms are formed with similar properties to W . The argument is very similar to the case (3) above. So this is straightforward. $\square$

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Pseudo-Developing Maps for Ideal Triangulations I: Essential Edges and Generalised Hyperbolic Gluing Equations

Henry Segerman and Stephan Tillmann

To Bus Jaco on the occasion of his seventieth birthday

ABSTRACT. Let N be a topologically finite, orientable 3-manifold with ideal triangulation. We show that if there is a solution to the hyperbolic gluing equations, then all edges in the triangulation are essential. This result is extended to a generalisation of the hyperbolic gluing equations, which enables the construction of hyperbolic cone-manifold structures on N with singular locus contained in the 1-skeleton of the triangulation.

1. Introduction

Neumann and Zagier [9] study the variation of the volume function on a cusped hyperbolic 3-manifold of finite volume using a decomposition of the manifold into hyperbolic ideal tetrahedra. This is based on a construction by Thurston [12], which associates to the underlying topological ideal triangulation a *parameter space* of shapes of ideal hyperbolic tetrahedra satisfying certain polynomial equations. Such a parameter space can be associated to any ideal triangulation of a non-compact 3-manifold with torus cusps, and several authors have studied this affine algebraic set in this generality (see, for instance, Yoshida [16], Francaviglia [4], Tillmann [15], Segerman [11]). A key step in many applications is the construction of a so-called pseudo-developing map for a given point on the parameter space in order to produce a representation of the fundamental group of the manifold into the group of orientation preserving isometries of hyperbolic 3-space. The map is called a pseudo-developing map rather than a developing map because it is not necessarily locally injective.

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More recently, Luo [6] initiated the study of the parameter space for arbitrary ideally triangulated manifolds. In Luo's setting, they arise from (possibly semi-simplicial) triangulations of closed 3-manifolds by removing the vertices. This raises the question of whether the existence of pseudo-developing maps, their associated holonomies and continuous extensions depend on topological hypotheses. This note considers the most general setting, and has the following theorem as its core result. The relevant definitions can be found in Sections 2 and 3.

THEOREM 1. *Let N be a topologically finite, orientable 3-manifold with ideal triangulation. If there is a solution to the hyperbolic gluing equations, then all edges in the triangulation are essential.*

COROLLARY 2. *Let N be a topologically finite, orientable 3-manifold with ideal triangulation. If there is a solution to the hyperbolic gluing equations, then associated to it are a pseudo-developing map $\tilde{N} \rightarrow \mathbb{H}^3$ and a representation of $\pi_1(N)$ into $PSL_2(\mathbb{C})$, which makes this map equivariant.*

Luo [6] also introduced *generalised hyperbolic gluing equations*, which state that the product of all shape parameters around an ideal edge is either $+1$ or -1 . The sign may be different at different edges. This can be generalised further by choosing, for each edge $e \in N^{(1)}$, an element $\xi_e \in S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$ and requiring the product of all shape parameters around e to equal ξ_e . The restriction of ξ_e to S^1 is natural: if the shape parameters of all tetrahedra have positive imaginary parts, then this gives a (possibly incomplete) hyperbolic cone manifold structure on N with cone angle $\arg(\xi_e) + 2\pi k_e$ (for some $k_e \in \mathbb{N}$) around e . The most interesting case arises when each ξ_e is a root of unity, but we will consider the general case throughout. These " S^1 -valued" gluing equations are determined by the vector $\xi = (\xi_e) = (\xi_e)_{e \in N^{(1)}}$, and are called the ξ -hyperbolic gluing equations, since they depend on the choice of ξ . Let $o = o(\xi) = (o(\xi_e))_{e \in N^{(1)}}$, where $o(\xi_e) \in \mathbb{N} \cup \{\infty\}$ is the order of ξ_e .

If one drills out the edges in N , one obtains a handlebody H with a natural epimorphism $\pi_1(H) \twoheadrightarrow \pi_1(N)$. We obtain a representation $\rho: \pi_1(H) \rightarrow PSL_2(\mathbb{C})$ for each solution to the ξ -hyperbolic gluing equations. If each order $o(\xi_e)$ is finite, then we obtain a natural branched (not necessarily finite) cover N_o of N corresponding to the kernel of ρ . The branch locus is contained in the 1-skeleton, and the cover has branch index $o(\xi_e)$ at e . If some order is infinite, one can still define N_o , but this will have an edge of infinite degree and the points on such an edge of infinite degree are not manifold points. If all orders are equal to 1, then $N = N_o$. Theorem 1 is thus a special case of the following result:

THEOREM 3. *Let N be a topologically finite, orientable 3-manifold with ideal triangulation. If there is a solution to the ξ -hyperbolic gluing equations for N , then all edges in the induced ideal triangulation of N_o are essential.*

Treating each ξ_e as an additional, circle-valued variable gives the *cone-hyperbolic gluing equations*. The existence of interesting solutions to them comes from the following observation:

OBSERVATION 4. *Given any ideally triangulated, topologically finite, orientable 3-manifold N , there is a complete, non-compact hyperbolic cone-manifold structure on N with singular locus contained in $N^{(1)}$ and having volume exactly the number of tetrahedra in N times the volume of the regular hyperbolic ideal tetrahedron.*

Outline: Triangulations, pseudo-manifolds and essential edges are discussed in Section 2. The deformation variety and its generalisations are defined in Section 3, and Yoshida's construction is recalled. The handlebody construction as well as the proofs of the main results can be found in Section 4. Examples are given in Section 5.

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2. Triangulations with essential edges

2.1. Ideal triangulation. An ideal triangulation $\mathcal{T}$ of the topologically finite 3-manifold N consists of a pairwise disjoint union of standard Euclidean 3-simplices, $\tilde{\Delta} = \cup_{k=1}^n \tilde{\Delta}_k$, together with a collection Φ of Euclidean isometries between the 2-simplices in $\tilde{\Delta}$, termed *face pairings*, such that $N = (\tilde{\Delta} \setminus \tilde{\Delta}^{(0)})/\Phi$. It is well-known that every non-compact, topologically finite 3-manifold admits an ideal triangulation.

Denote $P = \tilde{\Delta}/\Phi$ the associated *pseudo-manifold* (or *end-compactification* of N) with quotient map $p: \tilde{\Delta} \rightarrow P$. Let σ be a k -simplex in $\tilde{\Delta}$. Then $p(\sigma)$ may be a singular k -simplex in P . Denote by $P^{(k)}$ the set of all (possibly singular) k -simplices in P . An *ideal k -simplex* is a k -simplex with its vertices removed. The vertices of the k -simplex are termed the *ideal vertices* of the ideal k -simplex. Similarly for singular simplices. The standard terminology of (ideal) edges, (ideal) faces and (ideal) tetrahedra will be used for the singular simplices in N and P .

In this note, it is assumed throughout that N is oriented and that all singular simplices in N are given the induced orientations. It follows that the link of each vertex in P is an orientable surface.

The case in which each vertex link is a torus and N has a complete, hyperbolic structure supported by the ideal triangulation is the most common setting for the study of Thurston's hyperbolic gluing equations, see [2, 3, 4, 8, 9, 16]. The case in which each vertex link is a sphere and P is a closed hyperbolic 3-manifold is treated in [7]. This note will not make any of these additional assumption.

It is often convenient to start with the 3-dimensional, closed, orientable pseudo-manifold P with (possibly singular) triangulation $\mathcal{T}$. Then $N = P \setminus P^{(0)}$ is an ideally triangulated, non-compact, orientable, topologically finite 3-manifold.

2.2. Abstract edge-neighbourhood. The *degree of an edge e in P* , $\deg(e)$, is the number of 1-simplices in $\tilde{\Delta}$ which map to e . Given the edge e in P , there is an associated *abstract neighbourhood* $B(e)$ of e . This is a ball triangulated by $\deg(e)$ 3-simplices, having a unique interior edge $\tilde{e}$, and there is a well-defined simplicial quotient map $p_e: B(e) \rightarrow P$ taking $\tilde{e}$ to e . This abstract neighbourhood is obtained as follows.

If e has at most one pre-image in each 3-simplex in $\tilde{\Delta}$, then $B(e)$ is obtained as the quotient of the collection $\tilde{\Delta}_e$ of all 3-simplices in $\tilde{\Delta}$ containing a pre-image of e by the set Φ_e of all face pairings in Φ between faces containing a pre-image of e . There is an obvious quotient map $b_e: B(e) \rightarrow P$ which takes into account the remaining identifications on the boundary of $B(e)$.

If e has more than one pre-image in some 3-simplex, then multiple copies of this simplex are taken, one for each pre-image. The construction is modified accordingly,

so that $B(e)$ again has a unique interior edge and there is a well defined quotient map $b_e: B(e) \rightarrow P$. Complete details can be found in [14], Section 2.3.

2.3. End-compactification and null-homotopic edges. As above, let P be a 3-dimensional, closed, orientable pseudo-manifold with (possibly singular) triangulation $\mathcal{T}$, and $N = P \setminus P^{(0)}$. Let $E = \nu(P^{(0)})$ be an open regular neighbourhood of $P^{(0)}$ in P , chosen in such a way that ∂E meets each singular 3-simplex σ^3 in P in precisely four pairwise disjoint normal triangles, one at each of its corners. Hence $E \cup \partial E$ has the natural cone structure $(\partial E \times [0, 1]) / \sim$ where $(x, 1) \sim (y, 1)$ if x, y lie on a connected component of ∂E .

Then $C = P \setminus E$ is termed a *compact core* of N . It follows that P can be viewed as obtained from C by taking each connected component B of ∂C and either collapsing it to a point or by attaching the cone over B to a point. Neumann and Yang [10] call this the *end-compactification* of N . We have

$$C \subset N \subset P.$$

Let $\tilde{P}$ be the space obtained from the universal cover $\tilde{C}$ of C by attaching the cone over each connected boundary component to a point. We then have natural inclusions

$$\tilde{C} \subset \tilde{N} \subset \tilde{P},$$

and $\tilde{P}$ is termed the *end-compactification of $\tilde{N}$ with respect to N* . Note that $\tilde{P}$ is also simply connected since adding cones over connected spaces does not increase the fundamental group.

It is hoped that the notation and terminology does not lead to confusion. For instance, when N is hyperbolic, then $\tilde{N}$ is an open ball and the natural compactification of this open ball (without reference to N) is homeomorphic to the 3-ball, whilst $\tilde{P}$ is $\tilde{N}$ with countably many points added. Also, there are many examples where P is simply connected; for instance if C is the complement of a knot or link in S^3 .

The space $\tilde{P}$ has a natural decomposition into 3-simplices coming from the decomposition of $\tilde{C}$ into truncated 3-simplices and the coning construction. Lifting the ideal triangulation of N to $\tilde{N}$ gives a natural decomposition into ideal 3-simplices. It follows from the construction that the ideal triangulation of $\tilde{N}$ is precisely the restriction of the triangulation of $\tilde{P}$ to $\tilde{P} \setminus \tilde{P}^{(0)}$. In particular, we have a well-defined simplicial map $\tilde{P} \rightarrow P$.

A triangulation $\mathcal{T}$ of P is said to be *almost non-singular* if no 3-simplex has two of its edges identified. In this case, the only self-identifications of a 3-simplex are at the vertices. A triangulation $\mathcal{T}$ of P is *non-singular* if no 3-simplex has any self-identifications.

A triangulation $\mathcal{T}$ of P is said to be *virtually almost non-singular* if the induced triangulation of $\tilde{P}$ is almost non-singular, and *virtually non-singular* if the induced triangulation of $\tilde{P}$ is non-singular.

Note that a triangulation $\mathcal{T}$ of P is *virtually almost non-singular* if and only if every ideal 3-simplex in $\tilde{N}$ is embedded.

An edge $e \in P^{(1)}$ is *null-homotopic* if and only if there is a map $f: D^2 \rightarrow P$ such that $f(\partial D^2) = e$. Equivalently, e is null-homotopic in P if and only if it represents the trivial element in $\pi_1(P)$.

The intersection $\alpha = e \cap C$ is *homotopic into* ∂C if and only if there is an arc $\beta \subset \partial C$ such that α is homotopic to β by a fixed-endpoint homotopy.

To simplify terminology, we will say that the edge e in P is *essential* if $e \cap C$ is not homotopic into ∂C , and it is *not essential* otherwise. This reflects standard terminology for ideal edges in ideal triangulations.

LEMMA 5. *$\mathcal{T}$ is virtually non-singular if and only if every edge in P is essential.*

PROOF. Suppose the edge e is not essential. Then $e \cap C$ is homotopic into ∂C . The disc in C lifts to a disc in $\tilde{C}$ with part of its boundary on a lift $\tilde{e}$ of e and the remainder of its boundary on a connected component of $\partial\tilde{C}$. Hence $\tilde{e}$ is an edge in $\tilde{P}$ with both end-points at the same vertex. Thus, any tetrahedron containing $\tilde{e}$ is not embedded and therefore $\mathcal{T}$ is not virtually non-singular.

Conversely, suppose $\mathcal{T}$ is not virtually non-singular. There is a tetrahedron in $\tilde{P}^{(3)}$ with self-identifications and hence an edge e with both end-points at the same vertex. The end-points of $\alpha = e \cap \tilde{C}$ lie on the same boundary component of $\tilde{C}$, and hence can be connected by an arc $\beta \subset \partial\tilde{C}$. Since $\tilde{C}$ is simply connected, the loop $\alpha \cup \beta$ bounds an immersed disc in $\tilde{C}$. Let $p_C: \tilde{C} \rightarrow C$ be the covering map. It follows that $p_C(\alpha)$ is homotopic to $p_C(\beta)$, and hence the edge containing $p_C(\alpha)$ is not essential. ■

LEMMA 6. *If e is not essential in P , then e is null-homotopic in P .*

PROOF. If $\alpha = e \cap C$ is homotopic into ∂C , then both of its end-points lie on the same boundary component, B , of C . Hence the map $f: D^2 \rightarrow C$ can be extended to a continuous map $f_P: D^2 \rightarrow P$ by coning it over β to the vertex of P corresponding to B . ■

The converse of Lemma 6 is not true. For instance the ideal triangulation of a knot or link in S^3 gives rise to a simply connected pseudo-manifold, but there are many edges which are not homotopic into the boundary. For instance, Thurston's ideal triangulation of the figure eight knot complement yields a simply connected pseudo-manifold P with the property that every edge is null-homotopic and essential.

3. The deformation variety and its friends

3.1. Deformation variety. Let Δ^3 be the standard 3-simplex with a chosen orientation. Suppose the edges from one vertex of Δ^3 are labeled by z, z' and z'' so that the opposite edges have the same labeling. Then the cyclic order of z, z' and z'' viewed from each vertex depends only on the orientation of the 3-simplex. It follows that, up to orientation preserving symmetries, there are two possible labelings, and we fix one of these labelings. The labels are termed *shape parameters*.

Suppose $P^{(3)} = \{\sigma_1, \dots, \sigma_n\}$. For each $\sigma_i \in P^{(3)}$, fix an orientation preserving simplicial map $f_i: \Delta^3 \rightarrow \sigma_i$. Let $P^{(1)} = \{e_1, \dots, e_m\}$, and let $a_{ij}^{(k)}$ be the number of edges in $f_i^{-1}(e_j)$, which have label $z^{(k)}$.

For each $i \in \{1, \dots, n\}$, define

$$(1) \quad p_i = z_i(1 - z_i'') - 1, \quad p_i' = z_i'(1 - z_i) - 1, \quad p_i'' = z_i''(1 - z_i') - 1,$$

and for each $j \in \{1, \dots, m\}$, let

$$(2) \quad g_j = \prod_{i=1}^n z_i^{a_{ij}} (z'_i)^{a'_{ij}} (z''_i)^{a''_{ij}} - 1.$$

Setting $p_i = p'_i = p''_i = 0$ gives the *parameter relations*, and setting $g_j = 0$ gives the *hyperbolic gluing equations*. For a discussion and geometric interpretation of these equations, see [12, 9]. The parameter relations imply that $z_i \neq 0, 1$.

DEFINITION 7. *The deformation variety $\mathfrak{D}(\mathcal{T})$ is the variety in $(\mathbb{C} \setminus \{0\})^{3n}$ defined by the hyperbolic gluing equations together with the parameter relations.*

Thurston's original parameter space is obtained by choosing a coordinate from each coordinate triple in the above. In calculations, we will often use such a smaller coordinate system. Theorem 1 is equivalent to the statement that $\mathfrak{D}(\mathcal{T}) \neq \emptyset$ implies that all edges are essential.

3.2. Yoshida's construction. Let $C \subset N \subset P$, $\mathcal{T}$ and $\mathfrak{D}(\mathcal{T})$ be as defined above. Given $Z \in \mathfrak{D}(\mathcal{T})$, each ideal tetrahedron in N has edge labels which can be lifted equivariantly to $\tilde{N}$. Following [16], we would like to define a continuous map $\Phi_Z: \tilde{N} \rightarrow \mathbb{H}^3$, which maps every ideal tetrahedron σ in $\tilde{N}$ to an ideal hyperbolic 3-simplex $\Delta(\sigma)$, such that the labels carried forward to the edges of $\Delta(\sigma)$ correspond to the shape parameters of $\Delta(\sigma)$ determined by its hyperbolic structure; see [12] for the geometry of hyperbolic ideal tetrahedra. Thus,

we need to assume that $\mathcal{T}$ is virtually almost non-singular,

since no ideal hyperbolic simplex has edges identified with each other. It will be necessary to have a consistent choice of parameterisation for the maps from ideal tetrahedra in $\tilde{N}$ to $\mathbb{H}^3$; following Thurston [12], we ensure this by assuming that the map $\sigma \rightarrow \mathbb{H}^3$ always is a *straight map*. See [13, 4, 7] for the details, which play no role in the following.

Each ideal 3-simplex in $\tilde{N}$ inherits edge labels from Z . Choose a tetrahedron σ in $\tilde{N}$ and an embedding of σ into $\mathbb{H}^3$ of the specified shape. For each tetrahedron of $\tilde{N}$ distinct from σ and which has a face in common with σ , there is a unique embedding into $\mathbb{H}^3$ which coincides with the embedding of σ on the common face and which has the shape determined by Z . If the shape parameters of the new tetrahedron give an ideal hyperbolic tetrahedron of orientation opposite to that of the first one, then this map is not locally injective along the common face. Hence the map is called a *pseudo-developing map* rather than a developing map.

LEMMA 8. *Starting with an embedding of $\sigma \subset \tilde{N}$, there is a unique way to extend this to a well-defined, continuous map $\Phi_Z: \tilde{N} \rightarrow \mathbb{H}^3$, such that each ideal tetrahedron in $\tilde{N}$ is mapped to a hyperbolic ideal tetrahedron of the specified shape.*

PROOF. Since $\tilde{N}$ is simply connected and each abstract edge neighbourhood is embedded, it follows from the hyperbolic gluing equations that the map is well-defined. The reader may find pleasure in doing this exercise or consult [2] for a full treatment. ■

LEMMA 9. *If $\mathfrak{D}(\mathcal{T}) \neq \emptyset$ and $\mathcal{T}$ is virtually almost non-singular, then $\mathcal{T}$ is virtually non-singular (hence all edges in P are essential).*

PROOF. Let $Z \in \mathfrak{D}(\mathcal{T})$. The Yoshida map Φ_Z is well defined since $\mathcal{T}$ is virtually almost non-singular. Suppose the ideal tetrahedron σ in $\tilde{N}$ has two ideal vertices at the same vertex of $\tilde{P}$. Let t and t' denote the normal triangles in $\sigma \cap \partial\tilde{C}$ dual to these vertices. Since t and t' are in the link of the same vertex of $\tilde{P}$, there is a path in $\partial\tilde{C}$ from t to t' passing through finitely many normal triangles. This path corresponds to a finite sequence $\sigma_1, \dots, \sigma_k$ of ideal tetrahedra in $\tilde{N}$, and a corresponding sequence $v_1, \dots, v_k$ of ideal vertices of the σ_i , dual to the sequence of normal triangles. The map Φ_Z extends to map these ideal vertices into $\overline{\mathbb{H}}^3$, and as we develop along the sequence of tetrahedra, we see that $\Phi_Z(v_1) = \Phi_Z(v_2) = \dots = \Phi_Z(v_k)$. But v_1 and v_k are two distinct vertices of σ , so the image of σ cannot be an ideal hyperbolic 3-simplex, and thus the Yoshida map is not well-defined. ■

COROLLARY 10. *If the Yoshida map $\Phi_Z: \tilde{N} \rightarrow \mathbb{H}^3$ is well-defined, then it extends to a continuous map $\overline{\Phi}_Z: \tilde{P} \rightarrow \overline{\mathbb{H}}^3$.*

3.3. Representations. Suppose that the Yoshida map is well-defined and, in particular, that $\mathcal{T}$ is virtually almost non-singular. We will see (in Corollary 2) that these hypotheses hold if $\mathfrak{D}(\mathcal{T}) \neq \emptyset$.

For each $Z \in \mathfrak{D}(\mathcal{T})$, the Yoshida map Φ_Z can be used to define a representation $\rho_Z: \pi_1(N) \rightarrow PSL_2(\mathbb{C})$ as follows (see [16]). A representation into $PSL_2(\mathbb{C})$ is an action on $\mathbb{H}^3$, and this is the unique representation which makes Φ_Z $\pi_1(N)$ -equivariant: $\Phi_Z(\gamma \cdot x) = \rho_Z(\gamma)\Phi_Z(x)$ for all $x \in \tilde{N}$, $\gamma \in \pi_1(N)$. Thus, ρ_Z is well-defined up to conjugation, since it only depends upon the choice of the embedding of the initial tetrahedron σ . This yields a well-defined map $\chi_{\mathcal{T}}: \mathfrak{D}(\mathcal{T}) \rightarrow \overline{\mathfrak{X}}(N)$ from the deformation variety to the $PSL_2(\mathbb{C})$ -character variety. It is implicit in [8] that $\chi_{\mathcal{T}}$ is algebraic; see [2] for details using a faithful representation of $PSL_2(\mathbb{C}) \rightarrow SL(3, \mathbb{C})$. Note that the image of each peripheral subgroup under ρ_Z has at least one fixed point on the sphere at infinity.

REMARK 11. *The representation associated to a solution of the hyperbolic gluing equations may be reducible, or even trivial. For instance, the triangulation of S^3 with two tetrahedra obtained by identifying the boundary spheres of the tetrahedra in the natural way has a curve of solutions to the hyperbolic gluing equations, and the associated representations of the fundamental group of S^3 minus four points are all trivial.*

The representation arising from Yoshida's construction can be understood using *elementary face pairings* as follows. An elementary face pairing of the hyperbolic ideal tetrahedron Δ is an element of $PSL_2(\mathbb{C})$ taking one face of Δ to another. If the deck transformation $\gamma \in \pi_1(N)$ takes the ideal triangle τ in $\tilde{N}$ to the ideal triangle $\gamma \cdot \tau$, and $\sigma_0^3, \dots, \sigma_k^3$ is a sequence of tetrahedra with the property that $\tau \subset \sigma_0^3$, $\gamma \cdot \tau \subset \sigma_k^3$, and consecutive tetrahedra share a face distinct from τ and $\gamma \cdot \tau$, then there is an associated product of elementary face pairings, one for each $\Phi_Z(\sigma_j^3)$, such that $\rho_Z(\gamma)$ is their product. This fact is used in [2] to show that $\chi_{\mathcal{T}}$ is algebraic; it will be used below for a different purpose.

3.4. Generalisations. For each $e_j \in P^{(1)}$, we introduce a variable $\xi_{e_j} \in S^1$, and define

$$(3) \quad g_{e_j}(\xi_{e_j}) = \prod_{i=1}^n z_i^{a_{ij}} (z_i')^{a'_{ij}} (z_i'')^{a''_{ij}} - \xi_{e_j}.$$

Setting $g_{e_j}(\xi_{e_j}) = 0$ gives the *cone-hyperbolic gluing equation* for e_j . Since the product of all shape parameters associated to all edges equals 1, we have the following consequence:

$$\prod_{e \in P^{(1)}} \xi_e = 1.$$

We also define the *holonomy around e_j* to be

$$h(e_j) = \prod_{i=1}^n z_i^{a_{ij}} (z'_i)^{a'_{ij}} (z''_i)^{a''_{ij}}.$$

DEFINITION 12. *For a triangulation with n tetrahedra and m edges, the cone-deformation variety $\mathfrak{D}(\mathcal{T}; \star)$ is the variety in $(\mathbb{C} - \{0\})^{3n} \times (S^1)^m$ defined by the cone-hyperbolic gluing equations together with the parameter relations.*

The cone-deformation variety is non-empty for *any* triangulation. Indeed, if $z = \frac{1}{2}(1 + \sqrt{-3})$, then all of z , z' and z'' are roots of unity and specify the regular hyperbolic ideal 3-simplex. This is the unique hyperbolic ideal 3-simplex with the property that all shape parameters are roots of unity. Assigning values to all shape parameters in this way, one can solve $g_e(\xi_e) = 0$ for $\xi_e \in S^1$ uniquely for each e . It turns out that this solution has particularly nice properties, giving us Observation 4.

PROOF OF OBSERVATION 4. Identify each ideal 3-simplex in $\tilde{\Delta} \setminus \tilde{\Delta}^{(0)}$ with the regular ideal 3-simplex. Gluing faces in pairs by hyperbolic isometries according to the face pairings gives a hyperbolic structure on N minus edges. Around the edges there is no shearing, and the total angle around edge e is its degree times $\frac{\pi}{3}$. Hence there is a non-compact hyperbolic cone-manifold structure on N with singular locus contained in $N^{(1)}$. Near the ideal vertices of the tetrahedra, one can consistently choose horospherical triangles that match up and the structure is therefore complete. ■

Denote the projections to the factors

$$F_1: \mathfrak{D}(\mathcal{T}; \star) \rightarrow (\mathbb{C} - \{0\})^{3n} \text{ and } F_2: \mathfrak{D}(\mathcal{T}; \star) \rightarrow (S^1)^m.$$

Given $\xi \in (S^1)^m$, we let $\mathfrak{D}(\mathcal{T}; \xi) = F_2^{-1}(\xi)$. Then $F_2^{-1}((\xi_1, \dots, \xi_m))$ may be empty, and $\mathfrak{D}(\mathcal{T}) = \mathfrak{D}(\mathcal{T}, (1, \dots, 1))$.

4. The handlebody construction

4.1. The handlebody cover. Consider the compact manifold $H := C \setminus \nu(P^{(1)})$, where $\{\nu(e) \mid e \in P^{(1)}\}$ is a set of pairwise disjoint open tubular neighbourhoods of the edges $P^{(1)}$. The manifold H is a handlebody, and inherits a cell decomposition from $\mathcal{T}$ into *doubly truncated tetrahedra*: truncated at the vertices and at the edges. See Figure 1 for a picture of a doubly truncated tetrahedron. A doubly truncated tetrahedron has four *boundary hexagonal faces* on ∂C , six *rectangular faces* on $\partial H \setminus \partial C$, and four *interior hexagonal faces*.

Consider the decomposition of ∂H into the union of the *vertex boundary* $\partial H \cap \partial C$ (composed of boundary hexagonal faces) and the *edge boundary* $\partial H \setminus (\partial H \cap \partial C)$ (composed of rectangular faces). The edge boundary consists of a pairwise disjoint union of annulus components, one for each edge of C .

For any cover K of H , define a topological space K^* as follows. Lift the decomposition of H to K . The boundary of K decomposes into *vertex boundary* components made up out of boundary hexagonal pieces, and *edge boundary* components made up out of rectangular faces. For each edge boundary component of H , we fix a product structure which identifies it with $S^1 \times [0, 1]$. Then the edge boundary components of K are either of the form $S^1 \times [0, 1]$ or $\mathbb{R} \times [0, 1]$, and the product structure is preserved by the deck transformations. We form K^* by first collapsing each edge boundary component of ∂K by projection to the $[0, 1]$ factor, to form a topological space K' . Next we collapse each component of the boundary of K' to form K^* .

As in the construction of P from C , we can equivalently construct K^* from K by coning rather than collapsing. Note that $H^* = P$. Also note that K^* is a union of tetrahedra identified along faces. Unlike P or $\tilde{P}$, K^* can have edges incident with infinitely many tetrahedra. We will be interested in the universal cover $\tilde{H}$ of H , and in $\tilde{H}^*$. Since H is a handlebody, $\tilde{H}$ is homotopy equivalent to a tree. Notice that the construction gives a natural quotient map $\tilde{H}^* \rightarrow P$.

LEMMA 13. *Every tetrahedron in $\tilde{H}^*$ is non-singular.*

PROOF. If there is a tetrahedron with self-identifications of any kind then there will be some pair of its vertices that are identified, and thus an edge with its endpoints being the same point.

Suppose that we have such an edge e . Truncate all of the tetrahedra of $\tilde{H}^*$ at the vertices to form a topological space $\tilde{H}_C^*$ (a “core” of $\tilde{H}^*$, analogous to the compact core C of P). Both points $p_1, p_2 \in e \cap \partial \tilde{H}_C^*$ are on the same component of $\partial \tilde{H}_C^*$, so we can choose a path β contained in a single component of $\partial \tilde{H}_C^*$ whose endpoints are p_1 and p_2 . We may deform β slightly so that it does not pass through any edges other than e . We can now drill out the neighbourhoods $\nu(e_i)$ of the edges of $\tilde{H}_C^*$ to get back $\tilde{H}$, with the curve β contained in a component of $\partial \tilde{H} \cap \partial \tilde{C}$, and meeting $\partial \tilde{H} \setminus \partial \tilde{C}$ only at its endpoints, which are on the edge boundary near the two ends of e .

We may assume that the path β starts in some (doubly truncated) tetrahedron σ_0 at an intersection between a boundary hexagonal face and a rectangular face, traverses through the tetrahedra of $\tilde{H}$ along $\partial \tilde{H} \cap \partial \tilde{C}$, and then returns back to σ_0 , ending at the other end of the rectangular face.

Consider such a path β that visits a minimal number of tetrahedra. Since the tetrahedra form a tree, there must be at least one “leaf” tetrahedron σ in the path. That is, the path enters σ from one face gluing (at the interior hexagonal face f) and exits at the same face gluing. The path β is restricted to lie in $\partial \tilde{H} \cap \partial \tilde{C}$, and so it enters σ at the edge of one of the three boundary hexagonal faces of σ adjacent to f . There is no path within σ along $\partial \tilde{H} \cap \partial \tilde{C}$ from one of the boundary hexagonal faces to any of the others, so it must exit σ at the same edge. But then the part of β in σ could be homotoped away, and the path was not minimal. This gives a contradiction. $\blacksquare$

4.2. Representations of the handlebody group. Since there are no gluing consistency conditions to satisfy for the tetrahedra in $\tilde{H}^*$ (because every edge of $\tilde{H}^*$ is of infinite degree), we can assign *arbitrary* shape parameters $z \in \mathbb{C} \setminus \{0, 1\}$ to the tetrahedra of $\tilde{H}^*$ and define a pseudo-developing map $D: \tilde{H}^* \rightarrow \overline{\mathbb{H}}^3$. This follows

as in the proof of Lemma 13 from the fact that $\tilde{H}^*$ is homotopy equivalent to a tree. In particular, we can build the developing map by starting with a tetrahedron σ_0 of $\tilde{H}^*$ and any ideal hyperbolic tetrahedron in $\mathbb{H}^3$, and then developing along any non-backtracking path of tetrahedra in $\tilde{H}^*$ starting with σ_0 , using any non-degenerate shapes of ideal hyperbolic tetrahedra. As we develop, we never have any consistency conditions to satisfy, because the tetrahedra of $\tilde{H}^*$ form a tree.

LEMMA 14. *Suppose that the parameters for the tetrahedra in $\tilde{H}^*$ are lifts of the parameters of the tetrahedra of H . Then there exists a well-defined representation $\rho: \pi_1 H \rightarrow PSL_2(\mathbb{C})$. Moreover, this representation makes $D: \tilde{H}^* \rightarrow \overline{\mathbb{H}}^3$ equivariant if and only if for each edge e in P , the holonomy around e , $h(e)$ is an element of S^1 .*

PROOF. The map D has been defined as in the Yoshida construction, and we would like to define ρ in a similar fashion. Consider a triangle τ_0 of $\tilde{H}^*$, and its image $\gamma \cdot \tau_0$ under the deck transformation $\gamma \in \pi_1 H$. The image of the three vertices of τ_0 under D give a triplet of distinct points on $\partial \mathbb{H}^3$, and the three vertices of $\gamma \cdot \tau_0$ give another triplet. We define $\rho(\gamma)$ to be the unique element of $PSL_2(\mathbb{C})$ which maps the first triplet to the second. Since the parameters for the tetrahedra in $\tilde{H}^*$ are lifts of the parameters of the tetrahedra of H , this definition is independent of the choice of τ_0 . This proves the existence of ρ .

It follows from the construction that $D(\gamma \cdot x) = \rho(\gamma)D(x)$ for all $x \in \tilde{H}^*$ and $\gamma \in \pi_1(H)$ except possibly for those x that are contained in the 1-skeleton. In $\tilde{H}$, we have a product structure on the lifts of the edge boundary components of H , which is preserved by the deck transformations. Consider a deck transformation $\gamma \in \pi_1 H$, which preserves the edge boundary component B of $\tilde{H}$. Now in $\tilde{H}^*$, B is mapped to a 1-simplex, e' , and $l = D(e')$ is a geodesic in $\mathbb{H}^3$. Composing D with an isometry of $\mathbb{H}^3$, we may assume that $l = [0, \infty]$. Since γ preserves the product structure, $\gamma \cdot x = x$ for each $x \in e'$. Thus D is equivariant with respect to ρ if and only if $\rho(\gamma)$ acts on $\mathbb{H}^3$ by fixing l pointwise, i.e. acts as a (possibly trivial) rotation about l .

In $\tilde{H}^*$, γ acts as a translation on the set of all 3-simplices incident with e' . A connected fundamental domain for this action consists of a finite number of 3-simplices meeting in e' , and their number equals the degree of the corresponding edge e in P . Let σ be a 3-simplex in this fundamental domain. There is a unique isometry taking $D(\sigma)$ to $D(\gamma \cdot \sigma)$, and this is the rotation with eigenvalue precisely the product $h(e)$ of all shape parameters at e in P . Hence $\rho(\gamma)$ is a rotation if and only if $h(e)$ is an element in S^1 . $\blacksquare$

DEFINITION 15. *Given $\tilde{H}^* \rightarrow P$, $D: \tilde{H}^* \rightarrow \overline{\mathbb{H}}^3$ and $\rho: \pi_1 H \rightarrow PSL_2(\mathbb{C})$ as above, let $P_o = \tilde{H}^* / \ker(\rho)$ and $N_o = P_o \setminus P_o^{(0)}$. Since $\pi_1(H)$ acts simplicially, P_o and N_o have natural decompositions into simplices.*

If $o(\xi_e)$ is finite for each $e \in N^{(1)}$, then each edge in N_o has finitely many 3-simplices incident with it; namely if $N_o^{(1)} \ni \tilde{e} \rightarrow e \in N^{(1)}$, then $\deg(\tilde{e}) = o(\xi_e) \deg(e)$. In this case, the natural map $N_o \rightarrow N$ is a (not necessarily finite) branched cover, with branch locus contained in the 1-skeleton and group of deck transformations isomorphic to $\pi_1 H / \ker(\rho)$. If $o(\xi_e)$ is infinite for some e , then the points of N_o mapping to e are not manifold points.

PROOFS OF THEOREMS 1 AND 3. We first give a proof of Theorem 1, and then modify it for the general case. Note that the inclusion map $H \hookrightarrow C$ induces an epimorphism $\pi_1 H \twoheadrightarrow \pi_1 C$. The kernel of this map is generated by certain loops around the annuli in $\partial H \setminus (\partial H \cap \partial C)$. For each edge e we denote by γ_e a loop around the annulus corresponding to e , where γ_e is a level set in the product structure, with an arbitrary orientation.

Given Lemma 9, it suffices to assume for contradiction that there is a solution $Z \in \mathfrak{D}(\mathcal{T})$, but that $\mathcal{T}$ is not virtually almost non-singular. In particular, by Lemma 5, some edge is inessential. We work with the compact core C of N .

Give the tetrahedra in $\tilde{H}$ the shape parameters inherited from $Z \in \mathfrak{D}(\mathcal{T})$ (which are first inherited by H , then lifted to $\tilde{H}$ and finally inherited by $\tilde{H}^*$). Associated are a developing map $D : \tilde{H}^* \rightarrow \mathbb{H}^3$ and a representation $\rho : \pi_1(H) \rightarrow PSL_2(\mathbb{C})$. Recall that we specify a quotient map $\tilde{H} \rightarrow \tilde{H}^*$ by collapsing edge boundary components $\mathbb{R} \times [0, 1]$ to the second factor, then collapsing each vertex boundary component.

Now let e be an inessential edge in N . Consider a path $\alpha' : [0, 1] \rightarrow H$, which is the core curve of a rectangular face of a doubly truncated tetrahedron incident to the edge boundary annulus corresponding to e . Let $\alpha : [0, 1] \rightarrow H$ be the result of pushing α' slightly off the rectangle into the tetrahedron, keeping the endpoints on the boundary hexagonal faces. Then α is parallel to e , as in Figure 1.

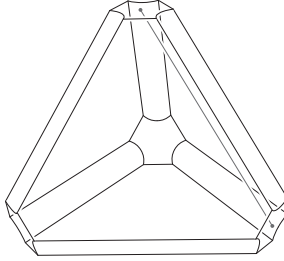


FIGURE 1. A doubly truncated tetrahedron of H , with the path α parallel to the boundary rectangle corresponding to an edge of P .

Since e is inessential in C , there is a homotopy of it into ∂C , fixing its endpoints. As α (viewed as a path in C since $H \subset C$) is parallel to e , we can use the same homotopy to homotope α into ∂C , fixing its endpoints, by first homotoping across the rectangle with α and e as one pair of opposite sides, and the other sides on ∂C in the obvious way. Viewing the homotopy as a map $D^2 = \mathbb{D} \rightarrow C$, we may deform it by a small amount to produce a homotopy h transverse to $P^{(1)}$, and moreover so that the intersection of $h(\mathbb{D})$ with $\partial(\nu(P^{(1)}))$ consists of a finite number of circles, each of which (pulled back through h) bounds a disk in $\mathbb{D}$ which is contained in a neighbourhood $\nu(e_i)$ and intersects e_i once, transversely. Thus each circle goes around the cylindrical part of ∂C corresponding to an edge of N , and is homotopic to a γ_e . In particular let $\beta : [0, 1] \rightarrow \partial C$ be the path after the homotopy, and by deforming if necessary we may assume that β in fact maps into $\partial H \cap \partial C$. See Figure 2.

We choose a base-point for $\pi_1(H)$ at $\alpha(0)$. Now $\mathbb{D} \cap h^{-1}(H)$ is a sphere with a number of boundary components, where $\partial \mathbb{D} = \alpha\beta^{-1}$ and all other boundary components correspond to circles $\gamma_k : [0, 1] \rightarrow H$, which are fibres in the product

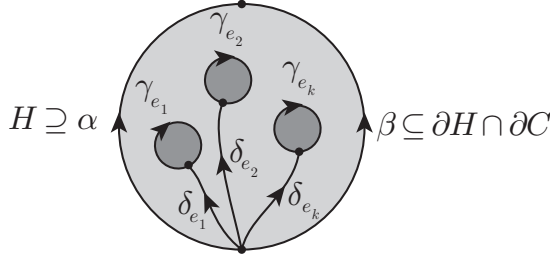


FIGURE 2. The homotopy $h : \bigcirc \rightarrow C$ between α and β . The parts mapping into H are shaded lightly, and the disks mapping into the neighbourhood of an edge in C shaded more darkly.

structure of the edge boundary. For each γ_k choose a non self-intersecting path δ_k on $\bigcirc \cap h^{-1}(H)$ from $\alpha(0)$ to $\gamma_k(0) = \gamma_k(1)$. We can choose these paths to be disjoint apart from at $\alpha(0)$. Then $\bigcirc \cap h^{-1}(H)$ gives the following relation between the elements in $\pi_1(H, \alpha(0))$ represented by these loops:

$$(4) \quad [\alpha\beta^{-1}] = \prod [\delta_k \gamma_k \delta_k^{-1}].$$

Choose a lift of $\alpha(0)$ to $\tilde{H}$ and lift α accordingly. Let $\tilde{\alpha}^*$ be the composition with the quotient $\tilde{H} \rightarrow \tilde{H}^*$. Then $D\tilde{\alpha}^* : [0, 1] \rightarrow \mathbb{H}^3$ is its image under the pseudo-developing map. Define $\tilde{\beta}$, $\tilde{\beta}^*$ and $D\tilde{\beta}^*$ similarly, choosing the lift $\tilde{\beta}$ to start from the same point as $\tilde{\alpha}$ does. Since the pseudo-developing map is well-defined and since the endpoints of e are on the same components of $\partial H \cap \partial C$ as the respective ends of α , $D\tilde{\alpha}^*(0)$ and $D\tilde{\alpha}^*(1)$ are distinct points on $\partial \mathbb{H}^3$. With our choice of base-points, $\rho([\alpha\beta^{-1}])$ is an isometry of $\mathbb{H}^3$ taking $D\tilde{\alpha}^*(0)$ to $D\tilde{\alpha}^*(1)$.

We now claim that for each term $\delta_k \gamma_k \delta_k^{-1}$ in the right hand side of relation (4), we have $\rho([\delta_k \gamma_k \delta_k^{-1}]) = 1$. Indeed, $\rho([\delta_k \gamma_k \delta_k^{-1}])$ can be expressed as a product of elementary face pairings. Now the products arising from δ_k and δ_k^{-1} are inverses. Hence $\rho([\delta_k \gamma_k \delta_k^{-1}]) = 1$ if and only if the product corresponding to γ_k is trivial. But γ_k gives a rotation with eigenvalue the product of all shape parameters around the edge corresponding to that loop. By hypothesis, this product equals 1, and hence the claim.

But then $\rho([\alpha\beta^{-1}]) = 1$, contradicting the fact that it acts non-trivially on $\partial \mathbb{H}^3$. This completes the proof of Theorem 1.

For the general case, the above proof can be applied using a punctured sphere in N_o . The curves γ_k can only correspond to edges of N_o which have finite degree, and the eigenvalue of the associated rotation is precisely of the form $\xi_e^{o_e} = 1$, giving the same contradiction as above. $\blacksquare$

REMARK 16. *The main restriction in using Yoshida's construction is the fact that ideal simplices in the universal cover are mapped to hyperbolic ideal simplices in hyperbolic 3-space. Daryl Cooper pointed out to us that the proofs in this section could be given using local arguments by subdividing the ideal tetrahedra in N . The authors feel that the approach using triangulations is more appropriate for this volume and the man it is dedicated to.*

5. A trip to the zoo

We give three examples of triangulations of S^3 with knotted or linked edges, which exhibit interesting features. Special hyperbolic cone-manifold structures on the first triangulation can be found in work by Boileau-Porti [1] and Hodgson [5], and the two concluding triangulations were provided by Bus Jaco.

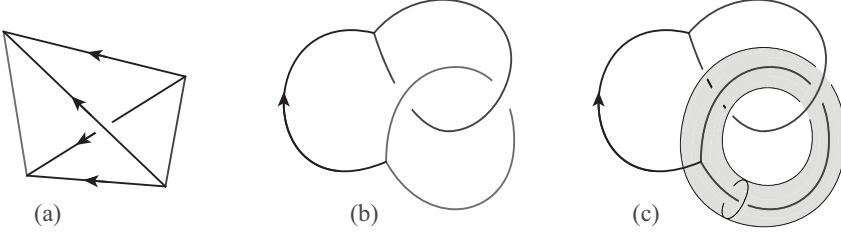


FIGURE 3. Triangulation of S^3 with the Hopf link as edges.

5.1. The Hopf link. Consider the one-tetrahedron three-edges triangulation of S^3 shown in Figure 3(a). We have $\mathfrak{D}(\mathcal{T}) = \emptyset$, since there are degree one edges. Giving the parameter z to the degree-one edges e_0 and e_1 , and letting e_2 be the degree-four edge, the cone-deformation variety is:

$$\mathfrak{D}(\mathcal{T}; \star) = \{(Z, \xi) = ((z, z', z''), (z, z, z^{-2})) \mid z \in S^1 \setminus \{1\}\} \cong S^1 \setminus \{1\}.$$

There is one ideal point (corresponding to $z \rightarrow 1$) and one flat solution (corresponding to $z = -1$). The degeneration $z \rightarrow 1$ corresponds to a normal surface which is a Heegaard torus in S^3 , and the flat solution will be analysed below using the face pairings.

The remaining structures come in pairs $(z, \bar{z})$ and it suffices to study the case $\Im(z) > 0$. Here, $|z| = 1$ implies that the arguments of z, z', z'' are the angles of an isosceles triangle. Letting α denote the argument of z , the angle around e_2 is $2(\pi - \alpha)$ and the angle around each of e_0 and e_1 is α . This gives hyperbolic cone-manifold structures, with respective cone angles $(\alpha, \alpha, 2(\pi - \alpha))$. Since the parameter z at the degree-one edges is an element of S^1 , it is easily verified that all of these hyperbolic cone-manifold structures are complete; see Figure 4.

In order to analyse the face pairings, we normalise the developing map so that the degree one edges are mapped to the geodesics $[1, \infty]$ and $[0, z'']$. The corresponding pairings for the faces incident with these edges are denoted γ_0 and γ_1 respectively. The holonomy around the third edge is denoted γ_2 . The images of the group elements under the holonomy representation are determined by the following Möbius transformations:

$$\begin{aligned} \rho_z(\gamma_0) &: [1, \infty, z''] \mapsto [1, \infty, 0], \\ \rho_z(\gamma_1) &: [0, z'', 1] \mapsto [0, z'', \infty], \\ \rho_z(\gamma_2) &: [0, \infty, 1] \mapsto [0, \infty, z^{-2}], \end{aligned}$$

where $\rho_z(\gamma_2) = \rho_z([\gamma_1, \gamma_0])$. This gives:

$$\rho_z(\gamma_0) = \frac{1}{\sqrt{z}} \begin{pmatrix} z & 1-z \\ 0 & 1 \end{pmatrix}, \quad \rho_z(\gamma_1) = \frac{1}{\sqrt{z}} \begin{pmatrix} 1 & 0 \\ -z & z \end{pmatrix}, \quad \rho_z(\gamma_2) = \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix}.$$

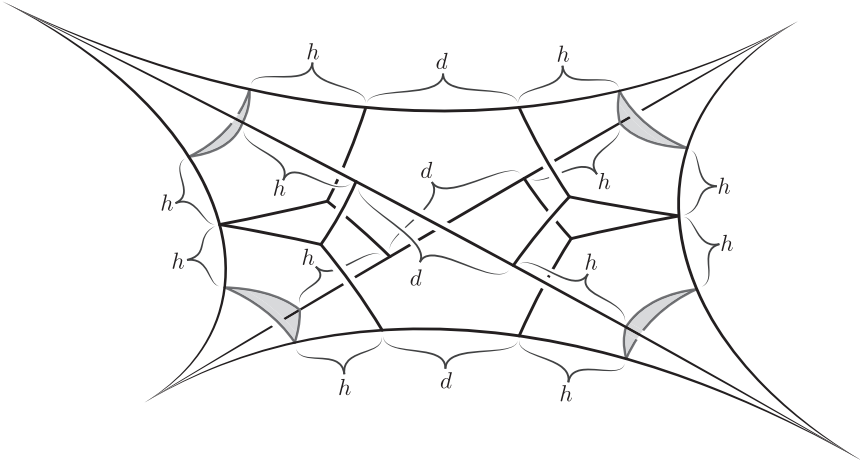


FIGURE 4. Geometry of the hyperbolic cone-manifolds for the Hopf link: The shown horospherical triangles give a cross-section of the cusps. The two face pairings are rotations about the left and right edges respectively.

Letting $z = e^{i\theta}$, we get the traces $(2 \cos(\theta/2), 2 \cos(\theta/2), e^{i\theta} + e^{-i\theta})$.

At $z = -1$, we have

$$\rho_{-1}(\gamma_0) = \begin{pmatrix} i & -2i \\ 0 & -i \end{pmatrix}, \quad \rho_{-1}(\gamma_1) = \begin{pmatrix} -i & 0 \\ -i & i \end{pmatrix}, \quad \rho_{-1}(\gamma_2) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Geometrically, as $z \rightarrow -1$, the fundamental domain degenerates to a quadrilateral, and the identification space is a sphere with cone angles $(0, 0, \pi)$. By construction, this is a hyperbolic 2-cone-manifold.

As $z \rightarrow 1$, the limiting representation is infinite cyclic:

$$\rho_1(\gamma_0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \rho_1(\gamma_1) = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

The fact that $\rho_1(\gamma_1)$ is the generator of the image corresponds to the chosen normalisation: The edge $[0, z'']$ pops off at infinity (since $z'' \rightarrow 0$ and $z \rightarrow 1$), giving $\rho_1(\gamma_0) = 1$.

5.2. The trefoil knot. Let $\mathcal{T}_{3_1}$ denote the minimal layered triangulation of S^3 with one degree-one edge e_1 and one degree-five edge e_2 , as shown in Figure 5. Here, e_1 is the trefoil knot. Again, $\mathfrak{D}(\mathcal{T}_{3_1}) = \emptyset$ and

$$\mathfrak{D}(\mathcal{T}_{3_1}; \star) = \{(Z, \xi) = ((z, z', z''), (z, z^{-1})) \mid z \in S^1 \setminus \{1\}\} \cong S^1 \setminus \{1\},$$

where z is the parameter given to e_1 . The angle at e_1 is α and the angle at e_2 is $2\pi - \alpha$.

As above, there is an ideal point (corresponding to $z \rightarrow 1$), and one flat solution (corresponding to $z = -1$). They will be analysed below using the face pairings. The degeneration $z \rightarrow 1$ corresponds to a thin edge-linking torus for e_K , and hence to the trefoil knot complement.

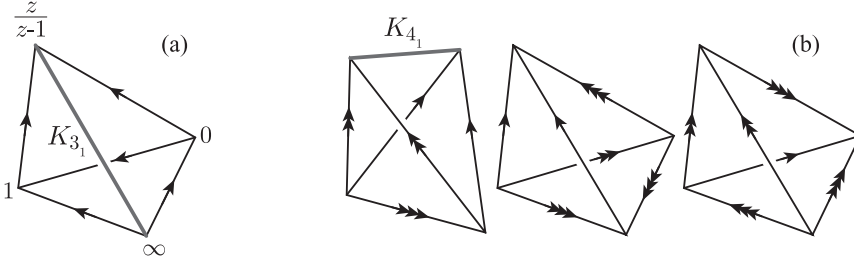


FIGURE 5. Triangulations of S^3 with knots as edges. In diagram (a) the edge K_{3_1} is a trefoil. In diagram (b) the edge K_{4_1} is the figure 8 knot.

The remaining structures come in pairs $(z, \bar{z})$ and it can again be verified directly that they all give complete hyperbolic cone-manifold structures with singular locus consisting of the three edges; see Figure 6.

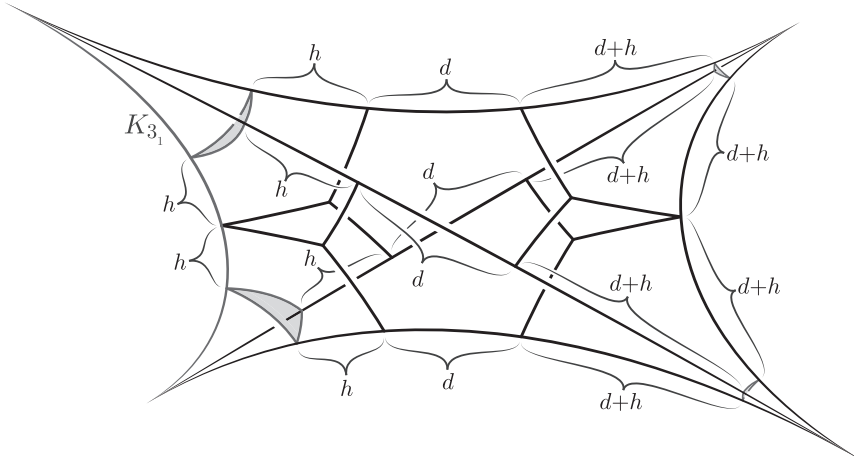


FIGURE 6. Geometry of the hyperbolic cone-manifolds for the trefoil: The shown horospherical triangles give a cross-section of the cusps. One face pairing is a rotation about the left-hand edge; the other identifies the faces incident with the right-hand edge with a twist.

The fundamental group is again generated by two face pairings, denoted γ_3 and γ_∞ . The holonomy images are:

$$\rho_z(\gamma_3) : [0, 1, \infty] \mapsto [1, \frac{z}{z-1}, 0], \quad \rho_z(\gamma_\infty) : [\infty, \frac{z}{z-1}, 0] \mapsto [\infty, \frac{z}{z-1}, 1].$$

These lead to representations:

$$\rho_z(\gamma_3) = \frac{1}{\sqrt{z}} \begin{pmatrix} 0 & z \\ -1 & z \end{pmatrix}, \quad \rho_z(\gamma_\infty) = \frac{1}{\sqrt{z}} \begin{pmatrix} 1 & z \\ 0 & z \end{pmatrix}.$$

Their product $\gamma_2 = \gamma_3\gamma_\infty$ is always of order two:

$$\rho_z(\gamma_2) = \begin{pmatrix} 0 & z \\ -z^{-1} & 0 \end{pmatrix}.$$

At $z \rightarrow 1$, we have:

$$\rho_1(\gamma_3) = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \quad \rho_1(\gamma_\infty) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \rho_1(\gamma_2) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

These generate the modular group, and the quotient of $\mathbb{H}^2$ is a sphere with 2, 3 and ∞ cone points. See Figure 7. The flat solution does not appear to have a nice interpretation.

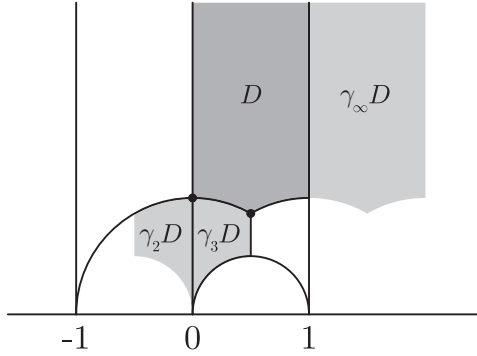


FIGURE 7. A fundamental domain of the action of ρ_1 for the trefoil on $\mathbb{H}^2$ is marked D .

5.3. The figure eight knot. In the previous examples, we have analysed structures arising at ordinary points of the cone-deformation variety as well as a single ideal point which corresponding to a 1-dimensional or 2-dimensional collapse. In the last example, we exhibit a surface of ideal points parameterising 3-dimensional structures. See Figure 5(b).

Label the tetrahedra z_0, z_1 and z_2 at their upper and lower edges in Figure 5. Then the generalised gluing equations are:

$$(5) \quad z_0 = \xi_K$$

$$(6) \quad \left(\frac{1}{z_0}\right) \left(\frac{1}{z_1}\right) \left(\frac{z_2 - 1}{z_2}\right) = \xi_1$$

$$(7) \quad \left(\frac{1}{z_0}\right) \left(\frac{z_1 - 1}{z_1}\right) \left(\frac{1}{z_2}\right) = \xi_2$$

$$(8) \quad z_0 \left(\frac{z_1^2}{1 - z_1}\right) \left(\frac{z_2^2}{1 - z_2}\right) = \xi_3$$

These simplify to:

$$(9) \quad z_0 = \xi_K$$

$$(10) \quad \xi_3 = \xi_K^{-1} \xi_1^{-1} \xi_2^{-1}$$

$$(11) \quad z_1 z_2 \xi_2 \xi_K = (z_1 - 1)$$

$$(12) \quad z_1 z_2 \xi_1 \xi_K = (z_2 - 1)$$

Hence $\mathfrak{D}(\mathcal{T}_4; \star)$ is parameterised by (ξ_K, z_1, z_2) , where $\xi_K \in S^1$ and $z_1, z_2 \in \mathbb{C} \setminus \{0, 1\}$, subject to $\frac{z_2''}{z_1} \in S^1$ from (12) and $\frac{z_1''}{z_2} \in S^1$ from (11) (note that $z_i'' = \frac{z_i - 1}{z_i}$).

Putting $z_1 = e^{-i\theta} z_2''$, where $\theta \in [0, 2\pi)$, the second condition gives $z_2' e^{i\theta} - z_2'' z_2' \in S^1$. Letting $z_2'' = x + iy$ for $x, y \in \mathbb{R}$, this yields a single equation in x, y, θ , and the space of solutions is a surface in $\mathbb{R}^2 \times S^1$ as shown in Figure 8. Now $\mathfrak{D}(\mathcal{T}_4; \star)$ is parameterised by (ξ_K, x, y, θ) , and since $\xi_K \in S^1$ is arbitrary, $\mathfrak{D}(\mathcal{T}; \star)$ is the product of this surface with S^1 .

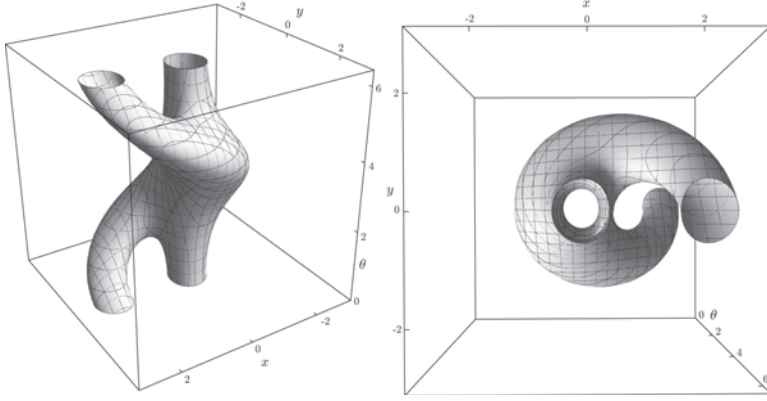


FIGURE 8. $\mathfrak{D}(\mathcal{T}_4; \star)$ is the product of this surface in $\mathbb{R}^2 \times S^1$ with S^1 .

The only ideal points are at $\xi_K = 1$, parameterised by $(1, x, y, \theta)$, and the associated normal surface is the thin edge linking torus around K , which splits the manifold into a solid torus and the figure 8 knot complement. Crushing along this surface identifies the single arrow edge with the double arrow edge, and gives us the canonical triangulation of the figure 8 knot complement. The equations become:

$$(13) \quad z_0 = 1$$

$$(14) \quad \frac{z_2 - 1}{z_1 z_2} = \xi_1$$

$$(15) \quad \frac{z_1 - 1}{z_1 z_2} = \xi_2$$

$$(16) \quad \frac{z_1^2 z_2^2}{(1 - z_1)(1 - z_2)} = \xi_3$$

To get the complete structure on the figure eight knot complement, we need $\xi_1 = 1 = \xi_2$, which implies that $z_1 = z_2''$ and $z_2 = z_1''$. Then $z_1 = e^{-i\theta} z_2'' = e^{-i\theta} z_1$, so $\theta = 0$, and we get $z_1 = z_2''$. We also have $z_2 = z_1''$, which implies that $z_2'' = z_1'$, so $z_1 = z_1'$, and z_1 is the shape parameter of the regular hyperbolic ideal tetrahedron. Similarly for z_2 . Note that solutions in a neighbourhood of this solution give 3-dimensional hyperbolic cone-manifold structures on the figure eight knot complement with singular locus contained in the ideal edges.

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A Generic Margulis Number for Hyperbolic 3-Manifolds

Peter B. Shalen

Dedicated to Bus Jaco on the occasion of his 70th birthday

ABSTRACT. We show that 0.29 is a Margulis number for all but finitely many hyperbolic 3-manifolds. The finitely many exceptions are all closed.

1. Introduction

If M is an orientable hyperbolic 3-manifold, we may write $M = \mathbb{H}^3/\Gamma$, where $\Gamma \cong \pi_1(M)$ is a discrete, torsion-free subgroup of $\text{Isom}_+(\mathbb{H}^3)$, uniquely determined up to conjugacy by the hyperbolic structure of M .

DEFINITION 1.1. Suppose that $M = \mathbb{H}^3/\Gamma$ is a non-elementary orientable hyperbolic 3-manifold. A *Margulis number* for M (or for Γ) is a number $\mu > 0$ such that:

- If $P \in \mathbb{H}^n$, and if x and y are elements of Γ such that $\max(d(P, x \cdot P), d(P, y \cdot P)) < \mu$, then x and y commute.

Here, and throughout the paper, d denotes hyperbolic distance on $\mathbb{H}^3$.

Note that if μ is a Margulis number for M then any number in the interval $(0, \mu)$ is also a Margulis number for M . Note also that if Γ is abelian then *any* positive number is a Margulis number for M .

The main result of this paper, which will be proved in Section 3, is:

THEOREM 1.2. *Up to isometry there are at most finitely many orientable hyperbolic 3-manifolds for which 0.29 is not a Margulis number. Furthermore, all such manifolds are closed.*

The first assertion of this theorem may be expressed as saying that 0.29 is a “generic Margulis number” for orientable hyperbolic 3-manifolds.

To put Theorem 1.2 in context, I will be quoting some material from [2], and to make the viewpoint consistent with that of [2] I will first discuss a generalization

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of the definition of Margulis number that I gave above. If M is a hyperbolic n -manifold, and if we write $M = \mathbb{H}^n/\Gamma$ where $\Gamma \leq \text{Isom}(\mathbb{H}^n)$ is discrete and torsion-free, one may define a Margulis number for M (or for Γ) to be a constant $\mu > 0$ such that for every $P \in \mathbb{H}^n$, the elements $x \in \Gamma$ such that $d(P, x \cdot P) < \mu$ generate a subgroup which is *elementary* in the sense that it has an abelian subgroup of finite index. If $n = 3$ and M is orientable, it follows from Proposition 2.1 below that every elementary subgroup of Γ is abelian. Hence this definition of Margulis number does generalize the one given above.

The *Margulis Lemma* [2, Chapter D] implies that there is a positive constant which is a Margulis number for every hyperbolic n -manifold. The largest such number, $\mu(n)$, is called the *Margulis constant* for hyperbolic n -manifolds.

Meyerhoff showed in [7] that $\mu(3) > 0.104$. Marc Culler has informed me that according to strong numerical evidence, 0.616 fails to be a Margulis number for the hyperbolic 3-manifold $\mathfrak{m}027(-4, 1)$, and hence $\mu(3) < 0.616$.

In [2, Chapter D] it is explained in detail how a Margulis number μ for a hyperbolic manifold M determines a canonical decomposition of M into a μ -thin part, consisting of cusp neighborhoods and tubes around closed geodesics, and a μ -thick part consisting of points where the injectivity radius is at least $\mu/2$. (In [2] the number μ is taken to be a Margulis constant, but the same arguments apply if μ is any Margulis number for M .)

There are only finitely many topological possibilities for the μ -thick part of M given an upper bound on the volume of M . In the case $n = 3$, the μ -thick part is a 3-manifold with torus boundary components, and M is diffeomorphic to the interior of a manifold obtained by Dehn fillings from its μ -thick part.

This makes estimation of the maximal Margulis number for M a major step in understanding the geometric structure of M . The larger μ is, the fewer possibilities there are for the μ -thick part.

The proof of Theorem 1.2 uses Proposition 8.5 of [3], which in turn depends in an essential way on Theorem VI.4.1 of the Memoir [4] by Bus Jaco and myself. This is an interesting example of how three-manifold topology can be applied to hyperbolic geometry. To deduce Theorem 1.2 from [3, Proposition 8.5] one needs results on algebraic convergence proved by T. Jorgensen, P. Klein and A. Weil. Many of the relevant definitions and results are reviewed in Section 2 below.

I am grateful to Dick Canary for crucial help in arriving at the algebraic convergence arguments needed for the proof of Theorem 1.2; to Dave Futer for helping me to clarify the logical relationship between Theorem 1.2 and the results of [3]; to Al Marden for helping me locate the reference [6]; and to Marc Culler for computing the example discussed above.

2. Preliminaries

As I mentioned in the introduction, a subgroup of $\text{Isom}_+(\mathbb{H}^3)$ is said to be *elementary* if it has an abelian subgroup of finite index.

In the case where Γ is discrete and torsion-free, I will say that the orientable hyperbolic 3-manifold $M = \mathbb{H}^3/\Gamma$ is *elementary* if Γ is elementary. If M has finite volume then M must be non-elementary.

The following easy result gives a more direct characterization of elementary orientable hyperbolic 3-manifolds.

PROPOSITION 2.1. *Every torsion-free, elementary, discrete subgroup of $\text{Isom}_+(\mathbb{H}^3)$ is abelian.*

PROOF. Suppose that $\Gamma \leq \text{Isom}_+(\mathbb{H}^3)$ is torsion-free, elementary, and discrete. For $1 \neq \gamma \in \Gamma$, the subset $\text{Fix}(\gamma)$ of the sphere at infinity is non-empty because $\langle \gamma \rangle$ is discrete and γ has infinite order, and consists of at most two points because $\gamma \neq 1$. For distinct elements $\gamma, \gamma' \in \Gamma$, we have $\text{Fix}(\gamma) = \text{Fix}(\gamma')$ if and only if γ and γ' commute. Now if x and y are non-trivial elements of Γ , the definition of an elementary group implies that x^m and y^n commute for some positive integers m and n . We have

$$\text{Fix}(x) = \text{Fix}(x^m) = \text{Fix}(y^n) = \text{Fix}(y),$$

so that x and y commute. □

We will need the following well-known consequence of the “half-lives, half-dies” principle:

PROPOSITION 2.2. *For any compact, connected, orientable 3-manifold N , the total genus of ∂N , i.e. the sum of the genera of its components, is bounded above by the first Betti number of N .*

PROOF. If one replaces the first Betti number of N , i.e. the dimension of the $\mathbb{Q}$ -vector space $H_1(N, \mathbb{Q})$, by the dimension of the $\mathbb{Z}_2$ -vector space $H_1(N, \mathbb{Z}_2)$, the statement become Lemma 7.3 of [1]. The proof of the latter result goes through without change if the coefficient field $\mathbb{Z}_2$ is replaced by $\mathbb{Q}$. □

The next result includes the second assertion of Theorem 1.2. The main ingredient in the proof is Proposition 8.5 of [3], which asserts that 0.292 is a Margulis number for every orientable 3-manifold M with $H_1(M; \mathbb{Q}) \neq 0$.

PROPOSITION 2.3. *Every non-compact, orientable hyperbolic 3-manifold admits 0.292 as a Margulis number.*

PROOF. Let M be a non-compact, orientable hyperbolic 3-manifold. Write $M = \mathbb{H}^3/\Gamma$ for some torsion-free discrete subgroup Γ of $\text{Isom}_+(\mathbb{H}^3)$. We must show that if x and y are non-commuting elements of Γ and P is a point in $\mathbb{H}^3$, we have

$$\max(d(P, x \cdot P), d(P, y \cdot P)) \geq 0.292.$$

Set $\widetilde{\Gamma} := \langle x, y \rangle$. Then $\widetilde{M} := \mathbb{H}^3/\widetilde{\Gamma}$ is a hyperbolic manifold which covers M , and is therefore non-compact. Since $\pi_1(\widetilde{M}) \cong \widetilde{\Gamma} = \langle x, y \rangle$ is finitely generated, $\widetilde{M}$ has a compact core N according to [9]. By definition, $N \subset \widetilde{M}$ is a compact submanifold, and the inclusion homomorphism $\pi_1(N) \rightarrow \pi_1(\widetilde{M})$ is an isomorphism.

Suppose that ∂N has a sphere component S . Since $\widetilde{M}$ is irreducible, S bounds a ball $B \subset \widetilde{M}$. If $B \supset N$ then $\widetilde{M}$ is simply connected, a contradiction since $\widetilde{\Gamma} \cong \pi_1(\widetilde{M})$ contains the non-commuting elements x and y . Hence $B \cap N = S$, and it follows that $B \cap N$ is a compact core for $\widetilde{\Gamma}$ with fewer boundary components than

N . Hence if we choose N among all compact cores so as to minimize its number of boundary components, then ∂N has no sphere component.

On the other hand, since $\widetilde{M}$ is not closed, we have $\partial N \neq \emptyset$. Hence the total genus of ∂N , i.e. the sum of the genera of its components, is strictly positive.

By Proposition 2.2, the first Betti number of N is bounded below by the total genus of ∂N . Hence $H_1(\widetilde{M}; \mathbb{Q}) \neq 0$. According to [3, Proposition 8.5], this implies that 0.292 is a Margulis number for $\widetilde{M}$. Since x and y do not commute, we therefore have

$$\max(d(P, x \cdot P), d(P, y \cdot P)) \geq 0.292.$$

□

If Γ is a group, we denote by $\text{Isom}_+(\mathbb{H}^3)^{(\Gamma)}$ the set of all set-theoretical mappings of Γ into $\text{Isom}_+(\mathbb{H}^3)$. Thus $\text{Isom}_+(\mathbb{H}^3)^{(\Gamma)}$ is the same as the product $\prod_{\gamma \in \Gamma} \text{Isom}_+(\mathbb{H}^3)$, where $G_\gamma = \text{Isom}_+(\mathbb{H}^3)$ for $\gamma \in \Gamma$. We shall always understand $\text{Isom}_+(\mathbb{H}^3)$ to have its usual Lie group topology, and $\text{Isom}_+(\mathbb{H}^3)^{(\Gamma)}$ to have the product topology. The set $\text{Hom}(\Gamma, \text{Isom}_+(\mathbb{H}^3))$ of all representations of Γ in $\text{Isom}_+(\mathbb{H}^3)$ is the subset of mappings $\rho \in \text{Isom}_+(\mathbb{H}^3)^{(\Gamma)}$ which satisfy $\rho(\gamma\gamma') = \rho(\gamma)\rho(\gamma')$ for all $\gamma, \gamma' \in \text{Isom}_+(\mathbb{H}^3)$. Hence $\text{Hom}(\Gamma, \text{Isom}_+(\mathbb{H}^3))$ is a closed subset of $\text{Isom}_+(\mathbb{H}^3)^{(\Gamma)}$. It will be given the subspace topology.

A sequence of representations $(\rho_j)_{j \in \mathbb{N}}$ in $\text{Hom}(\Gamma, \text{Isom}_+(\mathbb{H}^3))$ is said to *converge algebraically* to a representation $\rho_\infty \in \text{Hom}(\Gamma, \text{Isom}_+(\mathbb{H}^3))$ if it converges to ρ_∞ in the topology of $\text{Hom}(\Gamma, \text{Isom}_+(\mathbb{H}^3))$ described above.

I will denote by $D(\Gamma)$ the subset of $\text{Hom}(\Gamma, \text{Isom}_+(\mathbb{H}^3))$ consisting of all representations whose images are discrete and non-elementary, and by $DFC(\Gamma) \subset D(\Gamma)$ the subset of $\text{Hom}(\Gamma, \text{Isom}_+(\mathbb{H}^3))$ consisting of all faithful representations whose images are discrete and cocompact.

The following result summarizes the facts about algebraic convergence that will be needed in the next section.

THEOREM 2.4. *Let Γ be any group.*

- (1) *The set $D(\Gamma)$ is closed in $\text{Hom}(\Gamma, \text{Isom}_+(\mathbb{H}^3))$.*
- (2) *If a sequence $(\phi_j)_{j \in \mathbb{N}}$ in $D(\Gamma)$ converges algebraically to a representation ϕ_∞ , then for every sufficiently large j there is a homomorphism $\psi_j : \phi_\infty(\Gamma) \rightarrow \phi_j(\Gamma)$ such that $\psi_j \circ \phi_\infty = \phi_j$.*
- (3) *The set $DFC(\Gamma)$ is open in $\text{Hom}(\Gamma, \text{Isom}_+(\mathbb{H}^3))$.*

PROOF. In proving the first assertion we may assume that $D(\Gamma) \neq \emptyset$, so that Γ is isomorphic to a discrete subgroup of $\text{Isom}_+(\mathbb{H}^3)$ and is therefore countable. This implies that $\text{Isom}_+(\mathbb{H}^3)^{(\Gamma)}$ is a first-countable space, and hence that $\text{Hom}(\Gamma, \text{Isom}_+(\mathbb{H}^3))$ is also first-countable. It therefore suffices to prove that every algebraic limit of representations in $D(\Gamma)$ belongs to $D(\Gamma)$. But this is the first assertion of [6, p. 326, Theorem].

The second assertion of Theorem 2.4 is Theorem 2 of [5] (and is also the second assertion of [6, p. 326, Theorem]).

The third assertion is the case $G = \text{Isom}_+(\mathbb{H}^3)$ of the main theorem of [10]. □

3. Proof of the main theorem

The proof of Theorem 1.2 will occupy this section.

The second assertion of the theorem is immediate from Proposition 2.3.

To prove the first assertion, we argue by contradiction. Suppose that there is an infinite sequence $(M_j)_{j \geq 0}$ of pairwise non-isometric orientable hyperbolic 3-manifolds none of which admits 0.29 as a Margulis number. For each j write $M_j = \mathbb{H}^3/\Gamma_j$ for some torsion-free discrete subgroup Γ_j of $\text{Isom}_+(\mathbb{H}^3)$.

For each j , since 0.29 is not a Margulis number for M_j , there exist non-commuting elements $x_j, y_j \in \Gamma_j$ and a point $P_j \in \mathbb{H}^3$ such that

$$\max(d(P_j, x_j \cdot P_j), d(P_j, y_j \cdot P_j)) < 0.29.$$

After replacing each Γ_j by a suitable conjugate of itself in $\text{Isom}_+(\mathbb{H}^3)$, we may assume that the P_j are all the same point of $\mathbb{H}^3$, which I will denote by P . Thus for each j we have

$$(3.0.1) \quad \max(d(P, x_j \cdot P), d(P, y_j \cdot P)) < 0.29.$$

For each $j \in \mathbb{N}$, the group $\tilde{\Gamma}_j := \langle x_j, y_j \rangle$ is a subgroup of Γ_j , and is therefore discrete and torsion-free. Since x_j and y_j do not commute, it follows from Proposition 2.1 that $\tilde{\Gamma}_j$ is non-elementary. Hence $\tilde{M}_j := \mathbb{H}^3/\tilde{\Gamma}_j$ is a non-elementary hyperbolic manifold which covers M_j .

It follows from (3.0.1) that the x_j and y_j all lie in a compact subset of $\text{Isom}_+(\mathbb{H}^3)$. Hence after passing to a subsequence we may assume that $x_j \rightarrow x_\infty$ and $y_j \rightarrow y_\infty$ for some $x_\infty, y_\infty \in \text{Isom}_+(\mathbb{H}^3)$.

Let F_2 denote the free group on two generators x and y . For $1 \leq j \leq \infty$, let us define a homomorphism $\phi_j : F_2 \rightarrow \text{Isom}_+(\mathbb{H}^3)$ by $\phi_j(x) = x_j$ and $\phi_j(y) = y_j$. The sequence $(\phi_j)_{j \in \mathbb{N}}$ converges algebraically to ϕ_∞ . For each $j \in \mathbb{N}$, since $\tilde{\Gamma}_j$ is discrete and non-elementary, we have $\phi_j \in D(\Gamma)$ in the notation of Section 2. Applying Assertion (1) of Theorem 2.4, with $\Gamma = F_2$ and $\rho_j = \phi_j$, we deduce that $\phi_j \in D(\Gamma)$; that is, $\tilde{\Gamma}_\infty := \phi_\infty(F_2)$ is discrete and non-elementary. Furthermore, it follows from Assertion (2) of Theorem 2.4 that for sufficiently large j there is a homomorphism $\psi_j : \tilde{\Gamma}_\infty \rightarrow \tilde{\Gamma}_j$ such that $\psi_j \circ \phi_\infty = \phi_j$. After passing to a subsequence we may assume that such a ψ_j exists for every $j \in \mathbb{N}$.

We regard ψ_j as a representation of $\tilde{\Gamma}_\infty$ in $\text{Isom}_+(\mathbb{H}^3)$, and we let $\psi_\infty : \tilde{\Gamma}_\infty \rightarrow \text{Isom}_+(\mathbb{H}^3)$ denote the inclusion homomorphism. Since the sequence $(\phi_j)_{j \in \mathbb{N}}$ converges algebraically to ϕ_∞ , the sequence $(\psi_j)_{j \in \mathbb{N}}$ converges algebraically to ψ_∞ .

If $\gamma \in \tilde{\Gamma}_\infty$ has finite order, then for each $j \in \mathbb{N}$ we have $\psi_j(\gamma) = 1$, since $\tilde{\Gamma}_j$ is torsion-free. By algebraic convergence we have

$$\gamma = \psi_\infty(\gamma) = \lim_{j \rightarrow \infty} \psi_j(\gamma) = 1.$$

This shows that $\tilde{\Gamma}_\infty$ is torsion-free. Hence $\tilde{M}_\infty = \mathbb{H}^3/\tilde{\Gamma}_\infty$ acquires the structure of a hyperbolic 3-manifold in a natural way.

We now distinguish two cases; in each case we shall obtain a contradiction.

Case I: $\widetilde{M}_\infty$ is closed. In this case $\widetilde{\Gamma}_\infty$ is cocompact, so that in the notation of Section 2 we have $\psi_\infty \in DFC(\widetilde{\Gamma}_\infty)$. Since $(\psi_j)_{j \in \mathbb{N}}$ converges algebraically to ψ_∞ , it then follows from Assertion (3) of Theorem 2.4, with $\Gamma = \widetilde{\Gamma}_\infty$, that $\psi_j \in DFC(\widetilde{\Gamma}_\infty)$ for every sufficiently large j . After passing to a subsequence we may therefore assume that $\psi_j \in DFC(\widetilde{\Gamma}_\infty)$ for every $j \in \mathbb{N}$. In particular $\psi_j : \widetilde{\Gamma}_\infty \rightarrow \widetilde{\Gamma}_j$ is an isomorphism for each j ; and $\widetilde{M}_j = \mathbb{H}^3/\widetilde{\Gamma}_j = \mathbb{H}^3/\psi_j(\widetilde{\Gamma}_\infty)$ is closed, and is therefore a finite-sheeted covering of M_j . For each $j \in \mathbb{N}$, since $\widetilde{\Gamma}_j$ is isomorphic to $\widetilde{\Gamma}_\infty$, it follows from the Mostow rigidity theorem [2, Chapter C] that $\widetilde{M}_j$ is isometric to $\widetilde{M}_\infty$.

Let Δ denote the diameter of $\widetilde{M}_\infty$. Then since each M_j has a finite-sheeted covering isometric to $\widetilde{M}_\infty$, each M_j has diameter at most Δ .

Let v denote the infimum of the volumes of all closed hyperbolic 3-manifolds; we have $v > 0$, for example by [7, Theorem 1]. Since each M_j has volume at least v , diameter at most Δ , constant curvature -1 and dimension 3, it follows from the main theorem of [8] that the M_j represent only finitely many diffeomorphism types. By the Mostow rigidity theorem, they represent only finitely many isometry types. This gives the required contradiction in Case I.

Case II: $\widetilde{M}_\infty$ is not closed. In this case it follows from Proposition 2.3 that 0.292 is a Margulis number for $\widetilde{M}_\infty$. Since $\widetilde{\Gamma}_\infty = \langle x_\infty, y_\infty \rangle$ is non-elementary, x_∞ and y_∞ do not commute. We therefore have

$$\max(d(P, x_\infty \cdot P), d(P, y_\infty \cdot P)) \geq 0.292.$$

On the other hand, it follows from (3.0.1), upon taking limits as $j \rightarrow \infty$, that

$$\max(d(P, x_\infty \cdot P), d(P, y_\infty \cdot P)) \leq 0.29.$$

This gives the required contradiction in Case II.

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On Gradings in Khovanov homology and sutured Floer homology

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ABSTRACT. We discuss generalizations of Ozsváth-Szabó’s spectral sequence relating Khovanov homology and Heegaard Floer homology, focusing attention on an explicit relationship between natural $\mathbb{Z}$ (resp., $\frac{1}{2}\mathbb{Z}$) gradings appearing in the two theories. These two gradings have simple representation-theoretic (resp., geometric) interpretations, which we also review.

1. Introduction

Ozsváth-Szabó’s spectral sequence [26] relating Khovanov homology [17] and Heegaard-Floer homology [24] has inspired an abundance of work in these and related fields (see, e.g., [28], [29], [32], [21], [9], [11], [13], [37], [33], [2], [3], [6], [5], [38], [10], [8]), yielding applications in areas as disparate as contact geometry [3] and Dehn surgery [37]. A particularly stunning result with roots in the circle of ideas found in [26] is Kronheimer-Mrowka’s proof that Khovanov homology detects the unknot [20].

The Ozsváth-Szabó spectral sequence has been extended, by L. Roberts in [32] and the authors in [9, 8], to yield a relationship between the “sutured” versions of Khovanov homology [1, 19] and Heegaard Floer homology [14] one naturally associates to tangles in certain simple sutured manifolds. Applying Gabai’s sutured manifold technology [7] to these homology theories, an idea pioneered by Juhász [15], reveals that the Ozsváth-Szabó spectral sequence exhibits a great deal more algebraic structure than originally believed. See [10] as well as [9, 8] for more details.

Our goal in this short note is to establish an explicit relationship between certain natural gradings appearing in Khovanov homology and sutured Heegaard-Floer homology. This relationship is already implicit in the work of Plamenevskaya [28] and L. Roberts [32]. Our motivation here is simply to provide an easily-digestible account, complete with the appropriate representation-theoretic and geometric context, for the benefit of the Khovanov and Heegaard-Floer communities, for whom the result appears to be of independent interest.

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2. Background on Ozsváth-Szabó spectral sequence

Let $\mathbb{L} \subset S^3$ be a link, $\overline{\mathbb{L}} \subset S^3$ its mirror, and Y a closed, connected, oriented 3-manifold. In what follows, $\widehat{Kh}(\mathbb{L})$ will denote the reduced Khovanov homology of $\mathbb{L}$, defined in [18], and $\widehat{HF}(Y)$ will denote the (hat version of) the Heegaard-Floer homology of Y , defined in [24]. Furthermore, if $(B, \partial B) \subset (X, \partial X)$ is a properly-embedded codimension 2 submanifold of a manifold X , then $\Sigma(X, B)$ will denote the double-branched cover of X , branched over B , and $\widetilde{B}$ will denote its preimage, $p^{-1}(B)$, under the covering map $p : \Sigma(X, B) \rightarrow X$. Throughout the paper, all Khovanov and Heegaard-Floer homology theories will be considered with $\mathbb{F} = \mathbb{Z}/2\mathbb{Z}$ coefficients. Ozsváth-Szabó proved:

THEOREM 2.1. [26, Thm. 1.1] *Let $\mathbb{L} \subset S^3$ be a link. There is a spectral sequence whose E^2 term is $\widehat{Kh}(\overline{\mathbb{L}})$ and whose E^∞ term is $\widehat{HF}(\Sigma(S^3, \mathbb{L}))$.*

SKETCH OF PROOF. Ozsváth-Szabó built this spectral sequence by constructing a Floer-theoretic “cube of resolutions” complex for $\widehat{HF}(\Sigma(S^3, \mathbb{L}))$ that looks very much like the chain complex used to define the reduced Khovanov homology of $\overline{\mathbb{L}}$.

More precisely, recall that if we are given an n -crossing diagram for a link $\overline{\mathbb{L}} \subset S^3$ with crossings labeled $C_1, \dots, C_n$ along with a marked point $z \subset (\mathbb{L} - \bigcup_{i=1}^n N(C_i))$, where $N(C_i)$ is a small neighborhood of the preimage of C_i in $\mathbb{L}$, we can construct a chain complex computing $\widehat{Kh}(\mathbb{L})$ by the following procedure (see [17, 4, 18] for more details).

- (1) Associate to each of the 2^n vertices of an n -dimensional cube a so-called “complete resolution” of the diagram. Specifically, the vertices can be put in one-to-one correspondence with the elements of $\{0, 1\}^n$; to vertex $\mathcal{I} \in \{0, 1\}^n$ we then associate the resolved diagram whose crossings have been replaced by “0” and “1” resolutions (see Figure 1) according to the prescription $\mathcal{I}$.
- (2) Apply Khovanov’s $(1+1)$ -dimensional TQFT to this cube. This replaces the resolved diagram at vertex $\mathcal{I}$ with the graded vector space $V^{\otimes k}$, where $V = \mathbb{F}_{-1} \oplus \mathbb{F}_1$ (the subscripts correspond to the grading) and k is the number of unmarked components in the resolved diagram at $\mathcal{I}$. To the edges of the cube, Khovanov associates linear “multiplication” and “co-multiplication” maps.
- (3) This data defines a bigraded chain complex whose underlying vector space is the direct sum of all of the vector spaces at the vertices of the cube and whose differential is the sum of the linear maps along the edges of the cube. The homology of this chain complex is a link invariant.

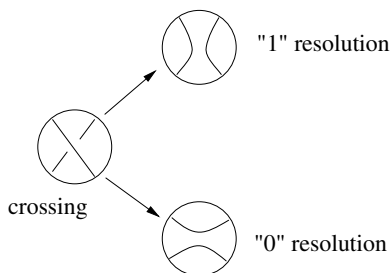


FIGURE 1. Replacing a crossing with a “0” or “1” resolution

On the Heegaard-Floer side, Ozsváth-Szabó had previously proved the existence of an exact sequence associated to a framed knot K in a closed, connected, oriented 3-manifold Y [23, Thm. 9.12]. Namely, there is a long exact sequence relating the triple

$$\{\widehat{HF}(K_\mu), \widehat{HF}(K_\lambda), \widehat{HF}(K_{\lambda+\mu})\},$$

where μ denotes the meridian of K in Y , λ denotes the framing on K , and K_α denotes the 3-manifold obtained by doing α -surgery on $K \subset Y$ for $\alpha = \mu, \lambda, \lambda + \mu$. Ozsváth-Szabó’s crucial observation is that the branched double cover of the 3-ball neighborhood $N(C_i)$ of each crossing C_i , branched along $L \cap N(C_i)$, is a solid torus, i.e., the neighborhood of a framed knot $K_i \subset \Sigma(S^3, \mathbb{L})$. Furthermore, replacing the crossing by the two different resolutions corresponds, in $\Sigma(S^3, \mathbb{L})$, to performing alternate Dehn surgeries on K_i , yielding two other 3-manifolds related to $\Sigma(S^3, \mathbb{L})$ by an appropriate long exact sequence for K_i .

Thus one can compute $\widehat{HF}(\Sigma(S^3, \mathbb{L}))$ using a filtered chain complex constructed as an iterated mapping cone associated to the link $K_1 \cup \dots \cup K_n \subset \Sigma(S^3, \mathbb{L})$ obtained by taking the preimage of the (neighborhoods of the) crossings $C_1, \dots, C_n$. More precisely, the existence of the long exact sequence for each K_i allows one to replace (up to homotopy) any chain complex $\widehat{CF}(\Sigma(S^3, \mathbb{L}))$ computing $\widehat{HF}(\Sigma(S^3, \mathbb{L}))$ with the mapping cone of a chain map

$$f_1 : \widehat{CF}(\Sigma(S^3, \mathbb{L}_{0,\infty,\dots,\infty})) \rightarrow \widehat{CF}(\Sigma(S^3, \mathbb{L}_{1,\infty,\dots,\infty}))$$

where $\mathbb{L}_{0,\infty,\dots,\infty}$ (resp., $\mathbb{L}_{1,\infty,\dots,\infty}$) represents the link obtained from $\mathbb{L}$ by replacing the crossing C_1 by its 0-resolution (resp., its 1-resolution). Recall that the mapping cone of a chain map $f : \mathcal{C} \rightarrow \mathcal{C}'$ between chain complexes $\mathcal{C}$ and $\mathcal{C}'$ is the chain complex with underlying vector space $\mathcal{C} \oplus \mathcal{C}'\{-1\}$ and differential, D , given by the sum of the internal differentials $\partial_{\mathcal{C}}, \partial_{\mathcal{C}'}$ and the chain map f :

$$D := \begin{pmatrix} \partial_{\mathcal{C}} & 0 \\ f & \partial_{\mathcal{C}'} \end{pmatrix}.$$

Iterating this process over all crossings, one obtains a cube of resolutions similar to the Khovanov cube described above (in particular, the vertices and edge maps can be seen to agree). However, there are additional higher-order differentials mapping along higher-dimensional faces of the cube. These arise, at least in part, because each of the three Heegaard-Floer chain complexes in the long exact sequence for a knot is only *chain homotopic* (not *chain isomorphic*) to the mapping cone of the chain map between the other two.

The filtered chain complex computing $\widehat{HF}(\Sigma(S^3, \mathbb{L}))$, described above, can be constructed explicitly using a so-called *Heegaard multi-diagram*. To understand this, recall that, given a closed, connected, oriented 3-manifold Y , one defines a chain complex computing $\widehat{HF}(Y)$ using the data of a generic self-indexing Morse decomposition of Y with a single critical point of index 0 and a single critical point of index 3. This yields a decomposition

$$Y = \mathcal{H}_\alpha \bigcup_{\Sigma_g} -\mathcal{H}_\beta$$

of Y into two solid handlebodies $\mathcal{H}_\alpha$ and $\mathcal{H}_\beta$. One can then choose a corresponding pointed Heegaard diagram $(\Sigma_g, \alpha, \beta, z)$, where

- Σ_g is a genus g Heegaard surface,
- $\alpha = \{\alpha_1, \dots, \alpha_g\}$ (resp., $\beta = \{\beta_1, \dots, \beta_g\}$) is a choice of g circles in Σ_g , linearly-independent in $H_1(\Sigma_g; \mathbb{Z})$, bounding a full set of compressing disks in $\mathcal{H}_\alpha$ (resp., $\mathcal{H}_\beta$), and
- $z \in \Sigma_g$ is a marked point in the complement of $\alpha \cup \beta$.

After picking a generic complex structure on Σ_g (see [24, Sec. 3]) and ensuring that the Heegaard diagram satisfies some additional technical assumptions (see [24, Sec. 4]), one then defines a chain complex from the data $(\Sigma_g, \alpha, \beta, z)$, whose

- generators are given by intersection points

$$\mathbb{T}_\alpha \cap \mathbb{T}_\beta = (\alpha_1 \times \dots \times \alpha_g) \cap (\beta_1 \times \dots \times \beta_g) \subset \text{Sym}^g(\Sigma_g),$$

where $\text{Sym}^g(\Sigma_g) := \Sigma^{\times g}/S_g$ is the space of unordered g -tuples of points in Σ_g , endowed with a nearly symmetric almost-complex structure, and

- differential is given by counting certain holomorphic maps of the disk into $\text{Sym}^g(\Sigma_g)$.

Now, if we have the data of a framed link $(L = K_1 \cup \dots \cup K_n) \subset Y$, we can construct a *Heegaard multi-diagram* compatible with L (and a corresponding filtered chain complex) by choosing

- a *bouquet*, B_L , for L [27, Defn. 4.1] and
- a self-indexing Morse function for $Y - N(B_L)$ with a single index 0 critical point and no index 3 critical points.

This specifies an “unbalanced” Heegaard decomposition of $Y - N(B_L)$ which can be completed to a balanced Heegaard decomposition (and corresponding diagram) for any closed 3-manifold obtained by Dehn filling L , just by appending the appropriate list of Dehn-filling curves $\beta_{g-n+1}, \dots, \beta_g$ to the list $\{\beta_1, \dots, \beta_{g-n}\}$ specifying $Y - N(B_L)$. A Heegaard multi-diagram $(\Sigma_g, \alpha, \beta_{\mathcal{D}}, z)$ compatible with a set $\mathcal{D}$ of Dehn fillings of a link, $L \subset Y$, then simultaneously encodes the data of a Heegaard diagram for all Dehn fillings in $\mathcal{D}$. In other words, one specifies a g -tuple, β_d , of β curves for each choice $d \in \mathcal{D}$, with the stipulation that $\{\beta_1, \dots, \beta_{g-n}\}$ agree for all β_d , as described above. In the case of interest to us, $Y = \Sigma(S^3, \mathbb{L})$, $N(L) = \bigcup_{i=1}^n N(C_i)$, and $\mathcal{D} = \{0, 1\}^n$. The corresponding filtered chain complex has

- generators given by the intersection points in $\bigcup_{d \in \mathcal{D}} \mathbb{T}_\alpha \cap \mathbb{T}_{\beta_d}$
- differential given by counting certain holomorphic polygons in $\text{Sym}^g(\Sigma_g)$, where

- 2-gons yield maps between generators associated to the same vertex of the cube of resolutions,
- 3-gons (triangles) yield maps along edges of the cube, and
- $(k+2)$ -gons yield maps along k -dimensional faces of the cube.

Please see [26] for more details.

The Floer-theoretic filtered chain complex described above gives rise to a spectral sequence whose E^2 term is $\widehat{Kh}(\mathbb{L})$ and whose E^∞ term is $\widehat{HF}(\Sigma(S^3, \mathbb{L}))$. $\square$

In [32], L. Roberts proved a refinement of the Ozsváth-Szabó spectral sequence for links in the solid torus complement of a standardly-imbedded unknot

$$U = (z - \text{axis} \cup \infty) \subset (S^3 = \mathbb{R}^3 \cup \infty).$$

Noting that this solid torus complement can be identified with the product sutured manifold $A \times I$ imbedded in S^3 as

$$A \times I := \{(r, \theta, z) \mid r \in [1, 2], \theta \in (-\pi, \pi], z \in [0, 1]\},$$

we reinterpreted his result in [8] using the language of sutured manifolds, as follows:

THEOREM 2.2. [32, Prop. 1.1], [8, Thm. 2.1] *Let $\mathbb{L} \subset A \times I$ be a link in the product sutured manifold $A \times I$. Then there is a spectral sequence whose E^2 term is $Kh^*(\mathbb{L})$ and whose E^∞ term is $SFH(\Sigma(A \times I, \mathbb{L}))$.*

Here, Kh^* is a version of Khovanov homology for links in thickened surfaces defined by Asaeda-Przytycki-Sikora in [1], and SFH is the version of Heegaard Floer homology for balanced, sutured manifolds defined by Juhász in [14]. The proof of Theorem 2.2 follows Ozsváth-Szabó's original proof closely. In fact, one sees that the chain complex computing $Kh^*(\mathbb{L} \subset A \times I)$ (resp., $SFH(\Sigma(A \times I, \mathbb{L}))$) is really just the original chain complex computing the ordinary Khovanov homology of the link $Kh(\mathbb{L} \subset S^3)$ (resp., a two-pointed version of the ordinary Floer homology of the double-branched cover, $\widehat{HF}(\Sigma(S^3, \mathbb{L})) \otimes \mathbb{F}^2$), where the chain complex has been equipped with an extra $\mathbb{Z}$ - (resp., $\frac{1}{2}\mathbb{Z}$ -) grading, and those components of the differential that do not respect this extra grading are set to zero.

There is a nice correspondence, detailed in Theorem 5.1, between the extra Heegaard-Floer $\frac{1}{2}\mathbb{Z}$ -grading, which we denote by $\mathbf{A}_S$, and the extra Khovanov $\mathbb{Z}$ -grading, which we denote by k . Furthermore, both gradings have natural interpretations, in terms of the geometry of vector fields on the Heegaard-Floer side (Section 3) and in terms of the representation theory of the quantum group $U_q(\mathfrak{sl}_2)$ on the Khovanov side (Section 4). This correspondence will be the focus of the remainder of the paper.

3. Sutured Floer grading

We begin by studying the $\frac{1}{2}\mathbb{Z}$ -grading on the sutured Floer homology of $\Sigma(A \times I, \mathbb{L})$, as described in [8], following [32].

In particular, we build a filtered chain complex whose homology is $SFH(\Sigma(A \times I, \mathbb{L}))$ from the data of a sutured Heegaard multi-diagram for $\Sigma(A \times I, \mathbb{L})$ and endow each generator, $\mathbf{x}$, of the complex with a $\frac{1}{2}\mathbb{Z}$ -grading, $\mathbf{A}_S(\mathbf{x})$. We furthermore describe how to understand this grading as an Euler number of a natural vector field associated to $\mathbf{x}$.

Recall that any sutured Floer chain complex $CFH(Y, \Gamma)$ for a balanced sutured manifold (Y, Γ) splits according to Spin^c structures. These are homology classes of

non-vanishing vector fields on Y with prescribed boundary behavior, and they form an affine set for the action of $H_1(Y; \mathbb{Z})$. Given a generator $\mathbf{x}$ of a sutured Floer complex $CFH(Y, \Gamma)$, one assigns a Spin^c structure to it as described in [14, Defn. 4.5]. See [35], [25, Sec. 3.2], [14, Sec. 4], [15, Sec. 3] for more details.

Given the data of a trivialization, t , of the restriction of a Spin^c structure, $\mathfrak{s}$, to ∂Y one can then define the first Chern class, $c_1(\mathfrak{s}, t) \in H^2(Y, \partial Y; \mathbb{Z})$, with respect to the trivialization ([15, Defn. 3.7]) as a relative Euler class of $\mathfrak{s}$. The added data of a properly-imbedded surface $(S, \partial S) \subset (Y, \partial Y)$ then endows $CFH(Y, \Gamma)$ with a $\frac{1}{2}\mathbb{Z}$ Alexander- S -grading ($\mathbf{A}_S$ -grading) as described in [10, Sec. 3], following [14].

Namely, given $\mathbf{x} \in CFH(Y, \Gamma)$, we define:

$$\mathbf{A}_S(\mathbf{x}) := \frac{1}{2} \langle c_1(\mathfrak{s}(\mathbf{x}), t), [S] \rangle.$$

Furthermore, suppose that $(S, \partial S) \subset (Y, \partial Y)$ is geometrically disjoint from a framed link $L \subset Y$. Then it is proved in [10, Sec. 3] that the entire filtered complex for $SFH(Y)$ associated to a Heegaard multi-diagram for the pair (Y, L) (as described in Section 2) can be endowed with $\mathbf{A}_S$ -gradings. Moreover [10, Lem. 3.12] implies that the filtered chain complex splits according to these gradings.

The case of interest to us is when $S \subset \Sigma(A \times I, \mathbb{L})$ is the preimage of a meridional disk in the solid torus $A \times I$ under the covering map $\pi : \Sigma(A \times I, \mathbb{L}) \rightarrow A \times I$. In order to study these $\mathbf{A}_S$ -gradings most easily, we construct a particularly convenient Heegaard multi-diagram which yields a filtered chain complex computing $SFH(\Sigma(A \times I, \mathbb{L}))$:

CONSTRUCTION 3.1. *Identifying*

$$A \times I := \{(r, \theta, z) \mid r \in [1, 2], \theta \in (-\pi, \pi], z \in [0, 1]\} \subset \mathbb{R}^2 \times \mathbb{R},$$

let $\gamma_\phi \subset A$ denote the arc $\{(r, \theta) \in \mathbb{R}^2 \mid \theta = \phi, r \in [1, 2]\}$ for fixed $\phi \in (-\pi, \pi]$, $\gamma_{[\phi_1, \phi_2]} \subseteq A$ denote the closed region $\{(r, \theta) \in \mathbb{R}^2 \mid \theta \in [\phi_1, \phi_2], r \in [1, 2]\}$, and $w(\mathbb{L})$ denote the minimal geometric intersection number, $|\mathbb{L} \cap (\gamma_0 \times I)|$, among all elements in the isotopy class of $\mathbb{L}$.

Choose an isotopy class representative of $\mathbb{L}$ satisfying:

- $\mathcal{P}(\mathbb{L})$ is an admissible projection (diagram) of $\mathbb{L}$ to the annulus in the sense of [8, Defn. 2.10], with small neighborhoods of the crossings $C_1, \dots, C_n$ labeled $N(C_1), \dots, N(C_n)$,
- the intersection of $\mathcal{P}(\mathbb{L})$ with $\gamma_{[-\frac{\pi}{4}, \frac{\pi}{4}]}$ in A is “standard” (see Figure 2),
- the complete list, $\vec{p}$, of intersection points of $\mathbb{L}$ with the half-level annulus $A \times \{\frac{1}{2}\}$ consists of
 - the $w(\mathbb{L})$ points along $\gamma_0 \times \{\frac{1}{2}\}$,
 - the $w(\mathbb{L})$ points along $\gamma_{\frac{\pi}{4}} \times \{\frac{1}{2}\}$, and
 - the four points $(\mathcal{P}(\mathbb{L}) \cap \partial(N(C_i))) \times \{\frac{1}{2}\}$ around each crossing for each $i \in 1, \dots, n$,
- $\mathbb{L}$ intersects $A \times [\frac{1}{2}, 1]$ in (and only in) the regions $\gamma_{[0, \frac{\pi}{4}]} \times [\frac{1}{2}, 1]$ and $N(c_i) \times [\frac{1}{2}, 1]$.

Note that if $\mathbb{L}$ is as above, then every complete resolution of the corresponding projection, $\mathcal{P}(\mathbb{L})$, represents a resolved link that is a union of simple “cups” and “caps” as in [9, Sec. 5.3]. We can therefore use the argument given in [8, Proof of Prop. 2.28] to construct a (not-yet-admissible) sutured Heegaard multi-diagram from which we can build the filtered chain complex computing $SFH(\Sigma(A \times I, \mathbb{L}))$.

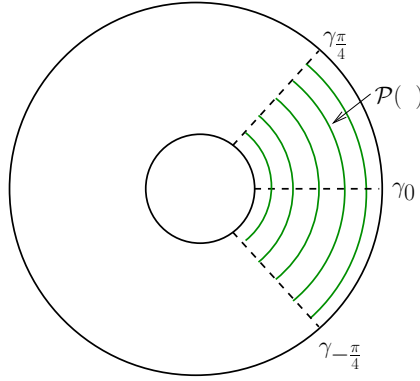


FIGURE 2. A “standard” projection, $\mathcal{P}(\mathbb{L})$, of $\mathbb{L}$ to $\gamma[-\frac{\pi}{4}, \frac{\pi}{4}]$

More precisely, recall that the set $\{0, 1\}^n$ indexes all possible complete resolutions of $\mathcal{P}(\mathbb{L})$ (see Section 2). Let $\vec{p} \subset (A \times \{\frac{1}{2}\})$ denote the set of intersection points (described above) of $\mathbb{L}$ with $A \times \{\frac{1}{2}\}$. Then the sutured Heegaard multi-diagram is given by $(\Sigma, \alpha, \beta_{\{0,1\}^n})$, where $\beta_{\{0,1\}^n}$ represents the list of elements of the set $\{\beta_{\mathcal{I}} \mid \mathcal{I} \in \{0, 1\}^n\}$ and

- $\Sigma := \Sigma(A \times \{\frac{1}{2}\}, \vec{p})$ (with covering projection $\pi : \Sigma(A \times \{\frac{1}{2}\}, \vec{p}) \rightarrow A$),
- $\alpha := \pi^{-1}(\mathcal{P}(\text{cups}))$, and
- $\beta_{\mathcal{I}} := \pi^{-1}(\mathcal{P}(\text{caps}_{\mathcal{I}}))$.

See Figure 3 for an example.

Let $\mathbb{L}_{\mathcal{I}}$ denote the complete resolution of $\mathbb{L}$ associated to the n -tuple $\mathcal{I} \in \{0, 1\}^n$. The construction of the sutured multi-diagram $(\Sigma, \alpha, \beta_{\{0,1\}^n})$ for $\Sigma(A \times I, \mathbb{L})$ given in Construction 3.1 is tailored to have nice properties with respect to $\{S_{\mathcal{I}}\}$, a particular collection of properly imbedded surfaces in the collection of sutured manifolds, $\{\Sigma(A \times I, \mathbb{L}_{\mathcal{I}})\}$. To explain what we mean, we must first recall a couple of definitions and facts.

Recall that if $F \subseteq (Y, \Gamma)$ is a *decomposing surface* [7, Defn. 3.1] in a balanced, sutured manifold satisfying some additional mild hypotheses, then Juhász proves, in [15, Prop. 4.4], that one can construct a balanced sutured Heegaard diagram Σ adapted to F ([15, Defn. 4.3]). The key feature of such a Heegaard diagram is that “most of” F sits as a subsurface of Σ with boundary a graph having certain important properties with respect to the curves $\alpha, \beta \subset \Sigma$. See Figure 4 for an illustration, and [15, Sec. 4] for more details.

In [10, Prop. 4.11], we generalized this to the case where $F \subseteq (Y, \Gamma)$ is a decomposing surface (again, satisfying some additional mild hypotheses) that is geometrically disjoint from a framed link $L \subset (Y, \Gamma)$. Letting $(Y_{\mathcal{I}}, \Gamma_{\mathcal{I}})$ denote the sutured manifold obtained by performing $\mathcal{I}$ -surgery along the link, we then see ([10, Defn. 3.10]) that F induces a compatible collection $\{F_{\mathcal{I}} \subset Y_{\mathcal{I}}\}$ of decomposing surfaces in the various surgeries on L . We can analogously construct ([10, Prop. 4.11]) a sutured multi-diagram adapted to the collection $\{F_{\mathcal{I}} \subset Y_{\mathcal{I}}\}$ in the sense of [10, Defn. 4.6].

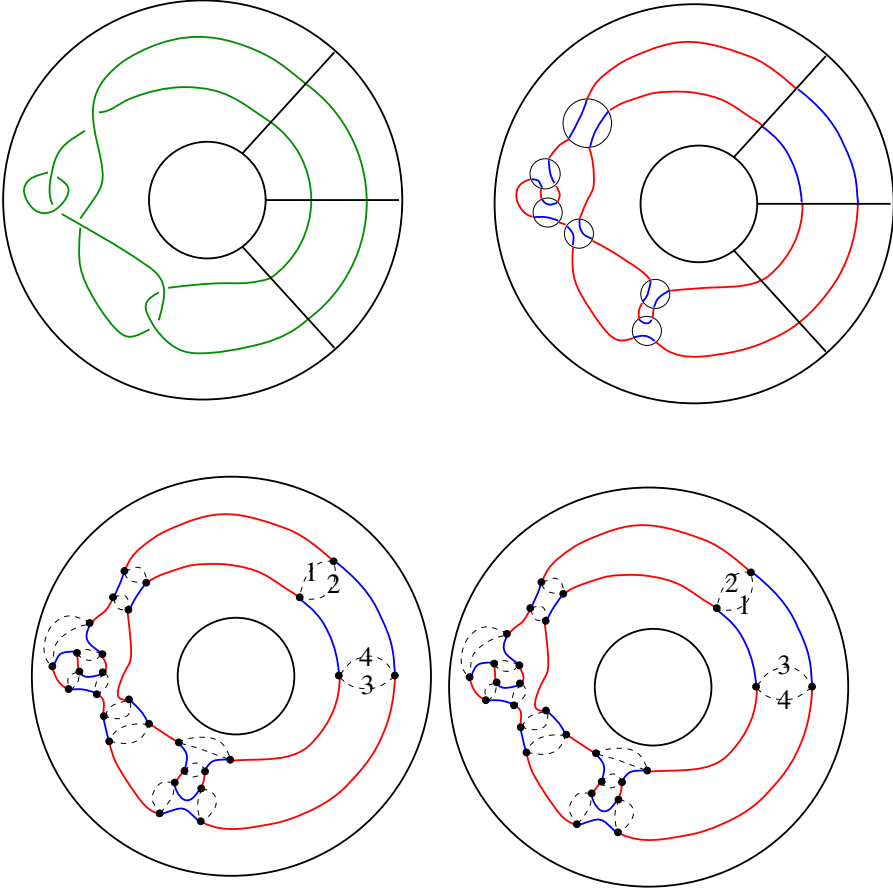


FIGURE 3. The figure at top left is an admissible projection (diagram) of a link $\mathbb{L}$ in the special form described in Construction 3.1. The figure at top right shows a diagram of a particular resolution, $\mathbb{L}_I$, of $\mathbb{L}$, with the regions corresponding to portions of $\mathbb{L}$ in the region $A \times [\frac{1}{2}, 1]$ (the “caps”) marked in dark gray, the region $A \times [0, \frac{1}{2}]$ (the “cups”) marked in light gray, and the neighborhoods of the crossings indicated by circles. The pair of figures in the second row above represent the two sheets of the double-branched cover of the resolution at top right, branched along the points of intersection of the resolved link with $A \times \{\frac{1}{2}\}$ (denoted by the bold black dots). The two-sheeted figure will therefore yield a Heegaard diagram for $\Sigma(A \times I, \mathbb{L}_I)$. The dotted lines are branch cuts, which are identified on the two sheets in the standard way, as indicated by the numbers assigned to the pair of branch cuts on the right. The light gray curves are the α circles, and the dark gray curves are the β circles.

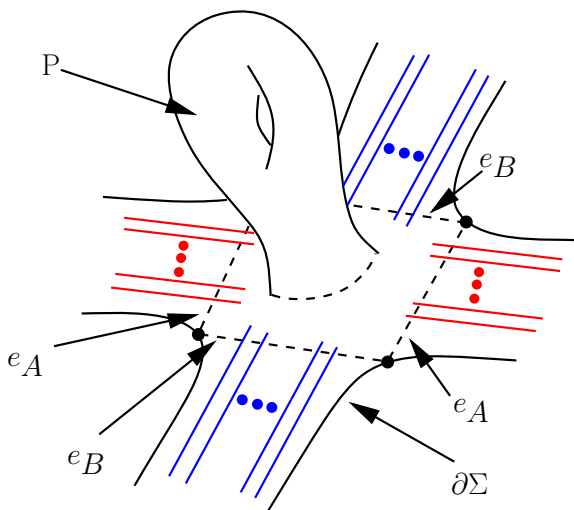


FIGURE 4. An illustration of a Heegaard diagram, Σ , adapted to a decomposing surface, F . The diagram Σ contains a distinguished subsurface, P , whose boundary is a disjoint union of finitely many cyclic graphs with the property that all vertices lie on $\partial\Sigma$ and the edges alternate between edges $\{e_A\}$ of type “A” and edges $\{e_B\}$ of type “B.” The “A” edges have trivial intersection with β (dark gray) and the “B” edges have trivial intersection with α (light gray). The decomposing surface F is then the surface (with corners) given by

$$F = P \cup \left(\{e_A\} \times \left[\frac{1}{2}, 1 \right] \right) \cup \left(\{e_B\} \times \left[0, \frac{1}{2} \right] \right).$$

In our situation, let $D \subset (A \times \{\frac{1}{2}\})$ be the bigon (quasi-polygon), pictured in Figure 5, with boundary graph given by the two labeled edges e_A, e_B . Note that, by the way we chose $\mathbb{L}$ in Construction 3.1, e_A only intersects the portion of $\mathcal{P}(\mathbb{L})$ projecting from $\mathbb{L} \cap (A \times [0, \frac{1}{2}])$ and e_B only intersects the portion of $\mathcal{P}(\mathbb{L})$ projecting from $\mathbb{L} \cap (A \times [\frac{1}{2}, 1])$. We now define the following piecewise smooth imbedded surface:

$$D' := D \cup \left(e_A \times \left[\frac{1}{2}, 1 \right] \right) \cup \left(e_B \times \left[0, \frac{1}{2} \right] \right) \subset A \times I.$$

Letting

- $\vec{q}$ denote the collection of intersection points $\vec{q} = (\mathbb{L} \cap D)$ of the special isotopy class representative $\mathbb{L}$ with $D \subset A \times \{\frac{1}{2}\}$ (which, by construction, occur along $\gamma_0 \times \{\frac{1}{2}\}$),
- and $\tilde{e}_A, \tilde{e}_B$ denote the lifts of e_A, e_B to $\Sigma(D, \vec{q})$,

we see that D' lifts, in the double-branched cover $\Sigma(A \times I, \mathbb{L})$, to a piecewise smooth surface,

$$S' := \Sigma(D, \vec{q}) \cup \left(\tilde{e}_A \times \left[\frac{1}{2}, 1 \right] \right) \cup \left(\tilde{e}_B \times \left[0, \frac{1}{2} \right] \right).$$

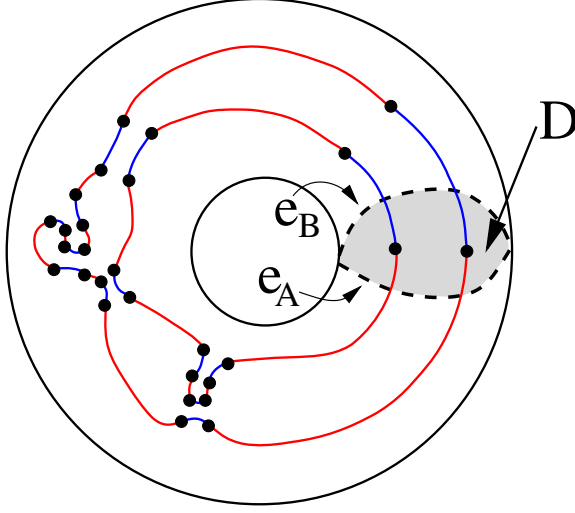


FIGURE 5. An illustration of the bigon $D \subset (A \times \{\frac{1}{2}\})$ that gives rise to the (compatible collection of) decomposing surface(s) in a Heegaard multi-diagram for $\Sigma(A \times I, \mathbb{L})$. As before, the black dots represent intersection points of the (resolved) link with $A \times \{\frac{1}{2}\}$, and the light gray (resp., dark gray) arcs are the projections of the portions of the resolution in the bottom (resp., top) half of $A \times I$.

In addition, the framed link $L_{\mathbb{L}} \subset \Sigma(A \times I, \mathbb{L})$ obtained as the lift of simple arcs associated to each of the crossings of $\mathcal{P}(\mathbb{L})$ (as described in Section 2) is geometrically disjoint from S' . Hence we have a compatible collection $\{S'_{\mathcal{I}} \subset \Sigma(A \times I, \mathbb{L}_{\mathcal{I}})\}$ of surfaces associated to each choice $\mathcal{I} \in \{0, 1\}^n$ of surgery on $L_{\mathbb{L}}$. Furthermore, each $S'_{\mathcal{I}}$ is isotopic to the preimage, in $\Sigma(A \times I, \mathbb{L}_{\mathcal{I}})$, of the meridional disk $\gamma_0 \times I \subset A \times I$. We see immediately:

LEMMA 3.2. *Let $\mathbb{L} \subset A \times I$ be a link. A Heegaard multi-diagram $(\Sigma, \alpha, \beta_{\{0,1\}^n})$ associated to $\Sigma(A \times I, \mathbb{L})$ via Construction 3.1 with the additional data of the sub-surface $P = \Sigma(D, \vec{q}) \subseteq \Sigma$ described above is a surface multi-diagram adapted to the collection $S'_{\mathcal{I}} \subset \Sigma(A \times I, \mathbb{L}_{\mathcal{I}})$.*

PROOF. This is essentially by definition; see [10, Defn. 4.6]. Note that the boundary of $\Sigma(D, \vec{q})$ decomposes as a union of edges $\tilde{e}_A$ and $\tilde{e}_B$. Since (downstairs in D) e_A only intersects the portions of the projection $\mathcal{P}(\mathbb{L})$ coming from $\mathbb{L} \cap (A \times [0, \frac{1}{2}])$ and e_B only intersects the portions of $\mathcal{P}(\mathbb{L})$ coming from $\mathbb{L} \cap (A \times [\frac{1}{2}, 1])$, the lifts, $\tilde{e}_A, \tilde{e}_B$ will satisfy the necessary condition

$$\tilde{e}_A \cap \beta_{\mathcal{I}} = \emptyset, \quad \tilde{e}_B \cap \alpha = \emptyset$$

for all $\mathcal{I} \in \{0, 1\}^n$.

Any isotopies needed to make $(\Sigma, \alpha, \beta_{\{0,1\}^n})$ admissible (see [9, Lem. 3.13]) can be chosen with support away from $\Sigma(D, \vec{q})$. $\square$

Since the Heegaard multi-diagram $(\Sigma, \alpha, \beta_{\{0,1\}^n})$ is adapted to the collection $S'_\mathcal{I}$, it is easy to compute the $\frac{1}{2}\mathbb{Z}$ -Alexander grading associated to any generator of the filtered chain complex for an appropriate trivialization, t . Namely, a version of the formula given in the proof of [12, Thm. 2.20] (see also [22, 15]), adjusted to our situation (our decomposing surface intersects the sutures in 4 points, not 2) tells us that, for some trivialization, t ,

$$\mathbf{A}_S(\mathbf{x}) = \frac{1}{2} (\chi(S) - 2 + 2 \cdot \#\{x_i \in \mathbf{x} \mid x_i \in P\}).$$

Noting that $\chi(S) = \chi(\Sigma(D, \vec{q})) = 2 - |\vec{q}|$ by the Riemann-Hurwitz formula, we obtain:

$$\mathbf{A}_S(\mathbf{x}) = -\frac{1}{2}|\vec{q}| + \#\{x_i \in \mathbf{x} \mid x_i \in P\}.$$

Since $\#\{x_i \in \mathbf{x} \mid x_i \in P\} \in \{0, \dots, |\vec{q}|\}$, we see that the Alexander gradings of generators range from $-\frac{1}{2}|\vec{q}|$ (when there are no intersection points in P) to $\frac{1}{2}|\vec{q}|$ (when there are the maximal number of intersection points in P).

In summary, the $\frac{1}{2}\mathbb{Z}$ grading, $\mathbf{A}_S(\mathbf{x})$, assigned to a generator, $\mathbf{x}$, is the Euler number of a vector field on $\Sigma(A \times I, \mathbb{L}_\mathcal{I})$ naturally defined using the data of $\mathbf{x}$. Furthermore, this Euler number can be computed explicitly by counting the number of intersection points of $\mathbf{x}$ that lie in the region, $P \subset \Sigma$, corresponding to the family of decomposing surfaces, $\{S_\mathcal{I}\}$.

4. Khovanov grading

4.1. Khovanov homology for links in thickened annuli. As before, let $\mathbb{L} \subset A \times I \subset S^3$ be a link in the product sutured manifold $A \times I$, where $A \times I$ has been imbedded in S^3 via the imbedding described before Theorem 2.2, and let $\mathcal{P}(\mathbb{L}) \subset A \subset \mathbb{R}^2$ be a diagram for $\mathbb{L}$ with n crossings. We similarly study a $\mathbb{Z}$ -grading on the Khovanov homology of $\mathbb{L}$, as described in [1] (see also [32], [8]).

If we forget about the imbedding $\mathbb{L} \subset A \times I$ and view $\mathbb{L}$ as a link in S^3 , we can define its ordinary (nonreduced) Khovanov homology in a manner very similar to the definition of the reduced Khovanov homology summarized during the proof of Theorem 2.1. In particular, the Khovanov homology of $\mathbb{L}$ is defined by associating to each complete resolution $\mathbb{L}_\mathcal{I}$ a graded vector space $V^{\otimes c}$, where c is the number of components in $\mathbb{L}_\mathcal{I}$, and V is a two-dimensional graded vector space spanned by two homogeneous elements v_- and v_+ of degrees -1 and $+1$, respectively. In fact, it is understood implicitly that we have chosen a numbering of the components of $\mathbb{L}_\mathcal{I}$ from 1 to c , so that the i th tensor factor of $V^{\otimes c}$ corresponds naturally to the i th component of $\mathbb{L}_\mathcal{I}$.

Let $\mathbb{L}_\mathcal{I}$ be a complete resolution of $\mathbb{L}$ with components $K_1, \dots, K_c$ as above. If we now remember the extra data of the imbedding $\mathbb{L} \subset A \times I \subset S^3$ (correspondingly, $\mathcal{P}(\mathbb{L}) \subset A \subset \mathbb{R}^2$), we can endow the vector space $V^{\otimes c}$ associated to $\mathbb{L}_\mathcal{I}$ with an additional grading, called the k -grading. Namely, we define the k -degree of a basis vector $v_{\epsilon_1} \otimes \dots \otimes v_{\epsilon_c} \in V^{\otimes c}$, for $\epsilon_i \in \{+, -\}$, by $k(v_{\epsilon_1} \otimes \dots \otimes v_{\epsilon_m}) := k_1(v_{\epsilon_1}) + \dots + k_m(v_{\epsilon_m})$, where

$$k_i(v_\pm) := \begin{cases} 0 & \text{if } [\mathcal{P}(K_i)] = 0 \in H_1(A; \mathbb{Z}), \\ \pm 1 & \text{if } [\mathcal{P}(K_i)] \neq 0 \in H_1(A; \mathbb{Z}). \end{cases}$$

It turns out that the differential in Khovanov's chain complex does not preserve the k -grading; however, it is non-increasing in the k -grading, which induces a

filtration on Khovanov’s chain complex. By forming the associated graded chain complex and taking its homology, one obtains an abelian group $Kh^*(\mathbb{L})$. This group is (up to an overall grading shift) the version of Khovanov homology for links in $A \times I$ that was introduced by Asaeda–Przytycki–Sikora [1]. $Kh^*(\mathbb{L})$ is equipped with three gradings: two gradings coming from the ordinary bigrading on Khovanov’s chain complex (in [17], these gradings were denoted by i and j), and the k –grading. See [32, 8] for more details.

The graded Euler characteristic of $Kh^*(\mathbb{L})$ is the Laurent polynomial

$$\chi_{q,t}(Kh^*(\mathbb{L})) := \sum_{i,j,k} (-1)^i q^j t^k \dim_{\mathbb{F}}(Kh^*(\mathbb{L})^{i,j,k}) \in \mathbb{Z}[q^{\pm 1}, t^{\pm 1}],$$

where $Kh^*(\mathbb{L})^{i,j,k}$ denotes the homogeneous component of $Kh^*(\mathbb{L})$ for the degrees (i, j, k) . We will henceforth abbreviate $SJ(\mathbb{L}) := \chi_{q,t}(Kh^*(\mathbb{L}))$. Note that $SJ(\mathbb{L})$ specializes to the Jones polynomial $J(\mathbb{L})$ if one sets t equal to 1. If $\mathbb{L}_{\mathcal{I}}$ is a complete resolution with c components, then $SJ(\mathbb{L}_{\mathcal{I}})$ is given by

$$(1) \quad SJ(\mathbb{L}_{\mathcal{I}}) := \chi_{q,t}(V^{\otimes c}) = (q + q^{-1})^u z^{c-u},$$

where $z := qt + (qt)^{-1}$, and u is the number of components of $\mathbb{L}_{\mathcal{I}}$ that are null-homologous in $A \times I$. In particular, since $SJ(\mathbb{L})$ is a linear combination of the $SJ(\mathbb{L}_{\mathcal{I}})$ with coefficients in $\mathbb{Z}[q^{\pm 1}]^{\times}$, this shows that $SJ(\mathbb{L})$ is actually contained in the subring $\mathbb{Z}[q^{\pm 1}][z] \subset \mathbb{Z}[q^{\pm 1}, t^{\pm 1}]$. In Subsection 4.6, we will discuss how this subring is related to the Kauffman bracket skein module of $A \times I$.

4.2. Enhanced resolutions. Let $\mathbb{L}$ be a link in $A \times I$ with diagram $\mathcal{P}(\mathbb{L})$, as before. An enhanced resolution of $\mathcal{P}(\mathbb{L})$ is an oriented 1–manifold, $\mathbb{S} \subset A$, whose underlying unoriented 1–manifold coincides with the projection, $\mathcal{P}(\mathbb{L}_{\mathcal{I}})$, of a complete resolution of $\mathbb{L}$. (In [36], the term “enhanced Kauffman state” was used with a similar meaning).

To obtain a more geometric interpretation of the k –grading, we will now reinterpret the generators of $Kh^*(\mathbb{L})$ in terms of enhanced resolutions. Namely, if $V^{\otimes c}$ denotes the vector space associated to a complete resolution $\mathbb{L}_{\mathcal{I}}$, then we will identify the basis vector $v_{\epsilon_1} \otimes \dots \otimes v_{\epsilon_c} \in V^{\otimes c}$ ($\epsilon_i \in \{+, -\}$) with the enhanced resolution $\mathbb{S} \subset A$ which is obtained by orienting the i th component of $\mathcal{P}(\mathbb{L}_{\mathcal{I}}) \subset A \subset \mathbb{R}^2$ clockwise if $\epsilon_i = -$, and counterclockwise if $\epsilon_i = +$.

Using this correspondence between basis vectors and enhanced resolutions, we can now express the k –grading as an algebraic intersection number

$$k(v_{\epsilon_1} \otimes \dots \otimes v_{\epsilon_c}) = [\gamma_0] \cdot [\mathbb{S}],$$

where $\mathbb{S}$ is the enhanced resolution corresponding to the basis vector $v_{\epsilon_1} \otimes \dots \otimes v_{\epsilon_c}$, and $\gamma_0 \subset A$ denotes the arc introduced in Construction 3.1, oriented outward (in the direction of increasing r). Moreover, Formula (1) can be rewritten as

$$(2) \quad SJ(\mathbb{L}_{\mathcal{I}}) = \sum_{\mathbb{S} \in E(\mathbb{L}_{\mathcal{I}})} q^{j(\mathbb{S})} t^{k(\mathbb{S})},$$

where $k(\mathbb{S}) := [\gamma_0] \cdot [\mathbb{S}]$, and $j(\mathbb{S})$ denotes the difference between the number of counterclockwise and clockwise circles in $\mathbb{S}$, and $E(\mathbb{L}_{\mathcal{I}})$ denotes the set of all enhanced resolutions of $\mathbb{L}$ whose underlying unoriented 1–manifold is equal to $\mathcal{P}(\mathbb{L}_{\mathcal{I}})$. By summing over all complete resolutions $\mathbb{L}_{\mathcal{I}}$, one then obtains a formula expressing $SJ(\mathbb{L})$ as a sum over all enhanced resolutions of $\mathbb{L}$.

4.3. Reshetikhin-Turaev invariant. As mentioned in the introduction, the integer k -grading defined above has a natural representation-theoretic interpretation in terms of weight space decompositions for the Reshetikhin-Turaev invariant.

Again, let $\mathbb{L} \subset A \times I \subset S^3$ be a link and $\mathcal{P}(\mathbb{L})$ a diagram of $\mathbb{L}$. In the following, we will assume that the intersection of $\mathcal{P}(\mathbb{L})$ with $\gamma_{[-\frac{\pi}{4}, \frac{\pi}{4}]} \subset A$ is “standard”, in the sense of Construction 3.1.

Let $\mathbb{T}'$ be the tangle in $D^2 \times I$ obtained by removing the neighborhood of a meridional disk in the solid torus $A \times I$:

$$\mathbb{T}' := \mathbb{L} \setminus \left(\gamma_{(-\frac{\pi}{4}, \frac{\pi}{4})} \times [0, 1] \right) \subset (A \times I) \setminus \left(\gamma_{(-\frac{\pi}{4}, \frac{\pi}{4})} \times [0, 1] \right).$$

For convenience, we now change the angular (θ) coordinate on $(A \times I) \setminus \left(\gamma_{(-\frac{\pi}{4}, \frac{\pi}{4})} \times [0, 1] \right)$ so that it is continuous on this region, by letting

$$\vartheta = \begin{cases} \pi - \theta & \text{if } \theta \in \left[\frac{\pi}{4}, \pi \right], \text{ and} \\ -\pi - \theta & \text{if } \theta \in \left(-\pi, -\frac{\pi}{4} \right]. \end{cases}$$

This allows us to easily identify $\mathbb{T}'$ with a tangle $\mathbb{T} \subset [1, 2] \times \left[-\frac{3\pi}{4}, \frac{3\pi}{4} \right] \times [0, 1] \subset \mathbb{R}^3$ by sending a point

$$(r, \theta, z) \in (A \times I) \setminus \left(\gamma_{(-\frac{\pi}{4}, \frac{\pi}{4})} \times [0, 1] \right)$$

to the point $(r, \vartheta, z) \in [1, 2] \times \left[-\frac{3\pi}{4}, \frac{3\pi}{4} \right] \times [0, 1]$. If the “standard” intersection of $\mathbb{L}$ with the region $\gamma_{(-\frac{\pi}{4}, \frac{\pi}{4})} \times [0, 1]$ consists of m arcs, then $\mathbb{T}$ has m top endpoints (with ϑ -coordinate equal to $\frac{3\pi}{4}$) and m bottom endpoints (with ϑ -coordinate equal to $-\frac{3\pi}{4}$).

To $\mathbb{T}$, the Reshetikhin-Turaev construction [31] for the quantum group $U_q(\mathfrak{sl}_2)$ associates a linear map

$$J(\mathbb{T}) : V_1^{\otimes m} \longrightarrow V_1^{\otimes m}$$

which intertwines the quantum group action. Here, V_1 is the two-dimensional fundamental representation of $U_q(\mathfrak{sl}_2)$, with underlying vector space $V_1 := \mathbb{C}(q)v_1 \oplus \mathbb{C}(q)v_{-1}$. The generators E, F, K of $U_q(\mathfrak{sl}_2)$ act by

$$Kv_1 = qv_1, \quad Kv_{-1} = q^{-1}v_{-1}, \quad Ev_1 = Fv_{-1} = 0, \quad Ev_{-1} = v_1, \quad Fv_1 = v_{-1}.$$

Note that when $m = 0$ (i.e., if $\mathbb{T}$ is a link), $J(\mathbb{T})$ is given by scalar multiplication by the Jones polynomial $J(\mathbb{L})$. In general, $J(\mathbb{T})$ can be described by a state sum formula, as we will now explain. See [16] for a reference.

As before, an enhanced resolution of $\mathbb{T}$ is an oriented 1-manifold $\mathbb{S} \subset [1, 2] \times \left[-\frac{3\pi}{4}, \frac{3\pi}{4} \right]$ whose underlying unoriented 1-manifold is the projection of a resolution $\mathbb{T}_{\mathcal{L}}$. Given an enhanced resolution $\mathbb{S}$ of $\mathbb{T}$, we can associate two sequences of up and down arrows: the bottom sequence $b(\mathbb{S}) \in \{\uparrow, \downarrow\}^m$, obtained by reading off the local orientations of the arc components of $\mathbb{S}$ near the m bottom endpoints; and the top sequence $t(\mathbb{S}) \in \{\uparrow, \downarrow\}^m$, obtained by reading off the local orientations of the arc components of $\mathbb{S}$ near the m top endpoints.

Each sequence $\mathbf{a} \in \{\uparrow, \downarrow\}^m$ determines a standard basis vector in $V_1^{\otimes m}$, namely the vector obtained by replacing each $\uparrow$ by $v_1 \in V_1$, and each $\downarrow$ by $v_{-1} \in V_1$. For example, the sequence $\mathbf{a} = (\uparrow\uparrow\downarrow\uparrow\downarrow) \in \{\uparrow, \downarrow\}^5$ corresponds to the standard basis vector

$$v_{\mathbf{a}} := v_1 \otimes v_1 \otimes v_{-1} \otimes v_1 \otimes v_{-1} \in V_1^{\otimes 5}.$$

The matrix entry $J(\mathbb{T}_{\mathcal{I}})_{\mathbf{a}, \mathbf{b}} := v_{\mathbf{a}}^*(J(\mathbb{T})(v_{\mathbf{b}}))$ of the linear map $J(\mathbb{T}_{\mathcal{I}}): V_1^{\otimes m} \rightarrow V_1^{\otimes m}$ can now be defined explicitly by

$$(3) \quad J(\mathbb{T}_{\mathcal{I}})_{\mathbf{a}, \mathbf{b}} := \sum_{\substack{\mathbb{S} \in E(\mathbb{T}_{\mathcal{I}}) \\ b(\mathbb{S}) = \mathbf{b}, t(\mathbb{S}) = \mathbf{a}}} q^{j(\mathbb{S})},$$

where $E(\mathbb{T}_{\mathcal{I}})$ is the set of all enhanced resolutions of $\mathbb{T}$ whose underlying unoriented 1-manifold is equal to $\mathcal{P}(\mathbb{T}_{\mathcal{I}})$, and $j(\mathbb{S})$ is defined as follows. Let $G^{\mathbb{S}}: \mathbb{S} \rightarrow S^1$ be the Gauss map which sends a point $p \in \mathbb{S} \subset [1, 2] \times [-\frac{3\pi}{4}, \frac{3\pi}{4}]$ to the positive unit tangent vector of $\mathbb{S}$ at p . If we assume that $\mathbb{S}$ is vertical near its endpoints, then the image of the fundamental class of $\mathbb{S}$ defines a relative homology class $G_*^{\mathbb{S}}[\mathbb{S}] \in H_1(S^1, \{\pm i\})$, where $i = \sqrt{-1} \in S^1$. Now $j(\mathbb{S})$ is the algebraic intersection number $j(\mathbb{S}) := [\{1\}] \cdot G_*^{\mathbb{S}}[\mathbb{S}]$.

4.4. Weight spaces. As a vector space, $V_1^{\otimes m}$ decomposes into weight spaces, i.e. into eigenspaces for the action of $K \in U_q(\mathfrak{sl}_2)$:

$$V_1^{\otimes m} = \bigoplus_{n=0}^m V_1^{\otimes m}[m - 2n]$$

Here, $V_1^{\otimes m}[\lambda]$ is the eigenspace for the eigenvalue q^{λ} for the K -action, and the generators E and F act by maps $E_{\lambda}: V_1^{\otimes m}[\lambda] \rightarrow V_1^{\otimes m}[\lambda + 2]$ and $F_{\lambda}: V_1^{\otimes m}[\lambda] \rightarrow V_1^{\otimes m}[\lambda - 2]$, respectively.

Since K acts on tensor products by $K(v \otimes w) := (Kv) \otimes (Kw)$, and since v_1 and v_{-1} are eigenvectors for the K -action with eigenvalues $q^{\pm 1}$, each standard basis vector $v_{\mathbf{a}}$ for $\mathbf{a} \in \{\uparrow, \downarrow\}^m$, is an eigenvector for the K -action. Explicitly, letting $\uparrow(\mathbf{a})$ (resp., $\downarrow(\mathbf{a})$) denote the number of up arrows in $\mathbf{a}$ (resp., down arrows in $\mathbf{a}$), we have:

$$Kv_{\mathbf{a}} = q^{k(\mathbf{a})}v_{\mathbf{a}}.$$

where

$$k(\mathbf{a}) = \uparrow(\mathbf{a}) - \downarrow(\mathbf{a}).$$

Thus, the weight space $V_1^{\otimes m}[\lambda]$ is the span of all $v_{\mathbf{a}}$ with $k(\mathbf{a}) = \lambda$. For the following discussion, it will be useful to observe that $k(\mathbf{a})$ can be identified with the algebraic intersection number $k(\mathbf{a}) = [\ell] \cdot [\mathbf{a}]$, where here the arrows in $\mathbf{a}$ are viewed as upward and downward oriented vertical lines, and ℓ is a horizontal line, oriented to the right, and intersecting each of the arrows in $\mathbf{a}$ in a single point.

4.5. $SJ(\mathbb{L})$ and quantum trace. If $\mathbb{S}$ is an enhanced resolution of $\mathbb{T}$ with $t(\mathbb{S}) = \mathbf{a}$ and $b(\mathbb{S}) = \mathbf{b}$ for $\mathbf{a}, \mathbf{b} \in \{\uparrow, \downarrow\}^m$, then $k(\mathbf{a}) = [\ell] \cdot [\mathbf{a}] = [\gamma_{\phi}] \cdot [\mathbb{S}] = [\ell] \cdot [\mathbf{b}] = k(\mathbf{b})$, where γ_{ϕ} is the rightward-oriented horizontal line $\gamma_{\phi} := [1, 2] \times \{\phi\}$ for arbitrary $\phi \in [-\frac{3\pi}{4}, \frac{3\pi}{4}]$. Hence it follows that $J(\mathbb{T}): V_1^{\otimes m} \rightarrow V_1^{\otimes m}$ preserves the weight space decomposition. (Alternatively, this follows from the fact that $J(\mathbb{T})$ intertwines the quantum group action).

Recalling that $[\gamma_{\phi}] \cdot [\mathbb{S}] = k(\mathbb{S})$ (where $k(\mathbb{S})$ is the k -degree defined immediately after equation (2)), we see that the restriction of $J(\mathbb{T})$ to the weight space $V_1^{\otimes m}[k]$ is given by summing over enhanced resolutions with $k(\mathbb{S}) = k$.

Now note that every enhanced resolution of $\mathbb{T}$ which satisfies $t(\mathbb{S}) = b(\mathbb{S})$ can be lifted to an enhanced resolution of $\mathbb{L}$. Hence we have a bijection

$$E(\mathbb{L}) \xrightarrow{1:1} \{\mathbb{S} \in E(\mathbb{T}) : t(\mathbb{S}) = b(\mathbb{S})\}.$$

where $E(\mathbb{L})$ (resp., $E(\mathbb{T})$) denotes the set of all enhanced resolutions of $\mathbb{L}$ (resp., $\mathbb{T}$).

It is fairly straightforward to check that $j(\mathbb{S}') = j(\mathbb{S}) + k$ whenever $\mathbb{S}'$ is an enhanced resolution of $\mathbb{L}$ corresponding via the above bijection to an enhanced resolution $\mathbb{S}$ of $\mathbb{T}$ with $t(\mathbb{S}) = b(\mathbb{S})$ and $k(\mathbb{S}) = k$. Hence

$$SJ(\mathbb{L}_{\mathcal{I}}) = \sum_{\mathbb{S}' \in E(\mathbb{L}_{\mathcal{I}})} q^{j(\mathbb{S}')} t^{k(\mathbb{S}')} = \sum_k (qt)^k \sum_{\substack{\mathbb{S} \in E(\mathbb{T}_{\mathcal{I}}) \\ t(\mathbb{S})=b(\mathbb{S}), k(\mathbb{S})=k}} q^{j(\mathbb{S})},$$

and thus we obtain

$$SJ(\mathbb{L}) = \sum_k (qt)^k \sum_{\substack{\mathbf{a} \in \{\uparrow, \downarrow\}^m \\ k(\mathbf{a})=k}} J(\mathbb{T}_{\mathcal{I}})_{\mathbf{a}, \mathbf{a}} = \sum_k (qt)^k \operatorname{tr}(J(\mathbb{T})|_{V_1^{\otimes m}[k]}).$$

where $\operatorname{tr}(-)$ stands for the ordinary trace for $\mathbb{C}(q)$ -linear endomorphisms. Setting t equal to 1, we recover the well-known fact that the Jones polynomial of $\mathbb{L}$ can be expressed as the quantum trace $\operatorname{tr}_q(J(\mathbb{T}))$, defined via the action of the generator K on $V_1^{\otimes m}$ by $\operatorname{tr}_q(J(\mathbb{T})) := \operatorname{tr}(K \circ J(\mathbb{T}))$:

$$J(\mathbb{L}) = \sum_k q^k \operatorname{tr}(J(\mathbb{T})|_{V_1^{\otimes m}[k]}) = \operatorname{tr}_q(J(\mathbb{T})).$$

4.6. Skein module interpretation of $SJ(\mathbb{L})$. Let a be a formal variable satisfying $q = -a^{-2}$, and let $\mathbb{L}_f$ denote the framed link $\mathbb{L}_f \subset A \times I$ obtained by equipping the (oriented) link $\mathbb{L} \subset A \times I$ with the framing induced by a compact oriented surface $F \subset A \times [0, 1]$ with $\mathbb{L} \subset \partial F$ and $\partial F \setminus \mathbb{L} \subset (\partial A) \times [0, 1]$.

Recall that the Kauffman bracket skein module of $A \times I$ is defined as the quotient

$$K_a(A \times I) := M/R,$$

where M is the free $\mathbb{Z}[a^{\pm 1}]$ -module generated by all isotopy classes of framed links in $A \times I$, and R is the submodule of M generated by all Kauffman bracket skein relations (cf. [30], [34]). Now it follows essentially from the definitions that $SJ(\mathbb{L})$ is equal to the element

$$\varphi^{-1}([\mathbb{L}_f]) \in \mathbb{Z}[q^{\pm 1}][z] \subset \mathbb{Z}[a^{\pm 1}][z]$$

where φ is the isomorphism

$$\varphi : \mathbb{Z}[a^{\pm 1}][z] \xrightarrow{\cong} K_a(A \times [0, 1])$$

given by sending z^n to a collection of n disjoint homologically nontrivial imbedded circles in $A \times \{1/2\}$, equipped with the framing induced by the surface $A \times \{1/2\}$.

5. The Relationship between the Khovanov and Sutured Floer Gradings

We have seen that if $\mathbb{L} \subset A \times I$ is a link in the thickened annulus and $\mathcal{P}(\mathbb{L}) \subset A$ is a diagram for $\mathbb{L}$ that has been placed in a “standard” form via Construction 3.1, we can construct from the data of $\mathcal{P}(\mathbb{L})$ either:

- (1) a filtered sutured Floer chain complex for $\Sigma(A \times I, \mathbb{L})$, as described in Sections 2 and 3, or
- (2) a Khovanov chain complex for $\overline{\mathbb{L}}$, the mirror of $\mathbb{L}$, as described in Section 4.

Furthermore, there is a bijection between the generators of the two chain complexes ([26, Prop. 6.2], [8, Prop. 2.28]). Each generator can therefore be assigned either:

- (1) a geometric $\frac{1}{2}\mathbb{Z}$ sutured Floer grading, called the $\mathbf{A}_S$ grading, or
- (2) a representation-theoretic $\mathbb{Z}$ Khovanov grading, called the k -grading.

We now show that these two gradings are closely related. Namely:

THEOREM 5.1. *Let $\mathbb{L} \subset A \times I$ be a link in the product sutured manifold $A \times I$, $\overline{\mathbb{L}}$ its mirror, and $\mathcal{P}(\mathbb{L})$ (resp., $\mathcal{P}(\overline{\mathbb{L}})$) a diagram for $\mathbb{L}$ (resp., $\overline{\mathbb{L}}$) produced by Construction 3.1. Then if $\mathbf{x}$ is a generator of the associated (filtered) chain complex, we have:*

$$k(\mathbf{x}) = -2\mathbf{A}_S(\mathbf{x}).$$

PROOF. We begin by noting that the generators of both the Khovanov and sutured Floer complex are in one-to-one correspondence with enhanced resolutions. On the Khovanov side, this correspondence is explained in Section 4.

On the sutured Floer side, recall that we have one generator for each d -tuple of intersection points between α and $\beta_{\mathcal{I}}$, for each $\mathcal{I} \in \{0, 1\}^\ell$. Furthermore, each connected component of the diagram, $\mathcal{P}(\mathbb{L}_{\mathcal{I}})$, of a resolution gives rise to a cyclic alternating chain of α and β circles in the Heegaard diagram for $\Sigma(A \times I, \mathbb{L}_{\mathcal{I}})$, each intersecting the next in a single point. Hence, once a single intersection point in a chain is chosen, the other intersection points on that chain are completely determined by the requirement that a d -tuple of intersection points yielding a Heegaard Floer generator must occupy each α circle (resp., each β circle) exactly once. There are then two choices associated to each such cyclic chain, corresponding to the two choices of orientation for that connected component in the relevant enhanced resolution.

Now let $\mathbb{S}$ be an enhanced resolution for $\mathcal{P}(\mathbb{L})$ (resp., $\mathcal{P}(\overline{\mathbb{L}})$) and $\mathbf{a} \in \{\uparrow, \downarrow\}^m$ the associated sequence of $\uparrow$'s and $\downarrow$'s seen along γ_0 , oriented outward. Section 4 tells us that

$$k(\mathbb{S}) = \uparrow(\mathbf{a}) - \downarrow(\mathbf{a}).$$

Now recall that $w(\mathbb{L}) = |\vec{q}|$ denotes the number of arcs of $\mathcal{P}(\mathbb{L})$ in the standard region $\gamma_{[-\frac{\pi}{4}, \frac{\pi}{4}]} \subset A$. On the sutured Floer side, we have $|\vec{q}|$ distinguished intersection points in the special subsurface, P , lying along the preimage of γ_0 in Σ . Furthermore, the correspondence between enhanced resolutions and generators of the sutured Floer chain complex guarantees that an intersection point is *occupied* (resp., *unoccupied*) iff the corresponding arrow for the enhanced resolution at that position is $\downarrow$ (resp., $\uparrow$). Hence, the end of Section 3 tells us that

$$\begin{aligned} \mathbf{A}_S(\mathbb{S}) &= -\frac{1}{2}|\vec{q}| + \downarrow(\mathbf{a}) \\ &= \frac{1}{2}(-\uparrow(\mathbf{a}) + \downarrow(\mathbf{a})) \\ &= -\frac{1}{2}k(\mathbb{S}), \end{aligned}$$

since $|\vec{q}| = \uparrow(\mathbf{a}) + \downarrow(\mathbf{a})$. We then see that the unique generator $\mathbf{x}$ associated to $\mathbb{S}$ satisfies

$$k(\mathbf{x}) = -2\mathbf{A}_S(\mathbf{x}),$$

as desired. □

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Hyperbolic Knots in Irreducible Heegaard Surfaces

Robert Myers

Dedicated to William Jaco in honor of his 70th birthday.

ABSTRACT. This paper gives a simple combinatorial proof that every irreducible Heegaard surface S of genus at least two in a closed orientable 3-manifold M contains a hyperbolic knot J ; moreover J can be chosen so that $S - J$ is incompressible in $M - J$.

1. Introduction

Let M be a closed, connected, orientable 3-manifold. A classical theorem of Bing [1] states that M is homeomorphic to S^3 if and only if every knot in M lies in a 3-ball in M . There have been many subsequent alternative proofs of Bing's theorem. Many of these proofs proceed by constructing a knot K in M which is irreducible in the sense that $M - K$ is an irreducible 3-manifold; the existence of such a knot clearly implies Bing's theorem. In particular Rieck [11] has shown that every Heegaard surface in M contains an irreducible knot.

In [10] the author showed, using a construction inspired by that of Bing, that M always contains a hyperbolic knot. There have been several much stronger versions of this result. In particular Kobayashi and Nishi [8] constructed for each $n \geq 2$ a hyperbolic knot K_n such that $M - K_n$ contains no closed incompressible surfaces of genus g with $2 \leq g \leq n$. Recently Campisi and Rathbun [2] have constructed for each $m \geq 2$ a knot K'_m whose exterior has a Heegaard splitting of Hempel distance [5] $d \geq m$. This generalizes results of Minsky, Moriah, and Schleimer [9] for knots in S^3 . It also, by a result of Hartshorn [3], implies the Kobayashi-Nishi result. A remarkable feature of the Campisi-Rathbun construction is that given any Heegaard surface S (of genus at least two) each K'_m can be chosen to lie on the Heegaard surface S' obtained by adding one trivial handle to S .

The constructions in [10] and [8] use certain tangles in 3-balls which have special properties whose verification is rather tedious. The construction of K'_m in [2] uses the theory of measured laminations and pseudo-Anosov homeomorphisms.

One of the key lemmas in Campisi and Rathbun's work [2, Lemma 5.2] states that every Heegaard surface S in M contains a knot γ with $S - \gamma$ incompressible in $M - \gamma$, thereby strengthening Rieck's result. They show this by first proving, using

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a combinatorial argument on Whitehead graphs, that every irreducible Heegaard splitting (S, V, W) of M having genus at least two has a pants decomposition $(\mathcal{P}, \mathcal{Q})$ such that every seam of $\mathcal{Q}$ is traversed by a curve of $\mathcal{P}$. They then construct a train track τ which contains $\mathcal{P}$ and traverses all the seams of $\mathcal{P}$; they complete the irreducible case by noting that τ carries knots with all weights arbitrarily high and that for all weights sufficiently high the resulting knot γ traverses all the seams of $\mathcal{P}$ and all the seams of $\mathcal{Q}$ and is thus disk busting in both handlebodies. Genus one splittings are an easy separate case. They then pass to the general case by composing the knots for the irreducible summands.

This paper proves the following generalization of the irreducible case of their lemma.

THEOREM 1.1. *Let S be an irreducible Heegaard surface of genus at least two in a closed, orientable 3-manifold M . Then S contains a hyperbolic knot J such that $S - J$ is incompressible in $M - J$.*

The paper is organized as follows. Section 2 contains definitions and background information. In particular it proves that if a knot J in S is disk and annulus busting in V and disk busting in W , then the exterior of J in M is atoroidal (every incompressible torus is ∂ -parallel). Section 3 gives the construction of J and the proof of Theorem 1.1. It follows the general pattern of Campisi and Rathbun's argument with two exceptions. Instead of appealing to the theory of measured laminations for the existence of sufficiently high weight knots it gives specific examples, and it applies a criterion of Kobayashi [7] which ensures that J is annulus busting in V and in addition that the exterior of J is also anannular (every incompressible annulus is ∂ -parallel) and hence that J is hyperbolic.

2. Preliminaries

For general 3-manifold terminology see [4] or [6]. For Heegaard splittings see [12]. In general, notation will follow that of Campisi and Rathbun [2]. For a properly embedded submanifold F of a manifold M the *exterior* $X_M(F)$ is the closure of the complement of a regular neighborhood of F in M . When F is 2-sided and has codimension one in M one calls $X_M(F)$ the manifold obtained by *splitting M along F* and denotes it by $M|F$. (This notation will sometimes be abused by ignoring the regular neighborhood and regarding M as being obtained by identifying the two copies of F in $M|F$.) A properly embedded submanifold G of M is in *minimal general position with respect to F* if it is in general position with respect to F and the number of components of $G \cap F$ is minimal among all such submanifolds isotopic to G .

All closed surfaces S and handlebodies V and W are assumed to have genus at least two.

A *pants decomposition* $\mathcal{P}$ of a closed, orientable surface S is a maximal set of disjoint, non-parallel, essential simple closed curves in S . Each component P of $S|\mathcal{P}$ is a *pair of pants*. A *seam* σ of P is a properly embedded arc in P which connects two distinct components of ∂P . A simple closed curve α in S *traverses* σ if α is in minimal general position with respect to $\mathcal{P}$ and some component of $\alpha \cap P$ is isotopic in P to σ . If α traverses every seam of every pair of pants in $S|\mathcal{P}$ then α *traverses all the seams of $\mathcal{P}$* .

A pants decomposition $\mathcal{P}$ is *good* if for each component p of $\mathcal{P}$ no component of $S|p$ is a punctured torus. If $\mathcal{P}$ is good then for each component p of $\mathcal{P}$ the surface

L_p obtained by splitting S along $\mathcal{P} - p$ is a sphere with four punctures. We will henceforth assume that all pants decompositions under consideration are good.

A *pants decomposition* $\mathcal{P}$ of a handlebody V is a pants decomposition of $S = \partial V$ such that each curve in $\mathcal{P}$ bounds a disk in V ; these disks can and will be assumed disjoint. Each component of V split along this collection of disks is a *solid pair of pants*. The pants decomposition in Figure 1 is the *standard* pants decomposition of V .

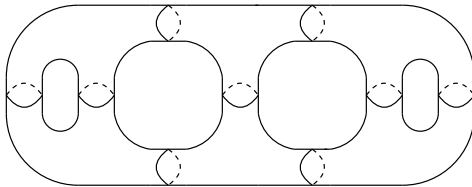


FIGURE 1

A *Heegaard splitting* of a closed 3-manifold M is a triple (S, V, W) such that $M = V \cup_S W$, where S is a closed, orientable surface and V and W are handlebodies with $\partial V = \partial W = S$. It is *reducible* if there is a 2-sphere Σ in M such that $\Sigma \cap S$ is an essential simple closed curve in S ; it is *irreducible* if it is not reducible. A *pants decomposition of a Heegaard splitting* (S, V, W) is a pair $(\mathcal{P}, \mathcal{Q})$, where $\mathcal{P}$ and $\mathcal{Q}$ are pants decompositions of V and W , respectively. It is of *full type* if for every seam of $\mathcal{Q}$ there is a curve of $\mathcal{P}$ which traverses it.

LEMMA 2.1 (Campisi-Rathbun). *If (S, V, W) is irreducible, then it has a pants decomposition $(\mathcal{P}, \mathcal{Q})$ which is of full type.*

PROOF. This is Lemma 3.6 of [2]. Note that their proof is constructive and that their $\mathcal{P}$ is standard. $\square$

A simple closed curve K in S is *good* (with respect to the good pants decomposition $\mathcal{P}$) if it is in minimal general position with respect to $\mathcal{P}$ and traverses every seam of $\mathcal{P}$; it is *very good* if for every component p of $\mathcal{P}$ every pair of distinct components of ∂L_p is joined by a subarc of K .

LEMMA 2.2 (Kobayashi). *Let $\mathcal{P}$ be a good pants decomposition of the handlebody V and let K be a simple closed curve in $S = \partial V$.*

- (1) *If K is good, then $S - K$ is incompressible in V .*
- (2) *If K is very good, then every properly embedded incompressible annulus A in $V - K$ is parallel in V to an annulus A^* in ∂V .*
- (3) *If K is very good, then V is not homeomorphic to an I -bundle for which $X_S(K)$ is the associated ∂I -bundle.*

PROOF. See Lemmas 6.1, 6.3, and 6.4 of [7]. $\square$

A curve K satisfying the first conclusion of the lemma above is said to be *disk busting* in V . A curve K satisfying the second conclusion is said to be *annulus busting* in V .

Kobayashi shows that if $(\mathcal{P}, \mathcal{Q})$ is a pants decomposition of the Heegaard splitting (S, V, W) of M and K is a knot in S which is very good with respect to both $\mathcal{P}$ and $\mathcal{Q}$, then $X_M(K)$ is irreducible and atoroidal [7, Theorem 3]. He then

shows that $X_M(K)$ is not Seifert fibered by noting that if it were, then the incompressibility of $S - K$ in $M - K$ would imply that $X_S(K)$ is incompressible in $X_M(K)$ and so by [6, Theorem VI.34] would be a horizontal surface in $X_M(K)$ and hence V would be an I -bundle as in Lemma 2.2. Since an irreducible, atoroidal 3-manifold with incompressible torus boundary which contains an incompressible non- ∂ -parallel annulus is Seifert fibered one has that $X_M(K)$ is anannular. He then quotes Thurston's hyperbolization theorem [13] to show that K is hyperbolic [7, Corollary 4] and gives an example [7, Example 5] of such a knot in the genus two Heegaard surface for S^3 .

Our approach is similar to that of Kobayashi in that our knot J is very good with respect to $\mathcal{P}$. It differs by not requiring it to be very good with respect to $\mathcal{Q}$. We merely require it to be disk busting in W , a property provided by the Campisi-Rathbun $(\mathcal{P}, \mathcal{Q})$. We then apply Lemma 2.3 below to conclude that $X_M(J)$ is atoroidal and then proceed as described above to conclude that J is hyperbolic.

LEMMA 2.3. *Suppose J is a knot in S which is disk busting and annulus busting in V and is disk busting in W . Then $X_M(J)$ is atoroidal.*

PROOF. Let T be an incompressible torus in $M - J$. Put T in minimal general position with respect to $S - J$. Since J is disk busting in V and W we have that $S - J$ is incompressible in $V - J$ and $W - J$. Hence no component of $T \cap V$ or $T \cap W$ is a disk. It follows that $T \cap V$ is the union $\mathcal{A}$ of annuli $A_1, \dots, A_n$ and $T \cap W$ is the union $\mathcal{A}'$ of annuli $A'_1, \dots, A'_n$. Each A_i is parallel in V to an annulus A_i^* in S . If for some i one has $J \cap A_i^* = \emptyset$ then there is an isotopy which removes at least ∂A_i from $T \cap S$. Thus $J \subseteq A_i^*$ and is a core of A_i^* . It follows that the A_i are nested, i.e. there is a solid torus Z in V such that, after possibly renumbering, $Z \cap S = A_1^*$, $(\partial Z) - S = A_1$, for each $i > 1$ one has A_i properly embedded in Z with $\partial A_i \subseteq \text{Int } A_1^*$, and A_i separates $A_1, \dots, A_{i-1}$ from $A_{i+1}, \dots, A_n$ in Z when $1 < i < n$. Note that $A_1^* \supseteq A_2^* \supseteq \dots \supseteq A_n^* \supseteq J$.

Let D be an essential disk in W which is in general position with respect to $\mathcal{A}'$ and meets it in a minimal number of components. Since $\mathcal{A}'$ is incompressible in W and W is irreducible no component of $D \cap \mathcal{A}'$ is a simple closed curve. Since each component of $\partial \mathcal{A}'$ is parallel in S to J and J is disk busting in W each component of $\partial \mathcal{A}'$ must meet ∂D . Let G be a component of $S| \partial \mathcal{A}'$. Then every component of $G \cap \partial D$ must be an essential arc in G . Otherwise there is an outermost bigon Γ in G with $\partial \Gamma = \varepsilon \cup \delta$ such that ε is a component of $G \cap \partial D$, δ is in ∂G , and $\Gamma \cap \partial D = \varepsilon$. There is an isotopy of D which moves ε across Γ to δ and then past δ . If the points of $\partial \varepsilon$ lie on different components of $D \cap \mathcal{A}'$ this has the effect of replacing them by a single component, thereby contradicting minimality. If the points of $\partial \varepsilon$ lie on the same component of $D \cap \mathcal{A}'$ this has the effect of replacing that arc by a simple closed curve which can then be removed by the incompressibility of $\mathcal{A}'$ in W and the irreducibility of W , again contradicting minimality.

Thus $A_1^* \cap \partial D$ is the union ω of spanning arcs $\omega_1, \dots, \omega_m$ each of which meets $\partial \mathcal{A}'$ in $2n$ points. Let $a_j = \omega_j \cap J$. Consider an outermost bigon Δ of D with $\partial \Delta = \beta \cup \gamma$, $\beta \subseteq \mathcal{A}'$ and $\gamma \subseteq \partial D$. Let A'_i be the component of $\mathcal{A}'$ containing β . There are only two possibilities for the position of Δ .

Case 1: There are distinct ω_j and ω_k which are adjacent on ∂D , and γ is an arc which joins an endpoint of one to an endpoint of the other. Note that $\gamma \cap J = \emptyset$ since $J \cap \partial D$ lies in ω .

Suppose β is inessential in A'_i . Then there is a bigon Δ' in A'_i with $\partial\Delta' = \beta \cup \delta$, where δ lies in a component of $\partial A'_i$. $\Delta \cup \Delta'$ is a properly embedded disk in W which misses J . Since J is disk busting in W this disk is inessential, so its boundary $\gamma \cup \delta$ bounds a disk Δ^* in $S - J$. Since A_i^* is incompressible in W we have that Δ^* cannot contain ∂A_i^* , and so Δ^* lies in a component G of $S \setminus \partial\mathcal{A}'$. But this implies that γ is inessential in G , a contradiction.

Thus β is essential in A'_i . A ∂ -compression of A'_i along Δ yields a properly embedded disk Δ' in W such that $\partial\Delta'$ lies in A_1^* and misses J . It follows that $\partial\Delta'$ bounds a disk Δ^* in $A_1^* - J$ and that $\Delta' \cup \Delta^*$ bounds a 3-ball B in W . The union of B and a regular neighborhood of Δ is a solid torus across which A'_i is parallel into $S - J$, contradicting the minimality of $T \cap (S - J)$.

Case 2: There is an ω_j which contains γ .

Suppose the endpoints of β lie on the same side of a_j . Then β cannot be inessential in A'_i since its endpoints lie in different components of $\partial A'_i$. As above we then get that A'_i is parallel in W to an annulus in $S - J$, contradicting the minimality.

Thus the endpoints of β lie on opposite sides of a_j . Since Δ is outermost we must have that γ meets no other components of $D \cap \mathcal{A}'$. Thus $\partial A'_i = \partial A_n$. It follows that $i = n = 1$ and that $T = A_1 \cup A'_1$. Now $A_1^* \cup A'_1$ is a torus in the handlebody W and so is the boundary of a 3-manifold U in W . Since Δ is a compressing disk for ∂U in U and β is parallel to γ across Δ we have that A'_1 is parallel to A_1^* in U . Since A_1 is parallel to A_1^* in V it follows that T is ∂ -parallel in $X_M(J)$. $\square$

LEMMA 2.4. *Let A be a properly embedded annulus in a handlebody W of genus at least two. Suppose $\partial A = \partial A'$ for an annulus A' in ∂W . Suppose A' is disk busting in W . then A is parallel to A' in W .*

PROOF. Let D be an essential disk in W such that $D \cap A$ is minimal. Since A' is disk busting in W we have that $D \cap A \neq \emptyset$ and A is incompressible in W . Since $D \cap A$ is minimal, A is incompressible, and W is irreducible we have that $D \cap A$ consists of arcs. Let α be a component of $D \cap A$ which is outermost on D . Let Δ be the bigon with boundary $\alpha \cup \beta$, where $\beta = (\partial\Delta) - \alpha$.

Case 1: β does not lie in A' .

Subcase (a): α is ∂ -parallel in A .

Then there is a bigon Δ' in A with $\partial\Delta' = \alpha \cup \gamma$, where $\gamma \subseteq \partial A$. Then $\Delta \cup \Delta'$ is a disk which is disjoint from $\text{int } A'$. Since A' is disk busting in W we have that $\partial(\Delta \cup \Delta') = \partial\Delta''$ for a disk Δ'' in $\partial W - \text{int } A$. Since W is irreducible $\Delta \cup \Delta' \cup \Delta'' = \partial B$ for a 3-ball B . It follows that there is an isotopy of Δ across B to Δ' and then off A which removes at least α from the intersection, a contradiction.

Subcase (b): α is not ∂ -parallel in A .

Then A is ∂ -compressible into $\partial W - \text{int } A'$. The result is a disk Δ' with $\partial\Delta' \subseteq \partial W - \text{int } A'$. Since A' is disk busting in W we have that $\partial\Delta' = \partial\Delta''$ for a disk Δ'' in $\partial W - \text{int } A'$. But then $\partial W - A'$ is an annulus A'' , so ∂W is a torus, contradicting the fact that W has genus at least two.

Case 2: β lies in A' .

Subcase (a): α is ∂ -parallel in A .

Then there is a bigon Δ' in A with $\partial\Delta' = \alpha \cup \gamma$, where $\gamma \subseteq \partial A$. So $\Delta \cup \Delta'$ is a disk with $\partial(\Delta \cup \Delta')$ in A' . Since A' is disk busting in W we have that A' is incompressible in W . Hence $\partial(\Delta \cup \Delta') = \partial\Delta''$ for a disk Δ'' in A . Then $\Delta \cup \Delta' \cup \Delta'' = \partial B$ for a 3-ball B . An isotopy removes at least α from $D \cap A$, a contradiction.

Subcase (b): α is not ∂ -parallel in A .

Then β is not ∂ -parallel in A' . Thus A is ∂ -compressible into A' . The result is a disk Δ' with $\partial\Delta' \subseteq A'$. So $\partial\Delta' = \partial\Delta''$ for a disk Δ'' in A' . $\Delta' \cup \Delta'' = \partial B$ for a 3-ball B . Thus $B \cup N$, where N is a regular neighborhood of Δ in $W|A$ is then a solid torus across which A is parallel to A' . $\square$

3. The construction and proof

Throughout this section we assume that the Heegaard splitting (S, V, W) of M is irreducible. As in Campisi-Rathbun [2] we have a pants decomposition $(\mathcal{P}, \mathcal{Q})$ of (S, V, W) of full type for which $\mathcal{P}$ is standard. We will define a train track τ on S which contains all the curves of $\mathcal{P}$ and traverses all the seams of $\mathcal{P}$.

The genus two and genus three cases use special constructions. The higher genus cases have a uniform construction. The proofs for genera four, five, and six are slightly different. The proofs of the cases above genus six then follow by a simple induction.

3.1. Building blocks. We will decompose S into annuli and pairs of pants. The annuli are regular neighborhoods of the elements p of $\mathcal{P}$. The pairs of pants are the remaining pieces. The train track τ meets each pair of pants in three arcs, with each pair of boundary components joined by one arc. It meets each annulus as in Figure 2.

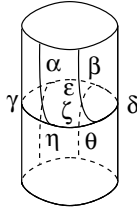


FIGURE 2

In Figures 3-10 we place weights on the branches of τ inside these cylinders. The boxed letter on the left is the label for this building block. The diagram on the right indicates how the endpoints of the arcs at one end of the cylinder join to the endpoints at the other end of the cylinder, with the meridional twisting around the cylinder and the distinction between front and back suppressed; the dot separates the endpoints into the two groups which will follow different branches into the adjacent pair of pants P and go to different components of ∂P . Since those two components will themselves be joined by arcs in P the verification that the final knot J is very good with respect to $\mathcal{P}$ consists in checking that each of the subintervals at each end of a diagram is connected by an arc to both of the subintervals at the other end of the diagram.

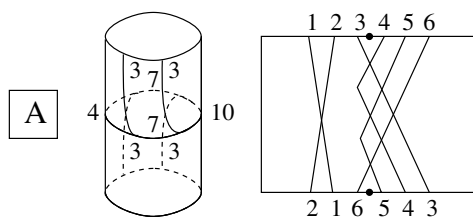


FIGURE 3

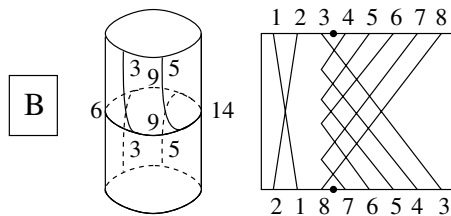


FIGURE 4

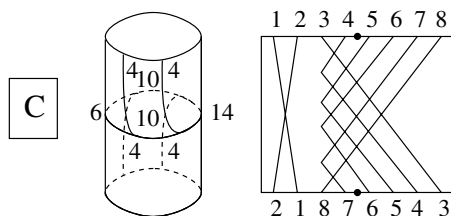


FIGURE 5

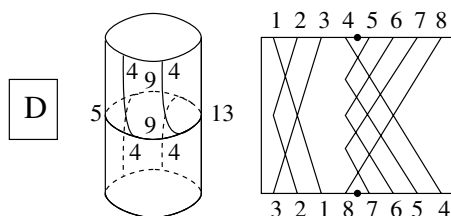


FIGURE 6

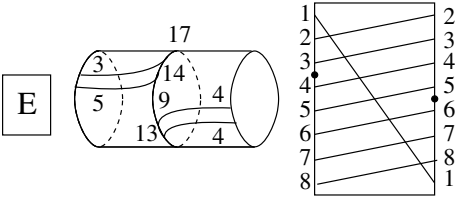


FIGURE 7

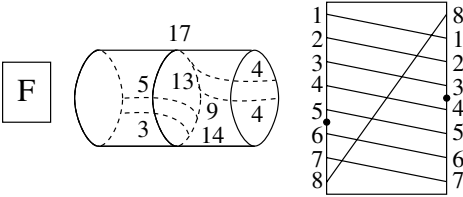


FIGURE 8

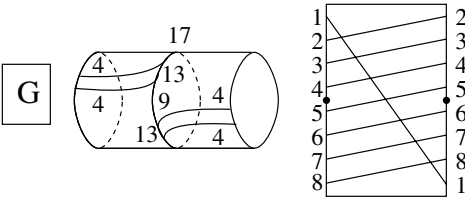


FIGURE 9

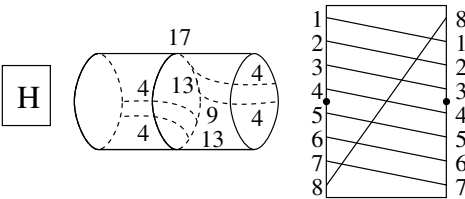


FIGURE 10

3.2. Genus two. We configure the train track as shown in Figure 11. Inserting the first A gives Figure 12. Inserting the second A glues arcs 6 and 4 together into a new arc which we call the new arc 4, discarding the label 6. For shorthand we write $6 = 4$. Similarly we get $5 = 3$, $c = 5 = 3$, $d = b$, $e = a$, and $f = 4$, giving Figure 13. Inserting the third A gives $3 = a = 2 = 4 = b = 1 = 3$, so J is connected.

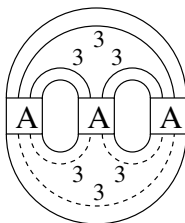


FIGURE 11

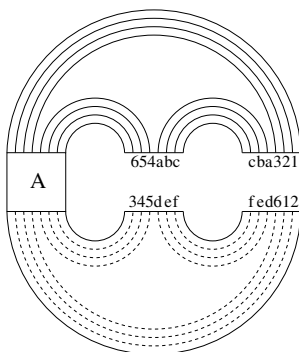


FIGURE 12

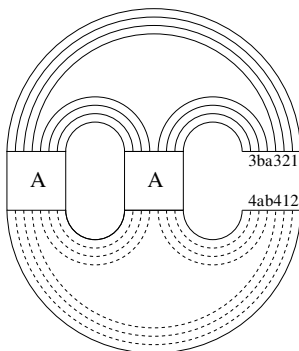


FIGURE 13

3.3. Genus three. We configure the train track as in Figure 14. Inserting A gives Figure 15. Then inserting B results in $6 = 4$, $5 = 3$, $j = 4$, $i = a$, $h = b$, $g = c$, $f = d$, and $e = 5$, giving Figure 16. Inserting E and F and the pairs of pants on their right gives Figure 17. Inserting C results in $3 = 2$, $4 = 1$, $r = d$, $q = c$, $p = k$, $o = \ell$, $m = b$, and $n = a$, giving Figure 18. Finally inserting D results in $a = k = 1 = b = c = 2 = \ell = d = a$, so J is connected.

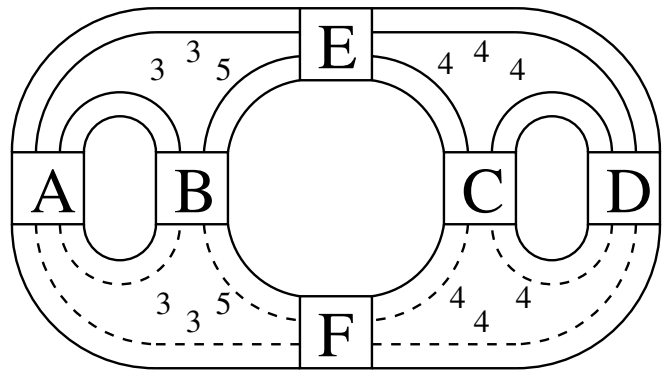


FIGURE 14

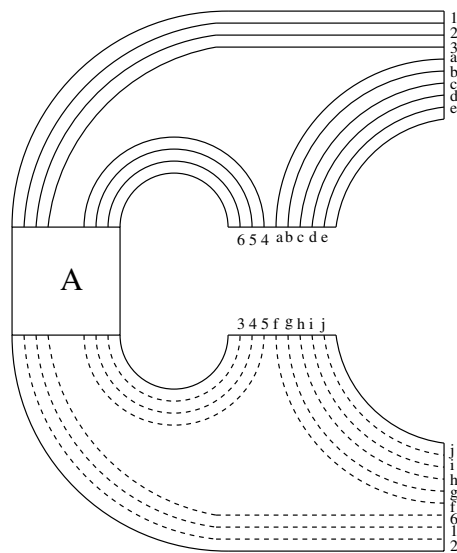


FIGURE 15

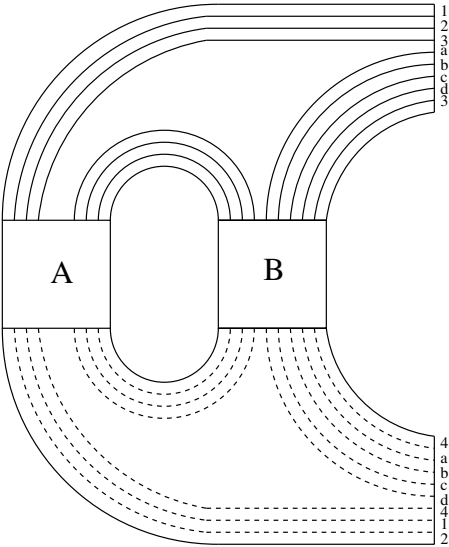


FIGURE 16

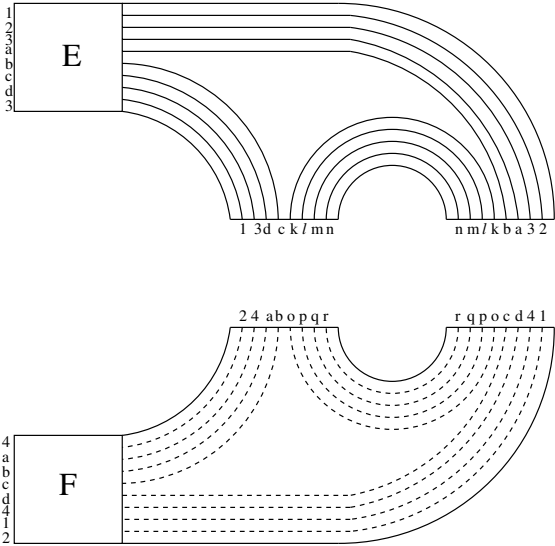


FIGURE 17

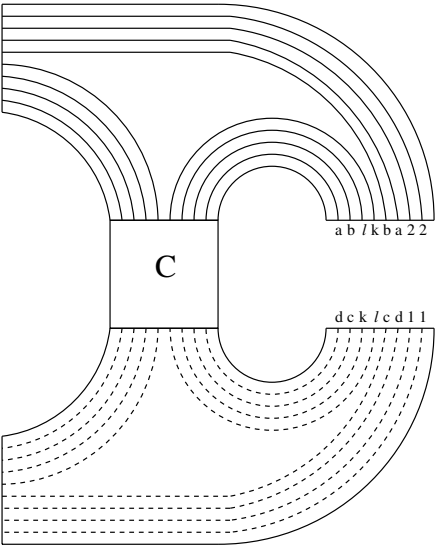


FIGURE 18

3.4. Genus four. We configure the train track as in Figure 19. Inserting A, B, E, F, and the first C gives the same configuration as in the last figure for the genus three case. Continuing this through G and H and the pairs of pants on their right gives Figure 20. Inserting the second C results in $d = 2$, $a = 1$, $z = b$, $y = \ell$, $x = s$, $w = t$, $u = k$, and $v = c$, giving Figure 21. Inserting D results in $c = s = 1 = \ell = k = 2 = t = b = c$, so J is connected.

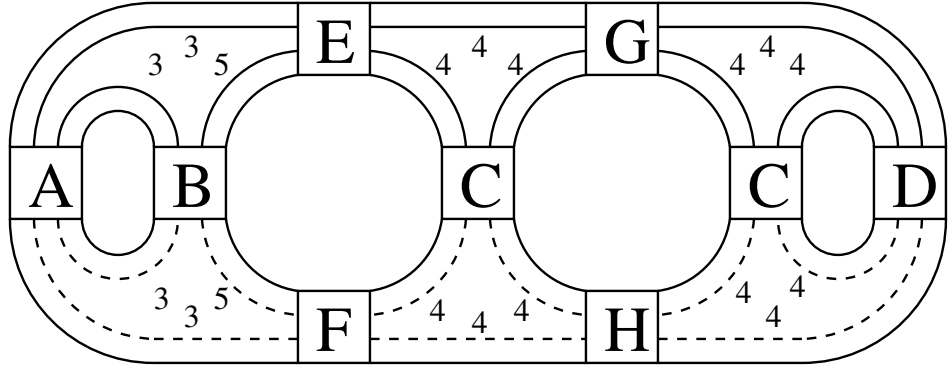


FIGURE 19

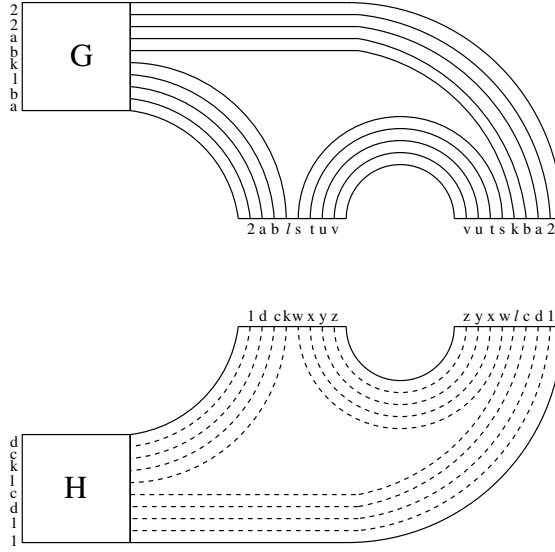


FIGURE 20

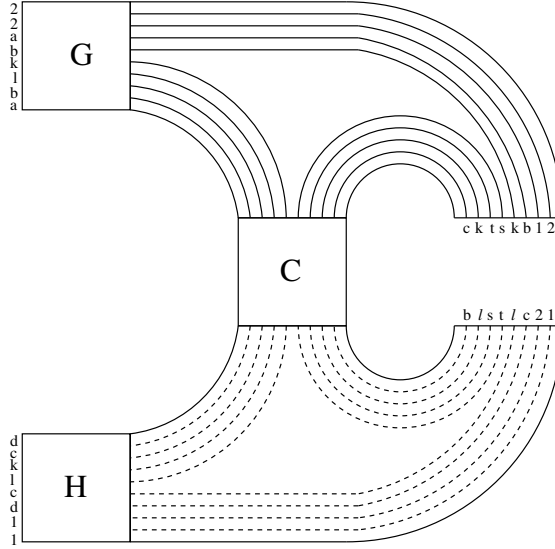


FIGURE 21

3.5. Genus five. From now on we proceed to the next genus by taking the previous construction and replacing D by the configuration consisting of G, H, C, and D and the two pairs of pants they meet. Absent the final C and the D we have Figure 22. Inserting the final C results in $b = 2$, $c = 1$, $p = k$, $o = t$, $n = w$, $m = x$, $y = s$, and $z = \ell$, giving Figure 23. We then regard the labels on the top and bottom of D as the output of the input consisting of the labels on the left sides of G and H. Inserting D gives $\ell = w = 2 = t = s = 1 = x = k = \ell$, so J is connected.

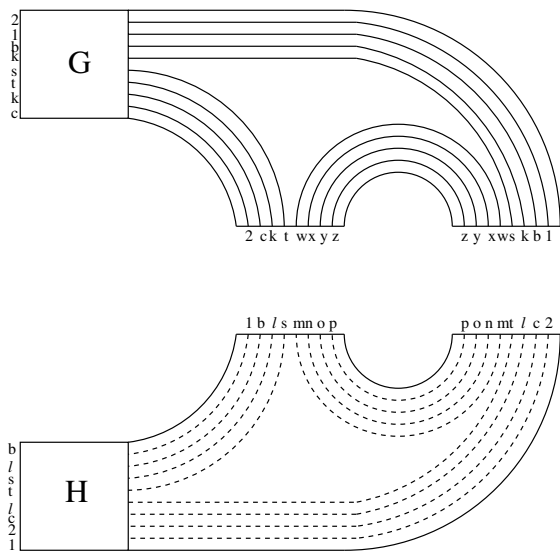


FIGURE 22

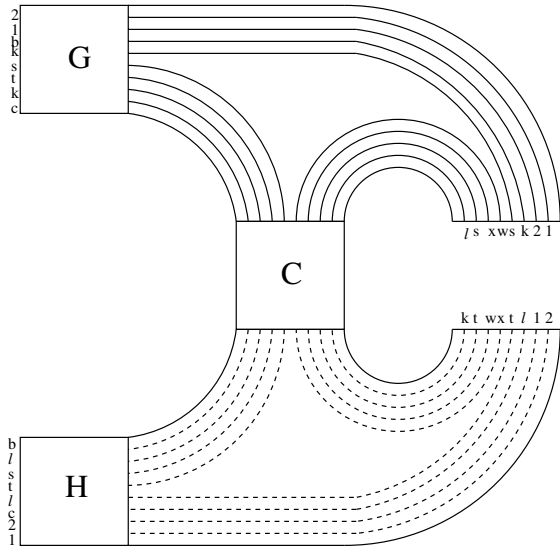


FIGURE 23

3.6. Genus six. We take the output of the previous step and use it as input for the next step. The result is Figure 24. Inserting the last C results in $k = 1$, $\ell = 2$, $p = s$, $o = x$, $n = y$, $m = z$, $u = w$, and $v = t$, giving Figure 25. Inserting D gives $t = y = 1 = x = w = 2 = z = s = t$, so J is connected.

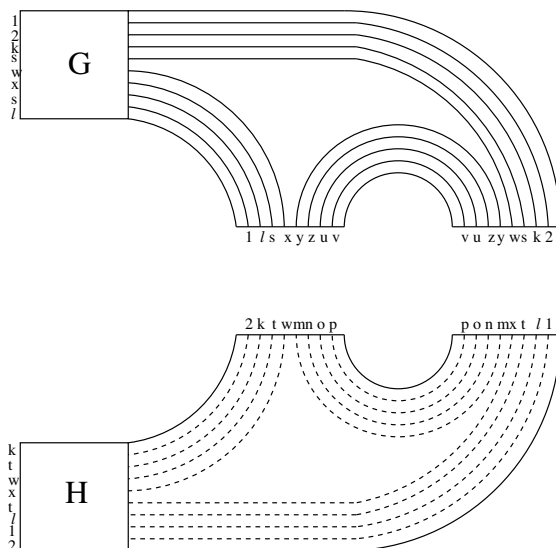


FIGURE 24

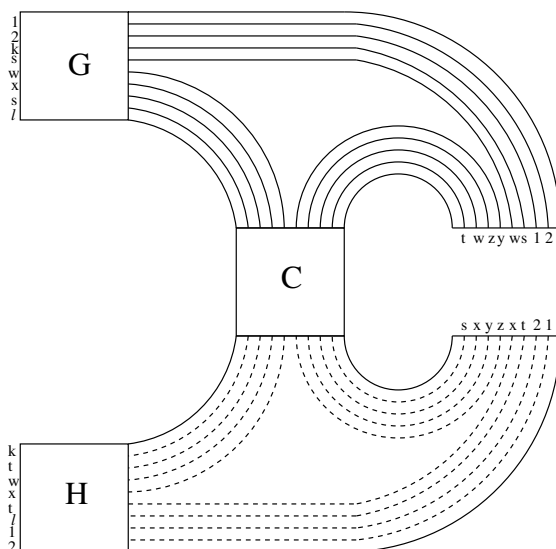


FIGURE 25

3.7. Higher genera. Reading the input of the last step from top to bottom gives the sequence $1, 2, k, s, w, x, s, \ell, k, t, w, x, t, \ell, 1, 2$. Making the substitutions that send the ordered set $(1, 2, k, s, w, x, \ell, t)$ to $(2, 1, s, w, y, z, t, x)$ gives the sequence $2, 1, s, w, y, z, w, t, s, x, y, z, x, t, 2, 1$, which is the output sequence. Thus by induction J will be connected for all higher genera.

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Stable W-length

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ABSTRACT. We study stable W -length in groups, especially for W equal to the n -fold commutator $\gamma_n := [x_1, [x_2, \cdots [x_{n-1}, x_n] \cdots]]$. We prove that in any perfect group, for any $n \geq 2$ and any element g , the stable commutator length of g is at least as big as 2^{2-n} times the stable γ_n -length of g . We also establish analogues of Bavard duality for words γ_n and for $\beta_2 := [[x, y], [z, w]]$. Our proofs make use of geometric properties of the asymptotic cones of verbal subgroups with respect to bi-invariant metrics. In particular, we show that for suitable W , these asymptotic cones contain certain subgroups that are *normed vector spaces*.

1. Introduction

Geometric group theory aims to produce functors from the algebraic category of groups and homomorphisms to geometric categories of spaces and structure-preserving maps. The category of metric spaces and isometries does not have enough morphisms for many applications, so one instead typically studies functors from the category of groups and homomorphisms, to the category of metric spaces and 1-Lipschitz (i.e. distance decreasing) maps.

A rich source of such functors arise in the theory of *bounded cohomology*, introduced systematically by Gromov in [12]. In that theory, the metric spaces are usually *normed vector spaces*, and the morphisms *bounded linear operators*. Another rich source of such examples comes from the study of (conjugation-invariant) *norms*; see e.g. [3] for a discussion, and [14] for an application to the theory of mapping class groups.

Such functors are often useful for the study of groups as dynamical objects, where the functor “geometrizes” the group action, and allows one to obtain *a priori* control of dynamical quantities from algebra. For example, *stable commutator length* (hereafter scl; see [6] or [1] for an introduction) has rich connections to 2-dimensional dynamics, symplectic geometry, hyperbolic geometry, and so on.

A natural class of characteristic norms (those invariant under any automorphism), with good monotonicity properties, arise from the theory of *words*. Given a subset W of a free group F , a W -word in G is the image of some $w \in W$ under some homomorphism $\phi : F \rightarrow G$. The W -words in G generate a so-called *verbal*

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subgroup G_W (see e.g. [17]), and the W -length of any $g \in G_W$ is defined to be the smallest number of W -words and their inverses in G whose product is equal to g . For example, many authors study *square length*, which is W -length in the case W consists of the single word x^2 . W -length (usually for W consisting of a single word w) has been intensively studied in finite groups, and recently some very strong theorems have been obtained by Shalev and his collaborators (e.g. [20, 21, 15]). See [19] for a survey, and an introduction to some of these methods. However, with some exceptions (notably [18]), W -length has not been widely studied in (general) infinite groups except in the special case of $W = \{[x, y]\}$ — i.e. commutator length. Part of the problem is that W -length seems to be such an unstructured quantity in general, and is exceedingly hard to compute, or even to estimate — even in finite groups! Therefore in this paper we propose to study a suitable “rationalization” of this quantity, namely *stable W -length*, where the stable W -length of an element $g \in G_W$ is defined to be the limit of the quantity (W -length of g^n divided by n) as $n \rightarrow \infty$. Our aim is to generalize (to the extent that it is possible) some of what is known about stable commutator length to more general classes of norms.

Perhaps the most significant difficulty in generalizing the theory of (stable) commutator length to more general stable lengths is that of *linearizing* the theory. The commutator calculus exhibits an intimate connection between the algebraic theory of commutators and the linear theory of (2-dimensional) homology. Commutators arise as boundaries in group homology, and one may obtain a duality (known as *Bavard Duality*; see [1] and [6], Chapter 2) between stable commutator length and certain natural group 1-cocycles called *homogeneous quasimorphisms*. There is no natural homology theory available for stable W -length for more general W , but one may obtain a generalization of the theory for commutators by a *geometric* construction. Given a group G , one considers the Cayley graph of $[G, G]$ taking as generators *all commutators in G* . This is a metric space (a graph), and the group $[G, G]$ acts on itself by metric isometries. Because every commutator has “length 1” in this graph, the action of the group $[G, G]$ on itself is “almost” commutative. When one replaces the Cayley graph by its asymptotic cone (an infinite re-scaled version), the limit becomes *exactly* commutative, and one obtains a normed vector space, which can be identified with one of the natural spaces obtained from the homological approach. It is this geometric construction that generalizes: given any W , one considers the Cayley graph of G_W with all W -words and their inverses as generators. Since W -words act with bounded length in the Cayley graph, the rescaled asymptotic cone obeys the “law” that all W -words are *trivial*. With more work, one can obtain a certain subset of the asymptotic cone where, for suitable W , the resulting group is actually *abelian*, and is in fact a normed vector space where one can establish the analogue of Bavard duality. This program is most successful when W is an n -fold commutator γ_n , in which case we are able to establish (for perfect groups G) a precise analog of Bavard duality, and to prove the existence of *two-sided* estimates of stable γ_n -length in terms of stable commutator length. This comparison theorem is a genuinely stable phenomenon, and does not hold for “naive” γ_n -length (for a precise statement of results, see § 1.1).

1.1. Statement of results. We now give a precise statement of the main results in the paper.

In § 2 we introduce terminology, and establish basic properties. We show that there are various inequalities relating W -length and stable W -length in various groups and for various different W . We use the notation (here and elsewhere) of $l(*|W)$ for W -length, and $\text{sl}(*|W)$ for stable W -length, with $\text{cl}(*)$ and $\text{scl}(*)$ for the special case of commutator length and stable commutator length respectively.

We give simple examples of groups where stable w -length is nontrivial for essentially all words (free groups, hyperbolic groups) and where stable w -length is trivial for all w ($\text{SL}(2, A)$ for certain rings of algebraic integers A). We also derive some nontrivial estimates (usually upper bounds) on sl in free groups. For example,

(COR. 2.17) $\text{sl}(w|w)$ INEQUALITY. *For any w there is an inequality*

$$1/2 \leq \frac{\text{scl}(w)}{\text{scl}(w) + 1/2} \leq \text{sl}(w|w) \leq 1$$

We strengthen the upper bound by an explicit construction, for certain natural classes of words. Let γ_n denote the n -fold commutator, so $\gamma_1 = x$, $\gamma_2 = [x, y]$, $\gamma_3 = [x, [y, z]]$ and so on. We obtain the estimate

(PROP. 2.28) γ_n INEQUALITY. *For any n there is an inequality*

$$\text{sl}(\gamma_n|\gamma_n) \leq 1 - 2^{1-n}$$

§ 3 is the heart of the paper. To any group G and any W , we study the asymptotic geometry of the Cayley graph C_W of G_W with all W -words as generators. A limit of rescalings of C_W gives rise to the *asymptotic cone* $\hat{A}_W$. The cone $\hat{A}_W$ is itself a group, with a bi-invariant metric, obeying a nontrivial law. It contains a canonical contractible subgroup A_W which metrically encodes the values of stable W -length on G_W . Our results are most definitive for certain specific W ; in particular,

(THM. 3.11). *Suppose A_W is nilpotent (for example, if $W = \gamma_n$ for some n). Then it is a normed vector space (in particular it is abelian).*

Moreover, if β_2 denotes the word $[[x, y], [z, w]]$,

(THM. 3.14). *Suppose in some group G that stable commutator length vanishes identically. Then A_W is abelian (and hence a normed vector space) for $W = \beta_2$.*

The fact that these asymptotic cones are normed vector spaces lets one study their geometry (and thereby stable W -length) dually, via 1-Lipschitz homomorphisms $A_W \rightarrow \mathbb{R}$. Motivated by Bavard duality, we say a function $\phi : G \rightarrow \mathbb{R}$ is a *weak γ_n -hoq* (“homogeneous quasimorphism”) if it is homogeneous (i.e. $\phi(g^n) = n\phi(g)$) and if there is a least non-negative real number $D(\phi)$ (called the *defect*) such that for any $g, h \in G$ there is an inequality

$$|\phi(g) + \phi(h) - \phi(gh)| \leq D(\phi) \min(l(g|\gamma_{n-1}), l(h|\gamma_{n-1}), l(gh|\gamma_{n-1}))$$

(THM. 3.23) γ_n -DUALITY THEOREM. *For any perfect group G , and any $g \in G$ there is an inequality*

$$\sup_{\phi} \phi(g)/D(\phi) \geq \text{sl}(g|\gamma_n) \geq \sup_{\phi} \phi(g)/2D(\phi)$$

where the supremum is taken over all weak γ_n -hoqs.

From this we derive the following rather surprising estimate, comparing stable commutator length with stable γ_n length in any perfect group:

(THM. 3.25) γ_n -COMPARISON THEOREM. *For any perfect group G , for any $n \geq 2$ and for any $g \in G$ there is an inequality*

$$2^{n-2}\text{scl}(g) \geq \text{sl}(g|\gamma_n) \geq \text{scl}(g)$$

Together the γ_n -Comparison theorem and the γ_n -Duality theorem show that stable commutator length in a perfect group G can be bounded from below by weak γ_n -hoqs. It would be interesting to try to find naturally occurring examples of such functions arising from realizations of G as a group of automorphisms of some geometric object.

In a similar vein, we say $\phi : G \rightarrow \mathbb{R}$ is a *weak β_2 -hoq* if it is homogeneous, and if there is a least non-negative real number $D(\phi)$ (called the *defect*) such that for any $g, h \in G$ there is an inequality

$$|\phi(g) + \phi(h) - \phi(gh)| \leq D(\phi) \min(l(g|\beta_2), l(h|\beta_2), l(gh|\beta_2))$$

(THM. 3.31) β_2 -DUALITY THEOREM. *For any perfect group G in which scl vanishes identically, and for any $g \in G$ there is an inequality*

$$\sup_{\phi} 2\phi(g)/D(\phi) \geq \text{sl}(g|\beta_2) \geq \sup_{\phi} \phi(g)/2D(\phi)$$

where the supremum is taken over all weak β_2 -hoqs.

Note that stable commutator length is known to vanish identically in many important classes of groups, including

- amenable groups
- lattices in higher rank Lie groups [4]
- groups of PL homeomorphisms of the interval [5]
- groups that satisfy a law [7]

It is not known whether stable W -length for various W vanish in these classes of groups. One of the aims of this paper is to develop tools to approach this question.

Finally, in § 4, we give an elementary construction, using hyperbolic geometry, of a class of elements w_n in free groups of various ranks for which $\text{scl}(w_n) = 1/2$ and $\text{sl}(w_n|\gamma_3) = 1$, but for which $\text{cl}(w_n) = 1$ and $l(w_n|\gamma_3) \geq 2n/3$.

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2. Basic properties

The purpose of this section is to standardize definitions and notation, and to survey some elementary constructions in the theory.

NOTATION 2.1. We use the notation $[x, y]$ for $xyx^{-1}y^{-1}$ and x^y for yxy^{-1} . With this convention, exponentiation obeys $(x^y)^z = x^{zy}$. We also use the notation x^* for some (unspecified) conjugate of x .

The following identities, though elementary, are useful.

LEMMA 2.2. *The following identities hold (the letters denote free generators):*

- (1) $[x, y] = [y^x, x^{-1}]$
- (2) $[x, y]^{-1} = [y, x] = [x^y, y^{-1}]$

- (3) $[x, yz] = [x, y][x, z]^y = [x, y][x, z][[z, x], y]$
- (4) $[xy, z] = [x, z][y, z]^{x^z} = [x, z][y, z][[z, y], x^z]$
- (5) $[[y, x], z^x][[x, z], y^z][[z, y], x^y] = 1$

Bullet (5) is known as the *Hall-Witt identity*.

2.1. Verbal subgroups.

DEFINITION 2.3 (*W-word*). Let F be a free group, and W a subset of F . A *W-word* in a group G is the image of some $w \in W$ under a homomorphism $F \rightarrow G$.

DEFINITION 2.4 (*verbal subgroup*). Let G be a group. A *verbal subgroup* of G is the subgroup generated by the W -words, for some $W \subset F$. The W -verbal subgroup of G is denoted G_W .

The group G is said to *obey the law W* if G_W is trivial. A group is said to *obey a law* if G_W is trivial for some nonempty W .

For any sets $V, W \subset F$ with $F_V = F_W$ (for example, if $W \subset V \subset F_W$) and for any group G , the verbal subgroups G_V and G_W are equal, though the V -words and the W -words in G might not be equal *as sets*.

Since the image of a W -word under any automorphism is also a W -word, verbal subgroups are characteristic (i.e. invariant under every automorphism). In particular, a conjugate of a W -word is a W -word. In a free group, it is tautologically true that characteristic subgroups are verbal.

In the sequel we are interested in verbal subgroups G_W where W is a single element of F . For the sake of legibility, we use a lower case w to denote a single element of F , and likewise denote the w -subgroup of G by G_w . In fact, we are interested in certain explicit words w . The main w of interest in this paper are the *n-fold commutators*.

DEFINITION 2.5. Here and in the sequel, we let γ_n denote the n -fold commutator of the generators of F_n . That is,

$$\gamma_1 = x, \quad \gamma_2 = [x, y], \quad \gamma_3 = [x, [y, z]], \quad \gamma_4 = [x, [y, [z, w]]]$$

and so on.

DEFINITION 2.6. A word $w \in F$ is *reflexive* if w^{-1} is a w -word in F .

LEMMA 2.7. *The words γ_n are all reflexive. Moreover, any nested bracket of n elements is a γ_n word.*

PROOF. The second statement means, for example, that an expression like $[[x, y], z]$ is a γ_3 word, that $[x, [[w, [u, v]], z], y]$ is a γ_5 word, and so on. This and the reflexivity of γ_n follow inductively from Lemma 2.2, bullet (2). $\square$

For any n and any G , the subgroups G_{γ_n} are usually denoted G_n and called the *lower central series*. The successive quotients G/G_n are (universal) $(n-1)$ -step nilpotent quotients of G (so that for instance G/G_2 is abelian, G/G_3 is 2-step nilpotent, and so on).

DEFINITION 2.8. Here and in the sequel, we define $\beta_1 = [x, y]$, and for $n > 1$ we let β_n denote the commutator of two copies of β_{n-1} (in different generating sets). Hence

$$\beta_1 = [x, y], \quad \beta_2 = [[x, y], [z, w]], \quad \beta_3 = [[[x, y], [z, w]], [[s, t], [u, v]]]$$

and so on.

LEMMA 2.9. *The words β_n are reflexive.*

PROOF. This follows inductively from Lemma 2.2, bullet (2). $\square$

The subgroups G_{β_n} are usually denoted $G^{(n)}$ and called the *derived series*. The successive quotients $G/G^{(n)}$ are (universal) n -step solvable quotients of G . Note that $\beta_1 = \gamma_2$.

2.2. W -length and w -length.

DEFINITION 2.10 (W -length). Let $W \subset F$ be given. Let G be a group, and let G_W be the subgroup generated by W -words. If $g \in G_W$, the W -length of g , denoted $l(g|W)$, is the minimum number of W -words and their inverses in G whose product is g , and the *stable W -length* of g , denoted $sl(g|W)$, is the limit

$$sl(g|W) = \lim_{n \rightarrow \infty} \frac{l(g^n|W)}{n}$$

If W consists of a single element, we usually denote it by a lower case w , and define w -length and stable w -length. We are typically concerned with this case in the sequel.

If w is reflexive (which shall usually be the case in the sequel), the inverse of a w -word is a w -word. By abuse of notation we will sometimes refer to both w -words and their inverses as w -words. The meaning should be clear from context in each case.

NOTATION 2.11. If we need to emphasize that g is in specific group G , we use the notation $l_G(g|W)$ and $sl_G(g|W)$.

In the special case that $w = [x, y]$ in the free group generated by x and y , we refer to w -length as *commutator length*, and stable w -length as *stable commutator length*, and denote them by $cl(\cdot)$ and $scl(\cdot)$ respectively.

NOTATION 2.12. We abbreviate the product of n (arbitrary) w -words by $w^{\{n\}}$ (or $W^{\{n\}}$ for W -words). This should not be confused with w^n , the n th power of a specific word w .

LEMMA 2.13. *Suppose w is not in $[F, F]$. Then $sl(\cdot|w)$ is identically zero in every group.*

PROOF. Since w is not in $[F, F]$, there is a surjective homomorphism $F \rightarrow \mathbb{Z}$ sending w to some nonzero n . Without loss of generality, we may assume $n > 0$. But then in any group $l(g^{nm}|w) \leq 1$ so $sl(g|w) = 0$. $\square$

LEMMA 2.14. *The functions $l(\cdot|W)$ and $sl(\cdot|W)$ are characteristic, and their values do not increase under homomorphisms.*

PROOF. The group $\text{Aut}(G)$ permutes the canonical generators of G_W , proving the first claim. The image of a W -word under any homomorphism is a W -word, proving the second claim. $\square$

By convention, we set $sl(g|W) = sl(g^n|W)/n$ whenever $g^n \in G_W$ (even if g is not necessarily in G_W).

2.3. Inequalities. If w and v are contained in free groups F and F' respectively, by abuse of notation we think of w and v as being contained in the single free group $F * F'$. By convention therefore we think of all abstract words as being contained in a single (infinitely generated) free group, which we denote hereafter by F .

LEMMA 2.15 (fundamental inequality). *For any $v, w \in F$ and any $g \in G$ there is an inequality*

$$l_G(g|v) \leq l_G(g|w)l_F(w|v)$$

PROOF. Write g minimally as a product of w -word or their inverses, and then rewrite each of these as a minimal product of v -words or their inverses. This gives an expression for g as a product of v -words and their inverses. $\square$

Note that for this lemma to make sense we must have $g \in G_w$ and $w \in F_v$. In the case $v = [x, y]$ we obtain $cl_G(g) \leq l_G(g|w)cl_F(w)$. In the case of stable commutator length, one obtains a better estimate

LEMMA 2.16 (scl inequality). *For any $w \in F$ and any $g \in G$ and $n \in \mathbb{Z}$ there is an inequality*

$$scl_G(g^n) \leq l_G(g^n|w)scl_F(w) + \frac{1}{2}(l_G(g^n|w) - 1)$$

Moreover,

$$scl_G(g) \leq sl_G(g|w)(scl_F(w) + \frac{1}{2})$$

PROOF. Express g^n as a product $w_1 \cdots w_m$ of w -words or their inverses, where $m = l_G(g^n|w)$. For any even k we have

$$g^{nk} = (w_1 \cdots w_m)^k = w_1^k \cdots w_m^k v$$

where v is a product of at most $(m-1)k/2$ commutators (see e.g. [6], Lemma 2.24 for the case $m = 2$, and apply induction). Taking the limit as $k \rightarrow \infty$ we obtain the first inequality.

Taking the limit $n \rightarrow \infty$ and stabilizing gives us the second inequality. $\square$

COROLLARY 2.17 ($sl(w|w)$ inequality). *For any $w \in F$ there is an inequality*

$$1/2 \leq \frac{scl(w)}{scl(w) + 1/2} \leq sl(w|w) \leq 1$$

PROOF. The first inequality follows from the fact that $scl(w) \geq \frac{1}{2}$ for any nontrivial w in a free group (see e.g. [6], Theorem 4.111). The second inequality follows from Lemma 2.16 by setting $g = w$. The last inequality follows from the tautological expression of w^n as a product of n copies of w . $\square$

A natural question to ask is the following:

QUESTION 2.18 (strict inequality). *For which $w \in F$ is there is a strict inequality*

$$sl(w|w) < 1 \text{ ?}$$

This question has a positive answer for several interesting classes of words.

EXAMPLE 2.19. For each g , let $x_1, y_1, \dots, x_g, y_g$ be free generators, and let $w_g = [x_1, y_1] \cdots [x_g, y_g]$. Since $\text{scl}(w_g) = g - \frac{1}{2}$ (see e.g. Bavard [1]) we obtain a lower bound $\text{sl}(w_g|w_g) \geq 1 - 1/2g$. On the other hand, an expression of w_g^n as a product of $n(g - 1/2) + o(n)$ commutators is also an expression as a product of $n(g - \frac{1}{2})/g + o(n)$ w_g -words. So

$$\text{sl}([x_1, y_1] \cdots [x_g, y_g] | [x_1, y_1] \cdots [x_g, y_g]) = 1 - 1/2g$$

From the definition, $\text{sl}(g|w) \leq \text{l}(g^n|w)/n$ for any n . On the other hand, this inequality can often be definitely improved:

LEMMA 2.20. *For any $w \in F$ and any $g \in G$, there is an inequality*

$$\text{sl}_G(g|w) \leq (\text{l}(g^n|w) - 1 + \text{sl}(w|w))/n$$

PROOF. Suppose $g^n = w_1 \cdots w_m$ where $m = \text{l}(g^n|w)$. For any k we can write

$$\begin{aligned} g^{nk} &= w_1 \cdots w_m w_1 \cdots w_m \cdots w_1 \cdots w_m \text{ (} mk \text{ terms)} \\ &= w_1^k w_1' w_2' \cdots w_{(m-1)k}' \end{aligned}$$

where each w_i' is a w -word (since $xy = y(y^{-1}xy)$, and any conjugate of a w -word or its inverse is a w -word or its inverse). In particular, if k is large, then w_1^k can be expressed as a product of $k \text{sl}(w|w) + o(k)$ w -words and therefore g^{nk} can be expressed as a product of $k \text{sl}(w|w) + (m-1)k + o(k)$ w -words. Taking k large, we obtain the desired inequality. $\square$

COROLLARY 2.21. *If $\text{l}(w^n|w) = m$ then $\text{sl}(w|w) \leq (m-1)/(n-1)$.*

PROOF. Substitute $w = g$ in Lemma 2.20. $\square$

We end this section with a couple of examples, illustrating the range of possibilities one can expect in certain specific classes of groups.

EXAMPLE 2.22 (Free group). If F is a free group, $\text{scl}(g)$ is positive for every $g \in [F, F]$ (in fact, $\text{scl}(g) \geq 1/2$). So for every w with $F_w \subset [F, F]$ we get $\text{sl}_F(g|w) \geq \text{scl}_F(g)/(\text{scl}_F(w) + 1/2) > 0$. In other words, for every w for which $\text{sl}(*|w)$ is not identically zero in every group, it is positive on every element in a free group. A similar phenomenon holds in groups for which scl is typically positive, such as hyperbolic groups (see e.g. [6], Chapter 3).

EXAMPLE 2.23 ($\text{SL}(2, A)$). Carter-Keller-Paige [23] Thm. 6.1. show that if A is the ring of integers in a number field K containing infinitely many units, and if T is an element of $\text{SL}(2, A)$ which is not a scalar matrix (i.e. not of the form $\lambda \cdot \text{id}$) then $\text{SL}(2, A)$ has a finite index normal subgroup which is boundedly generated by conjugates of T . That is, in this finite index normal subgroup, every element can be written as a bounded number of conjugates of T .

For any w , one can find T in $\text{SL}(2, A)$ which is a w -word but not a scalar matrix (to see this, observe that $\text{SL}(2, A)$ contains $\text{SL}(2, \mathbb{Z})$ which contains a free group, none of whose nontrivial elements is a scalar matrix). It follows that $\text{SL}(2, A)$ has a finite index subgroup in which every element has finite w -length. Consequently $\text{sl}(*|w)$ vanishes identically in $\text{SL}(2, A)$ for every w .

A similar phenomenon holds in groups such as $\text{SL}(n, \mathbb{Z})$ when $n \geq 3$.

2.4. Culler's identity. Culler [10], in his analysis of commutator length in free groups, discovered the beautiful identity

$$[x, y]^3 = [y^x, x^{y^{-1}} x^{-2}] [x^{y^{-1}}, y^2]$$

This identity expresses algebraically the geometric fact that a once-punctured torus admits an (irregular) cover of degree 3 with one boundary component. We can get a lot of mileage out of this identity; this is not really for a very specific reason, rather because this is the simplest identity which certifies that $\text{scl}([x, y]) < 1$, and we are able to “bootstrap” this identity to show $\text{sl}(w|w) < 1$ for many w .

EXAMPLE 2.24 ($[x, y^n]$). If we set $w = [x, y^n]$ then we obtain $\text{l}(w^3|w) \leq 2$ and therefore $\text{sl}(w|w) \leq \frac{1}{2}$. Together with Corollary 2.17 this implies

$$\text{sl}([x, y^n] | [x, y^n]) = \frac{1}{2}$$

EXAMPLE 2.25 ($[x, y]^2$). If we denote $a = [y^x, x^{y^{-1}} x^{-2}]$ and $b = [x^{y^{-1}}, y^2]$ then Culler's identity says

$$\begin{aligned} ([x, y]^2)^6 &= ([x, y]^3)^4 = abababab \\ &= aabbcaabbc \text{ for } c = [b^{-1}, b^{-1}a^{-1}b^{-1}] \\ &= a^2b^2c^2(a')^2(b')^2 \end{aligned}$$

for suitable commutators a', b' . Hence $\text{l}([x, y]^2 | [x, y]^2) \leq 5$ and therefore

$$\frac{2}{3} \leq \text{sl}([x, y]^2 | [x, y]^2) \leq \frac{4}{5}$$

2.5. γ_n inequalities. Generalizations of Culler's identity can be obtained from a uniform topological argument.

LEMMA 2.26 (generalized Culler identity). *For all x , for any n and any y , there is an identity of the form*

$$[x, y]^{2n+1} = [* , y^{n+1}]^* \cdot \text{product of } n \text{ words of the form } [* , y]^*$$

PROOF. Let $F = \langle x, y \rangle$. For each n , we construct a permutation representation $\rho_n : F \rightarrow S_{2n+1}$. We think of an element of S_{2n+1} as a bijective function from $\{0, 1, \dots, 2n\}$ to itself, and denote such a function by an ordered string of values. With this convention, we define

$$\rho_n(x) = 2n \ 2n-1 \ 2n-2 \ \cdots \ 1 \ 0$$

$$\rho_n(y) = 1 \ 2 \ 3 \ \cdots \ n \ 0 \ n+1 \ \cdots \ 2n$$

In words, $\rho_n(y)$ cycles the first $n+1$ elements and fixes the rest, while $\rho_n(x)$ is the involution which reverses the order.

If we think of F as the fundamental group of a once-punctured torus S , then x and y can be represented by embedded essential loops α and β intersecting in one point. Since β is homologically essential, it is non-separating. Let S' be the $2n+1$ -fold cover of S whose monodromy representation is ρ_n . By the definition of ρ_n , the permutation $\rho_n([x, y])$ is a single $2n+1$ -cycle, and therefore $\partial S'$ consists of a single boundary component, mapping to ∂S with degree $2n+1$ under the covering projection.

Likewise, β has $n+1$ preimages; one (call it β_0) maps to β with degree $n+1$. The others β_i for $1 \leq i \leq n$ map with degree 1. We claim that there are disjoint, embedded curves α_i in S' so that each α_i and β_i intersect transversely in one point, and α_i does not intersect β_j for $i \neq j$. Evidently the lemma follows from this.

To prove this claim we first show that $\cup\beta_i$ is non-separating. Let T be a component of $S' - \cup\beta_i$. Then T covers $S - \beta$, and consequently $T \cap \partial S'$ is nonempty. But this implies $T \cap \partial S' = \partial S'$, since $\partial S'$ is connected. Since T was arbitrary, every component of $S' - \cup\beta_i$ contains $\partial S'$, and therefore $S' - \cup\beta_i$ is connected and $\cup\beta_i$ is non-separating, as claimed.

It follows that $S' - \cup\beta_i$ is connected, and therefore (by reason of χ) is homeomorphic to a sphere with $2n+3$ holes. We can therefore find $n+1$ disjoint properly embedded arcs in $S' - \cup\beta_i$, each of which runs between the two boundary components corresponding to a single β_i . This proves the claim, and therefore the lemma. $\square$

As a corollary, we obtain the following inequality:

PROPOSITION 2.27 ($[x, w]$ inequality). *For any word w , the word $[x, w]$ (where x is an extra free generator) satisfies*

$$l([x, w]^{2n+1} | [x, w]) \leq n + l(w^{n+1} | w)$$

and consequently

$$\text{sl}([x, w] | [x, w]) \leq \frac{1 + \text{sl}(w | w)}{2}$$

PROOF. The proposition follows from Lemma 2.26, together with the observation that $[x, w^{n+1}]$ can be written as a product of $l(w^{n+1} | w)$ $[x, w]$ -words, by bullet (3) of Lemma 2.2. $\square$

As an important special case, one deduces:

PROPOSITION 2.28 (γ_n inequality). *For any n there is an inequality*

$$\text{sl}(\gamma_n | \gamma_n) \leq 1 - 2^{1-n}$$

PROOF. For $n = 1$ this follows from $\text{sl}(x | x) = 0$, a special case of Lemma 2.13. Then the result follows from Proposition 2.27 and induction on n . $\square$

QUESTION 2.29. *Is the estimate in Proposition 2.28 sharp?*

3. Geometry of G_W

In this section we begin a more systematic study of stable W -length, from a geometric point of view. The main tool is the geometry of Cayley graphs, and their asymptotic cones. A verbal subgroup of any group admits a tautological, characteristic, bi-invariant metric, and it is the asymptotic geometry of this metric that gives us insight into stable W -length.

3.1. Cayley graph C_W . Let G be a group, and G_W its verbal subgroup associated to some subset $W \subset F$. If w is a single element, we assume it is reflexive. Otherwise, we assume that W is symmetric; i.e. $W = W^{-1}$.

Let $C_W(G)$ (or just C_W or C if G or W are understood) denote the Cayley graph of G_W with all W -words in G as generators. We give C the structure of a (path) metric space, by declaring that each edge has length 1. This induces a metric on G_W which we call the W -metric, where the distance from g to h is the W -length of $g^{-1}h$.

LEMMA 3.1. *The W -metric is both left and right invariant for the action of G_W on itself. Furthermore, $\text{Aut}(G)$ acts on G_W by isometries, and any homomorphism $G \rightarrow H$ induces a 1-Lipschitz (simplicial) map from $C_W(G)$ to $C_W(H)$.*

PROOF. For any $f, g, h \in G_W$ we have

$$d(fg, fh) = l(g^{-1}h|W) = d(g, h)$$

and

$$d(gf, hf) = l((g^{-1}h)^{f^{-1}}|W) = l(g^{-1}h|W) = d(g, h)$$

The latter two properties are restatements of Lemma 2.14 in geometric language. $\square$

From the construction, the stable W -length of an element $g \in G_W$ is just the translation length $\tau(g)$ of g on G_W ; i.e. the limit $\tau(g) = \lim_{n \rightarrow \infty} d(\text{id}, g^n)/n$.

The large-scale geometry of commutator subgroups (i.e. the case $w = [x, y]$) was studied in [8].

3.2. Asymptotic cone. Though C_W is an interesting geometric object in its own right, stable W -length can be more easily studied in an object derived from C_W , namely the *asymptotic cone*; see e.g. Gromov [13] Chapter 2 (especially p. 36) for an introduction to asymptotic cones in group theory and their properties. The construction of an asymptotic cone depends (in general) on a highly non-constructive choice, namely the choice of a non-principal ultrafilter. However, we will shortly restrict attention to a subset of the asymptotic cone whose geometry does not depend on any choices, and therefore our discussion of such objects is very terse.

Essentially, a non-principal ultrafilter ω picks out a limit of every absolutely bounded sequence a_n , which we denote $\lim_{\omega} a_n$. We can think of $\lim_{\omega}$ as a function from ℓ^{∞} to $\mathbb{R}$; this function is a ring homomorphism. The “limit” $\lim_{\omega}$ is always contained between $\limsup$ and $\liminf$, and is equal to the honest limit of some infinite subsequence of the a_n (so, for instance, if $\lim_{n \rightarrow \infty} a_n$ exists it is equal to $\lim_{\omega} a_n$) but otherwise satisfies no *a priori* constraint. For more details, see Gromov *op. cit.*

Fix (once and for all) a non-principal ultrafilter ω .

DEFINITION 3.2. Given a subset $W \subset F$ and a group G , define the asymptotic cone $\hat{A}_W(G)$ (or just $\hat{A}_W$ or $\hat{A}$ if G is understood) to be the ultralimit of the sequence of metric spaces $C_W(G)$ with the W -metric rescaled by a factor of $1/n$.

Thus a point in $\hat{A}_W(G)$ is an equivalence class of sequence $\{a_n\}$ with $d(\text{id}, a_n) = O(n)$, where $d(\{a_n\}, \{b_n\}) = \lim_{\omega} d(a_n, b_n)/n$, and where $\{a_n\}$ and $\{b_n\}$ are in the same equivalence class (denoted $\{a_n\} \sim \{b_n\}$) iff $\lim_{\omega} d(a_n, b_n)/n = 0$.

PROPOSITION 3.3. *The asymptotic cone $\hat{A}_W$ has the structure of a group, with a bi-invariant metric, satisfying the law W .*

PROOF. The group structure on $\hat{A}$ is defined by taking $\{\text{id}\}$ (the constant equivalence class) as the identity, defining multiplication by $\{a_n\} \cdot \{b_n\} = \{a_n b_n\}$, and inverses by $\{a_n\}^{-1} = \{a_n^{-1}\}$. To see that this is well-defined, suppose $\{a_n\} \sim \{a'_n\}$ and $\{b_n\} \sim \{b'_n\}$. By definition, we have $a_n = w_n a'_n$ and $b_n = v_n b'_n$ where $l(w_n|W) = d(a_n, a'_n)$ and $l(v_n|W) = d(b_n, b'_n)$. Hence

$$a_n b_n = w_n a'_n v_n b'_n = w_n v_n^{a'_n} a'_n b'_n$$

so

$$\lim_{\omega} d(a_n b_n, a'_n b'_n)/n \leq \lim_{\omega} (d(a_n, a'_n) + d(b_n, b'_n))/n = 0$$

The bi-invariance of the metric on $\widehat{A}$ follows immediately from the definition of the multiplication, and the bi-invariance of the metric on C .

Finally, if $\{a_{1,n}\}\{a_{2,n}\}\cdots\{a_{m,n}\}$ is a W -word in $\widehat{A}$ (with its group structure), then $a_{1,n}a_{2,n}\cdots a_{m,n}$ is a W -word in G for each n , and therefore

$$\{a_{1,n}\}\{a_{2,n}\}\cdots\{a_{m,n}\} = \{a_{1,n}a_{2,n}\cdots a_{m,n}\} \sim \{\text{id}\}$$

□

3.3. The group B_W . We now relate $\widehat{A}_W$ to G_W (in fact, to a slightly larger group depending on G and W).

LEMMA 3.4. *Let $W \subset F$ be given, and let $g_1, g_2, \dots, g_m$ be in G_W . There is an estimate*

$$l((g_1^{a_1} g_2^{a_2} \cdots g_m^{a_m})^{-1} (g_1^{b_1} g_2^{b_2} \cdots g_m^{b_m}) | W) \leq \sum_i l(g_i^{|b_i - a_i|} | W)$$

PROOF. By expanding,

$$\begin{aligned} \text{LHS} &= g_m^{-a_m} g_{m-1}^{-a_{m-1}} \cdots g_2^{-a_2} g_1^{b_1 - a_1} g_2^{b_2} \cdots g_m^{b_m} \\ &= g_m^{-a_m} g_{m-1}^{-a_{m-1}} \cdots g_2^{b_2 - a_2} \cdots g_m^{b_m} (g_1^{b_1 - a_1})^* \\ &= \prod_i (g_i^{b_i - a_i})^* \end{aligned}$$

□

REMARK 3.5. It would be nice to generalize Lemma 3.4 to certain homogeneous expressions $g_1 g_2 \cdots g_m$ whose *product* is in G_W , even if the individual g_i are not. In fact, this can be done if $m = 2$, but *not* for higher m unless $G_W = [G, G]$ (i.e. unless every commutator is a product of W -words).

Suppose $gh \in G_W$. Then

$$\begin{aligned} g^n h^n &= (gh)^n [g^{n-1}, h]^* [g^{n-2}, h]^* \cdots [g, h]^* \\ &= (gh)^n [g^{n-1}, gh]^* \cdots [g, gh]^* \end{aligned}$$

where we use the identity $[a, b] = [a, ab]^*$. Since any commutator $[*, gh]$ is a product of at most $2 \cdot l(gh|W)$ W -words, we see that $l(g^n h^n | W) \leq 3n \cdot l(gh|W)$. Moreover,

$$(g^{n_1} h^{n_1})^{-1} (g^{n_2} h^{n_2}) = (g^{n_2 - n_1} h^{n_2 - n_1})^*$$

and we derive the inequality

$$l((g^{n_1} h^{n_1})^{-1} (g^{n_2} h^{n_2}) | W) \leq 3|n_2 - n_1| \cdot l(gh|W)$$

which is analogous to the estimate in Lemma 3.4.

On the other hand, in general the condition that xyz is in G_W does *not* imply that $x^n y^n z^n$ is in G_W for all n (in fact, for any n other than 0, 1), unless $G_W = [G, G]$. For example, if y, z are arbitrary and x is chosen so that $xyz \in G_W$, then

$$x^{-1} y^{-1} z^{-1} = x^{-1} z^{-1} y^{-1} [y, z]$$

Now, $x^{-1} z^{-1} y^{-1}$ is in G_W because xyz is, so the left hand side is in G_W iff $[y, z]$ is. But y and z are arbitrary, so this holds for all y, z if and only if $G_W = [G, G]$, as claimed.

Let B_W be the free group obeying the law W generated by the elements of G_W as a set, subject to the relation of homogeneity $g^n = g^{\star n}$ where $\star$ is the group operation in B_W . There is a natural inclusion $G_W \rightarrow B_W$ of sets, taking each element of G_W to the corresponding generator.

LEMMA 3.6. *There is a homomorphism $\rho : B_W \rightarrow \hat{A}_W$ defined by the formula*

$$\rho(g_1 \star g_2 \star \cdots \star g_m) = \{g_1^n g_2^n \cdots g_m^n\}$$

PROOF. From the definition of the group operation in $\hat{A}_W$, the right hand side is equal to the product $\{g_1^n\}\{g_2^n\} \cdots \{g_m^n\}$, so the map is a homomorphism. To see that it is well-defined, observe that all relations (i.e. W -words, and the homogeneity relations) are satisfied in $\hat{A}_W$ by Proposition 3.3, so the map is well-defined. $\square$

REMARK 3.7. It is not typically true that this homomorphism is an injection, even if G is free.

This homomorphism lets us map G_W to $\hat{A}_W$ by $g \rightarrow \{g^n\}$. From the definitions, $\text{sl}(g|W) = d(\{\text{id}\}, \{g^n\})$; in other words *stable w -length can be recovered from the geometry of $\hat{A}_W$ and the map $B_W \rightarrow \hat{A}_W$.*

3.4. Real structure. For any group G , let $B_W \otimes \mathbb{R}$ be the free group obeying the law W generated by expressions of the form g^t where $g \in G_W$ and $t \in \mathbb{R}$, subject to the relation of homogeneity $g^s \star g^t = g^{s+t}$. We topologize $B_W \otimes \mathbb{R}$ with the weakest topology for which multiplication and inverse are continuous, as well as each homomorphism $\mathbb{R} \rightarrow B_W \otimes \mathbb{R}$ of the form $t \rightarrow g^t$.

We extend $\rho : B_W \rightarrow \hat{A}_W$ to $\rho : B_W \otimes \mathbb{R}$ by defining

$$g^t \rightarrow \{g^{\lfloor tn \rfloor}\}$$

on the $\mathbb{R}$ subgroups, and extending to arbitrary products by using the group multiplication. Hence

$$g_1^{t_1} g_2^{t_2} \cdots g_m^{t_m} \rightarrow \{g_1^{\lfloor t_1 n \rfloor} g_2^{\lfloor t_2 n \rfloor} \cdots g_m^{\lfloor t_m n \rfloor}\}$$

Notice that although $g^{\lfloor tn \rfloor}$ is not necessarily the inverse of $g^{\lfloor -tn \rfloor}$ if tn is not an integer, nevertheless these elements are distance at most $2l(g|W)$ apart in C_W , and therefore $\{g^{\lfloor tn \rfloor}\}$ and $\{g^{\lfloor -tn \rfloor}\}$ are inverse in $\hat{A}_W$.

DEFINITION 3.8. The *real cone* A_W is the image of $B_W \otimes \mathbb{R}$ in $\hat{A}_W$.

For the remainder of the paper we restrict attention to A_W . The metric on $\hat{A}_W$ restricts to a (bi-invariant) metric on A_W , consequently giving it the structure of a topological group. In the sequel, we use the notation $\|g\|$ for $d(\text{id}, g)$, where $g \in A_W$.

The real structure on $B_W \otimes \mathbb{R}$ gives rise to a natural (multiplicative) $\mathbb{R}$ action on A_W .

LEMMA 3.9. *There is a continuous family of endomorphisms $\mathbb{R} \times A_W \rightarrow A_W$ with the following properties:*

- (1) *the action is multiplicative — i.e. $1 \times$ is the identity on A_W , and $\lambda \times (\mu \times g) = (\lambda\mu) \times g$ for $\lambda, \mu \in \mathbb{R}$ and $g \in A_W$; and*
- (2) *for any $g \in A_W$ and $\lambda \in \mathbb{R}^*$, we have $\|\lambda \times g\| = |\lambda| \cdot \|g\|$*

PROOF. Let B be the group freely generated by expressions of the form g^t with $g \in G$ and $t \in \mathbb{R}$, subject to the relations $g^t g^s = g^{t+s}$. There is a natural $\mathbb{R}$ action on G , given by $\lambda \times g^t = g^{\lambda t}$. This action evidently preserves the W -subgroup of B , and therefore descends to an automorphism of $B_W \otimes \mathbb{R}$. We claim that this action preserves the kernel of ρ , and thereby defines an action on A_W . In other words, we need to show that if $\|\rho(g_1^{t_1} \cdots g_m^{t_m})\| = 0$ then $\|\rho(g_1^{\lambda t_1} \cdots g_m^{\lambda t_m})\| = 0$ for any λ . But

$$\begin{aligned} \|\rho(g_1^{\lambda t_1} \cdots g_m^{\lambda t_m})\| &= d(\{\text{id}\}, \{g_1^{\lfloor \lambda t_1 n \rfloor} \cdots g_m^{\lfloor \lambda t_m n \rfloor}\}) \\ &= \lim_{\omega} \frac{1}{n} l(g_1^{\lfloor \lambda t_1 n \rfloor} \cdots g_m^{\lfloor \lambda t_m n \rfloor} | W) \\ &= \lambda \cdot \lim_{\omega} \frac{1}{n} l(g_1^{\lfloor t_1 n \rfloor} \cdots g_m^{\lfloor t_m n \rfloor} | W) \\ &= \lambda \cdot \|\rho(g_1^{t_1} \cdots g_m^{t_m})\| \end{aligned}$$

(where the third line follows from the second by an approximate change of variables and elementary estimates) thereby establishing both claims. $\square$

The existence of this $\mathbb{R}$ action has the following topological consequence.

LEMMA 3.10. *A_W is contractible and locally contractible.*

PROOF. The $\mathbb{R}$ action defines a deformation retraction of A_W to the identity element, where each element g moves along the path $(1-t) \times g$ where t goes from 0 to 1. This retraction takes the ball of radius r around the identity inside itself for all positive t , so A_W is locally contractible. $\square$

Since $\widehat{A_W}$ obeys the law W , so does A_W . However, under certain circumstances we can say much more about A_W . The main theorem of this section is the following:

THEOREM 3.11. *Suppose A_W is nilpotent. Then it is a normed vector space (in particular it is abelian).*

PROOF. If A_W is abelian, then the existence of the $\mathbb{R}$ action with the properties proved in Lemma 3.9 shows that A_W is a normed vector space, with norm $\|\cdot\|$. So it suffices to show A_W is abelian.

For legibility, denote the n th element of the lower central series of A_W by A_n , so that $A_1 = A_W$ and $A_n = 0$ for some n . Suppose we have shown for some integer k , for all real t , for all $h \in A_k$ and all $g \in G_W$, that the commutator $[\rho(g^t), h] = \text{id}$. Since A_W is generated by elements of the form $\rho(g^t)$ this implies that $A_{k+1} = 0$.

Let $h \in A_{k-1}$. Then

$$\begin{aligned} [\rho(g^t), h] &= [\rho(g^{t/2}), h][\rho(g^{t/2}), h][[h, \rho(g^{t/2})], \rho(g^{t/2})^h] && \text{by Lemma 2.2} \\ &= [\rho(g^{t/2}), h][\rho(g^{t/2}), h] && \text{because } A_{k+1} = 0 \\ &= [\rho(g^{t/2}), h^2][[h, \rho(g^{t/2})], h]^{-1} && \text{by Lemma 2.2} \\ &= [\rho(g^{t/2}), h^2] && \text{because } A_{k+1} = 0 \end{aligned}$$

By induction, $[\rho(g^t), h] = [\rho(g^{t/2^m}), h^{2^m}]$ for every positive integer m . Hence

$$\|[\rho(g^t), h]\| = \|[\rho(g^{t/2^m}), h^{2^m}]\| = \|g^{t/2^m} \cdot (g^{-t/2^m})^* \| \leq 2 \cdot 2^{-m} \|g^t\|$$

Since m is arbitrary, $[\rho(g^t), h] = \text{id}$ and therefore (assuming $k \geq 2$), $A_k = 0$. Since A_W is nilpotent by hypothesis, $A_n = 0$ for some n , and therefore A_W is abelian, as claimed. $\square$

As a corollary, one obtains the following propositions.

PROPOSITION 3.12. *Suppose A_W is locally compact. Then it is a (finite dimensional) normed vector space.*

PROOF. Since A_W is connected, locally path connected (by Lemma 3.10) and locally compact by hypothesis, the Gleason-Montgomery-Zippin theorem (i.e. the affirmative solution of Hilbert's fifth problem [11, 16]) shows that A_W is a Lie group. Since it obeys a law, it is necessarily solvable, since non-solvable Lie groups contain nonabelian free subgroups.

By Theorem 3.11, it suffices to show it is nilpotent. Equivalently, we need to show that for each $g \in G_W$ and $t \in \mathbb{R}$ the conjugation action of $\rho(g^t)$ on the derived subgroup A' is trivial.

Suppose not, so that $\rho(g^t)$ is a 1-parameter family of isometric automorphisms of A' . Since A' is finite-dimensional, the group of isometric automorphisms is compact. In particular, there are arbitrarily large values of t for which $\rho(g^t)$ is arbitrarily close to the identity in the isometry group of A' .

By the compactness of the group of isometric automorphisms of A' there is a constant C so that if ι is an automorphism with $d(h, \iota(h)) < \epsilon$ for all h in the ball about the identity in A' of radius $1/2$, then $d(h, \iota(h)) < C \cdot \epsilon$ for all h in the ball about the identity in A' of radius 1 (in fact, since the metric on A' is bi-invariant, we can take $C = 2$, but this is unnecessary for our argument).

Now, for any positive ϵ , there are arbitrarily large t such that $d(h, h^{\rho(g^t)}) < \epsilon$ for all h in the ball of radius 1 in A' . Consequently, for all $s \in [1/2, 1]$ and all h in the ball of radius 1 in A' ,

$$d(s \times h, s \times (h^{\rho(g^t)})) = d(s \times h, (s \times h)^{\rho(g^{ts})}) < s\epsilon$$

It follows that $d(h, h^{\rho(g^{ts})}) < s\epsilon C$ for all h in the ball of radius 1 in A' , and all $s \in [1/2, 1]$. It follows that $d(h, h^{\rho(g^{t'})}) < 2\epsilon C$ for all $t' \in [0, t/2]$.

Since ϵ is arbitrarily small, since C is fixed, and since t can be chosen arbitrarily large for each ϵ , it follows that the conjugation action of $\rho(g^t)$ is trivial for all $g \in G_W$ and $t \in \mathbb{R}$. Hence A' is central, and therefore A_W is nilpotent, and therefore abelian and a finite dimensional normed vector space, as claimed. $\square$

REMARK 3.13. In fact, we do not really need the full power of Gleason-Montgomery-Zippin. We only need to know that the group of isometries of A' is compact (which follows from Arzela-Ascoli and local compactness) to deduce that A is abelian; then one may appeal to Pontriagin's solution of Hilbert's fifth problem for locally compact abelian groups.

Recall the notation $\beta_2 := [[x, y], [z, w]]$ for free generators x, y, z, w .

THEOREM 3.14. *Suppose in some group G that stable commutator length vanishes identically. Then A_W is abelian (and hence a normed vector space) for $W = \beta_2$.*

PROOF. By Theorem 3.11 it suffices to show that A_W is nilpotent. We show it satisfies the law $[x, [y, z]] = 0$.

Let f, g, h be arbitrary elements of A_W , and suppose $f = \rho(f_1^{t_1} f_2^{t_2} \cdots f_m^{t_m})$. We write $g = \{g_n\}$, $h = \{h_n\}$ and $f = \{f_1^{\lfloor nt_1 \rfloor} f_2^{\lfloor nt_2 \rfloor} \cdots f_m^{\lfloor nt_m \rfloor}\}$. By hypothesis, the

commutator length of each expression $f_i^{\lfloor nt_i \rfloor}$ is $o(n)$. It follows that

$$[f, [g, h]] = \{[f_1^{\lfloor nt_1 \rfloor} f_2^{\lfloor nt_2 \rfloor} \cdots f_m^{\lfloor nt_m \rfloor}, [g_n, h_n]]\} = \{o(n) \beta_2 - \text{words}\} = 0$$

where we use Lemma 2.2 bullet (4) in the second step. Hence A_W is nilpotent and therefore abelian. $\square$

REMARK 3.15. Recall that $\beta_1 = [x, y]$, so that β_1 -length is just commutator length. A similar argument shows that if stable β_n -length vanishes identically in G , then for $W = \beta_{n+1}$ the group A_W obeys the law $[x, \beta_n]$.

3.5. γ_n -quasimorphisms. For the remainder of this section we typically specialize to the case that $w = \gamma_n$ (recall $\gamma_2 = [x, y]$, $\gamma_3 = [x, [y, z]]$ and so on). Inspired by the phenomenon of (generalized) *Bavard duality* in the theory of stable commutator length, we make the following definition:

DEFINITION 3.16. A *homogeneous W -quasimorphism* (hereafter *W -hoq*) is a Lipschitz homomorphism from A_W to $\mathbb{R}$.

If ϕ is a W -hoq, let $d(\phi)$ denote the optimal Lipschitz constant.

For general W , it is not clear whether any homomorphisms from A_W to $\mathbb{R}$ exist, let alone Lipschitz ones. But when A_W is a normed vector space (which holds, for instance, if $W = \gamma_n$, by Theorem 3.11) the Hahn-Banach theorem guarantees the existence of a rich supply of W -hoqs.

In fact, we have the following proposition:

LEMMA 3.17. *For any n and any $g \in G_n$, there is an equality*

$$\text{sl}(g|\gamma_n) = \sup_{\phi} \phi(\rho(g))/d(\phi)$$

where the supremum is taken over all γ_n -hoqs ϕ .

PROOF. This follows from the fact that $\|\rho(g)\| = \text{sl}(g|\gamma_n)$, together with Theorem 3.11 and the Hahn-Banach theorem. $\square$

Whenever A_W is a vector space, it is spanned by the vectors $\rho(g)$ for $g \in G_W$. Since a W -hoq ϕ is a homomorphism, such a function on A_W is determined by its values on $\rho(g)$. Therefore, by abuse of notation, we think of a W -hoq in this case as a function on G_W ; thus the equality from proposition 3.17 would be expressed in the form $\text{sl}(g|\gamma_n) = \sup_{\phi} \phi(g)/d(\phi)$. On the other hand, it is not evident from the definition how to recover (or to estimate) $d(\phi)$ directly from the values of ϕ on the elements of G_W . We now address this issue.

DEFINITION 3.18. A *weak γ_n -hoq* is a function $\phi : G \rightarrow \mathbb{R}$ for which there is a least non-negative real number $D(\phi)$ (called the *defect*) satisfying the following properties:

- (1) (homogeneity) for any $g \in G$ and any $n \in \mathbb{Z}$ there is an equality $\phi(g^n) = n\phi(g)$;
- (2) (quasimorphism) for any $g, h \in G$ there is an inequality

$$|\phi(g) + \phi(h) - \phi(gh)| \leq D(\phi) \min(l(g|\gamma_{n-1}), l(h|\gamma_{n-1}), l(gh|\gamma_{n-1}))$$

REMARK 3.19. For $n = 2$, this is precisely the classical definition of a homogeneous quasimorphism. For, $\gamma_1 = x$ and therefore $l(g|\gamma_1) = 1$ if $g \neq \text{id}$ and 0 otherwise.

From the definition one can deduce some basic properties of weak γ_n -hoqs.

LEMMA 3.20. *Suppose G is perfect. Then any weak γ_n -hoq $\phi : G \rightarrow \mathbb{R}$ satisfies the following properties:*

- (1) ϕ is a class function;
- (2) if h is a γ_n -word, then $\phi(h) \leq D(\phi)$;
- (3) for any $g \in G$ there is an estimate

$$\phi(g) \leq (2l(g|\gamma_n) - 1)D(\phi)$$

and consequently

$$sl(g|\gamma_n) \geq \phi(g)/2D(\phi)$$

PROOF. The hypothesis that G is perfect implies (inductively) that $l(g|\gamma_m)$ is finite for every $g \in G$ and every integer m . To see that ϕ is a class function, observe that for any elements $g, h \in G$ and any n one has

$$|\phi(hg^n h^{-1}) - \phi(h) - \phi(g^n) - \phi(h^{-1})| \leq 2D(\phi)l(g|\gamma_{n-1})$$

By homogeneity, the left hand side is equal to $n \cdot |\phi(hgh^{-1}) - \phi(g)|$ whereas the right hand side is a constant independent of n , thus proving the first claim.

Now suppose h is a γ_n -word; i.e. $h = [x, y]$ for some x and y where y is a γ_{n-1} -word. Then

$$|\phi(h) - \phi(xy x^{-1}) - \phi(y^{-1})| \leq D(\phi)l(y|\gamma_{n-1}) = D(\phi)$$

On the other hand, since ϕ is a class function and homogeneous, the left hand side is equal to $|\phi(h)|$ and the inequality is proved.

Finally, if we express $g = h_1 h_n \cdots h_m$ where each h_i is a γ_n -word, then by induction $\phi(g) \leq \sum \phi(h_i) + (m-1)D(\phi) \leq (2m-1)D(\phi)$. This proves bullet (3). $\square$

PROPOSITION 3.21. *The set of weak γ_n -hoqs on G is a real vector space, which we denote ${}_nQ(G)$, and $D(\cdot)$ is a semi-norm. If G is perfect, ${}_nQ(G)$ is a Banach space with the norm $D(\cdot)$.*

PROOF. The defining properties of a weak γ_n -hoq are an (infinite) system of linear equalities and inequalities. It follows by inspection that ${}_nQ(G)$ is a real vector space, and that $D(\cdot)$ is a semi-norm.

If G is perfect, then $l(g|\gamma_{n-1})$ is finite for any g , and therefore $D(\phi) = 0$ if and only if ϕ is a homomorphism to $\mathbb{R}$, which is necessarily trivial for G perfect. Thus $D(\cdot)$ is a norm, and it remains to show that ${}_nQ(G)$ is complete in this norm.

So, let ϕ_i be a sequence of weak γ_n -hoqs for which $D(\phi_i - \phi_j)$ is a Cauchy sequence; i.e. for all $\epsilon > 0$ there is an N so that $D(\phi_i - \phi_j) < \epsilon$ for $i, j > N$. By bullet (3) from Lemma 3.20 it follows that the values of the ϕ_i on any element g form a Cauchy sequence, and therefore the ϕ_i converge pointwise to a limit ϕ . A pointwise limit of homogeneous functions is homogeneous; moreover, for any g, h pointwise convergence implies

$$|(\phi - \phi_j)(g + h - gh)| \leq \limsup_i D(\phi_i - \phi_j) \min(l(g|\gamma_{n-1}), l(h|\gamma_{n-1}), l(gh|\gamma_{n-1}))$$

and therefore ϕ is in ${}_nQ(G)$ and $D(\phi_i - \phi) \rightarrow 0$, and the lemma is thereby proved. $\square$

The following lemma justifies the terminology “weak γ_n -hoq”.

LEMMA 3.22. *If G is perfect, every γ_n -hoq is a weak γ_n -hoq with $D(\phi) \leq d(\phi)$.*

PROOF. Let $\phi : A_{\gamma_n} \rightarrow \mathbb{R}$ be a $d(\phi)$ -Lipschitz homomorphism, and by abuse of notation we let $\phi : G \rightarrow \mathbb{R}$ be the function defined by $\phi(g) := \phi(\rho(g))$. We estimate

$$\begin{aligned}
|\phi(g) + \phi(h) - \phi(gh)| &= |\phi(\rho(g * h * (h^{-1}g^{-1})))| \\
&\leq d(\phi) \|\{g^m h^m h^{-1}g^{-1} \dots h^{-1}g^{-1}\}\| \\
&= d(\phi) \lim_{m \rightarrow \infty} \frac{1}{m} l(g^m h^m h^{-1}g^{-1} \dots h^{-1}g^{-1} | \gamma_n) \\
&\leq d(\phi) \lim_{m \rightarrow \infty} \frac{1}{m} l([g, h]^* [g^2, h]^* \dots [g^{m-1}, h]^* | \gamma_n) \\
&\leq d(\phi) l(h | \gamma_{n-1})
\end{aligned}$$

By symmetry, we obtain the estimate

$$|\phi(g) + \phi(h) - \phi(gh)| \leq d(\phi) \min(l(g | \gamma_{n-1}), l(h | \gamma_{n-1}), l(gh | \gamma_{n-1}))$$

so that ϕ is a weak γ_n -hoq with $D(\phi) \leq d(\phi)$, as claimed. $\square$

We therefore obtain the following generalization of Bavard duality.

THEOREM 3.23 (γ_n -Duality theorem). *For any perfect group G , and any $g \in G$ there is an inequality*

$$\sup_{\phi} \phi(g)/D(\phi) \geq \text{sl}(g | \gamma_n) \geq \sup_{\phi} \phi(g)/2D(\phi)$$

where the supremum is taken over all weak γ_n -hoqs.

PROOF. The upper bound follows from Lemma 3.22 and Lemma 3.17, and the lower bound from Lemma 3.20. $\square$

An interesting application of this duality theorem is an *a priori* estimate of the ratio of $\text{scl}(\ast)$ and $\text{sl}(\ast | \gamma_n)$ in any perfect group. This estimate follows from the surprising fact that one may bound from below the stable γ_{n-1} length of a word from the value of a weak γ_n -hoq, providing $n \geq 3$.

LEMMA 3.24. *Let G be a perfect group. Then any weak γ_n -hoq $\phi : G \rightarrow \mathbb{R}$ satisfies the following properties:*

- (1) *if h is a γ_{n-1} -word, then $\phi(h) \leq D(\phi)$*
- (2) *for any $g \in G$ there is an estimate*

$$\phi(g) \leq (2l(g | \gamma_{n-1}) - 1)D(\phi)$$

and consequently

$$\text{sl}(g | \gamma_{n-1}) \geq \phi(g)/2D(\phi)$$

PROOF. Let g be arbitrary, and let k be a γ_{n-2} -word. There is a constant C so that for any m we have $l(g^m k^m (gk)^{-m} | \gamma_n) \leq Cm$ and therefore by Lemma 3.20 we can estimate

$$|\phi(g^m k^m (gk)^{-m})| \leq C'm$$

for some constant C' depending only on g , k and $D(\phi)$.

On the other hand, we can write

$$\begin{aligned}
g^m k^m (gk)^{-m} &= g^m k^m k^{-1} g^{-1} k^{-1} g^{-1} \dots k^{-1} g^{-1} \\
&= g^m k^{m-1} g^{-1} k^{-1} \dots k^{-1} g^{-1} \\
&= g^m k^{m-2} g^{-1} [g, k] g^{-1} k^{-1} \dots k^{-1} g^{-1} \\
&= g^m k^{m-2} g^{-1} g^{-1} k^{-1} \dots k^{-1} g^{-1} [g, k]^* \\
&= g^m k^{m-3} g^{-1} g^{-1} g^{-1} k^{-1} \dots k^{-1} g^{-1} [g, k]^* [g, k]^* [g, k]^* \\
&= \text{product of } m(m-1)/2 \text{ conjugates of } [g, k]
\end{aligned}$$

By abuse of notation, for any r we denote a product of r conjugates of $[g, h]$ by $[g, h]^{*r}$. Hence we have $g^m k^m (gk)^{-m} = [g, k]^{*m(m-1)/2}$. On the other hand, for any r ,

$$|\phi([g, k]^{*r}) - \phi([g, k]^{*(r-1)}) - \phi([g, k])| \leq D(\phi)$$

by the defining property of an γ_n -hoq, since ϕ is a class function, and $l(k|\gamma_{n-2}) = 1$ implies $l([g, k]|\gamma_{n-1}) \leq 1$. By induction and the triangle inequality we obtain

$$|\phi([g, k]^{*r}) - r\phi([g, k])| \leq rD(\phi)$$

Hence we conclude

$$|\phi(g^m k^m (gk)^{-m}) - (m(m-1)/2)\phi([g, k])| \leq (m(m-1)/2)D(\phi)$$

Since $|\phi(g^m k^m (gk)^{-m})| \leq C'm$, dividing by $m(m-1)/2$ and taking $m \rightarrow \infty$ gives the estimate

$$|\phi([g, k])| \leq D(\phi)$$

Since g was arbitrary, and k is an arbitrary γ_{n-2} -word, we have proved bullet (1). Bullet (2) follows exactly as in Lemma 3.20: if $g = h_1 h_2 \dots h_m$ where each h_i is a γ_{n-1} -word, then by induction

$$\phi(g) \leq \sum \phi(h_i) + (m-1)D(\phi) \leq (2m-1)D(\phi)$$

This completes the proof of the lemma. $\square$

From this easily follows our Comparison Theorem:

THEOREM 3.25 (Comparison theorem). *For any perfect group G , for any $n \geq 2$ and for any $g \in G$ there is an inequality*

$$2^{n-2} \text{scl}(g) \geq \text{sl}(g|\gamma_n) \geq \text{scl}(g)$$

PROOF. From Lemma 3.24 and Theorem 3.23 we estimate

$$2 \text{sl}(g|\gamma_{n-1}) \geq \sup_{\phi} \phi(g)/D(\phi) \geq \text{sl}(g|\gamma_n)$$

where the supremum is taken over all weak γ_n -hoqs ϕ . By induction on n , we get

$$2^{n-2} \text{scl}(g) \geq \text{sl}(g|\gamma_n)$$

The second inequality in the theorem is trivial, since every γ_n -word is a γ_2 -word. $\square$

It follows that weak γ_n -hoqs can be used to estimate scl:

COROLLARY 3.26. *Let G be a perfect group, and ϕ a weak γ_n -hoq. Then for any $g \in G$ we have*

$$\text{scl}(g) \geq \phi(g)/2^{n-1}D(\phi)$$

Stable commutator length has many important applications to geometry, topology, dynamics, etc. In general, estimating scl is very difficult; lower bounds are usually obtained from (ordinary homogeneous) quasimorphisms (equivalently, weak γ_n -hoqs when $n = 2$), and by now many interesting constructions of such quasimorphisms are known; see e.g. [6], especially Chapters 3 and 5. In view of Corollary 3.26, it becomes interesting to ask whether there are any natural constructions of weak γ_n -hoqs with $n > 2$, arising from geometry or dynamics.

The statement of the Comparison Theorem is purely algebraic, and it is therefore natural to try to find a purely algebraic proof, bypassing the construction of asymptotic cones, the use of Hahn-Banach, etc. In fact, it is not too hard to translate the geometric argument into an algebraic one, though the geometric argument has its own charm. The case of γ_3 is especially straightforward; with the same amount of work, one proves a slightly stronger statement.

First, in any group G , let Γ_3 denote the set of words of the form $[x, y]$ where $y \in [G, G]$. The Γ_3 words generate the subgroup G_3 (as do the γ_3 words, which are special examples of Γ_3 words), but there is no uniform comparison between γ_3 length and Γ_3 length. If G is perfect, then $G = [G, G] = G_2$, so for such groups, Γ_3 words are nothing other than commutators, and $l(*|\Gamma_3) = \text{cl}(*)$.

PROPOSITION 3.27. *In any group G , for any $g \in G_3$ there is an inequality*

$$2\text{scl}(g|\Gamma_3) \geq \text{scl}(g|\gamma_3) \geq \text{scl}(g|\Gamma_3)$$

PROOF. Recall (from Notation 2.12) the notation $\gamma_3^{\{k\}}$ for an arbitrary product of γ_3 -words.

Express g^k as a product of commutators

$$g^k = [a_1, b_1][a_2, b_2] \cdots [a_m, b_m]$$

where the b_i are all in $[G, G]$, and where m/k is as close to $\text{scl}(g|\Gamma_3)$ as we like. Then we have

$$g^{2^n k} = ([a_1, b_1] \cdots [a_m, b_m])^{2^n} = [a_1, b_1]^{2^n} \cdots [a_m, b_m]^{2^n} \gamma_3^{\{m2^n\}}$$

Now, for each i we have $[a_i, b_i]^2 = [a_i^2, b_i]\gamma_3^{\{1\}}$ and therefore

$$[a_i, b_i]^{2^n} = [a_i^2, b_i]^{2^{n-1}} \gamma_3^{\{2^{n-1}\}} = [a_i^{2^n}, b_i] \gamma_3^{\{2^n-1\}}$$

and therefore we can write

$$g^{2^n k} = [a_1^{2^n}, b_1][a_2^{2^n}, b_2] \cdots [a_m^{2^n}, b_m] \gamma_3^{\{m2^n-m\}}$$

We can estimate $l([a_1^{2^n}, b_1][a_2^{2^n}, b_2] \cdots [a_m^{2^n}, b_m]|\gamma_3) \leq \sum_i l(b_i|\gamma_2)$, which is a constant independent of n . Since m/k is as close as we like to $\text{scl}(g)$, and n is arbitrary, we deduce $\text{scl}(g|\gamma_3) \leq 2\text{scl}(g)$ as claimed. $\square$

We do not know whether there is an *a priori* comparison between $\text{scl}(*|\gamma_3)$ and $\text{scl}(*)$ on elements of G_3 for an arbitrary group.

3.6. β_2 -quasimorphisms. By Theorem 3.14, if G is a perfect group in which scl vanishes identically, then A_{β_2} is a vector space, and $\text{scl}(g|\beta_2) = \sup_{\phi} \phi(g)/d(\phi)$ where the supremum is taken over all β_2 -hoqs ϕ . We would like to obtain a “weak” characterization of β_2 -hoqs on such groups, analogous to the definition of weak γ_n -hoqs.

DEFINITION 3.28. A *weak* β_2 -hoq is a function $\phi : G \rightarrow \mathbb{R}$ for which there is a least non-negative real number $D(\phi)$ (called the *defect*) satisfying the following properties:

- (1) (homogeneity) for any $g \in G$ and any $n \in \mathbb{Z}$ there is an equality $\phi(g^n) = n\phi(g)$
- (2) (quasimorphism) for any $g, h \in G$ there is an inequality

$$|\phi(g) + \phi(h) - \phi(gh)| \leq D(\phi) \min(l(g|\beta_2), l(h|\beta_2), l(gh|\beta_2))$$

The following is the analogue of Lemma 3.22:

LEMMA 3.29. *If G is perfect and scl vanishes identically, every β_2 -hoq is a weak β_2 -hoq with $D(\phi) \leq 2d(\phi)$.*

PROOF. As in Lemma 3.22, we must estimate the β_2 -length of $g^m h^m (gh)^{-m}$. As before, we have an identity

$$g^m h^m (gh)^{-m} = [g, h]^* [g^2, h]^* \cdots [g^{m-1}, h]^*$$

We can write $h = [a_1, b_1] \cdots [a_r, b_r]$ where each a_i, b_i is a commutator, and $r = l(h|\beta_2)$. Then

$$[g^j, h] = [g^j, [a_1, b_1]]^* [g^j, [a_2, b_2]]^* \cdots [g^j, [a_r, b_r]]^*$$

By the Hall-Witt identity (i.e. bullet (5) from Lemma 2.2) we can write

$$[g^j, [a_i, b_i]] = [a_i^*, [b_i^*, (g^j)^*]] [b_i^*, [a_i^*, (g^j)^*]]$$

and since a_i, b_i are both commutators by hypothesis, the right hand side is a product of two β_2 -words. Hence

$$l(g^m h^m (gh)^{-m} | \beta_2) \leq 2ml(h|\beta_2)$$

By symmetry, the lemma follows. $\square$

The following is the analogue of Lemma 3.20, though the proof is more circuitous, and has something in common with that of Lemma 3.24:

LEMMA 3.30. *Suppose G is perfect, and scl vanishes identically in G . Then any weak β_2 -hoq $\phi : G \rightarrow \mathbb{R}$ satisfies the following properties:*

- (1) ϕ is a class function;
- (2) if h is a β_2 -word, then $\phi(h) \leq D(\phi)$;
- (3) for any $g \in G$ there is an estimate

$$\phi(g) \leq (2l(g|\beta_2) - 1)D(\phi)$$

and consequently

$$\text{sl}(g|\beta_2) \geq \phi(g)/2D(\phi)$$

PROOF. To see that ϕ is a class function, observe as in the proof of Lemma 3.20 that for any g, h and any n ,

$$|\phi(hg^n h^{-1}) - \phi(h) - \phi(g^n) - \phi(h^{-1})| \leq 2D(\phi)l(h|\beta_2)$$

By homogeneity, and the fact that the right hand side is constant independent of n , we see that $\phi(hgh^{-1}) = \phi(g)$; i.e. ϕ is a class function.

Secondly, observe that if $l(g|\beta_2) = 1$ then $\phi([g, h]) \leq D(\phi)$. This is because

$$|\phi(hgh^{-1}h^{-1}) - \phi(g) - \phi(hg^{-1}h^{-1})| \leq D(\phi)l(g|\beta_2) = D(\phi)$$

and because ϕ is a homogeneous class function. If we let α denote the word $[[u, v], [[x, y], [z, w]]]$ then α is both a β_2 -word, and the commutator of something with a β_2 word. Thus by induction, $|\phi(g)| \leq 2D(\phi)l(g|\alpha)$. Since G is perfect, every element has a finite α -length, so for any g and h and any integer m we have $|\phi(g^m h^m (gh)^{-m})| \leq Cm$ for some constant C depending only on g, h and $D(\phi)$.

On the other hand, as in the proof of Lemma 3.24, we can write $g^m h^m (gh)^{-m}$ as a product of $m(m-1)/2$ conjugates of $[g, h]$. If g and h are both commutators, $[g, h]$ is a β_2 -word, and therefore

$$|\phi(g^m h^m (gh)^{-m}) - (m(m-1)/2)\phi([g, h])| \leq (m(m-1)/2)D(\phi)$$

and consequently $|\phi([g, h])| \leq D(\phi)$. Since g and h are arbitrary commutators, this shows that $|\phi(g)| \leq D(\phi)$ for any β_2 -word g , proving the second claim.

The third claim follows immediately from this, as in the proof of Lemma 3.20. $\square$

THEOREM 3.31 (β_2 -Duality theorem). *For any perfect group G in which scl vanishes identically, and for any $g \in G$ there is an inequality*

$$\sup_{\phi} 2\phi(g)/D(\phi) \geq \text{sl}(g|\beta_2) \geq \sup_{\phi} \phi(g)/2D(\phi)$$

where the supremum is taken over all weak β_2 -hoqs.

PROOF. The upper bound follows from Lemma 3.29, and the lower bound from Lemma 3.30. $\square$

We conclude this section by making a curious observation on the relation between $\text{sl}(*|\beta_2)$ and $\text{l}(*|\gamma_3)$, under the hypothesis that scl vanishes identically.

PROPOSITION 3.32. *Suppose G is perfect, and scl vanishes identically. If $\text{l}(g|\gamma_3) = 1$ then $\text{sl}(g|\beta_2) \leq 1$.*

PROOF. Since for any a, b, c we have $[a, b][a, c] = [a, bc][[a, c], b]^*$ by bullet (3) of Lemma 2.2, it follows that

$$[x, [y, z]][x, [y, z]^n] = [x, [y, z]^{n+1}][[x, [y, z]^n], [y, z]]^*$$

for any n and any x, y, z . Since $[[x, [y, z]^n], [y, z]]$ is a β_2 -word, it follows by induction that $[x, [y, z]]^n$ can be written as a product of $(n-1)$ β_2 -words with $[x, [y, z]^n]$. If scl vanishes identically, $[y, z]^n$ can be written as a product of $o(n)$ commutators. If G is perfect, x is a product of a finite number of commutators. Hence the β_2 -length of $[x, [y, z]^n]$ is $o(n)$, and therefore $\text{sl}([x, [y, z]]|\beta_2) \leq 1$ as claimed. $\square$

It seems hard to generalize Proposition 3.32 to estimate $\text{sl}(*|\beta_2)$ from $\text{l}(*|\gamma_3)$.

3.7. Perfectness and virtual perfectness. Throughout this section we have usually made the assumption that $G = G_W$. When $W = \gamma_n$ for some n , this is equivalent to the statement that G is perfect. However it is evident that in most arguments it is sufficient to replace any given element g with a (fixed) power g^m . In particular, the theorems in this section remain true under the weaker hypothesis that for every $g \in G$ there is a positive integer m such that $g^m \in G_W$; equivalently, the quotient G/G_W is torsion. For general W this is implied by, but weaker than, the condition that G_W has finite index in G , even if G is finitely generated. But for $W = \gamma_n$, the two conditions are equivalent when G is finitely generated.

LEMMA 3.33. *For any n , the quotient G/G_n is torsion if and only if G/G_2 is torsion.*

PROOF. One direction is obvious. We prove the other direction. Suppose G/G_2 is torsion, and suppose by induction we have shown G/G_k is torsion for some $k \geq 2$. Let $g \in G_k$, and write $g = [a_1, b_1][a_2, b_2] \cdots [a_r, b_r]$ where each $b_r \in G_{k-1}$. Let m be such that $b_i^m \in G_k$ for all i . Then

$$\begin{aligned} g^m &= [a_1, b_1]^m \cdots [a_r, b_r]^m \pmod{G_{k+1}} \\ &= [a_1, b_1^m] \cdots [a_r, b_r^m] \pmod{G_{k+1}} \\ &= 0 \pmod{G_{k+1}} \end{aligned}$$

By induction, the lemma is proved. $\square$

Thus, Theorem 3.25 remains true with “perfect” replaced by “virtually perfect” (or even by: “group whose abelianization is torsion”).

Given a group G , one may attempt to obtain a lower bound on $\text{sl}(*|\gamma_n)$ on elements of G_n by embedding G in a perfect group H , and using Theorem 3.23 applied to H together with monotonicity of $\text{sl}(*|*)$ under homomorphisms. It is sometimes easier to take H to be virtually perfect rather than perfect; we shall see an example in § 4.

4. Grotes

The purpose of this section is to show how hyperbolic geometry can be used to give straightforward lower bounds on $\text{l}(*|\gamma_3)$. We give some examples that show that the uniform comparisons in Theorem 3.25 and Proposition 3.27 for *stable* lengths do not have any analogue for *unstable* length.

The arguments depend on the geometry of certain objects called *grotes*. We do not discuss here the most general kind of grotes, but only the simplest nontrivial examples. For a general introduction to grotes, see § 13 of [9]. Informally speaking, grotes topologize the commutator calculus, and questions about expressing elements in groups as products of γ_n (or β_n) words can be translated into questions about the existences of maps of certain kinds of grotes to spaces. We also use some elementary facts from the theories of $\text{CAT}(-1)$ complexes and pleated surfaces. A basic reference for the first is [2], especially pp. 347–362. A basic reference for the second in [22], Chapter 8.

Let S_n be an oriented surface of genus n with one boundary component. Let $S_{n,1}$ be obtained by attaching a once-punctured torus to each of a maximal collection of pairwise disjoint homologically essential loops β_i in S_n (there are n such). We denote by $\partial S_{n,1}$ the boundary of S_n (contained in $S_{n,1}$).

Now, let G be a group, and let X be a space with $\pi_1(X) = G$. If $g \in G$ is given, let $\gamma : S^1 \rightarrow X$ be a loop whose free homotopy class corresponds to the conjugacy class of g . Observe that from the definitions, $\text{l}(g|\gamma_3) \leq n$ if and only if there is a map $f : S_{n,1} \rightarrow X$ such that $\partial f : \partial S_{n,1} \rightarrow X$ factors through a homeomorphism $h : \partial S_{n,1} \rightarrow S^1$ in such a way that $\gamma \circ h = \partial f$. Informally, the γ_3 -length of g is at most n if and only if there is a map from $S_{n,1}$ to X whose boundary wraps around γ .

PROPOSITION 4.1. *Let F be free on the generators $x_1, y_1, \dots, x_n, y_n, z$ and let $w_n = [z, [x_1, y_1][x_2, y_2] \cdots [x_n, y_n]]$. Then $\text{l}(w_n|\gamma_3) \geq 2n/3$.*

PROOF. We build a $K(F, 1)$ (called X) as follows. Start with a hyperbolic once-punctured torus S with totally geodesic boundary. Let α be an embedded geodesic in S representing the meridian. Take another hyperbolic surface S' of genus n with totally geodesic boundary of length $\text{length}(\alpha)$, and attach $\partial S'$ isometrically to α . The resulting space X is a $K(F, 1)$, and the conjugacy class of w_n is represented by the boundary circle ∂S , which by abuse of notation, we denote ∂X .

Since X is obtained by gluing convex hyperbolic 2-complexes along convex subsets, it is itself a $\text{CAT}(-1)$ 2-complex. If $l(w_n|\gamma_3) = m$, there is a map $f : S_{m,1} \rightarrow X$ sending $\partial S_{m,1}$ to ∂X . We homotop this map to a *pleated representative* in a special way. First we choose an ideal triangulation of S_m for which the geodesic representatives of the β_i (with notation as above) are contained in the pleating locus λ . The map f can be homotoped on S_m to take each leaf of λ to a geodesic in X , and to be 1-Lipschitz on each ideal triangle of $S_m - \lambda$ for some hyperbolic metric on S_m . Then f can be homotoped rel. $\cup_i \beta_i$ to a pleated representative with respect to some hyperbolic metric on each once-punctured torus component of $S_{m,1} - S_m$.

The key property of a pleated map is that it is area non-increasing. Moreover, it is surjective onto X , since for any point $p \in X$, the conjugacy class of w_n is not in $[\pi_1(X - p), [\pi_1(X - p), \pi_1(X - p)]]$. By Gauss–Bonnet, $\text{area}(S_{m,1}) = 2\pi \cdot (3m - 1)$ and $\text{area}(X) = 2\pi \cdot 2n$. Hence $3m - 1 \geq 2n$ and therefore $m \geq 2n/3$, as claimed. $\square$

On the other hand, $\text{sl}(w_n|\gamma_3) \leq 1$ for all n , by Proposition 3.27. This example shows that the comparison theorem (Theorem 3.25) has no analogue for unstable γ_n -lengths, even if $n = 3$.

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Turn Graphs and Extremal Surfaces in Free Groups

Noel Brady, Matt Clay, and Max Forester

ABSTRACT. This note provides an alternate account of Calegari's rationality theorem for stable commutator length in free groups.

1. Introduction

The purpose of this note is to provide an alternate account of Calegari's main result from [4], establishing the existence of extremal surfaces for stable commutator length in free groups, via linear programming. The argument presented here is similar to that given in [4], except that we avoid using the theory of branched surfaces. Instead, the reduction to linear programming is achieved directly, using the combinatorics of words in the free group. We note that the specific linear programming problem resulting from the discussion here essentially agrees with that described in Example 4.34 of [3].

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2. Preliminaries

We start by giving a working definition of stable commutator length. Propositions 2.10 and 2.13 of [3] show that it is equivalent to the basic definition in terms of commutators or genus.

DEFINITION 1. Let $G = \pi_1(X)$ and suppose $\gamma: S^1 \rightarrow X$ represents the conjugacy class of $a \in G$. The *stable commutator length* of a is given by

$$(1) \quad \text{scl}(a) = \inf_S \frac{-\chi(S)}{2n(S)}$$

where S ranges over all singular surfaces $S \rightarrow X$ such that

- S is oriented and compact with $\partial S \neq \emptyset$
- S has no S^2 or D^2 components

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- the restriction $\partial S \rightarrow X$ factors through γ ; that is, there is a commutative diagram:

$$\begin{array}{ccc} \partial S & \longrightarrow & S \\ \downarrow & & \downarrow \\ S^1 & \xrightarrow{\gamma} & X \end{array}$$

- the restriction of the map $\partial S \rightarrow S^1$ to each connected component of ∂S is a map of *positive degree*

and where $n(S)$ is the total degree of the map $\partial S \rightarrow S^1$ (of oriented 1-manifolds).

A surface S satisfying the conditions above is called a *monotone admissible surface* in [3], abbreviated here as an *admissible surface*. Such a surface exists if and only if $a \in [G, G]$. If $a \notin [G, G]$ then by convention $\text{scl}(a) = \infty$ (the infimum of the empty set).

A surface $S \rightarrow X$ is said to be *extremal* if it realizes the infimum in (1). Notice that if this occurs, then $\text{scl}(a)$ is a rational number.

3. Singular surfaces in graphs

Let X be a graph with oriented 1-cells $\{e_\nu\}$. These edges may be formally considered as a generating set for the fundamental groupoid of X based at the vertices. These generators also generate the fundamental group F of X . Note that F is free, but the groupoid generators are not a basis unless X has only one vertex. (The reader may assume this latter property with no harm, in which case the fundamental groupoid is simply the fundamental group.)

Let $\gamma: S^1 \rightarrow X$ be a simplicial loop with no backtracking. There is a corresponding cyclically reduced word $w = x_1 \cdots x_\ell$ in the fundamental groupoid generators and their inverses. This word w represents a conjugacy class in $\pi_1(X)$, which we assume to be in $[F, F]$. Finally, let $S \rightarrow X$ be an admissible surface for w , as in Definition 1.

We are interested in computing $\chi(S)$ and $n(S)$, to estimate $\text{scl}(w)$ from above. We are free to modify S if the resulting surface S' satisfies $\frac{-\chi(S')}{2n(S')} \leq \frac{-\chi(S)}{2n(S)}$, since this only strengthens the estimate.

Using transversality, the map $S \rightarrow X$ can be homotoped into a standard form, sometimes called a *transverse map* [2]. The surface is decomposed into pieces called 1-handles, which map to edges of X , and complementary regions, which map to vertices of X . Each 1-handle is a tubular neighborhood of a connected 1-dimensional submanifold, either an arc with endpoints on ∂S or a circle. The submanifold maps to the midpoint of an edge of X , and the fibers of the tubular neighborhood map over the edge, through its characteristic map. In particular, the boundary arcs or circles of the 1-handle (comprised of endpoints of fibers) map to vertices of X . A *transverse labeling* is a labeling of the fibers of 1-handles by fundamental groupoid generators, indicating which edge of X (and in which direction) the handle maps to. For more detail on putting maps into this form, see for instance [6, 7, 5, 1].

Let $S_0 \subset S$ be the codimension-zero submanifold obtained as the closure of the union of a collar neighborhood of ∂S and the 1-handles that meet ∂S . We will see that S_0 is the essential part of S , containing all of the relevant information. It is determined completely by ∂S , together with the additional data of which pairs

of edges in ∂S are joined by 1–handles. Note that ∂S_0 consists of ∂S together with additional components in the interior of S . These latter components will be called the *inner boundary* of S_0 , denoted $\partial^- S_0$. Let S_1 be the closure of $S - S_0$. Note that $\partial S_1 = S_1 \cap S_0 = \partial^- S_0$. Figure 1 shows an example of S_0 for the word $w = aba^{-1}b^{-1}$. (The “turns” mentioned there are discussed in the next section.)

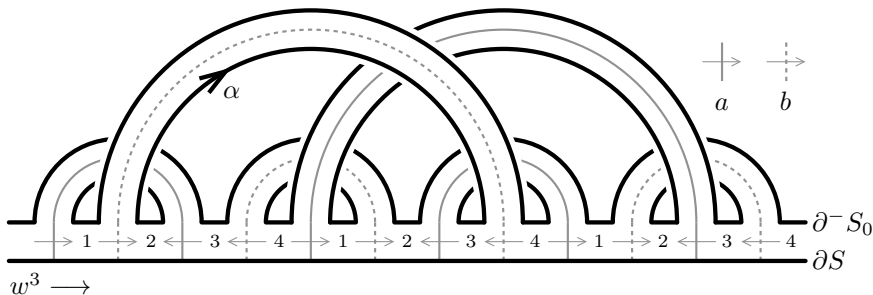


FIGURE 1. One possible S_0 for the word $w = aba^{-1}b^{-1}$, with outer boundary w^3 . There are four turns, indicated by the numbers, each occurring three times. The boundary arc α leads from turn 2 to turn 3. This surface leads to an estimate $\text{scl}(w) \leq 1/2$ (and in fact is extremal).

How large can $\chi(S)$ be? Note that $\chi(S) = \chi(S_0) + \chi(S_1)$ since S_0 and S_1 meet along circles. Also,

$$\chi(S_0) = \frac{-n(S)|w|}{2}$$

as can be seen by counting the 1–handles meeting ∂S : each 1–handle contributes -1 to $\chi(S_0)$ and occupies two edges in ∂S , of which there are $n(S)|w|$ in total. Finally, given S_0 , the quantity $\chi(S_1)$ is largest when S_1 is a collection of disks. The number of disks is simply the number of components of $\partial^- S_0$. We can always replace S_1 by disks, since each component of ∂S_1 maps to a vertex of X and disks can be mapped to vertices also. Thus, after this modification, we have

$$(2) \quad \chi(S) = \frac{-n(S)|w|}{2} + |\pi_0(\partial^- S_0)|,$$

and therefore an upper bound for $\text{scl}(w)$ is given by

$$(3) \quad \frac{-\chi(S)}{2n(S)} = \frac{|w|}{4} - \frac{|\pi_0(\partial^- S_0)|}{2n(S)}.$$

Indeed, $\text{scl}(w)$ is precisely the infimum of the right hand side of (3) over all surfaces S_0 arising as above. (Note that $n(S)$ is determined by S_0 .) Equation (3) essentially replaces the quantity $\chi(S)$ by the number of inner boundary components of S_0 in the computation of $\text{scl}(w)$.

4. The turn graph

To help keep track of the inner boundary $\partial^- S_0$, we define the *turn graph*. Consider the word $w = x_1 \cdots x_\ell$. A *turn* in w is a position between two letters of w considered as a cyclic word. Turns are indexed by the numbers 1 through ℓ , with turn i being the position just after the letter x_i . Each turn is labeled by the length

two subword $x_i x_{i+1}$ (or $x_\ell x_1$) of w which straddles the turn. Note that turns are not necessarily determined by their labels.

The *turn graph* $\Gamma(w)$ is a directed graph with vertices equal to the turns of w , and with a directed edge from turn i to turn j if $x_i^{-1} = x_{j+1}$. That is, if the label of a turn begins with the letter $a^{\pm 1}$, then there is a directed edge from this turn to every other turn whose label ends with $a^{\mp 1}$. Note that because w is cyclically reduced, $\Gamma(w)$ has no loops.

The turn graph has a two-fold symmetry, or duality: if $e \in \Gamma(w)$ is an edge from turn i to turn j , then one verifies easily that there is also an edge $\bar{e}$ from turn $j+1$ to turn $i-1$, and moreover $\bar{\bar{e}} = e$. Figure 2 shows a turn graph and a dual edge pair.

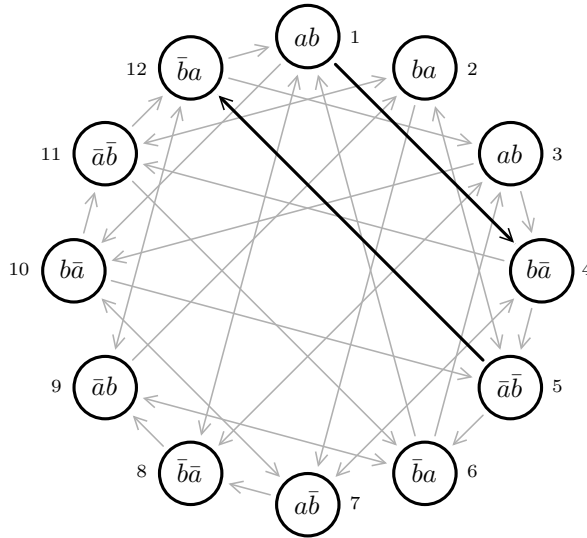
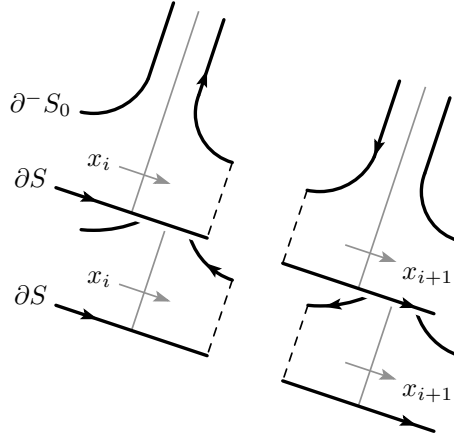


FIGURE 2. The turn graph for the word $w = abab\bar{a}b\bar{a}b\bar{a}b\bar{a}b$ (bar denotes inverse). The highlighted edges form a dual pair.

Turn circuits. Given the surface S_0 , each inner boundary component can be described as follows. Traversing the curve in the positively oriented direction, one alternately follows 1–handles and visits turns of w positioned along ∂S ; see again Figure 1 (this situation is the reason for the word “turn”). If a 1–handle leads from turn i to turn j , then the 1–handle bears the transverse label $x_i^{-1} = x_{j+1}$, and so there is an edge in $\Gamma(w)$ from turn i to turn j . The sequence of 1–handles traversed by the boundary component therefore yields a directed circuit in $\Gamma(w)$. In this way the inner boundary $\partial^- S_0$ gives rise to a finite collection (possibly with repetitions) of directed circuits in $\Gamma(w)$, called the *turn circuits* for S_0 .

Recall that ∂S is labeled by $w^{n(S)}$ (possibly spread over several components), so there are $n(S)$ occurrences of each turn on ∂S . The turn circuits do not contain the information of which particular instances of turns are joined by 1–handles.

Turn surgery. There is a move one can perform on S_0 which is useful. Given two occurrences of turn i in S_0 , cut the collar neighborhood of ∂S open along arcs positioned at the two turns, between the adjacent 1–handles; see Figure 3. These arcs both map to the same vertex of X . Now re-glue the four sides of the arcs,

FIGURE 3. Cutting along two instances of turn i .

switching two of them. There is one way to do this which preserves orientations of S and of $\partial^- S_0$. The new surface is still admissible (that is, after capping off $\partial^- S_0$) and $n(S)$ is preserved.

The move changes both ∂S and $\partial^- S_0$, in each case either increasing or decreasing the number of connected components by one. If both instances of the turn occupy the same component, then the move splits this component into two, with each occupied by one of the turns. Otherwise, the move joins the two components occupied by the turns into one.

DEFINITION 2. An admissible surface S is *taut* if every component of $\partial^- S_0$ visits each turn at most once. In terms of the turn graph, this means that each turn circuit for S_0 is embedded in $\Gamma(w)$ (though distinct circuits are allowed to cross). Let $\mathcal{T}(w)$ be the set of taut admissible surfaces for w .

Any admissible surface S can be made taut by performing a finite number of turn surgeries, each increasing the number of inner boundary components of ∂S_0 . Since $n(S)$ remains constant, the quantity (3) will only decrease. Hence we have the following result:

LEMMA 3. *There is an equality*

$$\text{scl}(w) = \inf_{S \in \mathcal{T}(w)} \frac{|w|}{4} - \frac{|\pi_0(\partial^- S_0)|}{2n(S)}.$$

5. Weight vectors and linear optimization

Let $\{\alpha_1, \dots, \alpha_k\}$ be the set of embedded directed circuits in $\Gamma(w)$. For each taut admissible surface S let $u_i(S)$ be the number of occurrences of α_i among the turn circuits of S_0 , and let $u(S) \in \mathbb{R}^k$ be the non-negative integer vector $(u_1(S), \dots, u_k(S))$. We call $u(S)$ the *weight vector* for S .

For each vertex v and edge e of $\Gamma(w)$, there are linear functions

$$F_v: \mathbb{R}^k \rightarrow \mathbb{R}, \quad F_e: \mathbb{R}^k \rightarrow \mathbb{R}$$

whose values on the i^{th} standard basis vector $(0, \dots, 0, 1, 0, \dots, 0)$ are given by the number of times α_i passes through the vertex v (respectively, over the edge e).

Since α_i is embedded, these numbers will be 0 or 1, although this is not important. For the taut surface S , if $e \in \Gamma(w)$ is an edge from turn i to turn j , then $F_e(u(S))$ counts the number of times $\partial^- S_0$ follows a 1-handle from turn i to turn j . Similarly, if $v \in \Gamma(w)$ is turn i , then $F_v(u(S))$ counts the number of occurrences of turn i on $\partial^- S_0$ (which is $n(S)$, as observed earlier).

REMARK 4. For taut admissible surfaces, the functions $|\pi_0(\partial^- S_0)|$ and $n(S)$ both factor as

$$\mathcal{T}(w) \xrightarrow{u} \mathbb{R}^k \longrightarrow \mathbb{R}$$

where the second map is linear, with integer coefficients. In the case of $|\pi_0(\partial^- S_0)|$ the second map is given by $(u_1, \dots, u_k) \mapsto \sum_i u_i$, and in the case of $n(S)$, the second map is simply F_v (for any vertex $v \in \Gamma(w)$).

By (2) it follows that the function $-\chi(S)$ also factors as above, through an integer coefficient linear function $\mathbb{R}^k \rightarrow \mathbb{R}$.

LEMMA 5. *Every weight vector $u(S)$ satisfies the linear equation*

$$F_e(u(S)) = F_{\bar{e}}(u(S))$$

for every dual pair $e, \bar{e}$ of edges in $\Gamma(w)$.

PROOF. Suppose e leads from turn i to turn j (so $\bar{e}$ leads from turn $j+1$ to turn $i-1$). If a 1-handle has a boundary arc representing e then the other side of the 1-handle represents $\bar{e}$. Hence both sides of the equation count the number of 1-handles of S_0 joining occurrences of x_i and x_{j+1} in ∂S . $\square$

This lemma has a converse:

PROPOSITION 6. *If $u \in \mathbb{R}^k - \{0\}$ has non-negative integer entries and satisfies the linear equations*

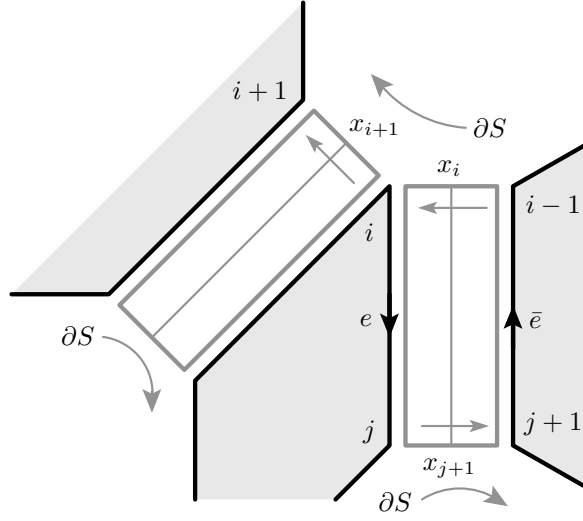
$$(4) \quad F_e(u) = F_{\bar{e}}(u) \quad \text{for all dual pairs } e, \bar{e}$$

then u is the weight vector of a taut admissible surface.

PROOF. Suppose $u = (u_1, \dots, u_k)$. For each i let D_i be a polygonal disk with $|\alpha_i|$ sides. Label the oriented boundary of D_i by the edges and vertices of α_i . That is, sides are labeled by edges of $\Gamma(w)$, and corners are labeled by turns. Note that there are no monogons, since $\Gamma(w)$ has no loops. To form the taut admissible surface S , take u_i copies of D_i for each i . For each dual edge pair $e, \bar{e}$ the total number of edges labeled e among the D_i 's will equal the number of edges labeled $\bar{e}$, by (4). Hence the sides of the disks can be joined in dual pairs to form a closed oriented surface.

However, this is not how S is formed. Instead, whenever two disks were to be joined along sides labeled e and $\bar{e}$, insert an oriented rectangle, with sides labeled by $e, x_{j+1}, \bar{e}, x_i$ (here, e leads from turn i to turn j , and $\bar{e}$ from turn $j+1$ to turn $i-1$). See Figure 4. The opposite sides labeled by e and $\bar{e}$ are joined to the appropriate sides of the disks, and the remaining two sides become part of the boundary of S . Each rectangle can be transversely labeled by a fundamental groupoid generator (equal to $x_i^{-1} = x_{j+1}$), and then the rectangles become 1-handles in the resulting surface S .

Note that the side of a rectangle labeled x_i has neighboring polygonal disk corners labeled $i-1$ and i . Following this edge along ∂S , the next edge must be labeled x_{i+1} (adjacent to i and $i+1$); see again Figure 4. Hence each component

FIGURE 4. Building S from $(u_1, \dots, u_k)$.

of ∂S is labeled by a positive power of w . There are no S^2 components since no component of S is closed, and no D^2 components, since an outermost 1-handle on such a disk would have to bound a monogon. The map $S \rightarrow X$ is defined on the rectangles according to the transverse labels (each maps to an edge of X) and the disks map to vertices. Now S is admissible, and by construction, the turn circuits will all be instances of the circuits α_i , so S is taut. $\square$

THEOREM 7 (Calegari). *If X is a graph and $a \in [\pi_1(X), \pi_1(X)]$ then there exists an extremal surface $S \rightarrow X$ for a . Moreover, there is an algorithm to construct S . In particular, $\text{scl}(a)$ is rational and computable.*

PROOF. Let w be the cyclically reduced word representing the conjugacy class of a , as defined in Section 3. By Remark 4 the function $\frac{-\chi(S)}{2n(S)}$ factors as

$$\mathcal{T}(w) \xrightarrow{u} \mathbb{R}^k \longrightarrow \mathbb{R}$$

where the second map is a ratio of linear functions $A(u)/B(u)$ with integer coefficients. Let $P \subset \mathbb{R}^k$ be the polyhedron defined by the (integer coefficient) linear equations (4) and the inequalities $B(u) > 0$ and $u_i \geq 0$, $i = 1, \dots, k$. Lemma 5 and Proposition 6 together imply that the image of $u: \mathcal{T}(w) \rightarrow \mathbb{R}^k$ is precisely $P \cap \mathbb{Z}^k$. Hence

$$\text{scl}(w) = \inf_{u \in P \cap \mathbb{Z}^k} A(u)/B(u).$$

Note that P and $A(u)/B(u)$ are projectively invariant. Normalizing $B(u)$ to be 1, we have

$$(5) \quad \text{scl}(w) = \inf_{u \in P' \cap \mathbb{Q}^k} A(u)$$

where P' is the rational polyhedron $P \cap B^{-1}(1)$. Note that P' is a closed set.

From Remark 4 and equation (2) the function A is given by

$$A(u) = \frac{|w|}{2} F_v(u) + \sum_i u_i,$$

which has strictly positive values on the standard basis vectors. Hence A achieves a minimum on P' , along a non-empty rational sub-polyhedron. The vertices of this sub-polyhedron are rational points realizing the infimum in (5). Hence there exist extremal surfaces for w , and $\text{scl}(w)$ is rational. An extremal surface can be constructed explicitly from a rational solution $u \in P' \cap \mathbb{Q}^k$, by first multiplying by an integer to obtain a minimizer for $A(u)/B(u)$ in $P \cap \mathbb{Z}^k$, and then applying the procedure given in the proof of Proposition 6. Lastly, we note that from the word w it is straightforward to algorithmically construct the turn graph $\Gamma(w)$, the equations (4), and the polyhedron P' . $\square$

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Kauffman brackets, character varieties and triangulations of surfaces

Francis Bonahon and Helen Wong

Dedicated to William ‘Bus’ Jaco

ABSTRACT. A Kauffman bracket on a surface is an invariant for framed links in the thickened surface, satisfying the Kauffman skein relation and multiplicative under superposition. This includes representations of the skein algebra of the surface. We show how an irreducible representation of the skein algebra usually specifies a point of the character variety of homomorphisms from the fundamental group of the surface to $\mathrm{PSL}_2(\mathbb{C})$, as well as certain weights associated to the punctures of the surface. Conversely, we sketch a proof of the fact that each point of the character variety, endowed with appropriate puncture weights, uniquely determines a Kauffman bracket. Details will appear elsewhere.

This article closely follows the talk given by one of us at the conference *Topology and geometry in dimension three: triangulations, invariants and geometric structures*. It turns out to feature all three of the scientific themes emphasized in the title of the conference. The main object under study consists of Kauffman brackets, which are invariants of links in 3-manifolds. We will connect these to hyperbolic metrics on a thickened surface, and finally use triangulations of surfaces as a tool in our constructions. The article describes a long-term program, and focusses more on statements than proofs. Details will appear elsewhere [4, 5, 6].

The two authors were greatly honored to be invited to this great celebration of 3-dimensional topology and of the achievements of one of its major contributors, and greatly enjoyed their participation. They are very pleased to dedicate this article to Bus Jaco, as a grateful acknowledgment of his impact on their field.

1. Kauffman brackets and the skein algebra

1.1. The classical Kauffman bracket. For a fixed number $A \in \mathbb{C}^*$, the classical *Kauffman bracket* is the unique map

$$\mathcal{K}: \{\text{framed links } K \subset S^3\} \rightarrow \mathbb{C}$$

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that satisfies the following four properties.

- (1) ISOTOPY INVARIANCE: If the framed links K_1 and K_2 are isotopic, then $\mathcal{K}(K_1) = \mathcal{K}(K_2)$.
- (2) SKEIN RELATION: Let the three framed links K_1 , K_0 and $K_\infty \subset S^3$ form a *Kauffman triple*, in the sense that the only place where they differ is in a small ball of S^3 , where they are as represented in Figure 1 and where the framings are all pointing towards the reader. Then

$$\mathcal{K}(K_1) = A^{-1}\mathcal{K}(K_0) + A\mathcal{K}(K_\infty).$$

- (3) SUPERPOSITION RELATION: Let $K_1 \cdot K_2$ be the framed link obtained by stacking K_2 on top of K_1 , as in Figure 2. Then

$$\mathcal{K}(K_1 \cdot K_2) = \mathcal{K}(K_1)\mathcal{K}(K_2).$$

- (4) NON-TRIVIALITY: There exists a non-empty link K such that $\mathcal{K}(K) \neq 0$.

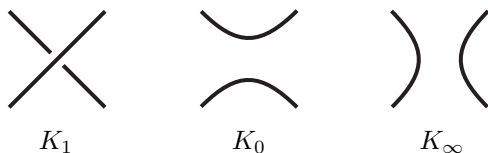


FIGURE 1. A Kauffman triple

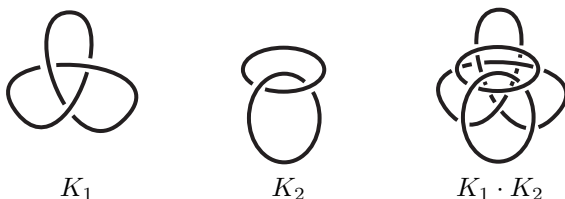


FIGURE 2. Superposing two links

Recall that a *framing* for a link K in S^3 , or more generally in any 3-dimensional manifold, is the choice at each $x \in K$ of a vector v_x that depends differentiably on x and is never tangent to K .

In general, it is customary to replace the Isotopy Invariance Condition and the Non-Triviality Condition by the condition that $\mathcal{K}(O) = -A^2 - A^{-2}$ when O is a circle in $\mathbb{R}^2 \subset \mathbb{R}^3$ with vertical framing. The two conventions are easily seen to be equivalent; see for instance [32, Lemmas 3.2 and 3.3].

L. Kauffman [31] proved the following fundamental result.

THEOREM 1. *For every $A \in \mathbb{C}^*$, there exists a unique Kauffman bracket*

$$\mathcal{K}: \{\text{framed links } K \subset S^3\} \rightarrow \mathbb{C}.$$

□

In addition, it easily follows from the skein relation and the invariance under isotopy that the Kauffman bracket $\mathcal{K}(K)$ of a framed link K is a Laurent polynomial in A with integer coefficients.

1.2. Kauffman brackets on a surface. The theory of knots and links in S^3 can be described from a purely 2-dimensional point of view, by considering their projections to the plane up to Reidemeister moves. In this regard, the existence and uniqueness of the Kauffman bracket can be considered as a property of the plane, or of the disk. It is then tempting to extend this property to more general surfaces.

More precisely, let S be a compact oriented surface, possibly with boundary. We can then consider “pictures of knots and links” on the surface S , considered up to the Reidemeister moves II and III. Such a link diagram in S determines an isotopy class of framed links in the thickened surface $S \times [0, 1]$, where the framing is vertical pointing upwards, namely parallel to the $[0, 1]$ factor and pointing in the direction of 1. (The Reidemeister move I is excluded as it would alter the framing.)

The Skein Relation and the Superposition Relation have automatic extensions to this context of framed links in the thickened surface $S \times [0, 1]$. More precisely, a *Kauffman triple* in $S \times [0, 1]$ consists of three framed links K_1, K_0, K_∞ that differ only above a small disk in S , where they are represented by the link diagrams of Figure 1 (with vertical framing pointing upwards).

Also, if K_1 and K_2 are two framed links in $S \times [0, 1]$, we can consider the framed links $K'_1 \subset S \times [0, \frac{1}{2}]$ and $K'_2 \subset S \times [\frac{1}{2}, 1]$ respectively obtained from K_1 and K_2 by rescaling in the $[0, 1]$ direction. Then the *superposition* of K_2 above K_1 is the framed link $K_1 \cdot K_2 = K'_1 \cup K'_2$. This is illustrated in Figure 3 in the case where the surface S is a twice punctured torus. Note that, in contrast with the case of framed links in S^3 , the superposition $K_2 \cdot K_1$ is not always isotopic to $K_1 \cdot K_2$ in $S \times [0, 1]$.

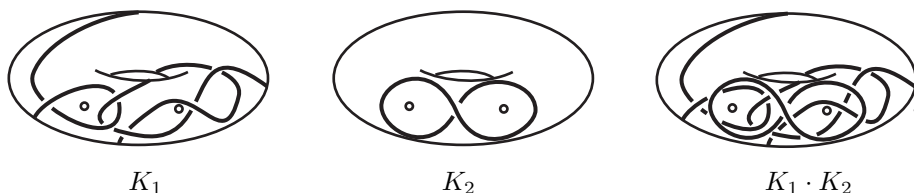


FIGURE 3. Superposition in $S \times [0, 1]$

It is now natural to introduce the following definition.

DEFINITION 2. For a fixed $A \in \mathbb{C}^*$, a *Kauffman bracket* on the surface S , valued in the algebra $\mathcal{A}$, is a map

$$\mathcal{K}: \{\text{framed links } K \subset S \times [0, 1]\} \rightarrow \mathcal{A}$$

satisfying the following four properties.

- (1) **ISOTOPY INVARIANCE:** If the framed links K_1 and K_2 are isotopic in $S \times [0, 1]$, then $\mathcal{K}(K_1) = \mathcal{K}(K_2)$.
- (2) **SKEIN RELATION:** If the three framed links K_1, K_0 and $K_\infty \subset S^3$ form a Kauffman triple in $S \times [0, 1]$, then

$$\mathcal{K}(K_1) = A^{-1}\mathcal{K}(K_0) + A\mathcal{K}(K_\infty).$$

- (3) **SUPERPOSITION RELATION:** If $K_1 \cdot K_2$ is the framed link obtained by stacking K_2 above K_1 in $S \times [0, 1]$, then

$$\mathcal{K}(K_1 \cdot K_2) = \mathcal{K}(K_1)\mathcal{K}(K_2).$$

- (4) **NON-TRIVIALITY:** There exists a non-empty framed link $K \subset S \times [0, 1]$ such that $\mathcal{K}(K) \neq 0$.

Again, the combination of the Isotopy Invariance Condition and the Non-Triviality Condition is essentially equivalent to the property that $\mathcal{K}(O) = -A^2 - A^{-2}$, when O is a circle embedded in $S \times \{\frac{1}{2}\} \subset S \times [0, 1]$ with vertical framing.

In view of Theorem 1, it is now natural to aim for the following

MAIN GOAL. *Classify all possible Kauffman brackets for the surface S .*

This goal will turn out to be too optimistic at this point, but at least we will construct many examples of interesting Kauffman brackets.

1.3. The skein algebra. We first rephrase the definition of Kauffman brackets, following [37, 40, 7, 9, 38, 41, 11].

DEFINITION 3. The *skein algebra* $\mathcal{S}^A(S)$ of the surface S is defined by considering the vector space freely generated (over $\mathbb{C}$) by the set of all framed links in $S \times [0, 1]$, and then by taking the quotient of this vector space by the subspace generated by elements of the following two types:

$$\begin{aligned} K_1 - K_2 & \text{ for every pair of isotopic framed links } K_1, K_2 \subset S \times [0, 1]; \\ K_1 - A^{-1}K_0 - AK_\infty & \text{ for every Kauffman triple } K_1, K_0, K_\infty \subset S \times [0, 1]. \end{aligned}$$

The multiplication of the algebra $\mathcal{S}^A(S)$ is induced by the superposition operation, with $[K_1][K_2] = [K_1 \cdot K_2]$ for any two framed links $K_1, K_2 \subset S \times [0, 1]$.

A *skein* is a class $[K] \in \mathcal{S}^A(S)$ represented by a framed link $K \subset S \times [0, 1]$. Note that the empty skein $[\emptyset]$ is a unit in the skein algebra $\mathcal{S}^A(S)$.

The definition of the skein algebra is specially designed so that the following holds.

FACT 4. *A Kauffman bracket valued in the algebra $\mathcal{A}$ is the same thing as an algebra homomorphism $\mathcal{K}: \mathcal{S}^A(S) \rightarrow \mathcal{A}$ that is non-trivial, in the sense that there exists a non-empty skein $[K] \in \mathcal{S}^A(S)$ with $\mathcal{K}([K]) \neq 0$.* $\square$

Note that, when $\mathcal{A}$ contains an idempotent element $\iota \in \mathcal{A}$ (for instance a unit element), the skein algebra always admits a *trivial* algebra homomorphism $\mathcal{S}^A(S) \rightarrow \mathcal{A}$ sending the unit $[\emptyset]$ to ι and every other skein $[K] \neq [\emptyset] \in \mathcal{S}^A(S)$ to 0.

Our Main Goal now becomes:

MAIN GOAL. *Classify all non-trivial algebra homomorphisms $\mathcal{K}: \mathcal{S}^A(S) \rightarrow \mathcal{A}$.*

One problem appears right away: there really exists a great variety of algebras $\mathcal{A}$. For instance, we could take $\mathcal{A}$ equal to the skein algebra $\mathcal{S}^A(S)$ and $\mathcal{K}$ equal to the identity map (and call it the universal Kauffman bracket!), but this would clearly have little impact on our understanding of links in $S \times [0, 1]$.

It therefore makes sense to restrict attention to a specific class of algebras, namely to algebras $\mathcal{A} = \text{End}(V)$ of linear endomorphisms of a finite dimensional vector space V . Choosing a basis for V , such an algebra is just an algebra of square matrices of a given size, and consequently is very explicit.

Thus, we henceforth limit our investigation to algebra homomorphisms $\mathcal{K}: \mathcal{S}^A(S) \rightarrow \text{End}(V)$, where V is a finite dimensional vector space. Such an algebra homomorphism is a *representation* of the algebra $\mathcal{S}^A(S)$, and we will reflect this change in emphasis by switching from the letter $\mathcal{K}$ (for “Kauffman bracket”) to ρ (for “representation”) in the notation.

We will also restrict our investigation to irreducible representations, as building blocks of more general representations. Recall that a representation $\rho: \mathcal{S}^A(S) \rightarrow \text{End}(V)$ is *irreducible* if there is no proper subspace $W \subset V$ that is invariant under the image of ρ .

REVISED GOAL. *Classify all irreducible representations $\rho: \mathcal{S}^A(S) \rightarrow \text{End}(V)$ of the skein algebra $\mathcal{S}^A(S)$.*

This goal now becomes much more realistic. Although we are not able to completely attain it at this point, we will make progress in this direction. We begin by discussing in §2 and §3 a few simple cases where the program has been completed. In §§4–5 we describe a conjectural classification of irreducible representations $\rho: \mathcal{S}^A(S) \rightarrow \text{End}(V)$, and state an existence result in Theorem 14. We conclude with a few indications on its proof in §6.

2. The case where $A = \pm 1$

Surprisingly enough, when $A = \pm 1$, Kauffman brackets are related to group homomorphisms from the fundamental group $\pi_1(S)$ to $\text{SL}_2(\mathbb{C})$ or $\text{PSL}_2(\mathbb{C})$.

It is an immediate consequence of the Skein Relation that the skein algebra $\mathcal{S}^{\pm 1}(S)$ is commutative. Elementary linear algebra then shows that every representation $\mathcal{S}^{\pm 1}(S) \rightarrow \text{End}(V)$ splits as a direct sum of irreducible representations, and that every irreducible representation has dimension 1. As a consequence, every representation is isomorphic to a direct sum of representations $\rho: \mathcal{S}^{\pm 1}(S) \rightarrow \text{End}(\mathbb{C}) = \mathbb{C}$.

2.1. The case where $A = -1$. Consider a group homomorphism $r: \pi_1(S) \rightarrow \text{SL}_2(\mathbb{C})$. A closed curve K in $S \times [0, 1]$ determines a conjugacy class in $\pi_1(S)$, so that the element $r(K) \in \text{SL}_2(\mathbb{C})$ is well-defined up to conjugation. In particular, the trace $\text{Tr } r(K) \in \mathbb{C}$ is uniquely determined.

The corresponding trace map $K \mapsto \text{Tr } r(K)$ depends only on the class of r in the *character variety*

$$\mathcal{R}_{\text{SL}_2(\mathbb{C})} = \{\text{group homomorphisms } r: \pi_1(S) \rightarrow \text{SL}_2(\mathbb{C})\} // \text{SL}_2(\mathbb{C})$$

where $\text{SL}_2(\mathbb{C})$ acts on such group homomorphisms by conjugation. Here the double bar indicates that the quotient has to be taken in the sense of geometric invariant theory [35]. In practice, because the trace function Tr generates all conjugation invariant functions on $\text{SL}_2(\mathbb{C})$, two group homomorphisms $r, r': \pi_1(S) \rightarrow \text{SL}_2(\mathbb{C})$ represent the same element of the character variety $\mathcal{R}_{\text{SL}_2(\mathbb{C})}$ if and only if $\text{Tr } r'(K) = \text{Tr } r(K)$ for every closed curve $K \subset S \times [0, 1]$. In the generic case where r is irreducible, namely where its image in $\text{SL}_2(\mathbb{C})$ leaves no line in $\mathbb{C}^2$ invariant, this is equivalent to the property that r and r' are conjugate by an element of $\text{SL}_2(\mathbb{C})$. See for instance [14].

The trace function provides a one-to-one correspondence between Kauffman brackets $\mathcal{S}^{-1}(S) \rightarrow \mathbb{C}$ and elements of the character variety $\mathcal{R}_{\text{SL}_2(\mathbb{C})}$.

THEOREM 5 ([7, 38, 39, 19, 20, 27, 13, 24, 34]). *A group homomorphism $r: \pi_1(S) \rightarrow \text{SL}_2(\mathbb{C})$ defines a non-trivial homomorphism*

$$\rho_r: \mathcal{S}^{-1}(S) \rightarrow \mathbb{C}$$

by the property that

$$\rho_r([K]) = -\text{Tr } r(K)$$

for every framed knot $K \subset S \times [0, 1]$.

Conversely, every non-trivial homomorphism $\mathcal{S}^{-1}(S) \rightarrow \mathbb{C}$ is associated in this way to a homomorphism $r: \pi_1(S) \rightarrow \mathrm{SL}_2(\mathbb{C})$, and the class of r in $\mathcal{R}_{\mathrm{SL}_2(\mathbb{C})}$ is unique. $\square$

The first half of the statement is a crucial observation of D. Bullock [7] that, if we associate to each link $K \subset S \times [0, 1]$ with components $K_1, \dots, K_n$ the number

$$\mathcal{K}_r(K) = (-1)^n \prod_{i=1}^n \mathrm{Tr} r(K_i) \in \mathbb{C},$$

then $\mathcal{K}_r$ satisfies the skein relation with $A = -1$. This property is an easy consequence of the classical trace formula in SL_2 , which states that

$$\mathrm{Tr}(MN) + \mathrm{Tr}(MN^{-1}) = \mathrm{Tr}(M)\mathrm{Tr}(N)$$

for every $M, N \in \mathrm{SL}_2(R)$ (for an arbitrary commutative unit ring R).

The converse statement, which can be traced back to H. Vogt and R. Fricke [39, 19, 20] and was apparently first proved by H. Helling [27], amounts to characterizing which functions on a group G can be realized as the trace function of a homomorphism $G \rightarrow \mathrm{SL}_2(\mathbb{C})$. See also [13, 38, 34, 24].

2.2. The case where $A = +1$. One can go from $A = -1$ to $A = +1$ by a construction of J. Barrett [1], which associates an isomorphism $\mathcal{S}^A(S) \rightarrow \mathcal{S}^{-A}(S)$ to each spin structure on S . See [38, §2] for a proof that Barrett's linear isomorphism is in fact an algebra isomorphism.

Let $\mathrm{Spin}(S)$ denote the space of isotopy classes of spin structures on S . Recall that the difference between two elements of $\mathrm{Spin}(S)$ is measured by an obstruction in $H^1(S; \mathbb{Z}_2)$. As a consequence, $\mathrm{Spin}(S) \cong H^1(S; \mathbb{Z}_2)$ once we have chosen a base spin structure, and $H^1(S; \mathbb{Z}_2)$ acts freely and transitively on $\mathrm{Spin}(S)$.

The cohomology group $H^1(S; \mathbb{Z}_2)$ also acts on $\mathcal{R}_{\mathrm{SL}_2(\mathbb{C})}(S)$ by the property that, if $\alpha \in H^1(S; \mathbb{Z}_2)$ and $r \in \mathcal{R}_{\mathrm{SL}_2(\mathbb{C})}(S)$,

$$\alpha r(\gamma) = (-1)^{\alpha(\gamma)} r(\gamma) \in \mathrm{SL}_2(\mathbb{C})$$

for every $\gamma \in \pi_1(S)$. The quotient space $\mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}/H^1(S; \mathbb{Z}_2)$ is the subset $\mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^0$ of

$$\mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})} = \{\text{group homomorphisms } r: \pi_1(S) \rightarrow \mathrm{PSL}_2(\mathbb{C})\} // \mathrm{PSL}_2(\mathbb{C})$$

consisting of those homomorphisms $r: \pi_1(S) \rightarrow \mathrm{PSL}_2(\mathbb{C})$ that admit a lift $\pi_1(S) \rightarrow \mathrm{SL}_2(\mathbb{C})$. Note that $\mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^0$ is equal to $\mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}$ when $\partial S \neq \emptyset$, and is one of the two components of $\mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}$ when S is closed [23].

We now consider the twisted product

$$\begin{aligned} \mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^{\mathrm{Spin}}(S) &= \mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^0(S) \tilde{\times} \mathrm{Spin}(S) \\ &= (\mathcal{R}_{\mathrm{SL}_2(\mathbb{C})}(S) \times \mathrm{Spin}(S)) / H^1(S; \mathbb{Z}_2). \end{aligned}$$

An element of $r \in \mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^{\mathrm{Spin}}(S)$ associates a well-defined trace to each framed knot $K \subset S \times [0, 1]$. Indeed, if we represent r by a pair $(\hat{r}, \sigma)$ consisting of a group homomorphism $\hat{r}: \pi_1(S) \rightarrow \mathrm{SL}_2(\mathbb{C})$ and of a spin structure $\sigma \in \mathrm{Spin}(S)$, we can consider the trace

$$\rho_r(K) = (-1)^{\sigma(K)} \mathrm{Tr} \hat{r}(K),$$

where $\sigma(K) \in \mathbb{Z}_2$ is the monodromy of the framing of K with respect to the spin structure on $S \times [0, 1]$ defined by σ . This clearly depends only on the class r of $(\hat{r}, \sigma)$ in $\mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^{\mathrm{Spin}}(S)$.

The isomorphism $\mathcal{S}^{-1}(S) \rightarrow \mathcal{S}^1(S)$ associated to σ by Barrett [1] similarly involves a factor $(-1)^{\sigma(K)}$. Combining [1] with Theorem 5 immediately gives:

THEOREM 6. *An element $r \in \mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^{\mathrm{Spin}}(S)$ determines a non-trivial algebra homomorphism*

$$\rho_r: \mathcal{S}^1(S) \rightarrow \mathbb{C}$$

associating to a connected skein $[K] \in \mathcal{S}^1(S)$ the trace $\rho_r(K)$ defined above.

Conversely, every non-trivial homomorphism $\mathcal{S}^1(S) \rightarrow \mathbb{C}$ is associated in this way to a unique $r \in \mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^{\mathrm{Spin}}(S)$. $\square$

Note that the spaces $\mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^{\mathrm{Spin}}(S)$ and $\mathcal{R}_{\mathrm{SL}_2(\mathbb{C})}(S)$ are very similar, since they both are coverings of the component $\mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^0(S)$ of the character variety $\mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}(S)$, with fiber $\mathrm{Spin}(S) \cong H^1(S; \mathbb{Z}_2)$.

We will refer to the elements of $\mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^{\mathrm{Spin}}(S)$ as *spinned homomorphisms* from $\pi_1(S)$ to $\mathrm{PSL}_2(\mathbb{C})$, as they consist of a group homomorphism $r: \pi_1(S) \rightarrow \mathrm{PSL}_2(\mathbb{C})$ together with some additional spin information specifying how to lift r to $\mathrm{SL}_2(\mathbb{C})$.

Hyperbolic geometries are actually quite familiar with spinned homomorphisms. Indeed, suppose that we are given a hyperbolic metric on the thickened surface $S \times (0, 1)$, not necessarily complete. The holonomy of this metric gives a homomorphism $r: \pi_1(S) \rightarrow \mathrm{PSL}_2(\mathbb{C})$. What is less well-known is that this r comes with additional spin information. More precisely, the choice of a spin structure $\sigma \in \mathrm{Spin}(S)$ specifies a preferred lift of r to $\hat{r}_\sigma: \pi_1(S) \rightarrow \mathrm{SL}_2(\mathbb{C})$, and a different choice for the spin structure σ leaves the class of $(\hat{r}_\sigma, \sigma)$ in $\mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^{\mathrm{Spin}}(S)$ unchanged; see for instance [4, Proposition 10]. This proves:

PROPOSITION 7. *A hyperbolic metric on $S \times (0, 1)$ uniquely determines a spinned homomorphism $r \in \mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^{\mathrm{Spin}}(S)$.* $\square$

Theorems 5 and 6 enable one to consider the skein algebra $\mathcal{S}^A(S)$ as a quantization of the character varieties $\mathcal{R}_{\mathrm{SL}_2(\mathbb{C})}(S)$ and $\mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^{\mathrm{Spin}}(S)$. See [40, 9, 11].

3. Small surfaces

We now return to our Revised Goal for arbitrary values of A . The irreducible representations of the skein algebra $\mathcal{S}^A(S)$ have been completely classified for a few surfaces with relatively simple topological type.

3.1. The sphere with ≤ 3 punctures. When S is a sphere with at most three punctures, the skein algebra $\mathcal{S}^A(S)$ is commutative for all values of A . This is because any two simple closed curves on S are disjoint after isotopy. By commutativity, every irreducible representation of $\mathcal{S}^A(S) \rightarrow \mathrm{End}(V)$ then has dimension 1, so that canonically $\mathrm{End}(V) \cong \mathbb{C}$.

An immediate extension of Kauffman's uniqueness result for the Kauffman bracket (Theorem 1) then gives the following result. See for instance [32].

THEOREM 8. *Let S be a sphere with at most three punctures, the skein algebra $\mathcal{S}^A(S)$ is commutative and:*

- (1) If S is the sphere or the disk, then $\mathcal{S}^A(\mathbb{C}) \cong \mathbb{C}$; as a consequence, it admits a unique irreducible representation.
- (2) If S is the annulus, then $\mathcal{S}^A(S) \cong \mathbb{C}[X]$ where X is represented by a simple closed curve going around the annulus, and two irreducible representations $\mathcal{S}^A(S) \rightarrow \text{End}(V) \cong \mathbb{C}$ are isomorphic if and only if they assign the same number $x \in \mathbb{C}$ to the generator X .
- (3) If S is the three-puncture sphere, then $\mathcal{S}^A(S) \cong \mathbb{C}[X, Y, Z]$ where the generators X, Y, Z are represented by simple closed curves parallel to the boundary components; as a consequence, two irreducible representations $\mathcal{S}^A(S) \rightarrow \text{End}(V) \cong \mathbb{C}$ are isomorphic if and only if they assign the same numbers $x, y, z \in \mathbb{C}$ to the generators X, Y, Z . $\square$

3.2. The torus with ≤ 1 puncture. The algebraic structure of the skein algebra $\mathcal{S}^A(S)$ is much more interesting when S is the torus with 0 or 1 puncture, although still manageable.

THEOREM 9 ([12]).

- (1) When S is the one-puncture torus, the skein algebra $\mathcal{S}^A(S)$ admits a presentation with three generators X_1, X_2, X_3 and the three relations

$$AX_iX_{i+1} - A^{-1}X_{i+1}X_i = (A^2 - A^{-2})X_{i+2}$$

for $i = 1, 2, 3$.

- (2) When S is the torus (with no puncture), the skein algebra is isomorphic to the quotient of the above algebra by the central element

$$A^2X_1^2 + A^{-2}X_2^2 + A^2X_3^2 - AX_1X_2X_3 - 2A^2 - 2A^{-2}. \quad \square$$

Bullock and Przytycki [12] give similar presentations for the skein algebras of the 4-puncture sphere and the 2-puncture torus.

It turns out that the skein algebra of the one-puncture torus, as described by Theorem 9(1), is isomorphic to a certain quantum deformation $U'_q(\mathfrak{so}_3)$ of $\mathfrak{so}_3$ introduced in [22]. In particular, its representation theory has been completely analyzed in [25, 26]. The irreducible representations of $\mathcal{S}^A(S)$ in this case fall into two general categories, each subdivided into a few subcases. The first category arises for all values of A , whereas the second category is restricted to the case where A is a root of unity. The representation theory for the unpunctured torus easily follows from the one-puncture case by Theorem 9(2).

In the rest of the article, we will consider the case of a general surface S , when A is a root of unity. The representations of the skein algebra that we will encounter are very similar to the cyclic representations of $U'_q(\mathfrak{so}_3)$ that occur in [26].

4. Invariants of invariants

Recall that our (optimistic) goal is to classify all irreducible representations of the skein algebra $\mathcal{S}^A(S)$. Usually, the way one classifies a mathematical object is by extracting invariants, and by showing that these determine the object up to isomorphism. In our case, we need invariants of representations of the skein algebra, namely invariants of Kauffman brackets, namely invariants of link invariants!

We will need A to be a root of unity. Consequently, we assume that A is a primitive N -root of unity, namely that $A^N = 1$ and that N is minimum for this property. In addition, we require that N is odd.

4.1. Chebyshev polynomials. Recall that the (normalized) n -th Chebyshev polynomial (of the first type) is the polynomial $T_n(x)$ defined by the property that

$$\mathrm{Tr} M^n = T_n(\mathrm{Tr} M) \text{ for every } M \in \mathrm{SL}_2(\mathbb{C}).$$

In particular, $T_0(x) = 2$, $T_1(x) = x$, $T_2(x) = x^2 - 2$, $T_3(x) = x^3 - 3x$, and $T_n(x)$ is more generally defined by the recurrence relation

$$T_{n+1}(x) = xT_n(x) - T_{n-1}(x).$$

LEMMA 10. *Suppose that A is a primitive N -root of unity with N odd. Let K be a framed knot in $S \times [0, 1]$ that is represented by a simple closed curve in S , with vertical framing. Then, the evaluation $T_N([K]) \in \mathcal{S}^A(S)$ of the N -th Chebyshev polynomial is central in the skein algebra $\mathcal{S}^A(S)$.*

SKETCH OF PROOF. One easily reduces the problem to the case of the one-puncture torus. The result then follows from the combination of [12] and of the construction of central elements of $U'_q(\mathfrak{so}_3)$ in [26]. The more geometrically inclined reader may find the brute force computation of [26] (where the connection with Chebyshev polynomials is not explicit, but was later observed by V. Fock) somewhat frustrating, and may prefer the more geometric argument provided by the Product-to-Sum Formula of [21, Theorem 4.1]. See also [5] for a different proof. $\square$

4.2. The classical shadow. Suppose that we are given an irreducible representation $\rho: \mathcal{S}^A(S) \rightarrow \mathrm{End}(V)$. If $K \subset S \times [0, 1]$ is represented by a simple closed curve in S , with vertical framing, Lemma 10 asserts that $T_N([K])$ is central in $\mathcal{S}^A(S)$. As a consequence, Schur's lemma implies that the image of $T_N([K])$ under the irreducible representation ρ is a multiple of the identity, and there exists a number $\kappa_\rho(K) \in \mathbb{C}$ such that

$$\rho(T_N([K])) = \kappa_\rho(K) \mathrm{Id}_V.$$

LEMMA 11. *The above map $K \mapsto \kappa(K)$ induces a non-trivial algebra homomorphism*

$$\kappa_\rho: \mathcal{S}^{+1}(S) \rightarrow \mathbb{C}.$$

SKETCH OF PROOF. This essentially is equivalent to the property that κ_ρ satisfies the skein relation for $A = +1$. Again, one first reduces the problem to the case of the one-puncture torus (but this here requires using a formula such as [32, Lemma 14.1] and the hypothesis that N is odd), and then applies an analog of the Product-to Sum Formula of [21, Theorem 4.1]. See [5] for details. $\square$

Theorem 6 associates to the homomorphism κ_ρ of Lemma 11 an element $r_\rho \in \mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^{\mathrm{Spin}}(S)$. By definition, this spinned homomorphism $r_\rho: \pi_1(S) \rightarrow \mathrm{PSL}_2(\mathbb{C})$ is the *classical shadow* of the representation

$$\rho: \mathcal{S}^A(S) \rightarrow \mathrm{End}(V).$$

4.3. Puncture invariants. When S has non-empty boundary, the skein algebra $\mathcal{S}^A(S)$ contains central elements that are more obvious than those defined using Chebyshev polynomials. Indeed, if P_i is a curve parallel to the i -th boundary component of S and is endowed with the vertical framing, the skein $[P_i] \in \mathcal{S}^A(S)$ is clearly central in $\mathcal{S}^A(S)$.

As before, an irreducible representation $\rho: \mathcal{S}^A(S) \rightarrow \text{End}(V)$ provides a number $p_i \in \mathbb{C}$ such that

$$\rho([P_i]) = p_i \text{Id}_V.$$

This number $p_i \in \mathbb{C}$ is the i -th *puncture invariant* of the irreducible representation ρ .

There clearly should be a relationship between these puncture invariants and the classical shadow r_ρ , involving the Chebyshev polynomial T_N . Indeed:

LEMMA 12. *If $r_\rho \in \mathcal{R}_{\text{PSL}_2(\mathbb{C})}^{\text{Spin}}(S)$ is the classical shadow of the irreducible representation $\rho: \mathcal{S}^A(S) \rightarrow \text{End}(V)$,*

$$T_N(p_i) = \kappa_\rho([P_i]) = \text{Tr } r_\rho(P_i)$$

where $\text{Tr } r_\rho(P_i) = (-1)^{\sigma(P_i)} \text{Tr } \widehat{r}_\rho(P_i)$ is defined as in §2.2. □

Namely, the puncture invariant p_i belongs to the finite set $T_N^{-1}(\text{Tr } r_\rho(P_i))$, and is therefore determined by the classical shadow up to finitely many choices.

5. The conjectural classification of irreducible representations

Remember that we are restricting attention to the case where the number A intervening in the definition of the skein algebra $\mathcal{S}^A(S)$ is a primitive N -root of unity, with N odd.

We then associated to each irreducible representation $\rho: \mathcal{S}^A(S) \rightarrow \text{End}(V)$ a spinned homomorphism $r_\rho \in \mathcal{R}_{\text{PSL}_2(\mathbb{C})}^{\text{Spin}}(S)$ and puncture invariants $p_i \in T_N^{-1}(-\text{Tr } r_\rho(P_i))$.

CONJECTURE 13. *Suppose that we are given:*

- (1) *a spinned homomorphism $r: \pi_1(S) \rightarrow \text{PSL}_2(\mathbb{C})$;*
- (2) *for each boundary component of S , a number $p_i \in T_N^{-1}(\text{Tr } r(P_i))$*

Then, up to isomorphism, there is a unique irreducible representation $\rho: \mathcal{S}^A(S) \rightarrow \text{End}(V)$ whose invariants are r and the p_i .

This conjecture is probably not quite true as stated, but should be “essentially correct”. It should also fit in a larger picture, valid for all values of A .

More precisely, recall that the (finite-dimensional) irreducible representations of the quantum group $U_q(\mathfrak{sl}_2)$ fall into two wide categories: the *generic representations*, which occur for all values of q , and the *cyclic representations* restricted to the case where q is a root of unity. The generic representations are essentially rigid, depending only on the dimension and on a finite amount of additional information, but there are whole moduli spaces of cyclic representations, which depend on finitely many continuous choices of parameters. See for instance [30, §VI.5].

Each of these (types of) representations of $U_q(\mathfrak{sl}_2)$ should induce a representation of the skein algebra $\mathcal{S}^A(S)$ when $A = q^{\frac{1}{2}}$, by an extension of the framework of [4] discussed below (see also [10] for more ideas), and we conjecture that all irreducible representations of $\mathcal{S}^A(S)$ should be obtained in this way.

In this framework, the generic representations of $U_q(\mathfrak{sl}_2)$ should essentially give unique representations of $\mathcal{S}^A(S)$.

When A is a root of unity, the cyclic representations of $U_q(\mathfrak{sl}_2)$ should give the representations of $\mathcal{S}^A(S)$ discussed in Conjecture 13 (with minor variations when A is a primitive N -root of unity with N even). The classical shadow is here a

way to keep track of the moduli parameters describing the cyclic representations appearing in the construction.

We conjecture that, conversely, every irreducible representation of the skein algebra is obtained in this way, namely is associated by such a construction to irreducible representations of the quantum group $U_q(\mathfrak{sl}_2)$.

This conjectural picture will clearly need to be adjusted as our understanding progresses. At this point, however, we are able to prove the existence part of Conjecture 13.

THEOREM 14. *Suppose that A is a primitive N -root of unity with N odd, and that we are given:*

- (1) *a spinned homomorphism $r: \pi_1(S) \rightarrow \mathrm{PSL}_2(\mathbb{C})$;*
- (2) *for each boundary component of S , a number $p_i \in T_N^{-1}(\mathrm{Tr} r(P_i))$*

Then there exists an irreducible representation $\rho: \mathcal{S}^A(S) \rightarrow \mathrm{End}(V)$ whose classical shadow is $r \in \mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^{\mathrm{Spin}}(S)$ and whose puncture invariants are equal to the p_i .

In addition, our construction is natural with respect to the action of the mapping class group of S on all the objects considered.

The rest of this article is devoted to a discussion of the proof of Theorem 14. Details will appear elsewhere [5, 6].

6. Triangulations

For simplicity, we will focus attention on the case where the surface is closed, namely without punctures. This also turns out to be the harder case.

We follow a strategy proposed by other authors [36, Chap. 3], and drill many holes in S . More precisely, let S_0 be a surface obtained by removing finitely many points $v_1, v_2, \dots, v_p$ from S .

Choose an ideal triangulation T for S_0 , namely a triangulation of the closed surface S whose vertices are exactly the punctures $v_1, v_2, \dots, v_p$. We require that the end points of each edge are distinct, which is easily achieved by removing additional punctures if necessary.

We now associate an algebraic object to this triangulation. Its representation theory is relatively simple, and will help us understand that of the skein algebra.

6.1. The Chekhov-Fock algebra of a train track. In each triangle of the triangulation T , join the midpoints of the edges by three arcs with disjoint interiors, meeting orthogonally the edges as in Figure 4.

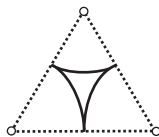


FIGURE 4. A triangle and its train track

These arcs fit together to form a train track τ on the surface S_0 , as illustrated in Figure 5.

To keep in line with the themes of the conference, this train track τ is traditionally used to specify the 1-dimensional submanifolds of S_0 that are normal with

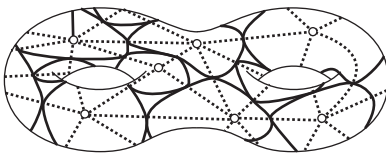


FIGURE 5. A triangulation and its associated train track

respect to the triangulation. More precisely, these normal 1-submanifolds are in one-to-one correspondence with systems of non-negative integer edge weights for τ satisfying the *Switch Condition* that, at each switch, the weights of the two edges incoming on one side of the switch add up to the sum of the weights of the two edges outgoing on the other side.

Let $\mathcal{W}(\tau; \mathbb{Z})$ be the set of integer weight systems for τ that satisfy the Switch Condition. Note that we are not any more restricting these weights to be non-negative. The set $\mathcal{W}(\tau; \mathbb{Z})$ is an abelian group under addition, and is easily computed to be isomorphic to $\mathbb{Z}^{6g+3p-6}$.

The space $\mathcal{W}(\tau; \mathbb{Z})$ is equipped with a very natural antisymmetric bilinear form, the *Thurston intersection form*

$$\omega: \mathcal{W}(\tau; \mathbb{Z}) \times \mathcal{W}(\tau; \mathbb{Z}) \rightarrow \mathbb{Z}.$$

The Thurston form is essentially an algebraic intersection number. To define ω , perturb τ to a train track τ' that is transverse to τ . Each intersection point x of τ with τ' comes with a sign: indeed, any choice of local orientation of τ near x specifies a local orientation of τ' , if the train tracks are sufficiently C^1 -close; reversing the local orientation of τ reverses the local orientation of τ' , and consequently does not change the algebraic sign of the intersection of τ with τ' at x . Then, given $\alpha, \beta \in \mathcal{W}(\tau; \mathbb{Z})$, the number $\omega(\alpha, \beta)$ counts these intersection signs when the edges of τ are equipped with the multiplicity α and the edges of τ' with the multiplicity β .

We now define an algebra $\mathcal{T}_\tau^A$ as follows. As a vector space, $\mathcal{T}_\tau^A$ is freely generated by the elements of $\mathcal{W}(\tau; \mathbb{Z})$; namely, $\mathcal{T}_\tau^A$ consists of all formal linear combinations $\sum_{i=1}^j a_i \alpha_i$ where $a_i \in \mathbb{C}$ and $\alpha_i \in \mathcal{W}(\tau; \mathbb{Z})$. If we defined a multiplication on $\mathcal{T}_\tau^A$ by the group law of $\mathcal{W}(\tau; \mathbb{Z})$, namely by the property that $\alpha \cdot \beta = (\alpha + \beta) \in \mathcal{W}(\tau; \mathbb{Z})$ for $\alpha, \beta \in \mathcal{W}(\tau; \mathbb{Z})$, we would obtain the group algebra $\mathbb{C}[\mathcal{W}(\tau; \mathbb{Z})]$. Instead we twist this group law by the constant A and by the Thurston form ω . More precisely, we define the multiplication of two elements $\alpha, \beta \in \mathcal{W}(\tau; \mathbb{Z})$ as

$$\alpha \cdot \beta = A^{\frac{1}{2}\omega(\alpha, \beta)}(\alpha + \beta)$$

and linearly extend this multiplication to $\mathcal{T}_\tau^A$.

This algebra $\mathcal{T}_\tau^A$ is the *Chekhov-Fock algebra* of the train track τ . The reader will recognize here one of the many avatars of the *quantum Teichmüller space* [18, 15, 29, 3, 33]. If we restrict attention to those edge weight systems $\alpha \in \mathcal{W}(\tau; \mathbb{Z})$ where, at each switch, the weights of the two edges incoming on one side have the same parity, the corresponding subalgebra of $\mathcal{W}(\tau; \mathbb{Z})$ is exactly the Chekhov-Fock algebra of the ideal triangulation T , as defined in [3] and for $q = A^{-2}$. Therefore, $\mathcal{T}_\tau^A$ is obtained from the Chekhov-Fock algebra of T by augmenting it with certain square roots of generators. The Chekhov-Fock algebra $\mathcal{T}_\tau^A$ appears in a different presentation in [4, §2.3], where it is called $\mathcal{Z}_T^{A^{-\frac{1}{2}}}$.

6.2. From the skein algebra to the quantum Teichmüller space. The algebraic structure of the Chekhov-Fock algebra $\mathcal{T}_\tau^A$ is remarkably simple. A key step in our analysis connects it to the skein algebra $\mathcal{S}^A(S_0)$.

THEOREM 15 ([4]). *There exists a natural embedding $\mathcal{S}^A(S_0) \rightarrow \mathcal{T}_\tau^A$.* $\square$

When $A = 1$, the homomorphism $\mathcal{S}^1(S) \rightarrow \mathcal{T}_\tau^1$ of Theorem 15 has a geometric interpretation. Using the technique of pleated surfaces, a system of complex weights associated to the edges of T determines a homomorphism $r: \pi_1(S_0) \rightarrow \mathrm{PSL}_2(\mathbb{C})$. Choosing suitably compatible square roots for these complex numbers even determines a spinned homomorphism $r \in \mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^{\mathrm{Spin}}(S_0)$. The homomorphism $\mathcal{S}^1(S) \rightarrow \mathcal{T}_\tau^1$ then associates to a framed link $K \subset S \times [0, 1]$ the Laurent polynomial in these square roots that gives $\mathrm{Tr} r(K)$ for the associated $r \in \mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^{\mathrm{Spin}}(S)$. See [4, §1]. The general homomorphism $\mathcal{S}^A(S_0) \rightarrow \mathcal{T}_\tau^A$ can be seen as a non-commutative deformation of this formula. The challenge is to make it natural, namely well-behaved with respect to changing the ideal triangulation T of S_0 (and therefore with respect to the action of the mapping class group of S_0).

Theorem 15 was conjectured in [16]. See also [17, 28] for partial results in this direction.

The Chekhov-Fock algebra $\mathcal{T}_\tau^A$ is an example of a quantum torus and, as such, its irreducible representations are easy to classify. In fact, when A is a primitive N -root of unity, they are classified by elements of $\mathcal{W}(\tau; \mathbb{C})$, namely complex edge weight systems, together with choices of N -roots for certain complex numbers associated to the boundary components of S_0 by the complex edge weights.

Interpreting these complex edge weights as shear-bend parameters for pleated surfaces provides Proposition 16 below. We need to introduce some definitions.

A *peripheral subgroup* of $\pi_1(S_0)$ is one associated to a puncture, namely generated by a loop P_i going around a puncture of S_0 . Note that the same puncture defines many peripheral subgroups, each associated to a path γ_i joining P_i to the base point used in the definition of the fundamental group $\pi_1(S)$. In particular, $\pi_1(S)$ acts on its peripheral subgroups by conjugation.

Two peripheral subgroups are *connected by the edge e* of the triangulation S if they are respectively defined by loops P_i, P_j and paths γ_i, γ_j such that the path $\gamma_i \gamma_j^{-1}$ is homotopic to a path in this edge e by a homotopy keeping path endpoints in $P_i \cup P_j$.

PROPOSITION 16 ([3]). *There is a one-to-one correspondence between isomorphism classes of irreducible representations $\rho: \mathcal{T}_\tau^A \rightarrow \mathrm{End}(V)$ and sets of data consisting of:*

- (1) *a spinned homomorphism $r_0: \pi_1(S_0) \rightarrow \mathrm{PSL}_2(\mathbb{C})$, namely an element $r_0 \in \mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^{\mathrm{Spin}}(S_0)$;*
- (2) *for each peripheral subgroup $\pi \subset \pi_1(S)$, a line $\xi_\pi \subset \mathbb{C}^2$ such that*
 - (a) *ξ_π is respected by the subgroup $r_0(\pi) \subset \mathrm{SL}_2(\mathbb{C})$;*
 - (b) *$\xi_{\gamma\pi\gamma^{-1}} = r_0(\gamma)(\xi_\pi)$ for every $\gamma \in \pi_1(S_0)$;*
 - (c) *$\xi_\pi \neq \xi_{\pi'}$ when the parabolic subgroup π and π' are connected by an edge of the triangulation;*
- (3) *for each puncture, a number $h_i \in \mathbb{C}$ whose power h_i^N is determined by the following property: if $r_0 \in \mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^{\mathrm{Spin}}(S)$ is represented by $\hat{r}_0: \pi_1(S) \rightarrow \mathrm{SL}_2(\mathbb{C})$ and by a spin structure $\sigma \in \mathrm{Spin}(S)$, if $P_i \in \pi_1(S)$ is represented*

by a small loop going counterclockwise around the i -th puncture, and if π_i is the peripheral subgroup generated by P_i , then $\hat{r}_0$ acts by multiplication by $(-1)^{\sigma(P_i)} h_i^N$ on the line $\xi_{\pi_i} \subset \mathbb{C}^2$.

Proposition 16 is an easy extension of some arguments of [3] to the square root context considered here. The harder part of [3] is to prove that Proposition 16 is well-behaved under change of the ideal triangulation T , but we will not need this property here.

6.3. Constructing representations of the skein algebra. Let us repeat the statement we intend to prove.

We are restricting attention to the case where the surface is closed, so that we do not have to worry about puncture invariants; this is only for the sake of exposition, and the general case is very similar. We also assume that A is a primitive N -root of unity with N odd, a critical hypothesis at this point. We are given a spinned homomorphism $r \in \mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^{\mathrm{Spin}}(S)$, and we want to construct an irreducible representation $\rho: \pi_1(S) \rightarrow \mathrm{End}(V)$ with classical shadow r .

Choose a triangulation T for S . We require that the end points of each edge are distinct. Let $v_1, v_2, \dots, v_p$ be the vertices of this triangulation, and consider the punctured surface $S_0 = S - \{v_1, v_2, \dots, v_p\}$. The spinned homomorphism $r \in \mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^{\mathrm{Spin}}(S)$ gives by restriction a spinned homomorphism $r_0 \in \mathcal{R}_{\mathrm{PSL}_2(\mathbb{C})}^{\mathrm{Spin}}(S_0)$.

To apply Proposition 16 to construct a representation $\mathcal{T}_\tau^A \rightarrow \mathrm{End}(V)$, we need to choose suitable invariant lines $\xi_\pi \subset \mathbb{C}^2$ and numbers $h_i \in \mathbb{C}$. In our case, the restriction of r_0 to each peripheral subgroup is trivial. This makes the choice of the lines ξ_π particularly easy, since Condition (2a) of Proposition 16 is automatically satisfied. In fact, there is a whole $2p$ -dimensional family of possible choices for the lines ξ_π . Similarly, the data (3) is simplified by this property of r_0 , and just means that h_i is an N -root of unity.

Having chosen lines ξ_π and weights h_i as above, Proposition 16 provides a representation $\mathcal{T}_\tau^A \rightarrow \mathrm{End}(V)$. Combining it with the homomorphism $\mathcal{S}^A(S_0) \rightarrow \mathcal{T}_\tau^A$ of Theorem 15, we now have a representation $\rho_0: \mathcal{S}^A(S_0) \rightarrow \mathrm{End}(V)$. This is beginning to look like the statement we are trying to prove for Theorem 15, but many things need to be checked:

- (1) Is the classical shadow of ρ_0 equal to r_0 ?
- (2) Is ρ_0 irreducible?
- (3) Does $\rho_0: \mathcal{S}^A(S_0) \rightarrow \mathrm{End}(V)$ induce a representation $\rho: \mathcal{S}^A(S) \rightarrow \mathrm{End}(V)$ of the skein algebra of S ? Namely is it true that $\rho_0([K]) = \rho_0([K'])$ when the framed links $K, K' \subset S_0 \times [0, 1]$ are isotopic in $S \times [0, 1]$, by an isotopy that is allowed to cross the punctures?
- (4) Is ρ independent of the choices that we have made, and in particular of the lines $\xi_\pi \subset \mathbb{C}^2$ and the triangulation of S ?

These issues are then dealt with in the following way.

- (1) The answer to (1) is yes, by the occurrence of many “miraculous cancellations” in the computation of Chebyshev polynomials $T_N(\rho_0([K])) \in \mathrm{End}(V)$.
- (2) The answer to (2) is a definite no. However, we can bypass this difficulty by restricting ρ_0 to an irreducible component.

- (3) After this irreducible reduction, the answer to (3) then remarkably becomes yes, provided that we chose all N -roots of unity h_i to be equal to $-A^{-2}$, and that we restricted ρ_0 to one very specific irreducible component.
- (4) It then turns out that the answer to (4) is also positive.

Proving these statements completes the proof of Theorem 14. Details, which are occasionally very elaborate, will appear elsewhere [5, 6].

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Problems at the Jacofest

J. Hyam Rubinstein

Dedicated to Bus Jaco for his 70th birthday.

Problem 1 Given conformal structures on surfaces $\Sigma, -\Sigma$ there is an essentially unique hyperbolic I -bundle on $\Sigma \times [0, 1]$. Assume that $\partial M \neq \emptyset$ and there is an ∞ volume hyperbolic structure on $\text{int}M$. Is there a unique hyperbolic structure on $M \times [0, 1]$ for a given conformal anti-homeomorphism from $M \rightarrow -M$? What about $\Sigma \times I \times I$ as a special case to study first? The idea is to find an analogue of the deformation theory of Kleinian groups in dimension 4.

Problem 2 (G. Mess) Can a closed surface bundle over a closed surface have;

- a faithful representation of the fundamental group of the base surface into the mapping class group of the fiber surface so that the image is entirely pseudo Anosov elements
- a word hyperbolic fundamental group
- a Riemannian metric with strictly negative sectional curvature
- a hyperbolic metric.

Problem 3 Is there an integer homology n -sphere for $n > 5$ which is aspherical?

Problem 4 Is there an efficient algorithm to compute efficient position for curves relative to train tracks?

Problem 5 Consider all triangulations of 3-manifolds with n tetrahedra and let $\mathcal{P}$ be the projective solution space of normal surfaces and V be the number of embedded vertex normal surfaces in $\mathcal{P}$. Then experimentally, V does not seem to grow with n . Find an explanation of this? Similarly what can be said about the growth of the number of topological types of such vertex normal surfaces?

Problem 6 Find a way of ‘navigating’ in the space of triangulations for a fixed 3-manifold, where two triangulations are connected by an edge if they are related by a Pachner move. (Recall that Matveev has shown that if two triangulations have the same number of vertices then only $2/3$ and $3/2$ moves are required.)

2000 *Mathematics Subject Classification*. Primary 57M50.

Problem 7 Are random 3-manifolds Haken? This depends on the way 3-manifolds are constructed and the measure used. Generalize Dunfield-Thurston, by gluing a handlebody to a compression body so that the Alexander polynomial is monic and find the measure of the set of fibered examples.

Problem 8 Same as Problem 7 but for random 3-manifolds with a lower bound on the first eigenvalue of Δ , i.e Cheeger constant. (cf. Cannon-Floyd).

Problem 9 Compare measures given by different random models of 3-manifolds -e.g triangulations or Heegaard splittings.

Problem 10 Are geometric triangulations minimal, i.e can one find geometric triangulations amongst minimal ones? Find additional combinatorial properties of minimal triangulations.

Can a 0-efficient triangulation of S^3 have an unknotted edge? What conditions on a triangulation imply that the edges are knotted/unknotted?

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This volume contains the proceedings of a conference held from June 4–6, 2010, at Oklahoma State University, in honor of William (Bus) Jaco's 70th birthday. His contributions to research in low dimensional geometry and topology and to the American mathematical community, especially through his work for the American Mathematical Society, were recognized during the conference.

The focus of the conference was on triangulations and geometric structures for three-dimensional manifolds. The papers in this volume present significant new results on these topics, as well as in geometric group theory.

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